LM7 PUSH: arithmetic sections and profinite images

Research manuscript with an explicit statement register

AI-produced by GPT-6 Astra (OpenAI, ChatGPT Pro mode)

13 November 2026

Abstract

We compute the intrinsic class-two section obstruction for the Fermat cubic with its nine coordinate cusps removed, including its integral residue complex, path origins, restricted convergence, and lifting fibers. We construct an actual obstruction invisible locally and at every finite coefficient level; refute the two proposed class-two inertia restrictions; and prove the pro-three cyclotomic centralizer criterion. Finite arithmetic field and cocycle constructions give the exact full, metabelian and marked-depth images and all their fibers, with compatibility for arbitrary ramification support. An explicit section at seventeen has the literal local cusp restriction and a global class-three lift, but no class-four lift. We identify the inherited quotient maps and every marked coefficient layer integrally. Over every number field we give the complete arithmetic obstruction test, restriction and descent maps, and the class-two germ classification, including finite algebraic formulas for every point's central coordinates. The three requested full local profinite fixed groups are computed through an explicit valuation and stabilizer comparison. Parts A, B, C and F are complete. The remaining full profinite image questions in Parts D and E are under investigation; the verified degree-five braid refinement retains all twenty-four CM candidates. This working manuscript is **INCOMPLETE**; the task register identifies every unresolved item. We claim no priority.

Contents

1 A. The intrinsic class-two problem **3**
- 1.1 The rank-forty-four residue complex . . . 3
- 1.2 An actual intrinsic obstruction invisible at all finite levels . . . 6
- 1.3 Two ramification assertions disproved . . . 8
- 1.4 The lift torsor and a formula retaining all path origins . . . 9
- 1.5 An exact two-prime decomposition at seven . . . 17
- 1.6 A strict restriction on ramification in the higher quotients . . . 22
- 1.7 Complete global images and fibers for one auxiliary prime . . . 28
- 1.8 Cofinal higher quotients and the first further coefficient . . . 68
- 1.9 Explicit nine-torsion covers at the first auxiliary Frobenius . . . 71

2 B. Geometric classes and centralizers **75**
- 2.1 Geometric classes and the cyclotomic centralizer . . . 75

3 C. Restriction and germs **102**
- 3.1 Restriction, descent, and persistent obstruction germs . . . 102
- 3.2 Algebraic evaluation of the point image . . . 109
- 3.3 Separating the point germs by the nine tangent lines . . . 114
- 3.4 Finite arithmetic matrices over an arbitrary number field . . . 117

4 D. Profinite leading terms **121**
4.1 Sources, comparison maps, and the two remaining image problems 121
4.2 What the linear and literature checks establish 122
4.3 A complete finite pentagon comparison in weight nine 124
4.4 The arithmetic lower bound with its exact coordinate comparison 128
4.5 The full-source symmetry restriction in weight five 130
4.6 What a finite leading congruence detects 132

5 E. The complex-multiplication coset **133**
5.1 The finite CM calculation and its independent verification 133
5.2 The homology refinements and an additional finite quotient 135
5.3 Finite tests from the full GT distribution theorem 137
5.4 The requested S_3-core refinement 138
5.5 A finite exclusion of all twelve recorded extra shadows 139
5.6 The exponent-sixteen fibres and the two-adic lifting argument 147
5.7 A four-strand pentagon constraint on the CM fibres 152
5.8 Arbitrary relative kernels supported away from two and three 157
5.9 A joint Ferrari test retaining cubic powers and fourth-degree commutators . . 159

6 F. Derived limits and applications **165**
6.1 The residue complex and the derived-limit test 165
6.2 Sections over the Laurent-series field 170
6.3 The fixed-lift compactness question in Task F.4 173

A Coverage and remaining questions **193**

Conventions and scope

All cohomology of profinite groups with finite discrete coefficients is continuous. For a free finite-rank $\mathbb{Z}_\ell$-module with continuous action, H^i_{cts} is the cohomology of continuous inhomogeneous cochains. Restriction, inflation, quotient, and comparison maps are identified in each statement. Section sets are taken modulo conjugation by the specified geometric kernel. Arithmetic Frobenius is used in residue complexes. We work with

$$E:\ x^3+y^3+z^3=0, \qquad U=E\setminus D,$$

where D is the set of nine coordinate cusps. Put $K_0=\mathbb{Q}(\zeta)$, $\zeta^2+\zeta+1=0$, and $\lambda=1-\zeta$. Our cusp labels are

	rational cusp	first conjugate	second conjugate
$x=0$	$P=[0:-1:1]$	$A=[0:-\zeta:1]$	$\bar{A}=[0:-\zeta^2:1]$
$y=0$	$Q_c=[-1:0:1]$	$B=[-\zeta:0:1]$	$\bar{B}=[-\zeta^2:0:1]$
$z=0$	$O=[1:-1:0]$	$C=[1:-\zeta:0]$	$\bar{C}=[1:-\zeta^2:0]$.

The geometric group N is the free pro-three group of rank ten of U, based at the unit tangent $s=z/x$ at O. Its abelianization and the proper elliptic quotient are $V=T_3J_{E,D}$ and $T=T_3E$. The cusp submodule is $W=\mathbb{Z}_3(1)[D]/\mathbb{Z}_3(1)\delta$, with $\delta=\sum_{d\in D}d$; its inertia vectors are e_d. Let $I=\ker(N\to T)$ and

$$N^{\mathrm{cc}}=N/[N,I], \qquad \mathcal{K}=\ker(\textstyle\bigwedge^2 V\to\bigwedge^2 T).$$

Thus $\mathcal{K}$ has rank 44 and is the central kernel of $N/\gamma_3 N \to N^{\mathrm{cc}}$. The marked subgroup I_{mark} is the closed normal subgroup generated by inertia at Q_c. A section class over L is the continuous nonabelian H^1 of G_L with the displayed geometric quotient. The full cyclotomic character, with target $\widehat{\mathbb{Z}}^\times$, is used in the centralizer statements; its three-adic component acts on the coefficient lattices.

The inherited group H has underlying set $\mathbb{Z}_3^3$ and law

$$(u, v, w)(u', v', w') = (u + u', v + v', w + w' + \tfrac{9}{2}(uv' - vu')),$$

with arithmetic weights $(\chi_3, \chi_3, \chi_3^2)$. Its base-origin comparison is made explicitly in B.9. The use of the cuspidally central lifting theorem over $\mathbb{Q}$ is Borne–Emsalem–Stix, Theorem 5.10 [BES15], applied to the nine torsion cusps and followed by the displayed pro-three quotient. Its proof uses vanishing of relative Brauer groups in all section neighbourhoods. The point O gives a retraction on the Brauer group of E itself; it does not supply those retractions for every such neighbourhood. Part C consequently retains the additional arithmetic terms over a general number field. For the profinite problems, $\widehat{F}_2$ is the full free profinite group on two peripheral generators; the special outer group and its $\mathfrak{S}_3$ centralizer $\widehat{\mathrm{GT}}_0$ use all finite quotients, with the latter defined by the two hexagons. The subgroup $\widehat{\mathrm{GT}}$ also satisfies the pentagon. Parts D and E specify every comparison to a pro-ℓ, graded, or finite quotient.

The background labels "LM7", "CK", and "F17" refer to the attached previous returns, and are checked only at the steps actually used. The available manuscript source is the supplied transcription of *La Longue Marche*, cote 140-4. The authoritative scans were not attached. Our manuscript citations identify their section, folios, scans, and formulas without asserting a fresh comparison to the scans.

1 A. The intrinsic class-two problem

1.1 The rank-forty-four residue complex

We put $K_0 = \mathbb{Q}(\zeta)$, reserving $\mathcal{K}$ for the rank-forty-four commutator lattice. Set $S = \{3, \infty\}$ and $X = \operatorname{Spec} \mathbb{Z}[1/3]$. Arithmetic Frobenius is used throughout. For each $q \neq 3$ choose a decomposition group and write

$$A_q = \mathcal{K}(-1), \qquad D_q = F_q - 1 \in \mathrm{End}_{\mathbb{Z}_3}(A_q).$$

The subscript records the local Galois action; it does not change the underlying lattice. Define

$$\mathcal{R}(A) = \{(x_q)_{q\neq 3} : x_q \in A_q, \quad \forall n \quad x_q \in 3^n A_q \text{ for almost all } q\}, \qquad D(x)_q = D_q x_q.$$

All quotients by $D\mathcal{R}(A)$ below use the actual image.

Statement A.1 (Proved). The intrinsic coefficient module and its residue complex. *The finite locally constant sheaves $\mathcal{K}_n = \mathcal{K}/3^n$ on X satisfy*

$$H^i(X, \mathcal{K}_n) = 0 \quad (i \geq 2), \qquad |H^1(X, \mathcal{K}_n)| = 3^{23n} |H^0(X, \mathcal{K}_n)|. \tag{A.1.1}$$

Their degree-one cohomology groups are finite. For every finite set J of primes different from three there is a compatible distinguished triangle in the derived category of abelian groups

$$R\Gamma(X, \mathcal{K}_n) \longrightarrow R\Gamma(X \setminus J, \mathcal{K}_n) \longrightarrow \bigoplus_{q\in J} R\Gamma(\mathbb{F}_q, \mathcal{K}_n(-1))[-1] \longrightarrow R\Gamma(X, \mathcal{K}_n)[1]. \tag{A.1.2}$$

Every D_q is injective and has finite cokernel. Localization and continuous cochains induce isomorphisms

$$H^2_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathcal{K}) \simeq \mathcal{R}(A)/D\mathcal{R}(A), \qquad H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathcal{K}) \simeq H^1_{\mathrm{cts}}(G_S, \mathcal{K}). \tag{A.1.3}$$

The latter group is finitely generated of rank 23*; its torsion is* $H^0(G_S, (\mathcal{K} \otimes \mathbb{Q}_3)/\mathcal{K})$*. This is an exact residue calculation, including the contribution over* $\mathbb{Z}[1/3]$ *in the degrees relevant to lifting sections.*

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. Every coordinate cusp of the Fermat cubic is a flex. Its tangent line cuts out three times that cusp; the ratio of two tangent lines has divisor three times their difference. The classifying homomorphism from degree-zero cusp divisors to E therefore has image in $E[3]$. The extension $0 \to W \to V \to T \to 0$ is killed by three, and

$$V \otimes \mathbb{Q}_3 \simeq (W \otimes \mathbb{Q}_3) \oplus (T \otimes \mathbb{Q}_3) \tag{A.1.4}$$

as continuous Galois representations. The comparison is the splitting of the pullback along $[3] : E \to E$ followed by division by three on the Tate module. Thus the integral lattice is not being replaced without its explicit isogeny comparison.

Over K_0 all nine points of $E[3]$ are the nine cusps. Hence both $W/3$ and $T/3$ have trivial action over K_0. The action on $V/3$ is an extension of trivial modules, so its finite image is a three-group. The congruence kernel in $\mathrm{GL}(V)$ is pro-three. Consequently the image on every $\mathcal{K}_n$ over $G_{K_0,S}$ is a finite three-group. Any finite module for a finite three-group has a composition series whose factors are trivial $\mathbb{F}_3$ modules: a nonzero invariant exists on every nonzero $\mathbb{F}_3$ representation, and induction applies to its quotient.

The ring $\mathbb{Z}[\zeta]$ is norm-Euclidean. Its units are μ_6 and its unique prime above three is $\lambda = 1 - \zeta$. Kummer theory therefore gives

$$H^1(G_{K_0,S}, \mathbb{F}_3) = \langle \zeta, \lambda \rangle \simeq \mathbb{F}_3^2.$$

Indeed valuations outside λ of an allowed Kummer element are multiples of three, and unique factorization removes them. The two remaining classes are independent, first by valuation and then by the unit group. Milne's Euler characteristic, I.5.1 [Mil06], is $1/3$ for this imaginary quadratic field and the trivial module $\mathbb{F}_3$. Since H^0 has order three and H^1 order nine, H^2 has order one. Devissage along the composition series just proved gives $H^2(G_{K_0,S}, \mathcal{K}_n) = 0$. The order-two quotient $\mathrm{Gal}(K_0/\mathbb{Q})$ has exact invariants on three-primary modules, so Hochschild–Serre gives the same vanishing over G_S. Milne I.4.10(c) gives vanishing in degrees at least three, and I.4.15 gives finiteness. Proposition II.2.9 identifies these Galois groups with the cohomology of the locally constant sheaves on X. In particular no inference from the cohomological dimension of a maximal pro-three quotient is needed here.

Complex conjugation has signs $(4, 6)$ on V: its permutation signs on $\mathbb{Z}_3[D]/\mathbb{Z}_3\delta$ are $(5, 3)$, the toric twist interchanges them, and its signs on T are $(1, 1)$. The signs on $\bigwedge^2 V$ are $(21, 24)$ and the quotient $\bigwedge^2 T = \mathbb{Z}_3(1)$ removes one minus sign. Thus the signs on $\mathcal{K}$ are $(21, 23)$. Milne I.5.1 over $\mathbb{Q}$ now gives (A.1.1). The same computation, passage to the inverse limit, and $H^0(G_S, \mathcal{K} \otimes \mathbb{Q}_3) = 0$ give the asserted rank and torsion description. Finite generation follows, for example, from the finite number of generators of the relative pro-three Galois group used in LM7 E.34; its cocycles are a closed submodule of a finite power of $\mathcal{K}$. Alternatively the finite groups in (A.1.1), the vanishing of H^2, and completeness give the same conclusion by successive lifting of a finite basis modulo three.

The localization sequence and purity give (A.1.2). At a strict Henselian trait of residue characteristic $q \neq 3$, the wild inertia is pro-q and tame three-inertia is $\mathbb{Z}_3(1)$; its degree-one cohomology is the twist $\mathcal{K}_n(-1)$ and its higher cohomology vanishes. This computes the purity shift and the residue-field action, rather than merely naming a local support group.

Take the direct limit over finite J. A continuous cochain with finite coefficients factors through a finite quotient and is unramified outside finitely many primes. The generic term is

therefore $R\Gamma(G_{\mathbb{Q}}, \mathcal{K}_n)$. At each prime the residue complex is

$$[A_q/3^n \xrightarrow{D_q} A_q/3^n]$$

in degrees zero and one. The transitions of both direct-sum terms are surjective. Their derived inverse limit is consequently the literal complex $[\mathcal{R}(A) \xrightarrow{D} \mathcal{R}(A)]$. Jannsen (1.6) and (2.2) [Jan88] identify the other derived limits with continuous cochains; the finite groups $H^1(X, \mathcal{K}_n)$ have zero first derived limit. Thus the contribution of X in degrees two and three vanishes.

For injectivity of D_q, write $P_D = \mathbb{Z}_3[D]/\mathbb{Z}_3\delta$. The rational graded pieces of $\mathcal{K}(-1)$ are

$$\mathbb{Q}_3(1) \otimes \bigwedge^2 (P_D \otimes \mathbb{Q}_3), \qquad (T \otimes \mathbb{Q}_3) \otimes (P_D \otimes \mathbb{Q}_3).$$

The first has Frobenius eigenvalues of complex absolute value q; the second has absolute value $\sqrt{q}$. The cusp permutation has finite order and the elliptic Frobenius polynomial is $X^2 - a_q X + q$. Neither list contains one. An integral endomorphism invertible over $\mathbb{Q}_3$ is injective with finite cokernel. The degree-two and degree-one parts of the limiting triangle now give (A.1.3). The torsion statement follows from the exact sequence $0 \to \mathcal{K} \to \mathcal{K} \otimes \mathbb{Q}_3 \to (\mathcal{K} \otimes \mathbb{Q}_3)/\mathcal{K} \to 0$. □

Statement A.2 (Proved). The exact first derived limit and Lead 5. *Jannsen's degree-two sequence for $\mathcal{K}$ is*

$$0 \longrightarrow \frac{\mathcal{R}(A) \cap \prod_{q \neq 3} D_q A_q}{D\mathcal{R}(A)} \longrightarrow \frac{\mathcal{R}(A)}{D\mathcal{R}(A)} \longrightarrow \varprojlim_n \bigoplus_{q \neq 3} \operatorname{coker}(D_q \bmod 3^n) \longrightarrow 0. \tag{A.2.1}$$

Its left term is $\varprojlim_n{}^1 H^1(G_{\mathbb{Q}}, \mathcal{K}_n)$ and is nonzero. It is torsion-free as a $\mathbb{Z}_3$-module. In particular Lead 5 is true.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. The finite-level localization sequence from A.1 gives an exact sequence

$$0 \to H^1(G_S, \mathcal{K}_n) \to H^1(G_{\mathbb{Q}}, \mathcal{K}_n) \to \bigoplus_q \ker(D_q \bmod 3^n) \to 0.$$

The first groups are finite. Taking first derived limits identifies the desired group with the first derived limit of the last tower. Apply inverse limits to the kernels, images, and cokernels of the surjective-term complexes used in A.1. The image towers have surjective transitions. Their limits are precisely $\mathcal{R}(A) \cap \prod_q D_q A_q$, which proves (A.2.1).

Choose for each j a new prime q_j splitting in the finite Galois extension cut out by $\mathcal{K}(-1)/3^j$. Chebotarev [Mil20b, V.3.23] supplies infinitely many choices. Then $D_{q_j} A_{q_j} \subset 3^j A_{q_j}$. Choose a primitive $v_j \in A_{q_j}$ and let $z_{q_j} = D_{q_j} v_j$, with all other entries zero. The family z is restricted and pointwise in the image of D. Its only preimage is v_j at the chosen primes, since each D_q is injective; this preimage is not restricted. Hence $[z] \neq 0$ in the left term. For torsion-freeness, write a left-term representative as $z = Dv$ with $v_q \in A_q$. A nonzero scalar $m \in \mathbb{Z}_3$ annihilates its class only when mv is restricted. Multiplication by a fixed nonzero scalar changes all valuations by a fixed integer, so then v is restricted. □

1.2 An actual intrinsic obstruction invisible at all finite levels

Let c_d denote the central inertia generators in $A = I/[N, I] = \mathbb{Z}_3(1)[D]$, and let e_d be their images in W. The rational cusps O and P define a continuous equivariant homomorphism

$$\ell : \mathbb{Z}_3(1) \longrightarrow N^{\mathrm{cc}}, \qquad z \longmapsto c_O^z c_P^z. \tag{A.3.1}$$

Its image is central. Pull back the central quotient $N/\gamma_3 N \to N^{\mathrm{cc}}$ along ℓ and call the resulting geometric group E_w, where $w = e_O + e_P$.

Statement A.3 (Proved). A linear family inside the nonabelian lifting problem. *The pullback is an exact sequence of continuous Galois lattices*

$$0 \longrightarrow \mathcal{K} \longrightarrow E_w \longrightarrow \mathbb{Z}_3(1) \longrightarrow 0. \tag{A.3.2}$$

Choose an integral lift l of the standard generator, and after twisting put

$$u_q = F_q(l \otimes 1) - l \otimes 1 \in A_q. \tag{A.3.3}$$

For $a = (a_p)_p \in \mathcal{R}_{\mathbb{P}}(\mathbb{Z}_3)$, the section $\ell \circ \kappa(a)$ of $N^{\mathrm{cc}} \rtimes G_{\mathbb{Q}}$ has obstruction represented, under (A.1.3), by

$$(a_q u_q)_{q \neq 3}. \tag{A.3.4}$$

Its unique possible local corrections are $-a_q D_q^{-1} u_q$. Thus it lifts precisely when these corrections are integral and form a restricted family. Changing l changes (A.3.4) by D of a restricted family. For the families used below, $a_3 = 0$. In E24's pro-three coordinates and the E36 generator $k = -\delta$, their actual torsor coordinates are

$$\alpha : \quad (r_1, r_2; A, B, C) = (0, -\kappa(a); -\kappa(a), -\kappa(a), -\kappa(a)), \qquad a_{\mathrm{E36}} = -a. \tag{A.3.5}$$

The three K_0 coordinates use the inclusion of rational Kummer classes; the other torsion, framed-λ, and divisible coordinates are zero. This specifies the pairs themselves in the requested convention.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. The abelian image of a lift of z is zw. The commutator of two lifts is therefore $(zw) \wedge (z'w) = 0$ in $\bigwedge^2 V$. The kernel $\mathcal{K}$ is central, so E_w is abelian. It is compact, torsion-free, and finitely generated over $\mathbb{Z}_3$; the quotient is free of rank one. This proves (A.3.2) and supplies an integral lift. All actions factor through G_S by smooth reduction of the pointed curve outside three.

The obstruction is now the ordinary connecting homomorphism $H^1(G_{\mathbb{Q}}, \mathbb{Z}_3(1)) \to H^2(G_{\mathbb{Q}}, \mathcal{K})$ for (A.3.2). On local tame inertia, $\kappa(a)(\tau_q) = a_q$. In $E_w(-1)$ the failure of its chosen lift to be Frobenius invariant is $a_q u_q$. Equivalently, apply the connecting morphism to the two-term residue complexes of A.1; the degree-zero element a_q in the trivial quotient has boundary $a_q u_q$ in the degree-one kernel. This proves (A.3.4) at the cochain level and hence before passage from the restricted quotient to a product of local groups. The bounded integral vectors u_q and restricted scalars a_q give a restricted representative. The final assertion follows from $u_q \mapsto u_q + D_q v$ for a fixed integral change of lift, with $(a_q v)_q$ restricted. For (A.3.5), in the non-O cusp basis the vector w has P-coordinate zero and all other coordinates minus one. Formula (E24.4) sends it to $(0, -1)$; (E24.5) gives the three equal negative rational Kummer classes. Its lift with O-coordinate zero differs from $c_O + c_P$ by δ. Since $k = -\delta$, the E36 parameter is $-a$, as asserted. □

Statement A.4 (Proved). A nonzero intrinsic obstruction in the first derived limit. *There is a continuous section of $N^{\mathrm{cc}} \rtimes G_{\mathbb{Q}}$ whose obstruction to $N/\gamma_3 N$ is nonzero in $\varprojlim{}^1_n H^1(G_{\mathbb{Q}}, \mathcal{K}_n)$. Its restriction to every local field is zero, and its image in every finite-coefficient obstruction group is zero. The construction below specifies such a section as $\ell \circ \kappa(a)$, hence specifies an actual pair of the torsor coordinates of R5 after the chosen origins are fixed. Its obstruction has infinite additive order and survives restriction to every number field.*

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. We first prove that the extension (A.3.2) has a nonzero unipotent action in the kernel of the action on $\mathcal{K}(-1)$. Let s_P be the rational tangential section at P, with parameter x/z and tangent one. Write $s_P = b_P s_O$ and also b_P for its abelian cocycle in V. For an inertia generator transported along the chosen path,

$$g(c_P) = b_P(g)^{-1} c_P^{\chi(g)} b_P(g), \qquad g(c_O) = c_O^{\chi(g)}. \tag{A.4.1}$$

The abelian tangential path is the Kummer class of the framed line bundle $z_P = \mathcal{O}(P - O)$ whose frames are $(x/z)^{-1}$ at P, z/x at O, and the canonical section at the other cusps. Indeed the Abel–Jacobi divisor $u - u_0$, as both points approach the indicated unit tangents, has exactly these two regularized frames. The Kummer comparison follows by pulling back multiplication by 3^n on the generalized Jacobian, which realizes the geometrically abelian covering with group $V/3^n$. This identifies the based path cocycle, not just its projection to the proper curve.

For completeness the needed framed cube is calculated directly. The function $f = (y+z)/(x+y)$ has divisor $3P - 3O$. Its regularized values are $1/3$ at P, 3 at O, and -1 at Q_c. Evaluating at the three quadratic cusp pairs and dividing by the value at O gives

$$3z_P = (3^{-2}, -3^{-1}, \lambda^{-1}, \lambda^{-2}, \zeta^2\lambda^{-3}). \tag{A.4.2}$$

Here the entries correspond to P, Q_c and the pairs denoted $A, \bar{A}$, $B, \bar{B}$, $C, \bar{C}$ in LM7 E.24. The identity $\lambda^2 = -3\zeta$ verifies the values and their conjugates. These are the origin terms needed in this proof.

The action on $V \otimes \mathbb{Q}_3$ over K_0 is abelian. The toric action is cyclotomic; on the elliptic factor the endomorphism induced by a coordinate multiplication by ζ makes T_3E a rank-one module over the discrete valuation ring $\mathbb{Z}_3[\zeta]$. The Galois action commutes with this endomorphism and thus lies in its commutative unit group. The splitting (A.1.4) proves the assertion for V; since the integral lattice embeds in its rationalization, commutators act trivially on the integral lattice too.

Choose $\tau \in \ker \chi_3$ with $\kappa_3(\tau)$ a unit. Such an element exists: $\mathbb{Q}(\sqrt[3]{3})$ is not contained in $\mathbb{Q}(\mu_{3^\infty})$, since every finite subfield of this abelian extension is Galois over $\mathbb{Q}$, whereas the real irreducible pure cubic is not normal. Choose $\sigma \in G_{K_0}$ with $\chi_3(\sigma) = 4$, and put $g = [\sigma, \tau]$. The Kummer homomorphism into $\mathbb{Z}_3 \rtimes \mathbb{Z}_3^\times$ gives

$$\chi_3(g) = 1, \qquad \kappa_3(g) = 3\kappa_3(\tau). \tag{A.4.3}$$

Thus g fixes V and $\mathcal{K}(-1)$. On this subgroup, $\kappa_\zeta(g) = 0$ and $\kappa_\lambda(g) = \kappa_3(g)/2$; the same holds for the conjugate of λ. Evaluating (A.4.2) on g, where coboundaries in V vanish, gives

$$3b_P(g) = \kappa_3(g) v_0, \qquad v_0 = -2e_P - e_{Q_c} - \tfrac{1}{2}(e_A + e_{\bar{A}}) - (e_B + e_{\bar{B}}) - \tfrac{3}{2}(e_C + e_{\bar{C}}). \tag{A.4.4}$$

The notation is in $W \otimes \mathbb{Q}_3$; all denominators in v_0 are units in $\mathbb{Z}_3$. Take $l = \log c_O + \log c_P$ in the class-two Lie lattice, whose Baker–Campbell–Hausdorff denominator two is a three-adic unit. Equation (A.4.1) gives

$$u_g = -b_P(g) \wedge e_P = -\kappa_3(\tau) v_0 \wedge e_P. \tag{A.4.5}$$

The coefficient of $e_P \wedge e_{Q_c}$ in $v_0 \wedge e_P$ is one. The eight non-O cusp classes form a basis of W, and W is a primitive sublattice of V. Therefore (A.4.5) is primitive in $\mathcal{K}$. A change of based path contributes $(g-1)v = 0$; a change of tangent at P adds a multiple of e_P to b_P and has zero wedge with e_P. This verifies the exact ambiguity.

Apply Chebotarev [Mil20b, V.3.23] to the finite representation $E_w(-1)/3^j$. Choose recursively a new prime q_j whose Frobenius belongs to the conjugacy class of g in this representation. Then

$$D_{q_j} \equiv 0 \pmod{3^j}, \qquad u_{q_j} \notin 3A_{q_j}. \tag{A.4.6}$$

Conjugacy preserves these properties: the restriction to the kernel is the identity modulo 3^j, and conjugation carries the off-diagonal vector through an invertible integral matrix. Define the finite integer

$$e_j = \min\{e \geq 0 : 3^e D_{q_j}^{-1} u_{q_j} \in A_{q_j}\}. \tag{A.4.7}$$

It exists by A.1, and (A.4.6) implies $e_j \geq j$. Set $a_{q_j} = 3^{e_j}$ and set every other a_p to zero. This is a restricted Kummer family. Its residues are pointwise in $D_q A_q$; their unique corrections $-3^{e_j} D_{q_j}^{-1} u_{q_j}$ are primitive by the minimality in (A.4.7). They do not tend to zero. A.3 and A.2 prove that its obstruction is a nonzero first-derived-limit class.

At each coefficient level only finitely many residue entries survive. Each is a boundary, so the finite-coefficient obstruction is zero. At every local field, the finite degree-one cohomology groups satisfy Mittag–Leffler; continuous degree-two cohomology therefore injects into the limit of finite degree-two groups. This proves local vanishing also at three. Real three-primary cohomology vanishes by averaging. Finally A.2 proves infinite additive order. Restriction followed by corestriction over a finite extension $L/\mathbb{Q}$ is multiplication by $[L : \mathbb{Q}]$ on continuous cohomology, by the finite coset cochain formula. Hence no restriction can kill this obstruction. □

1.3 Two ramification assertions disproved

Statement A.5 (Refuted). Lead 2: mixed tame inertia already occurs at the prime two. *Lead 2 is false. The section*

$$s = \ell \circ (3^{23}\kappa_2) : G_{\mathbb{Q}} \longrightarrow N^{\mathrm{cc}} \rtimes G_{\mathbb{Q}} \tag{A.5.1}$$

lifts to $N/\gamma_3 N$. Its abelian tame-inertia vector at two is $3^{23}(e_O + e_P)$, which is not a multiple of any cusp vector. More generally, for each rational prime $q \neq 3$, every multiple of $3^{v_3 \det D_q}\kappa_q$ in the line (A.3.1) lifts.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. For a fixed q the only nonzero residue of this family is a scalar multiple of u_q. The adjugate identity shows that $3^{v_3 \det D_q} D_q^{-1}$ is integral. Its local correction has finite support and is therefore restricted. A.3 proves the general assertion.

At two the cusp permutation has signs $(5, 3)$ on P_D. Thus its exterior square has signs $(13, 15)$, and the determinant of $F_2 - 1$ on $\bigwedge^2 W(-1)$ has three-adic valuation 15. The cubic has the three projective points $[0:1:1], [1:0:1], [1:1:0]$ over $\mathbb{F}_2$, so its elliptic Frobenius polynomial is $X^2 + 2$. On each of the eight copies of T, twisted by a cusp permutation sign, the determinant of $F_2 - 1$ is three. The remaining contribution is eight. Consequently

$$v_3 \det D_2 = 15 + 8 = 23.$$

This determinant calculation uses the rational filtration and is valid on the actual integral lattice because determinants are unchanged by rational change of basis.

To check the cusp claim integrally, eliminate e_O by the relation $\sum_d e_d = 0$. In the basis of the other eight cusps, $e_O + e_P = (0,-1,-1,-1,-1,-1,-1,-1)$. It is proportional neither to a basis vector nor to $e_O = (-1,-1,-1,-1,-1,-1,-1,-1)$. Multiplication by a nonzero three-adic scalar preserves this fact. The exact finite calculations are independently executed in `code/a_residue_checks.py`. □

Statement A.6 (Refuted). Lead 3: infinitely ramified noncuspidal class-two sections. *Lead 3 is false. Enumerate the rational primes different from three as $q_1, q_2, \ldots$, put $m_j = v_3 \det D_{q_j}$, and set*

$$a_{q_j} = 3^{m_j + j}, \qquad a_3 = 0. \tag{A.6.1}$$

The section $\ell \circ \kappa(a)$ lifts to $N/\gamma_3 N$ and every such lift is ramified at all the q_j. None is cuspidal. The same conclusion holds after replacing these coefficients by any nonzero coefficients satisfying $v_3(a_{q_j}) - m_j \longrightarrow +\infty$.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. The family is restricted. By the adjugate identity, $a_{q_j} D_{q_j}^{-1} u_{q_j} \in 3^j A_{q_j}$, so the unique corrections are integral and restricted. A.3 proves existence of a continuous lift. Its image in V has nonzero tame-inertia vector $a_{q_j}(e_O + e_P)$ at every q_j; thus geometric conjugation cannot remove the ramification. A cuspidal section at a rational cusp has all abelian tame-inertia vectors in the line generated by that cusp. A.5 proves that no one of the displayed vectors lies in any such line. The nonrational cusps have decomposition groups whose image in $G_{\mathbb{Q}}$ has index two, so cannot support a section over $\mathbb{Q}$. This proves noncuspidality and the more general assertion with the stated convergence bound. □

1.4 The lift torsor and a formula retaining all path origins

Statement A.7 (Proved). The integral group acting on every nonempty fibre. *There is an isomorphism of finitely generated $\mathbb{Z}_3$-modules*

$$H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathcal{K}) \simeq \mathbb{Z}_3^{23} \oplus (\mathbb{Z}/3)^9 \oplus (\mathbb{Z}/9)^5. \tag{A.7.1}$$

Every nonempty fibre of $\mathrm{Sec}(N/\gamma_3 N) \to \mathrm{Sec}(N^{\mathrm{cc}})$ is a torsor under this group. The ambiguity in (A.7.1) is the choice of a Smith basis and of a free direct summand. No further geometric conjugations identify members of the torsor.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. We give the integral lattice and the finite calculation determining its torsion. Write $\mathcal{O}_3 = \mathbb{Z}_3[\zeta]$ and identify T with $\mathcal{O}_3$ so that P is the class of 1 in $T/3T$. The automorphism $[x : y : z] \mapsto [x : y : \zeta z]$ fixes O and acts on its tangent by ζ. It is the chosen complex multiplication. Its orbit of P is P, A_{bar}, A. The eight nonzero cusp labels in the basis $(P, \zeta P)$ are

d	P	Q_c	A	$\bar{A}$	B	$\bar{B}$	C	$\bar{C}$
a_d	1	−1	−1	0	1	0	−1	1
b_d	0	0	−1	1	1	−1	1	−1

(A.7.2)

For example the line through P and $-\zeta P$ has third intersection $[1 : -\zeta : 0] = C$, so $C = -(P - \zeta P)$. This determines the only labels not obtained immediately by multiplication by ζ and negation.

Let Ψ be the 8×2 integer matrix with row $(b_d, -a_d)$. In the rational representation $W_{\mathbb{Q}_3} \oplus T_{\mathbb{Q}_3}$ the lattice V is

$$L_\Psi = \{(w + \Psi t/3, t) : w \in W,\ t \in T\}. \tag{A.7.3}$$

Here is the comparison proving this formula. Pullback of the semiabelian extension $J_{E,D}$ along $[3] : E \to E$ is split, since its class is killed by three. The splitting is unique because $\mathrm{Hom}(E, T_D) = 0$. Thus it is defined over $\mathbb{Q}$. The resulting isogeny $T_D \times E \to J_{E,D}$ has kernel the graph of a homomorphism $E[3] \to T_D[3]$; its projection to E is $[3]$. Evaluation of the graph on a degree-zero cusp character gives the Weil pairing with that cusp difference, by the defining Poincare biextension of the generalized Jacobian. In the bases in (A.7.2), this pairing is $t \mapsto \Psi t$ modulo three. One overall sign changes Ψ to $-\Psi$ and conjugates all the matrices by $(-1)_W \oplus 1_T$, so it does not affect their invariant factors. The Tate lattice of the quotient is obtained by adding $(\Psi t/3, t/3)$ to $W \oplus T$ and then using $[3]$ on the second coordinate. This is precisely (A.7.3), including its integral comparison with the geometric Tate module.

For a Galois element with matrices R on W and T_g on T, the matrix in the basis in (A.7.3) is therefore

$$G_V = \begin{pmatrix} R & (R\Psi - \Psi T_g)/3 \\ 0 & T_g \end{pmatrix}. \tag{A.7.4}$$

The numerator is divisible by three by equivariance of the Weil pairing. Over K_0, $R = \chi(g) I_8$. Complex conjugation has $R = -P_c$, where P_c fixes P, Q_c and interchanges the three quadratic cusp pairs, and

$$T_c = \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}.$$

The exact Tate image over K_0 is $1 + 3\mathcal{O}_3$. Containment follows from rationality of $E[3]$ and complex multiplication. To prove equality, count the Fermat cubic over $\mathbb{F}_7$: it has nine points, so its Frobenius polynomial is $X^2 + X + 7$. Its roots in $\mathcal{O}_3$ are $\alpha = 1 + 3\zeta$ and $\bar{\alpha} = -2 - 3\zeta$. Frobenius at a prime above seven gives one root; conjugation gives the other. The three-adic logarithm identifies $1 + 3\mathcal{O}_3$ with $3\mathcal{O}_3$. The vectors $\log(\alpha)/3$ and $\log(\bar{\alpha})/3$ reduce to ζ and $-1 - \zeta$ modulo three, a basis of $\mathcal{O}_3/3\mathcal{O}_3$. Their closed $\mathbb{Z}_3$-span is therefore all $\mathcal{O}_3$. This proves equality without an unspecified index from a complex multiplication reciprocity theorem. The matrices on T are

$$T_\alpha = \begin{pmatrix} 1 & -3 \\ 3 & -2 \end{pmatrix}, \qquad T_{\bar{\alpha}} = \begin{pmatrix} -2 & 3 \\ -3 & 1 \end{pmatrix}. \tag{A.7.5}$$

The determinants are seven. Equations (A.7.4)–(A.7.5) and c therefore topologically generate the actual image on V.

Take exterior squares in (A.7.4) and omit the basis wedge of the two T lifts. The remaining forty-four wedges are a basis of $\mathcal{K}$, since that omitted wedge maps to a generator of $\bigwedge^2 T$. Let the three resulting matrices be $G_\alpha, G_{\bar{\alpha}}, G_c$. The integer matrix

$$B = \begin{pmatrix} G_\alpha - 1 \\ G_{\bar{\alpha}} - 1 \\ G_c - 1 \end{pmatrix} : \mathbb{Z}_3^{44} \longrightarrow \mathbb{Z}_3^{132} \tag{A.7.6}$$

has three-adic Smith diagonal

$$\underbrace{1, \ldots, 1}_{30}, \quad \underbrace{3, \ldots, 3}_{9}, \quad \underbrace{9, \ldots, 9}_{5}. \tag{A.7.7}$$

The program `code/a_k_lattice.py` constructs every entry from (A.7.2)–(A.7.6), checks $c^2 = 1$ and $cG_\alpha c = G_{\bar{\alpha}}$, and performs elementary row and column operations over $\mathbb{Z}/3^8$. Each operation is invertible and every pivot has valuation at most two. Thus no factor divisible by 3^8 remains, and this calculation determines the exact three-adic Smith exponents. The full matrices and the output are included. As an independent consistency check, the same calculation on V gives seven unit factors and three factors three, recovering $J_{E,D}(\mathbb{Q})[3^\infty] = (\mathbb{Z}/3)^3$.

An invariant in $(\mathcal{K} \otimes \mathbb{Q}_3)/\mathcal{K}$ is exactly an x with Bx integral, modulo integral x. A Smith factor 3^e contributes $\mathbb{Z}/3^e$. A.1 and (A.7.7) prove (A.7.1). Finally fix a cocycle representing a section of N^{cc}. Central lifts differ by $\mathcal{K}$-valued cocycles and central conjugation gives coboundaries. A geometric element of N^{cc} stabilizing that section has invariant image in V, which is zero. It is then in the central kernel $\mathbb{Z}_3(1)$ and is invariant there, hence zero again. There is no additional stabilizer quotient. This proves the exact fibre statement. □

Statement A.8 (Proved). The full class-two residue criterion in based cusp-path coordinates. *Choose geometric paths from the specified tangent at O to a Galois-equivariant family of tangents at the nine cusps. Let c_d be the corresponding inertia elements in $N/\gamma_3 N$ and $b_{g,d} \in V$ their abelian comparison cochains, with convention*

$$g(\log c_d) = \chi(g)\big(\log c_{gd} - b_{g,d} \wedge e_{gd}\big). \tag{A.8.1}$$

For a continuous representative s of a class in $\mathrm{Sec}(N^{\mathrm{cc}})$*, write*

$$\log s(\tau_q) = \sum_{d \in D} t_{q,d} c_d, \quad v_q = \alpha(\phi_q), \quad w_q = \sum_d t_{q,d} e_d \in W. \tag{A.8.2}$$

Here $s(\tau_q)$ means its geometric cocycle value, the chosen arithmetic action is unramified at q, and $\phi_q \tau_q \phi_q^{-1} = \tau_q^q$. Then its full intrinsic obstruction is represented by the restricted family

$$r_q = v_q \wedge w_q - \sum_d t_{q,d}\, b_{\phi_q,d} \wedge e_{\phi_q d} \in \mathcal{K}(-1). \tag{A.8.3}$$

The section lifts exactly when

$$-D_q^{-1} r_q \in A_q \quad \textit{for all } q \neq 3, \qquad (-D_q^{-1} r_q)_q \in \mathcal{R}(A). \tag{A.8.4}$$

The nonempty fibres are exactly the torsors in A.7.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. The comparison cochains exist by transport of the inertia loops along the chosen paths. Composition gives $b_{gh,d} = b_{g,hd} + g b_{h,d}$ in V. Formula (A.8.1) is the class-two expansion of conjugation and includes its sign convention. It specifies the action on the cusp lifts themselves, including the path origins.

The cocycle restriction to local inertia is a homomorphism, since the original action is unramified there. Wild inertia maps to zero and the prime-to-three tame factors also map to zero. The elliptic projection t_T satisfies $F_q t_T = q t_T$. The Frobenius eigenvalues on T have absolute value $\sqrt{q}$, so $t_T = 0$. Hence the logarithm of tame inertia lies in the central lattice $A = \mathbb{Z}_3(1)[D]$ in N^{cc}, proving (A.8.2). The relation also gives $t_{q,\phi_q d} = t_{q,d}$.

Lift this inertia logarithm to $L_q = \sum_d t_{q,d} \log c_d + k_q$ in the full class-two lattice, where $k_q \in \mathcal{K}$. Lift the Frobenius cocycle arbitrarily; its abelian coordinate is v_q. In a class-two group conjugation by that lift changes a logarithm of abelian coordinate $q w_q$ by $q v_q \wedge w_q$.

The logarithm of the tame-relation defect, divided by q to identify inertia homomorphisms with the twist $\mathcal{K}(-1)$, is therefore

$$D_q k_q + v_q \wedge w_q - \sum_d t_{q,d} b_{\phi_q,d} \wedge e_{\phi_q d}.$$

The central coordinate of the Frobenius lift cancels. This is exactly the cochain-level localization boundary: it measures the failure of a chosen inertia splitting to be invariant under Frobenius. At every finite coefficient level the action and the section are unramified at almost all primes, so these choices give a direct-sum residue cochain. The inverse-limit family is restricted; concretely every term contains one of the restricted coefficients $t_{q,d}$ and all other vectors are integral. A.1 now identifies the global obstruction with $[r_q]$, before any passage to a product of local groups. Injectivity of every D_q proves (A.8.4), with the stated convergence condition. A.7 supplies the geometric-conjugacy fibre. □

Coordinates of the tame coefficients. The formulas can be connected to the chosen E24 coordinates without suppressing the central parameter. Separate those coordinates into the entries supported at three, including μ_3^3, the framed point z', and the noncanonical divisible $\mathbb{Z}_3$ parameter, and the prime valuations outside three. The elliptic projection is always unramified outside three: the same weight calculation used in A.8 makes every local integral elliptic tame vector zero. The surjectivity $H^1(G_S, V) \to H^1(G_S, T)$ follows from $H^2(G_S, W) = 0$, proved by the same composition-series argument as A.1. Thus origins can be chosen by a full unramified lift for the first set of parameters, as constructed in LM7 E.34, and by the central splitting $W \to A$ with O-coordinate zero for the remaining toric coordinates. We make this choice agree with the E36 convention as follows. For each elliptic class β, choose an unramified section c_β of C; take c_0 to be the unit tangent at O, and take the classes of $P - O$ and $Q_c - O$ to the images of their unit tangents. The reference over every α is (c_β, α) in $C \times_T V$. The prescribed unramified reference itself has a full unramified lift by the free pro-three embedding argument. The E36 central generator is $k = [a, b] = -\delta$, so the exact coordinate comparison is $a_\delta = -a_{\mathrm{E36}}$. In the formulas below a always means a_{E36}. This fixes both the origin and its sign; it agrees with the packets in B.3.

Write $r_{1,q}, r_{2,q}$ for the valuations of the two rational toric coordinates, and $\nu_{A,\mathfrak{q}}$, $\nu_{B,\mathfrak{q}}$, $\nu_{C,\mathfrak{q}}$ for those of the three K_0 toric coordinates. Put $N\nu_{A,q} = \sum_{\mathfrak{q}|q} f_{\mathfrak{q}/q} \nu_{A,\mathfrak{q}}$, and likewise for B, C. In this chosen origin, for $q \neq 3$,

$$\begin{aligned} t_{q,O} &= -a_q, \\ t_{q,P} &= r_{1,q} + N\nu_{C,q} - N\nu_{A,q} - a_q, \\ t_{q,Q_c} &= r_{2,q} + N\nu_{C,q} - N\nu_{B,q} - a_q, \\ t_{q,d} &= \nu_{A,\mathfrak{q}(d)} - a_q \qquad\qquad (d \in \{A, \bar{A}\}), \end{aligned} \tag{A.8.5}$$

with identical last formulas for the B and C pairs. At an inert prime the two valuations in a pair coincide and their norm contribution is twice that valuation. At a split prime each has residue degree one. These formulas are the valuation form of the actual embedding (E24.5), followed by minus the central diagonal $\mathbb{Z}_3(1)\delta$; the diagonal is the kernel of $A \to W$.

The cochains $b_{g,d}$ in (A.8.3) are abelian data. Let z_d be the framed tangential divisor $d - O$ in $J_{E,D}$ and choose compatible division points $z_{d,n}$ with $3^n z_{d,n} = z_d$. Then $g z_{d,n} - z_{gd,n} \in J_{E,D}[3^n]$ gives their reduction modulo 3^n, with the chosen path convention. Their origin relations are obtained by the tangent-line functions f_d with $\operatorname{div} f_d = 3(d - O)$, regularized at the indicated tangents. Formula (A.4.2) is the complete P instance and is the one needed for the proved counterexamples. Statements A.9 and A.10 below supply the complete regularized origin table and fix the extra unramified generator. Evaluating a Frobenius in that free

pro-three Galois group is part of the arithmetic input; a finite-query program on arbitrary completed families cannot decide the final convergence condition.

Statement A.9 (Proved). All nine tangential origin data, including roots of unity. *Order the non-O cusps as $P, Q_c, A, \bar{A}, B, \bar{B}, C, \bar{C}$. Use x/z at the first and third/fourth cusps, y/z at the second and fifth/sixth cusps, and z/x at the last two; all tangents are one. For $z_d = \mathcal{O}(d - O)$ with these regularized frames, write $3z_d = h_d \in T_D(K_0)$. Its O-frame is one. The other entries are $\pm\zeta^{Z_{d,e}}\lambda^{L_{d,e}}$, with matrices*

$$L = \begin{pmatrix} -4 & -2 & -1 & -1 & -2 & -2 & -3 & -3 \\ -2 & -4 & -2 & -2 & -1 & -1 & -3 & -3 \\ -1 & -2 & -4 & -1 & -2 & -2 & -3 & -3 \\ -1 & -2 & -1 & -4 & -2 & -2 & -3 & -3 \\ -2 & -1 & -2 & -2 & -4 & -1 & -3 & -3 \\ -2 & -1 & -2 & -2 & -1 & -4 & -3 & -3 \\ -3 & -3 & -3 & -3 & -3 & -3 & -6 & -3 \\ -3 & -3 & -3 & -3 & -3 & -3 & -3 & -6 \end{pmatrix}, \qquad Z = \begin{pmatrix} 2 & 1 & 0 & 1 & 0 & 2 & 2 & 1 \\ 1 & 2 & 0 & 2 & 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 1 & 0 & 2 & 0 & 1 & 2 \\ 0 & 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 2 & 2 & 0 & 0 \\ 2 & 1 & 2 & 2 & 1 & 1 & 0 & 0 \end{pmatrix}. \tag{A.9.1}$$

The signs, recorded in the exact program output, have zero pro-three Kummer cocycle, using the rational compatible odd-power roots of -1. No ζ term may be discarded. The row at O is the identity. These data are equivariant under complex conjugation.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. For $d = [x_d : y_d : z_d]$, put $L_d = x_d^2 x + y_d^2 y + z_d^2 z$ and $f_d = L_d/(x+y)$. The flex property proves $\operatorname{div} f_d = 3d - 3O$. At O, write $s = z/x$ and $y/x = -(1+s^3)^{1/3}$. The regularized value of f_d is

$$c_O(d) = \lim_{s\to 0} f_d s^3 = -3(x_d^2 - y_d^2).$$

For $d = [0 : -\eta : 1]$ or $[-\eta : 0 : 1]$, with $\eta^3 = 1$, the regularized value at d is $1/(3\eta)$, by expanding the nonzero coordinate as $-\eta(1+s^3)^{1/3}$. For $d = [1 : -\eta : 0] \neq O$ it is $-1/(3(1-\eta))$. At every other cusp e the ordinary value $f_d(e)$ applies. Division of all these values by $c_O(d)$ gives $h_{d,e}$. This proves the claimed framed relation over K_0. All arithmetic in the evaluation is in $\mathbb{Q}(\zeta)$; `code/a_cusp_origins.py` evaluates the formulas as rational pairs in the basis $1, \zeta$, factors each nonzero value uniquely as $\pm\zeta^a\lambda^b$, and verifies conjugation entry by entry. It also verifies the complete P row against (A.4.2). This gives (A.9.1), not a numerical approximation to its entries. □

Statement A.10 (Proved). An explicit choice of the extra unramified cohomology generator. *Let P_3 be the maximal pro-three quotient of $G_{K_0,S}$ and let $\Gamma = P_3 \rtimes \langle c \rangle$ be the relative quotient of G_S. Choose u to be Frobenius at the prime $(7, \zeta - 2)$, with a decomposition group fixed once and for all, and put $v = cuc$. Then u, v are a free pro-three basis of P_3, and c interchanges them. In the representation (A.7.3) let $\widetilde{t}_1 = (\Psi(1,0)/3, (1,0)) \in V$. The prescriptions*

$$\alpha_\infty(c) = 0, \qquad \alpha_\infty(u) = \widetilde{t}_1 \tag{A.10.1}$$

define a continuous V-valued cocycle on Γ and fix a splitting of the final $\mathbb{Z}_3$ coordinate in (E24.1). Together with the specified prime generators and the framed points, this fixes the formerly noncanonical cohomology coordinate.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. The vanishing $H^2(G_{K_0,S}, \mathbb{F}_3) = 0$ was proved in A.1. Inflation injects $H^2(P_3, \mathbb{F}_3)$ into this group: an invariant nonzero character of the kernel of $G_{K_0,S} \to P_3$ would give a central extension of P_3 by $\mathbb{F}_3$, a pro-three quotient contradicting maximality. The five-term exact sequence proves the injection. The minimal free pro-three presentation of P_3 has two generators, by the Kummer calculation in A.1; its relation module has dual $H^2(P_3, \mathbb{F}_3) = 0$. A nontrivial closed relation subgroup in the Frattini subgroup has a nonzero first term in the lower exponent-three filtration, so the relation module cannot vanish. Thus P_3 is free on two generators.

At the chosen prime over seven, finite-field Kummer theory gives the evaluations $(2, 0)$ on the classes (ζ, λ): $\zeta \mapsto 2$, $\lambda \mapsto -1$ in $\mathbb{F}_7$, and their squares are ζ^2 and one. At the conjugate Frobenius the values are $(2, 2)$. They are independent over $\mathbb{F}_3$. Consequently u, cuc generate the Frattini quotient of P_3; the pro-three basis theorem and freeness make them a free basis. Complex conjugation is an actual order-two element of G_S, so its relations are exactly $c^2 = 1$, $cuc = v$, and $cvc = u$.

For each finite three-primary quotient, arbitrary cocycle value on u, its prescribed conjugate value on v, and value zero on c respect all these relations. The free pro-three universal property gives a cocycle; taking the compatible inverse limit gives (A.10.1). On T the normalized cocycles are T modulo $(u - 1)T^c$. Now $T^c = \mathbb{Z}_3(1, 0)$ and $(T_\alpha - 1)(1, 0) = 3(0, 1)$, so

$$H^1(G_S, T) = \mathbb{Z}_3(1, 0) \oplus (\mathbb{Z}/3)(0, 1).$$

The projection of (A.10.1) is the free generator. The finite summand is the Kummer class of P, and filling the cusps identifies the free quotient with the Tate module of the divisible part used in (E24.9). Thus the given lift defines the asserted splitting. The Frobenius scalar at $(7, \zeta - 2)$ is $1 + 3\zeta$, rather than its conjugate: its reduction has zero differential at that prime, which selects the root divisible by the prime. This verifies the matrix used in (A.10.1). □

Statement A.11 (Proved). Executed finite and certified-tail evaluations. *The program* `code/a_criterion.py` *implements (A.8.4) for finite-support exact residue matrices and for the two infinite families A.4 and A.6 with their proved tail certificates. It gives the following verified results:*

1. *the E.28 quotient of* (κ_6, κ_{109}) *has correction* $-1/12$ *at* 109*, of three-adic valuation* -1*; all* 729 *central lift pairs modulo* 27 *fail the tame relation;*

2. *the E.31 origin retains* $\kappa_3(\zeta)$*, its local Frobenius and inertia values are* 2 *and* 1*, and all nine central lift pairs fail; the entire coset* $\alpha + 3\mathcal{K}_0$ *has the same excluded image;*

3. *all the off-three residues vanish for every section unramified outside three;*

4. *the E.35 norm polynomial is* $X^3 + AY^3 + A^2Z^3 - 3AXYZ$*, its example* $A = 2, b = 3$ *has witness* $(1, 1, 0)$*, and its finite conjugation calculation has span-rank histogram* $(1, 32, 48)$ *and* 153 *orbits over all* 81 *abelian assignments and* 729 *generator assignments.*

The infinite family A.6 is certified by correction valuations at least j*, whereas A.4 is certified by primitive corrections at every selected prime. These two conclusions use their complete tail proofs, with no inference from a finite sample.*

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. For exact finite-support data, the program calculates $-D_q^{-1} r_q$ by rational linear algebra and tests the three-adic valuation of every coordinate. All unlisted corrections are zero, so this is exactly (A.8.4). The two family constructors apply the adjugate bound of A.6

and the minimal-denominator proof of A.4, respectively. Their output records these bounds explicitly.

The comparison for E.28 is the actual covering homomorphism $N \to \widehat{F}_2^{(3)}$ followed by its class-two quotient. The two coordinate inertia exponents multiply by three, so the bracket multiplier is nine. At 109, choose the residue generator 6^4 of the three-primary roots of unity; the Kummer value of six is one. The resulting differential is $109 - 1 = 108$ and the residue is nine, giving $-9/108 = -1/12$. The program also enumerates both central coordinates modulo 27 and checks the same nonzero defect for all 27^2 possibilities.

For E.31 use the actual function $h = (x+y)/(x+\zeta y)$ on the cubic, with the same tangent at O. Writing $s = z/x$, its leading constant is $-1/(3\lambda) = \zeta\lambda^{-3}$. At $\mathfrak{q} = (3+\zeta)$ over seven, $\zeta = 4$, $\lambda = 4$ and $\bar{\pi} = 2 - \zeta = 5$. At the specified torus point the total h unit reduces to three, whose square is two, the square of $\zeta = 4$. Thus the Kummer Frobenius exponent is two and the other coordinate has inertia exponent one. The commutator is nonzero in the order-27 Heisenberg quotient. The morphism on abelian cohomology to these two $\mathbb{F}_3$ coordinates is additive, with the fixed affine origin just computed; adding $3\mathcal{K}_0$ leaves it unchanged. This proves the whole coset assertion, rather than a list of sampled coset elements.

For an unramified section $t_{q,d} = 0$ for every $q \neq 3$; every term of (A.8.3) is zero. The program verifies the corresponding polynomial identity. The existence of lifts to the full N uses the free pro-three argument, not this finite computation.

Finally the inner action of E.31 is removed by the explicit arithmetic-group isomorphism $(n, g) \mapsto (nq_g, g)$, $q_g = (\kappa_3(\zeta)(g), 0, 0)$. It sends the abelian pair (a, b) to $(A, b) = (\zeta a, b)$. The central extension then has cup obstruction $\kappa_3(A) \cup \kappa_3(b)$. The crossed-product comparison [Mil20b, IV.3.14] represents this class by the cyclic algebra $L_A \oplus L_A U \oplus L_A U^2$, with $U\ell = \sigma(\ell)U$ and $U^3 = b$. It splits exactly when $b = N_{L_A/K_0}(w)$: a norm witness gives the representation $U = w\sigma$ on the three-dimensional K_0-space L_A; a splitting conversely gives that form for U on a one-dimensional L_A-module. Its norm multiplication matrix on $1, t, t^2$ is

$$\begin{pmatrix} X & AZ & AY \\ Y & X & AZ \\ Z & Y & X \end{pmatrix}.$$

The determinant is the displayed polynomial. The finite group law $\frac{1}{2}(uv' - vu')$ gives the central change $rv - su$ under conjugation. Thus the complete nonempty fibre is $K_0^\times / (K_0^{\times 3} \langle A, b \rangle)$. The program checks the determinant identity, the norm witnesses, and every finite conjugation orbit. This confirms the exact quotient calculation used in E.35; it does not assert that a lift in this finite quotient already lifts to $N/\gamma_3 N$. □

Statement A.12 (Refuted). A universal finite-precision decision program. *There is no program which, for every completed Kummer input in A.3 supplied through finite-coefficient queries, terminates and decides whether its continuous intrinsic obstruction vanishes. The same failure holds for the full inputs of A.8. Finite-support inputs and inputs furnished with proved tail certificates have the exact evaluator in A.11. There is also a concrete family of inputs described by finite programs, each returning the actual R4/E36 finite-coefficient cocycles and each with the uniform known tail bound $v_3(a_j) \geq j$, on which the vanishing decision is undecidable. Thus replacing an oracle input by a finite program for its coefficients does not provide a total evaluator on this explicitly specified class. The task's phrase "given (α, a)" does not itself specify an input encoding; these two precise input classes state the scope of the refutation, while A.11 states the implemented domain and proves all four requested reproductions.*

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. Run such a program on the zero input. Its terminating computation uses finitely many coefficient levels, with maximum 3^n. Discard the first n terms of the nonzero obstruction constructed in A.4. The resulting input is zero modulo 3^n, since its remaining coefficients have valuations at least their indices. Every query in the alleged computation therefore receives exactly the same answer. Nevertheless its corrections remain primitive at infinitely many primes, so A.3 proves that its obstruction is nonzero. The program gives the same answer on two inputs requiring different answers, a contradiction. The counterexample is a tail of the actual intrinsic family, rather than an arbitrary formal sequence outside the section problem. This establishes the stated precise computational limit and explains the role of the tail certificates in A.11.

For the finite-program assertion, first make the family A.4 effective. At level 3^r, the representation $E_w(-1)/3^r$ is determined by the 3^r-division points of the explicit generalized Jacobian and the 3^r-division torsor of the framed point $P - O$. The latter point and its torus multiple are the actual framed point of A.9 and (E24.6). Multiplication by 3^r is a finite algebraic morphism on this semiabelian variety in characteristic zero. Its fiber equations, the elliptic division polynomials, and the torus root equations give a finite collection of algebraic numbers and their exact Galois action. These are effective finite algebraic operations: roots are represented by polynomials and isolating data, and their equalities, embeddings, and permutations are decided by exact algebraic arithmetic. The exterior-square map and the cochain $b_{g,P} \wedge e_P$ then give the finite representation used in A.4, in the fixed existing coordinates.

We can consequently enumerate primes and test the finite conditions $D_q \equiv 0 \pmod{3^j}$ and $u_q \not\equiv 0 \pmod 3$ by reduction in that explicitly constructed finite Galois extension. Take q_j to be the first unused larger prime passing these tests. A.4 and Chebotarev prove termination of each search. Put $m_j = v_3(\det D_{q_j})$. The matrix D_{q_j} is effective from the integral Frobenius matrices, and is invertible over $\mathbb{Q}_3$ by the weight calculation. Its inverse has denominators dividing 3^{m_j}. The minimal exponent e_j such that $3^{e_j} D_{q_j}^{-1} u_{q_j}$ is integral is therefore determined by the finite representation modulo 3^{m_j+1}; it satisfies $j \le e_j \le m_j$. This constructs uniformly computable (q_j, e_j) and all finite cocycles of the actual A.4 input, rather than a formal sequence of unrelated residue classes.

Now fix an ordinary finite Turing-machine description T. Define its input coefficients by the bounded calculation

$$a_{T,q_j} = \begin{cases} 3^{e_j}, & T \text{ has not halted within } j \text{ steps}, \\ 0, & T \text{ has halted within } j \text{ steps}, \end{cases} \qquad a_{T,q} = 0 \quad (q \notin \{q_j\}). \tag{A.12.1}$$

Every coefficient query requires only a bounded simulation. To return the entire input modulo 3^n, the program constructs q_j, e_j for $j < n$, performs those finitely many bounded simulations, and returns the finite Kummer product with the surviving exponents 3^{e_j}. Every omitted coefficient is divisible by 3^n. Thus $j \ge n$ is one uniform convergence modulus, independent of whether T halts.

The input section is the actual equivariant line $\ell\kappa(a_T)$ of A.3. Its R4 coordinates are $(0, -\kappa(a_T); -\kappa(a_T), -\kappa(a_T), -\kappa(a_T))$ and its E36 parameter is $-\kappa(a_T)$, with the same fixed origins as A.8. The finite Kummer product and its algebraic roots supply the corresponding finite cocycle; the preceding division-point construction supplies every required origin cochain at the requested level. Hence these programs give legitimate inputs in the existing coordinate system and include their finite cochain data.

If T halts, (A.12.1) has finite support. All of its local corrections are integral by the definition of e_j, so its obstruction vanishes. If T does not halt, (A.12.1) is the A.4 phantom input and its obstruction is nonzero. A total vanishing evaluator for these finite program descriptions would therefore decide whether an arbitrary Turing machine halts. The halting

theorem proves the second asserted impossibility. This conclusion concerns the precise uniform family (A.12.1); it does not assign an unstated encoding to all possible symbolic descriptions of completed inputs. □

Statement A.13 (Proved). The full finite-S complex, with its integral differential. *With the basis and the Frobenius u in A.7 and A.10, put $M = \mathcal{K}$, $M^+ = M^{c=1}$ and $d = (G_\alpha - 1)|_{M^+}$. There are compatible explicit quasi-isomorphisms*

$$R\Gamma(\mathbb{Z}[1/3], M/3^n) \simeq [M^+/3^n M^+ \xrightarrow{d} M/3^n M]$$

in degrees zero and one, and

$$R\Gamma_{\mathrm{cts}}(G_S, M) \simeq [M^+ \xrightarrow{d} M]. \tag{A.13.1}$$

The first lattice has rank 21 *and the second rank* 44. *The twenty-one nonzero Smith factors of d are seven units, nine factors three, and five factors nine. In particular*

$$\begin{aligned} H^0(G_S, M/3^n) &\simeq (\mathbb{Z}/3)^9 \oplus (\mathbb{Z}/3^{\min(n,2)})^5, \\ H^1(G_S, M/3^n) &\simeq (\mathbb{Z}/3^n)^{23} \oplus (\mathbb{Z}/3)^9 \oplus (\mathbb{Z}/3^{\min(n,2)})^5. \end{aligned} \tag{A.13.2}$$

The full integral matrices are printed by `code/a_k_lattice.py`*; thus (A.13.1) specifies the complex and not only its Euler characteristic.*

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. Every finite-coefficient cocycle factors through Γ. On $G_{K_0,S}$ its graph in the semidirect product of the coefficient three-group and the action image is a homomorphism to a three-group, so it kills the kernel defining P_3. This gives the required factorization; it is compatible with c. Inflation is therefore an isomorphism in degrees zero and one. Both groups have vanishing higher three-primary cohomology for these coefficients, by A.1 and the free pro-three kernel of Γ with its order-two quotient. Thus inflation is a quasi-isomorphism.

Normalize a one-cocycle on Γ to vanish on c, using averaging over its group of order two. Its value on the free generator u is arbitrary in $M/3^n$, and uniquely determines its value on cuc and on the entire group. The remaining coboundaries come precisely from $M^{c=1}/3^n$ and have value $(G_\alpha - 1)m$ at u. This construction is a cochain map from the displayed two-term complex to continuous cochains: in degree zero it is inclusion of the c-invariants; in degree one it sends a vector to the unique normalized cocycle. It induces the stated cohomology isomorphisms, hence is the claimed quasi-isomorphism. Reduction and inverse limits commute with these maps. The invariant lattice commutes with reduction because two is a unit.

For the matrix calculation, the columns of $1 + G_c$ span M^+. The program selects the twenty-one columns independent modulo three; Nakayama's lemma makes them a $\mathbb{Z}_3$-basis. It applies $G_\alpha - 1$ and performs the same certified Smith elimination as in A.7. The resulting valuations are seven zeros, nine ones, and five twos. Kernel and cokernel modulo 3^n give (A.13.2). The matrix has trivial integral kernel, consistent with $H^0(G_S, M) = 0$, and its integral cokernel is (A.7.1). □

1.5 An exact two-prime decomposition at seven

Put $\mathfrak{q} = (2 - \zeta)$ and $\overline{\mathfrak{q}} = (3 + \zeta)$ in $\mathbb{Z}[\zeta]$. Their residue fields are $\mathbb{F}_7$, with $\zeta = 2$ at $\mathfrak{q}$ and $\zeta = 4$ at $\overline{\mathfrak{q}}$. Write

$$S_7 = \{\lambda, \mathfrak{q}, \overline{\mathfrak{q}}, \infty\}, \quad G = G_{K_0,S_7}, \quad P_7 = G(3), \quad D_{\mathfrak{p}} = G_{K_{0,\mathfrak{p}}}(3).$$

Here (3) denotes the maximal pro-three quotient. Choose an embedding of the decomposition group at $\mathfrak{q}$ into G, and choose the embedding at $\bar{\mathfrak{q}}$ by conjugating it with the fixed complex conjugation c. The star in the following statement is the free product in the category of pro-three groups.

Statement A.14 (Proved). The two local factors generate the entire global pro-three group. *The natural morphism*

$$D_{\mathfrak{q}} *_3 D_{\bar{\mathfrak{q}}} \longrightarrow P_7 \tag{A.14.1}$$

is an isomorphism. Each factor has the explicit pro-three presentation

$$D_{\mathfrak{q}} = \langle \phi, \tau \mid \phi\tau\phi^{-1} = \tau^7 \rangle_3 \simeq \mathbb{Z}_3(1) \rtimes \mathbb{Z}_3, \tag{A.14.2}$$

where ϕ is arithmetic Frobenius and τ is tame inertia. The relative pro-three quotient of $G_{\mathbb{Q},\{3,7,\infty\}}$ over $\mathbb{Q}(\zeta)$ is therefore

$$\Gamma_7 \simeq (D_{\mathfrak{q}} *_3 D_{\bar{\mathfrak{q}}}) \rtimes \langle c \rangle, \qquad c^2 = 1, \tag{A.14.3}$$

with c exchanging the conjugately embedded factors. In particular the local morphisms in (A.14.1) are injective; this is part of the conclusion, not an assumption about realization of local extensions.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379); the one-prime higher-lifting problem in B.4.

Proof. We first prove exactly the comparison with a maximal pro-three quotient that will be used. Let H be any profinite group, $H \twoheadrightarrow H(3)$ its maximal pro-three quotient, and J its kernel. A nonzero invariant character $\eta : J \to \mathbb{F}_3$ would have a closed kernel normal in H. The resulting quotient $H/\ker\eta$ would be an extension of the pro-three group $H(3)$ by the group $\mathbb{F}_3$, hence a pro-three group. The defining maximality would make its map from H kill J, a contradiction. Thus

$$H^1(J, \mathbb{F}_3)^{H(3)} = 0, \qquad H^2(H(3), \mathbb{F}_3) \hookrightarrow H^2(H, \mathbb{F}_3). \tag{A.14.4}$$

The second assertion is the inflation map in the five-term Hochschild–Serre sequence. No assertion about all higher cohomology of this quotient is needed.

Let $U = \operatorname{Spec} \mathbb{Z}[\zeta, 1/(3 \cdot 7)]$. The ring is a localization of a principal ideal domain, so $\operatorname{Pic}(U) = 0$. The Kummer sequence, the fixed isomorphism $\mathbb{F}_3 \simeq \mu_3$ sending 1 to ζ, and Milne II.2.9 [Mil06] identify

$$H^1(G, \mathbb{F}_3) = \langle \zeta, \lambda, \pi, \bar{\pi} \rangle_{\mathbb{F}_3}, \qquad \pi = 2 - \zeta, \quad \bar{\pi} = 3 + \zeta, \tag{A.14.5}$$

and identify $H^2(G, \mathbb{F}_3)$ with $\operatorname{Br}(U)[3]$. Milne II.2.1 and Remark II.2.2(a) give its explicit invariant description

$$\operatorname{Br}(U)[3] = \{(x_\lambda, x_{\mathfrak{q}}, x_{\bar{\mathfrak{q}}}) \in (\tfrac{1}{3}\mathbb{Z}/\mathbb{Z})^3 : x_\lambda + x_{\mathfrak{q}} + x_{\bar{\mathfrak{q}}} = 0\}. \tag{A.14.6}$$

Consequently restriction from the full group G to the two local groups at seven is an isomorphism in degree two with these coefficients. Its inverse on invariants is $(x, y) \mapsto (-x - y, x, y)$. In particular it is injective. Combined with (A.14.4) and the natural local inflation maps, this proves the injection

$$H^2(P_7, \mathbb{F}_3) \hookrightarrow H^2(D_{\mathfrak{q}}, \mathbb{F}_3) \oplus H^2(D_{\bar{\mathfrak{q}}}, \mathbb{F}_3). \tag{A.14.7}$$

Indeed a class restricting to zero in both pro-three local groups inflates to a full global class with zero restrictions, and is therefore zero by (A.14.6) and (A.14.4).

For degree one, the two local Kummer groups each have dimension two: one uniformizer class and one residue-unit class. The unramified Kummer values of ζ and λ are

$$\begin{array}{c|cc} & \kappa_\zeta & \kappa_\lambda \\ \hline \phi_{\mathfrak{q}} & 2 & 0 \\ \phi_{\bar{\mathfrak{q}}} & 2 & 2 \end{array} \quad \text{in } \mathbb{F}_3. \tag{A.14.8}$$

These values follow by taking the square of the residue and expressing it as a power of the residue of ζ. For example $\lambda = -1$ at $\mathfrak{q}$, while $\lambda = 4 = \zeta$ at $\bar{\mathfrak{q}}$. The determinant of (A.14.8) is $4 = 1$ modulo three. The valuations of π and $\bar{\pi}$ at the two places form the identity matrix. Thus the restriction

$$H^1(P_7, \mathbb{F}_3) \longrightarrow H^1(D_{\mathfrak{q}}, \mathbb{F}_3) \oplus H^1(D_{\bar{\mathfrak{q}}}, \mathbb{F}_3) \tag{A.14.9}$$

is an isomorphism. With Frobenius lifts fixing the respective uniformizer roots, the complete matrix, in row order $(\phi_{\mathfrak{q}}, \tau_{\mathfrak{q}}, \phi_{\bar{\mathfrak{q}}}, \tau_{\bar{\mathfrak{q}}})$ and column order $(\zeta, \lambda, \pi, \bar{\pi})$, is

$$\begin{pmatrix} 2 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 2 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{A.14.10}$$

Here both tame generators are oriented using the same global ζ. Hence $c\tau_{\mathfrak{q}}c^{-1} = \tau_{\bar{\mathfrak{q}}}^{-1}$. Using the conjugately transported second generator instead negates the last row; it does not change invertibility. This specifies the orientation in the factor exchange in (A.14.3).

For completeness we give the group-theoretic detection argument. Let $B = D_{\mathfrak{q}} *_3 D_{\bar{\mathfrak{q}}}$ and $f : B \to P_7$ the natural morphism. Its image is compact and therefore closed. A proper closed subgroup of a pro-three group is contained in an open subgroup of index three after passing to a finite quotient. Such a subgroup would yield a nonzero character of P_7 trivial on the image of f, contrary to (A.14.9). Thus f is surjective.

Restriction $H^2(B, \mathbb{F}_3) \to \bigoplus_{\mathfrak{p}|7} H^2(D_{\mathfrak{p}}, \mathbb{F}_3)$ is injective. To see this directly, a degree-two class is represented by a central extension of B by $\mathbb{F}_3$. Zero restriction provides homomorphic splittings on the two factors. The extension is a pro-three group, so the universal property of the free pro-three product combines these into a splitting on B. Equation (A.14.7) in particular shows that $f^* : H^2(P_7, \mathbb{F}_3) \to H^2(B, \mathbb{F}_3)$ is injective. Put $R = \ker f$. Its five-term sequence and (A.14.9) now give $H^1(R, \mathbb{F}_3)^{P_7} = 0$.

A nontrivial closed normal subgroup R of a pro-three group B always has a nonzero invariant character to $\mathbb{F}_3$. For a proof, choose a finite three-group quotient B_0 in which its image R_0 is nontrivial. The nonzero vector space $R_0/\Phi(R_0)$ is a finite module for the three-group B_0. Its coinvariants are nonzero: the augmentation ideal of $\mathbb{F}_3[B_0]$ is nilpotent, so equality of that ideal times the module with the whole module would force the module to be zero. A nonzero linear functional on the coinvariants gives an invariant character of R, a contradiction. Thus $R = 1$ and (A.14.1) is proved.

At residue characteristic seven every finite extension of three-power degree is tame. The maximal pro-three extension is obtained from the unramified three-power tower and compatible three-power roots of a uniformizer. Its tame inertia is $\mathbb{Z}_3(1)$, its unramified quotient is $\mathbb{Z}_3$, and arithmetic Frobenius acts on inertia by seventh powers. Choosing Frobenius to fix the uniformizer roots gives exactly (A.14.2). Finally the kernel of $G \to P_7$ is characteristic in G and stable under c. The chosen order-two element c therefore splits the relative quotient over $\mathbb{Q}$ and exchanges the conjugately chosen local factors, proving (A.14.3).

The elementary residue and matrix computations, including the orientation change, are reproduced in `code/a_two_prime.py` and its output file. □

Statement A.15 (Proved). The exact local equation for every section supported at three and seven. *Let Q be any of the geometric pro-three quotients of N under consideration, with the action defined by the unit tangential section at O. Write F for the action of $\phi_{\mathfrak{q}}$ on Q. Sections unramified outside $\{3,7,\infty\}$ have the exact parameterization*

$$\mathrm{Sec}_{\{3,7\}}(Q) \simeq \{(a,b) \in Q^2 : aF(b)a^{-1} = b^7\}/Q^c, \tag{A.15.1}$$

where

$$h \cdot (a,b) = (h^{-1}aF(h),\ h^{-1}bh) \quad (h \in Q^c). \tag{A.15.2}$$

The parameterization is natural under all the quotient maps. For a surjective equivariant morphism $R \twoheadrightarrow Q$ among these geometric pro-three groups, a class represented by (a,b) lifts exactly when there are lifts $A, B \in R$ satisfying

$$AF(B)A^{-1} = B^7. \tag{A.15.3}$$

The same statement holds for $R = N$. Moreover a fixed class-two section supported at these primes lifts to N exactly when it lifts to every lower-central quotient $N/\gamma_r N$. The equivalence includes compatible full lifts; no compatibility of separately chosen representatives is assumed.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379); the one-prime higher-lifting problem in B.4.

Proof. The action on N defined by the tangential basepoint is unramified away from three. Over K_0, its action on $N/\Phi(N) = V/3$ has three-group image, as proved in A.1. The kernel of $\mathrm{Aut}(N) \to \mathrm{Aut}(N/\Phi(N))$ is pro-three. One can see this on the finite lower-exponent-three quotients: successive kernels embed into homomorphism groups between finite $\mathbb{F}_3$-vector spaces, and are three-groups; their inverse limit is the stated kernel. Hence the action image over K_0 on N, and on each Q, is pro-three. The graph of any Q-valued cocycle on G_{K_0,S_7} is a homomorphism into the pro-three semidirect product of Q with this action image. It factors through P_7 by its defining maximality. Thus the sections in question factor through the exact group Γ_7 of A.14.

Every cocycle on $\langle c \rangle$ with values in a pro-three group is a coboundary. For finite three-groups this follows inductively along c-stable central elementary abelian subgroups: the induction leaves an abelian cocycle, and division by two solves it. For a pro-three group apply the finite assertion to c-stable finite quotients; the nonempty closed sets of conjugating elements have the finite-intersection property, so compactness supplies a common solution. We may therefore represent any section by a cocycle β with $\beta(c) = 1$. The remaining conjugacies are precisely by Q^c.

The restriction of a normalized cocycle to the first local factor is determined by $a = \beta(\phi_{\mathfrak{q}})$ and $b = \beta(\tau_{\mathfrak{q}})$. Tame inertia acts trivially on Q, because the reference action is unramified at seven. The single relation in (A.14.2) therefore becomes $aF(b)a^{-1} = b^7$. Conversely any pair satisfying this relation gives a continuous splitting over the local pro-three group: send the two generators to (a,ϕ) and (b,τ) in $Q \rtimes D_{\mathfrak{q}}$ and use its pro-three presentation. On the conjugate factor define

$$\beta(cdc^{-1}) = c(\beta(d)) \quad (d \in D_{\mathfrak{q}}).$$

The free-product property extends these local splittings uniquely over P_7, and the displayed identity extends them with $\beta(c) = 1$ over Γ_7. Conjugation of the two generator values by $h \in Q^c$ gives exactly (A.15.2). This proves (A.15.1) and its naturality.

The fixed-point map $R^c \to Q^c$ is surjective. Indeed, for a lift r of a fixed element of Q, the discrepancy $r^{-1}c(r)$ is a cocycle in the pro-three kernel of $R \to Q$. The preceding coprime argument removes it by an element of that kernel. Thus changes of normalized representatives

can be lifted, and (A.15.3) gives the stated criterion for the conjugacy class, not only for an individually chosen pair.

Finally fix the pair (a, b) in the class-two quotient. In the compact space of its lifts to N^2, the condition

$$AF(B)A^{-1}B^{-7} \in \gamma_r N$$

defines a closed subset. Lifting at level r makes this set nonempty, by lifting the two elements from $N/\gamma_r N$ to N; the representative issue has just been resolved. These sets are nested as r increases. Compactness provides a pair belonging to all of them. The intersection $\bigcap_r \gamma_r N$ is trivial: any nontrivial element survives in a finite three-group quotient, and every such quotient is nilpotent. The resulting pair satisfies the full equation in N, proving the last assertion. □

Equations (A.15.1)–(A.15.3) remove the additional global embedding problem at the specified prime seven. They do not evaluate the full local equation for every ramified mixed inertia vector. In particular A.14–A.15 alone do not settle the all-nilpotent assertion in B.4 or the general ramified metabelian classification in A.6.

Statement A.16 (Proved). A positive-density set with the same exact one-prime reduction. *Define the following explicit set of rational primes:*

$$\Sigma = \{q : q \equiv 4 \text{ or } 7 \pmod 9, \quad 3^{(q-1)/3} \not\equiv 1 \pmod q\}. \tag{A.16.1}$$

It has natural density $2/9$. *For every* $q \in \Sigma$, *write* $(q) = \mathfrak{q}\overline{\mathfrak{q}}$ *in* $\mathbb{Z}[\zeta]$. *The natural local morphisms give an isomorphism*

$$G_{K_0,\{\lambda,\mathfrak{q},\overline{\mathfrak{q}},\infty\}}(3) \simeq D_{\mathfrak{q}} *_3 D_{\overline{\mathfrak{q}}}, \quad D_{\mathfrak{q}} = \langle \phi, \tau \mid \phi\tau\phi^{-1} = \tau^q \rangle_3. \tag{A.16.2}$$

Its relative quotient over $\mathbb{Q}$ *is the semidirect product by complex conjugation exchanging the factors. The exact section parameterization and lifting assertions of A.15 hold with* 7 *replaced throughout by* q. *In particular the defining equation is* $aF_q(b)a^{-1} = b^q$, *with the same action of* Q^c *on pairs. The primes in* Σ *up to* 1000 *are*

$$\begin{gathered} 7, 13, 31, 43, 79, 97, 139, 157, 211, 223, 229, 241, \\ 277, 283, 313, 331, 337, 349, 373, 409, 421, 457, 463, \\ 571, 601, 607, 673, 691, 709, 733, 751, 769, 823, 859, 877, 907. \end{gathered} \tag{A.16.3}$$

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379); the one-prime higher-lifting problem in B.4.

Proof. Let $q \equiv 1 \pmod 3$ be a rational prime different from three, choose one residue root of ζ at $\mathfrak{q}$, and write

$$a = \kappa_\zeta(\phi_{\mathfrak{q}}) = \frac{q-1}{3} \bmod 3, \qquad b = \kappa_\lambda(\phi_{\mathfrak{q}}) \in \mathbb{F}_3.$$

Conjugation sends ζ to ζ^{-1} and λ to $-\zeta^2\lambda$. Its action on the Kummer coefficients μ_3 is multiplication by minus one. Thus the conjugately chosen Frobenius at $\overline{\mathfrak{q}}$ has Kummer vector $(a, a-b)$, with both characters measured using the same global ζ. In particular the determinant of the two unramified-unit rows is

$$\det \begin{pmatrix} a & b \\ a & a-b \end{pmatrix} = a(a+b) \quad \text{in } \mathbb{F}_3. \tag{A.16.4}$$

The identity $\lambda^2 = -3\zeta$ gives $\kappa_3 = 2b - a = -a - b$. Hence (A.16.4) is nonzero precisely when $a \neq 0$ and the residue of 3 is not a cube. These are exactly the two elementary conditions in (A.16.1).

Since $\mathbb{Z}[\zeta]$ is principal, choose a generator π of $\mathfrak{q}$ and its conjugate for $\bar{\mathfrak{q}}$. The four S-unit Kummer classes are $\zeta, \lambda, \pi, \bar{\pi}$. Their inertia submatrix is the identity, while (A.16.4) is their unramified-unit submatrix. For $q \in \Sigma$ this proves the degree-one isomorphism used in A.14. The Brauer invariant proof is unchanged, since exactly three finite places are inverted. All group-theoretic maps and kernel arguments in A.14 consequently apply to these actual local morphisms and prove (A.16.2). The local tame presentation has exponent q; $q \equiv 1 \pmod 3$ ensures that its Frobenius action on three-inertia factors through $\mathbb{Z}_3$. The construction of normalized cocycles and the compactness argument in A.15 now give all the stated section assertions.

For the density assertion put

$$L = K_0(\zeta_9, \sqrt[3]{\lambda}).$$

The Kummer classes of ζ and λ are independent: valuation at λ first forces the latter exponent to be zero modulo three, and ζ is not a cube in K_0, as its unit group is μ_6. Hence $[L : K_0] = 9$ and its Galois group is $\mathbb{F}_3^2$, with the two displayed Kummer coordinates. The relation $\bar{\lambda} = -\zeta^2\lambda$ makes L stable under complex conjugation: a conjugate cube root is $-\zeta_9^2\sqrt[3]{\lambda}$. Thus $L/\mathbb{Q}$ is Galois of degree eighteen. The chosen complex conjugation acts on the normal subgroup by

$$(a, b) \longmapsto (a, a - b).$$

The admissible vectors are exactly $(1, 0), (1, 1), (2, 0), (2, 2)$; this four-element subset is stable under conjugation in the full group. A rational prime outside three belongs to Σ exactly when its Frobenius class belongs to this subset. Chebotarev [Mil20b, V.3.23] therefore gives natural density $4/18 = 2/9$. The exhaustive elementary check through 1000, including both residue roots, the conjugation identity, the κ_3 identity and the determinant test, is in `code/a_two_prime.py` and its output. □

1.6 A strict restriction on ramification in the higher quotients

Statement A.17 (Proved). Tame abelian inertia in a metabelian section lies on one cusp line. *Let k be a finite extension of $\mathbb{Q}_q$ with $q \neq 3$, and let*

$$s : G_k \longrightarrow (N/N'') \rtimes G_k$$

be a continuous section. For a generator of tame three-inertia, write t for its geometric value and w for its image in $V = N^{\mathrm{ab}}$. Then

$$w = 0 \quad \text{or} \quad w \in \mathbb{Z}_3 e_d \setminus \{0\} \text{ for a unique cusp } d \in D(k). \tag{A.17.1}$$

In particular the assertion T1 of Lead 2, refuted for class two by A.5, holds for the tame abelian inertia of every metabelian section through the actual quotient $N/N'' \twoheadrightarrow N/\gamma_3 N$.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. Put $Q = N/N''$ and let Q_k denote the size of the residue field of k. The good marked model and its integral tangent make the reference action of inertia on N trivial. Consequently the restriction of the cocycle of s to inertia is a homomorphism; wild inertia and the tame factors prime to three have trivial image in Q. For an arithmetic Frobenius lift $F = s(\phi_k)$ the exact relation in the arithmetic quotient is

$$FtF^{-1} = t^{Q_k}. \tag{A.17.2}$$

It follows on abelianization that w is in W: its image in T_3E is a Q_k-eigenvector, and that Tate module has Frobenius eigenvalues of complex absolute value $\sqrt{Q_k}$ and is torsion-free.

For $n \geq 1$ define the actual subgroups of the full geometric group

$$M_n = \ker(N \to V/3^n) = N'N^{3^n}, \qquad H_n = \{g \in N : g \bmod M_n \in \langle w \bmod 3^n \rangle\}. \quad \text{(A.17.3)}$$

Here products and powers denote closed subgroups. The group H_n is open and normal in N, contains N', and is stable under the action induced by the local section. Indeed inertia acts trivially on V, while Frobenius sends the generating vector w to $Q_k w$, and Q_k is a three-adic unit. Crucially, the inclusion $N' \subset H_n$ gives

$$N'' = [N', N'] \subset [H_n, H_n], \qquad H_n^{\mathrm{ab}} \xrightarrow{\sim} (H_n/N'')^{\mathrm{ab}}. \quad \text{(A.17.4)}$$

The displayed isomorphism is induced by the quotient map; its kernel is explicitly $N''[H_n, H_n]/[H_n, H_n] = 0$.

The subgroup $(H_n/N'')s(G_k)$ of $Q \rtimes G_k$ is open. Pull it back through the actual morphism $N \rtimes G_k \to Q \rtimes G_k$. The resulting arithmetic subgroup defines a finite etale cover $Y_n \to U_k$, with geometric group H_n. This cover has the smooth proper tame compactification needed for the weight argument in B.7. We spell out why that argument applies to the present metabelian section, without assuming a full section lift. The cocycle values on arithmetic inertia are powers of t and therefore lie in H_n/N''. Thus the pulled-back subgroup contains the original arithmetic inertia. Original inertia centralizes N, so its action on every coset in N/H_n is trivial. Prime-to-q specialization therefore descends the cover to the good marked model. At each boundary point its compactification is the tame normalization $z^e = x$ after an unramified extension, with e prime to q; it is smooth there and etale away from the boundary. This is the specialization comparison used in B.7, with the same hypotheses.

Choose any lift $\widetilde{t} \in H_n$ of t, and any lift $\widetilde{F}$ of the Frobenius value in this arithmetic cover subgroup. Relation (A.17.2) says

$$\widetilde{F}\widetilde{t}\widetilde{F}^{-1}\widetilde{t}^{-Q_k} \in N''.$$

By (A.17.4), the class of $\widetilde{t}$ in H_n^{ab} is therefore an exact Q_k-eigenvector. Let X_n denote the smooth proper compactification of Y_n. Its boundary sequence is the integral exact sequence

$$0 \longrightarrow \mathbb{Z}_3(1)[X_n \setminus Y_n]/\mathbb{Z}_3(1)\delta_n \longrightarrow H_n^{\mathrm{ab}} \longrightarrow T_3 J_{X_n} \longrightarrow 0. \quad \text{(A.17.5)}$$

Frobenius on the last term has complex absolute values $\sqrt{Q_k}$, by the smooth proper finite-field weight theorem used in B.7. It has no Q_k-eigenvector even integrally, because the lattice is torsion-free. Thus the class of $\widetilde{t}$ is in the actual boundary submodule in (A.17.5).

The finite cyclic three-group H_n/M_n is generated by the image of $\widetilde{t}$. The boundary submodule therefore maps onto this group. At least one boundary inertia subgroup maps onto it: otherwise all its images would be contained in the unique maximal subgroup of this cyclic group. Such a boundary generator is conjugate in N to c_d^e for some cusp $d \in D$ and some positive integer e. Its image in $V/3^n$ is ee_d. The assertion that this element generates H_n/M_n means that w mod 3^n is a scalar multiple of e_d mod 3^n. If the cyclic group is trivial, the same conclusion holds with scalar zero.

There are only nine cusps. One cusp occurs for an unbounded subsequence of n, and its line $\mathbb{Z}_3 e_d$ is closed in V. Equivalently, e_d is primitive and the quotient of V by that line is a free $\mathbb{Z}_3$-module, so congruence membership for unbounded n gives exact membership. This proves $w \in \mathbb{Z}_3 e_d$. Distinct cusp lines have zero intersection: in $\mathbb{Z}_3[D]/\mathbb{Z}_3\delta$ an equality $ae_d = be_{d'}$ with $d \neq d'$ has zero coefficients at the other seven cusps, forcing the diagonal coefficient and then a, b to be zero. Thus the cusp is unique when $w \neq 0$. Frobenius sends e_d to $Q_k e_{\phi_k d}$. Together with $F(w) = Q_k w$, this forces $\phi_k d = d$. The cusp scheme is unramified at q, so this is precisely $d \in D(k)$, proving (A.17.1). $\square$

Statement A.18 (Proved). Class-two mixed examples have no metabelian or marked-depth lift. *Let I_{mark} be the closed normal subgroup generated by the inertia at Q_c, so that the marked depth quotient in the task is $N/\gamma_3(I_{\mathrm{mark}})$. The class-two sections of A.5 and A.6 have no lift to either*

$$N/N'' \quad or \quad N/\gamma_3(I_{\mathrm{mark}}). \tag{A.18.1}$$

This holds for every class-two lift of their specified N^{cc} sections, and continues to hold after every finite number-field extension. Thus their section germs also have no lift to these two quotients.

More generally, a local section of the marked depth quotient over a finite extension $k/\mathbb{Q}_q$, $q \neq 3$, has tame abelian inertia w whose image in $V/\mathbb{Z}_3 e_{Q_c}$ is zero or lies on a single cusp line belonging to $D \setminus \{Q_c\}$ in that quotient. The images of both section maps in (A.18.1) to class two are therefore proper; the entire unramified family of A.20 still belongs to both images.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. The two comparison maps to class two are genuine quotient maps. Indeed $N'' \subset \gamma_4 N \subset \gamma_3 N$, and $\gamma_3(I_{\mathrm{mark}}) \subset \gamma_3 N$ because $I_{\mathrm{mark}} \subset N$. At each prime used in A.5 or A.6, the specified abelian tame inertia is a nonzero scalar multiple of

$$w_0 = e_O + e_P.$$

It is on no cusp line. In the basis of W given by the eight cusps other than O, its coefficient at P is zero and its coefficients at the other seven cusps are all minus one. This vector is neither a multiple of a basis vector nor of e_O, whose eight coefficients are all minus one. Multiplication by a nonzero three-adic scalar does not change this conclusion. A.17 consequently rules out every metabelian lift.

For the marked quotient use the explicit filling morphisms

$$N/\gamma_3(I_{\mathrm{mark}}) \twoheadrightarrow N/I_{\mathrm{mark}} \xrightarrow{\sim} \pi_1(E_{\overline{\mathbb{Q}}} setminus(D \setminus \{Q_c\}))^{(3)}. \tag{A.18.2}$$

The second map is induced by filling the puncture Q_c. It is an isomorphism because a finite three-group etale cover of U extends etale across that puncture exactly when its inertia there is trivial. Normal closure kills precisely that inertia condition, and the finite covers on the two sides are thereby identified. The same map on abelianizations is

$$V \longrightarrow V/\mathbb{Z}_3 e_{Q_c},$$

as follows from the boundary exact sequence, with the single summand at Q_c removed. These morphisms preserve the arithmetic action and the tangent at O.

The resulting open elliptic curve has eight cusps and good marked reduction at $q \neq 3$. The proof of B.7 applies verbatim to its full geometric pro-three group: it uses only the finite-cover boundary sequence, smooth proper weights, and compactness of the lifted cusp set. Hence every nonzero tame inertia of the projected full section lies in one cusp inertia subgroup. Its abelian image is zero or on that cusp line, which proves the more general assertion of the statement.

In the filled curve the images of the seven cusps other than O, Q_c form a basis of its boundary lattice. The image of w_0 has coefficient zero at P and minus one at the other six basis cusps. It is not zero and is on no cusp line by the same coordinate comparison. Thus (A.18.2) rules out the marked-depth lifts of A.5 and A.6 as well.

Under a finite extension of local fields the tame inertia generator maps to a nonzero integer multiple of the old one, with that integer equal to the ramification index after compatible choices. The new vector is consequently a nonzero scalar multiple of the same mixed vector; changes of decomposition group merely conjugate and permute its two distinct cusps. The two line exclusions remain valid over every such extension, including those of degree divisible by three. A hypothetical lift over a finite number field would give a forbidden local lift at a place above one of the specified ramified primes. This proves the finite-extension and germ assertions. Finally the full unramified lifts of A.20 project through both quotient maps, while A.5 supplies a class outside both images. The images are therefore nonempty and proper. The integral line exclusions before and after filling are independently certified by unit two-by-two minors in `code/a_higher_checks.py` and its output. □

Statement A.19 (Proved). The complete local metabelian image at every good prime. *Let $k/\mathbb{Q}_q$ be finite with $q \neq 3$, put $Q = N/N''$, and retain the unit tangential origin at O. Every continuous section of $Q \rtimes G_k$ is either unramified or contained in the decomposition group of a unique lifted k-rational cusp. It is unramified exactly when its abelianization is unramified.*

More precisely, with F denoting the reference arithmetic Frobenius action and Q_k the residue-field cardinality, the local section set is the disjoint union

$$\mathrm{Sec}_k(Q) = Q/\{a \sim h^{-1}aF(h) : h \in Q\} \;\sqcup \coprod_{d \in D(k)} \{u \in k^{\times \wedge_3} : v_k(u) \neq 0\}. \tag{A.19.1}$$

The first term consists of unramified sections, with a their geometric Frobenius value. The term at d uses its actual unit tangent section $s_{d,1}$ and the map $g \mapsto c_d^{\kappa(u)(g)} s_{d,1}(g)$, followed by the quotient to Q. Its parameter group is

$$k^{\times \wedge_3} \simeq \mathbb{Z}_3 \oplus \mu_{3^{v_3(Q_k - 1)}}(k).$$

The morphism $\mathrm{Sec}_k(N) \to \mathrm{Sec}_k(Q)$ is surjective and is bijective on the ramified parts. The ramified image of $\mathrm{Sec}_k(Q) \to \mathrm{Sec}_k(N/\gamma_3 N)$ is exactly the full cuspidal packet of B.8, with the same parameters, including the central inertia and Frobenius coordinates. In particular no extra local metabelian section remains over a class-two section with a noncuspidal central coordinate.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. We give the inverse-limit and integral arguments that strengthen the line condition of A.17 to an actual cusp subgroup. Put $M = Q' = N'/N''$. Let $t \in Q$ be the tame inertia value, $w \in V$ its abelianization, and $\widetilde{t} \in N$ a lift. Use the groups M_n, H_n of (A.17.3), and set

$$H_\infty = \bigcap_n H_n = \{g \in N : g^{\mathrm{ab}} \in \mathbb{Z}_3 w\}. \tag{A.19.2}$$

All these groups contain N'. The same good-reduction cover argument as in A.17 makes the image of $\widetilde{t}$ in every H_n^{ab} lie in its integral boundary submodule B_n.

First we justify the precise inverse limits. For any descending sequence of closed subgroups of a profinite group, their images in a fixed finite quotient eventually stabilize to the image of their intersection. This follows by compactness: a compatible nonempty sequence of fibers over the fixed finite quotient has nonempty intersection. The images of their closed commutator subgroups therefore stabilize to the commutator of that intersection's image. Applying this to the H_n and testing in every finite quotient of N proves

$$\bigcap_n H_n' = H_\infty', \qquad H_\infty^{\mathrm{ab}} \xrightarrow{\sim} \varprojlim_n H_n^{\mathrm{ab}}. \tag{A.19.3}$$

For surjectivity of the second map, representatives of a compatible family define nested compact cosets of H'_n; their intersection gives a representative in H_∞. Thus the image of $\widetilde{t}$ lies in $\varprojlim B_n \subset H_\infty^{\mathrm{ab}}$.

Let $C_n(e)$ be the geometric cusps of the cover Y_n above a base cusp e. Since H_n contains N', these finite sets and their inertia exponents are explicitly

$$C_n(e) = V/(\mathbb{Z}_3 w + \mathbb{Z}_3 e_e + 3^n V), \qquad j_n(e) = \min\{j \geq 1 : j e_e \in \mathbb{Z}_3 w + 3^n V\}. \tag{A.19.4}$$

The minimum is a positive power of three. A basis vector at a cusp in $C_n(e)$ represents a conjugate of $c_e^{j_n(e)}$. For the map from level $n+1$ to level n, the boundary-module transition is the pushforward on cusp sets multiplied by $j_{n+1}(e)/j_n(e)$. This follows directly from the map on tame inertia generators, so the transition is specified integrally, including ramification. There is an exact sequence of compact modules

$$0 \to \mathbb{Z}_3(1)\delta_n \to \bigoplus_{e \in D} \mathbb{Z}_3(1)[C_n(e)] \to B_n \to 0. \tag{A.19.5}$$

The diagonal transition multiplies by $[H_n : H_{n+1}]$: over each cusp, the number of its preimages times their ramification index equals the cover degree.

For every inverse sequence of compact abelian groups the first derived limit is zero. Indeed, in the standard difference equation $x_n - f_n(x_{n+1}) = y_n$, any finite set of equations can be solved backwards. Their solution sets are closed in the compact product of the groups and have the finite-intersection property, which solves all equations simultaneously. This applies to the diagonal groups in (A.19.5), without asserting a Mittag–Leffler property for their underlying discrete groups.

Suppose first that $w = 0$. Then $H_\infty = N'$, $j_n(e) = 3^n$ for every cusp, and every cusp-module transition in (A.19.5) has a factor three. Its inverse limit is zero: each coefficient at a fixed level is divisible by arbitrarily high powers of three. The diagonal transitions have the same vanishing inverse limit, and the preceding derived-limit argument gives $\varprojlim B_n = 0$. By (A.19.3) the image of $\widetilde{t}$ in $N'/N'' = M$ is zero. Hence

$$w = 0 \quad \Longrightarrow \quad t = 1. \tag{A.19.6}$$

Now suppose $w \neq 0$. By A.17 there is a unique $d \in D(k)$ and $a \in \mathbb{Z}_3 \setminus \{0\}$ with $w = a e_d$. Write $a = 3^s u$, $u \in \mathbb{Z}_3^\times$. For $n > s$ the exponents in (A.19.4) are

$$j_n(d) = 3^s, \qquad j_n(e) = 3^n \quad (e \neq d). \tag{A.19.7}$$

For the second equality, distinct cusp vectors form a primitive independent pair in V: this is immediate in $\mathbb{Z}_3[D]/\mathbb{Z}_3\delta$ by choosing a third cusp to eliminate. Thus no smaller power of e_e can belong to the line generated by $a e_d$ modulo 3^n. The transition degree is 3^9 for $n > s$. Consequently the diagonal inverse limit is zero, all summands at $e \neq d$ have zero inverse limit, and the remaining summand gives the exact isomorphism

$$\varprojlim_n B_n \simeq \mathbb{Z}_3(1)[[A_d]], \qquad A_d = V/\mathbb{Z}_3 e_d. \tag{A.19.8}$$

Here the right side means the completed permutation module $\varprojlim_n \mathbb{Z}_3(1)[A_d/3^n A_d]$, with pushforward transition maps. Its basis measure at a coset represented by $g \in N$ maps to the class of $g c_d^{3^s} g^{-1}$ in H_∞^{ab}. The isomorphism follows from (A.19.5), including its diagonal term and the vanishing of that term's first derived limit; no closure or rationalization replaces the integral boundary module.

The actual Frobenius action on this module is Q_k times an affine permutation of A_d. To specify its origin, let $v_F \in V$ be the geometric Frobenius value of the given section and use

the cusp-path cocycle of A.8, so that reference Frobenius sends c_d to $b_{F,d}^{-1}c_d^{Q_k}b_{F,d}$. The affine map on the boundary cosets is

$$\Psi(x) = F_V x + \overline{v_F - b_{F,d}} \quad (x \in A_d). \tag{A.19.9}$$

The bar means the quotient by $\mathbb{Z}_3 e_d$. This proves the sign and the tangential origin of the translation: the section contributes left translation by v_F, and the reference path contributes $-b_{F,d}$. Changing the based path conjugates this affine description by a translation and leaves the underlying boundary action unchanged.

We use the following elementary integral-measure fact. For a finite free $\mathbb{Z}_3$-module A, let $\Psi(x) = Bx + v$ be an affine automorphism with $B^m - 1$ invertible over $\mathbb{Q}_3$ for every positive integer m. Then

$$\mathbb{Z}_3[[A]]^{\Psi} = \begin{cases} \mathbb{Z}_3\delta_{x_0}, & x_0 = (1-B)^{-1}v \in A, \\ 0, & x_0 \notin A. \end{cases} \tag{A.19.10}$$

Here is a proof with the necessary uniformity. The sole possible periodic point in $A \otimes \mathbb{Q}_3$ is x_0, since translation by x_0 turns Ψ into B and $B^m - 1$ is invertible. A positive power $\Theta = \Psi^e$ has pro-three closure in the affine automorphism group: take e to kill its finite action on $A/3A$. Thus every orbit of Θ on a finite quotient $A/3^nA$ has three-power length. Fix a residue cell in A not containing x_0, and let the corresponding orbit at level n have length 3^r. For each h, there is a level $m \geq n$ at which no point of the cell is fixed modulo 3^m by $\Theta^{3^{r+h}}$. Otherwise the nested nonempty compact solution sets would give an actual periodic point in that cell. Every orbit at level m meeting this cell therefore has length divisible by 3^{r+h+1}. An invariant coefficient is constant on such an orbit, and its pushforward to the chosen level-n cell is multiplied by the ratio of orbit lengths, divisible by 3^{h+1}. Hence that level-n coefficient is divisible by every power of three and is zero. All cells away from x_0 have zero coefficient. If $x_0 \notin A$ every coefficient is zero. If $x_0 \in A$, compatibility leaves exactly one constant mass on its residue cells, giving (A.19.10).

Apply (A.19.10) to (A.19.9). The eigenvalues of F_V on $A_d \otimes \mathbb{Q}_3$ have complex absolute values Q_k or $\sqrt{Q_k}$; the rational isogeny comparison is (A.1.4), followed by the quotient of its boundary part by e_d. Hence no positive power has eigenvalue one, as required. The class of $\widetilde{t}$ is a Q_k-eigenvector in H_∞^{ab}, so its measure in (A.19.8), after removing the Tate twist, is invariant under Ψ. Its total mass is u, because every boundary basis measure has abelian image $3^s e_d$ whereas $\widetilde{t}$ has image ae_d. Thus this mass is nonzero. Formula (A.19.10) proves that $x_0 \in A_d$ and that the measure is exactly $u\delta_{x_0}$. Choose $g \in N$ representing x_0. Then

$$\widetilde{t} \equiv g c_d^a g^{-1} \pmod{H'_\infty}. \tag{A.19.11}$$

This congruence gives actual conjugacy in the metabelian group. The image J of H_∞ in Q fits into $1 \to M \to J \to \mathbb{Z}_3 ae_d \to 1$ and splits by the subgroup generated by t, or by c_d^a. Since M is abelian, its commutator subgroup is

$$J' = (T-1)M, \tag{A.19.12}$$

where T is conjugation by any element with abelian image ae_d. The image is already closed, as the continuous image of the compact group M. Write $h = gc_d^a g^{-1}$ in Q. Equations (A.19.11)–(A.19.12) give $th^{-1} = (T-1)m$ for some $m \in M$, in additive notation for M. The right side equals $m^{-1}hmh^{-1}$. Hence $t = m^{-1}hm$ in Q: the tame inertia lies in one actual cusp inertia subgroup of the metabelian group.

We recall and justify the centralizer calculation needed to conclude for the whole section. For every nonzero integer b,

$$C_Q(c_d^b) = \overline{\langle c_d \rangle}. \tag{A.19.13}$$

This is Yamaguchi, Proposition 1.1.6 [Yam23], with the full class of finite three-groups and $m = 2$. Each c_d is a free generator in some basis of N: in the surface presentation one eliminates a different cusp, which is available because there are nine cusps. For a direct proof in this case, commutation in class two first makes the abelian image of any centralizer element a multiple of e_d, since $\bigwedge^2 V$ is torsion-free. Remove that power of c_d to obtain an element of M. The completed abelian-cover chain complex of a graph with ten free generators embeds

$$M \hookrightarrow \mathbb{Z}_3[[V]]^{10}, \qquad \partial(f_i) = \sum_i (T_i - 1) f_i.$$

One obtains this injection by taking the inverse limit of the degree-one chain sequences of its finite abelian covers; (A.19.3) with $H_\infty = N'$ identifies the limiting first homology with M. This is also Proposition 1.1.2 of the cited paper. The ring $\mathbb{Z}_3[[V]] \simeq \mathbb{Z}_3[[X_1, \ldots, X_{10}]]$ is an integral domain, and $T_d^b - 1$ is nonzero. Its multiplication is therefore injective on M, which proves (A.19.13).

After a geometric conjugation take $t = c_d^a$. Its closed generated subgroup equals that generated by $c_d^{3^s}$. Compare the actual Frobenius value with the Frobenius value of the unit tangent section at d. Both conjugate t to t^{Q_k}, so their geometric ratio centralizes a nonzero integral power of c_d. By (A.19.13) that ratio lies in its inertia group. Frobenius and inertia generate the local arithmetic image topologically; hence the whole section is contained in the cusp decomposition group. The cusp lift is unique: distinct base cusps have distinct abelian lines, and for two lifts of the same cusp any conjugator identifying a nonzero inertia element belongs to (A.19.13).

Sections in this decomposition group are exactly $H^1(G_k, \mathbb{Z}_3(1)) = k^{\times \wedge_3}$ with the specified tangential origin. A nonzero valuation is exactly nontrivial tame inertia. Their conjugacies in Q are exactly their conjugacies inside the cusp decomposition group, by the same uniqueness and centralizer argument. These packets therefore give the second, disjoint term in (A.19.1). For unramified sections the cocycle factors through $G_k/I_k = \widehat{\mathbb{Z}}$. A geometric value $a \in Q$ at arithmetic Frobenius determines it and always defines a continuous section: the closed procyclic subgroup generated by $(a, 1)$ in $Q \rtimes \widehat{\mathbb{Z}}$ maps isomorphically onto $\widehat{\mathbb{Z}}$. Indeed the map from $\widehat{\mathbb{Z}}$ sending 1 to this element composes with projection to the identity. Conjugation is precisely the relation in the first term of (A.19.1).

Finally choose any lift of an unramified Frobenius value to N and repeat this procyclic construction. Every cuspidal packet already comes from the actual full cusp decomposition group. Thus all local metabelian sections lift to N. B.7 and the injective packet parameterization give bijectivity on the ramified parts. Projecting the same actual packets to class two gives exactly B.8; its injective parameters also show that both the central inertia and the Frobenius coordinate are retained. This proves all claims. □

1.7 Complete global images and fibers for one auxiliary prime

Put $A = N/\gamma_3 N$ and, for a geometric quotient B mapping to A, put $J_B = \ker(B \to A)$. For $q \neq 3$ let $\mathscr{L}_q \subset \mathrm{Sec}_{\mathbb{Q}_q}(A)$ be the union of the unramified local section classes and the ramified cuspidal packets in B.8. Thus this subset uses the actual class-two inertia and Frobenius values and their unit-tangent origins, including all central coordinates.

Statement A.24 (Proved). The complete one-prime metabelian image and its fibers for $q \in \Sigma$. *For each $q \in \Sigma$ of A.16, put $S_q = \{3, q, \infty\}$. Then*

$$\mathrm{Sec}_{S_q}(N) \longrightarrow \mathrm{Sec}_{S_q}(N/N'') \quad \textit{is surjective.} \tag{A.24.1}$$

For either $B = N$ or $B = N/N''$, the exact image in class two is

$$\mathrm{im}\big(\mathrm{Sec}_{S_q}(B) \to \mathrm{Sec}_{S_q}(A)\big) = \mathrm{res}_q^{-1}(\mathscr{L}_q). \tag{A.24.2}$$

All fibers are as follows. A target outside the displayed subset has empty fiber. A target unramified at q has precisely the unramified fiber of (A.20.2); every lift in the present fiber is unramified outside three. For a target whose restriction at q is ramified and belongs to $\mathscr{L}_q$, choosing one normalized lift gives a bijection of its fiber with the set of right cosets

$$J_B/J_B^c. \tag{A.24.3}$$

Here J_B^c is the fixed subgroup of the actual complex conjugation, and it is not asserted to be normal. The same description for the ramified fibers of (A.24.1) is $N''/(N'')^c$.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379); Tasks A.6 and B.4 for this explicit infinite set of primes.

Proof. Use the exact relative free product of A.16 and normalize all cocycles to have value one on c. A normalized target section is determined by its pair (a, b) on the chosen local factor at q. A local full lift of its conjugacy class can be made a lift of this exact pair: lift a conjugator through the surjection of geometric groups and conjugate the local lift. A.15 then extends that pair, together with its c-conjugate pair and value one at c, to a global section over $G_{\mathbb{Q},S_q}$. A.19 supplies such a local full lift for every local metabelian section. This proves (A.24.1). B.7–B.8 and A.19 give exactly the local subset $\mathscr{L}_q$, so the same construction proves both directions of (A.24.2).

If the target is unramified at q, its abelian inertia is zero. The inertia of every full or metabelian lift is then trivial, by B.7 or (A.19.6), respectively. Such a lift factors through $G_{\mathbb{Q},\{3,\infty\}}$: the kernel of the map from $G_{\mathbb{Q},S_q}$ to this group is the closed normal subgroup generated by the inertia groups above q, and the reference action is also trivial on these groups. Thus its entire fiber is exactly the one already determined in A.20.

Now fix a ramified target in the image and fix a full or metabelian global lift $\widetilde{\beta}$ with $\widetilde{\beta}(c) = 1$ and exact projection β. Such an exact representative exists because the fixed-point map $B^c \to A^c$ is surjective, as proved in A.15. The ramified local packet injects into class two by B.8 and A.19. Consequently all local B-section lifts of the target pair are conjugate in B.

The target local section has trivial centralizer in A. Indeed a centralizer element has abelian image fixed by the twisted Frobenius, whose action on V has no eigenvalue one. It therefore lies in $\bigwedge^2 V$; the Frobenius eigenvalues there also have absolute value strictly greater than one, so that central element is zero. The local section centralizer in B is likewise trivial: any such element centralizes its nonzero cusp inertia and so lies in that cusp group, by B.4 or (A.19.13); Frobenius multiplies this group by q, with no nonzero fixed element. Hence every local pair lifting the exact target pair is uniquely of the form

$$\widetilde{\beta}|_{D_q}^{j} \quad (j \in J_B),$$

where the superscript means geometric conjugation by j, with cocycle convention $\beta^j(g) = j^{-1}\beta(g)g(j)$. The conjugator belongs to J_B, since its projection centralizes the target local section.

Extend each such local pair to a global normalized cocycle by A.15. Every element of the global fiber is obtained this way. A conjugation between two resulting global normalized cocycles belongs to B^c and its projection centralizes β; the latter centralizer is already trivial after restriction at q. Thus the conjugator lies in J_B^c. Composition of conjugations is $(\widetilde{\beta}^{\cdot j})^h = \widetilde{\beta}^{\cdot jh}$, so the identifications are exactly right multiplication by J_B^c. This proves (A.24.3). For the map to the metabelian quotient the same argument uses its trivial ramified local section centralizer and kernel N'', giving the last assertion. □

Statement A.25 (Proved). The relative arithmetic presentation at every inert prime $q \equiv 2, 5 \pmod 9$. *Put*

$$\Xi = \{q \text{ prime} : q \equiv 2 \text{ or } 5 \pmod 9\}. \tag{A.25.1}$$

This set has natural density $1/3$ and contains 2. For $q \in \Xi$ let $\mathfrak{q} = (q)$ be the unique prime of K_0 above q, and write $P_q = G_{K_0,\{\lambda,\mathfrak{q},\infty\}}(3)$. After conjugating the local embedding compatibly with the fixed global complex conjugation, there is an isomorphism

$$P_q \simeq D_{q^2} *_3 \mathbb{Z}_3^{\langle z \rangle}, \qquad D_{q^2} = \langle \phi, \tau \mid \phi\tau\phi^{-1} = \tau^{q^2} \rangle_3, \tag{A.25.2}$$

and its relative arithmetic quotient is

$$\Gamma_q \simeq (D_{q^2} *_3 \mathbb{Z}_3^{\langle z \rangle}) \rtimes \langle c \rangle, \qquad c\phi c^{-1} = \phi, \quad c\tau c^{-1} = \tau^{-1}, \quad czc^{-1} = z^{-1}. \tag{A.25.3}$$

Here $c^2 = 1$. Thus $\sigma = zc$ is another involution, with no further relation beyond (A.25.3). The source category in these presentations consists of profinite groups whose kernel over $\langle c \rangle$ is pro-three; the kernel free product is precisely the pro-three free product, as displayed.

For an inert prime $q \equiv 8 \pmod 9$, the two unit Frobenius values used in this construction are both zero modulo three. In particular $q = 17$ is the smallest prime not covered by $\Sigma \cup \Xi$; (A.25.2) is not asserted there. The union $\Sigma \cup \Xi$ has natural density $5/9$.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. The three S-unit Kummer characters over K_0 are $\kappa_\zeta, \kappa_\lambda, \kappa_q$. At the inert prime the residue field is $\mathbb{F}_{q^2}$. Write $a = (q^2 - 1)/3 \bmod 3$. The two unramified values are

$$(\kappa_\zeta(\phi), \kappa_\lambda(\phi)) = (a, 2a). \tag{A.25.4}$$

Indeed $\lambda^{q-1} = -\zeta^2$ in this residue field, because Frobenius sends ζ to ζ^2 and $\bar{\lambda} = -\zeta^2\lambda$. Consequently $\lambda^{(q^2-1)/3} = \zeta^{2(q+1)/3}$. The sign is one for odd q, since $(q+1)/3$ is even, and is one in characteristic two as well. Since $q - 1 = 1$ modulo three, this exponent is $2a$. The value a is nonzero exactly for $q \equiv 2, 5 \pmod 9$. The character κ_q gives the independent tame-inertia valuation.

Thus restriction in degree one from P_q to the local group D_{q^2} has rank two, with a one-dimensional kernel. In degree two the argument of A.14 applies with just two finite places in S: the Brauer invariants at λ and $\mathfrak{q}$ sum to zero, so restriction to $\mathfrak{q}$ is injective. Inflation from the maximal pro-three quotient is injective by (A.14.4). Choose one element complementing the local image in the global Frattini quotient. The resulting map $D_{q^2} *_3 \mathbb{Z}_3 \to P_q$ is an isomorphism by the five-term kernel proof of A.14. In particular the local morphism is injective before any assertion about an equivariant complement is made.

The relative local group over $\mathbb{Q}_q$ is $D_{q^2} \rtimes C_2$. It is explicitly obtained from the tower of unramified extensions with degrees $2 \cdot 3^n$ and compatible three-power roots of the uniformizer q. Its order-two unramified element acts on tame three-inertia by inversion and commutes with the pro-three unramified generator ϕ; the latter acts on inertia by q^2. Its morphism to the global relative quotient is injective, since it is injective on the pro-three kernel just proved and on the quotient C_2. Its order-two complement is conjugate to the fixed global c by an element of P_q, using the coprime complement argument of A.15. Conjugate the local embedding accordingly. This yields the first two action formulas in (A.25.3) with the actual chosen c.

It remains to produce the equivariant free generator. On the three global Kummer characters, conjugation is

$$c(\kappa_\zeta) = \kappa_\zeta, \quad c(\kappa_\lambda) = \kappa_\zeta - \kappa_\lambda, \quad c(\kappa_q) = -\kappa_q. \tag{A.25.5}$$

This uses conjugation on the fixed coefficient μ_3 as well as on the S-units. The Frattini quotient therefore has signs $(1,2)$, whereas the local subgroup has signs $(1,1)$ by its explicit ϕ,τ action. Over $\mathbb{F}_3$, division by two gives an invariant complement, and that one-dimensional complement has minus sign. Choose an element $x \in P_q$ lifting a nonzero vector in it and put

$$z = x\,c(x)^{-1}.$$

Then $c(z) = z^{-1}$ exactly, while its Frattini class is twice that chosen vector. It and the local Frattini image span the whole quotient. The map with this new generator in place of the initial complement is again an isomorphism by the same degree-one and degree-two argument. This proves both (A.25.2) and the full equivariant presentation (A.25.3). In terms of $\sigma = zc$, the last relation is exactly $\sigma^2 = 1$; substitution $z = \sigma c$ gives the reverse morphism of presentations, proving that no additional relation was omitted.

Chebotarev in $\mathbb{Q}(\zeta_9)/\mathbb{Q}$ assigns density $1/6$ to each invertible residue class modulo nine, so (A.25.1) has density $1/3$. It consists of inert primes, whereas Σ consists of split primes. Their densities add to $1/3+2/9 = 5/9$. Formula (A.25.4) gives zero for both units at $q \equiv 8 \pmod 9$; thus the local degree-one restriction at 17 has rank one, not two. This identifies the precise failure of the present free-product construction there. The exact residue-field computations, involution matrices, and all primes through 1000 are reproduced in `code/a_inert_prime.py` and its output. The list has 58 primes in Ξ in that range. □

Statement A.26 (Proved). The complete one-prime metabelian image and fibers for $q \in \Xi$. *For every $q \in \Xi$ and $S_q = \{3, q, \infty\}$, the conclusions (A.24.1)–(A.24.2) hold. Thus every metabelian section supported at these primes lifts to N, and for $B = N$ or N/N'' the exact image in class two is*

$$\operatorname{im}\bigl(\mathrm{Sec}_{S_q}(B) \to \mathrm{Sec}_{S_q}(A)\bigr) = \mathrm{res}_q^{-1}(\mathscr{L}_q). \tag{A.26.1}$$

For targets unramified at q, the fiber is exactly the unramified fiber of A.20. For a ramified target in this image, choose a normalized lift and let X_0 be its cocycle value at the involution $\sigma = zc$ of A.25. On J_B put

$$\Theta = \mathrm{Ad}(X_0) \circ \sigma.$$

This is an involution. The fiber is in bijection with either of the following explicitly identified sets:

$$\{j \in J_B : j\Theta(j) = 1\} \simeq J_B^{\Theta}\backslash J_B, \qquad h \longmapsto h^{-1}\Theta(h). \tag{A.26.2}$$

The quotient in the right member is a set of left cosets, without an assertion of normality. The same ramified-fiber formula for the surjection from full to metabelian sections uses $J_B = N''$ and the involution supplied by the chosen full lift.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379); Tasks A.6 and B.4, including the prime two.

Proof. Normalize the cocycle at c to one. Presentation (A.25.3), with z replaced by σc, gives the following exact data for a section with values in a geometric quotient B:

$$\begin{gathered} a \in B^c, \qquad c(b) = b^{-1}, \qquad aF_{q^2}(b)a^{-1} = b^{q^2}, \\ X \in B, \qquad X\sigma(X) = 1. \end{gathered} \tag{A.26.3}$$

The first row is its section on the relative local group at q, with values at ϕ,τ. The second is its value at the additional involution. The remaining conjugations are by B^c, acting on the first row as in (A.15.2) and on the second row by $X \mapsto h^{-1}X\sigma(h)$. The pro-three

universal property in (A.25.2) proves that these data define continuous sections, and the given presentation proves that they exhaust them.

For every equivariant pro-three surjection $R \twoheadrightarrow B$, the second-row data lift. Indeed $H^1(\langle\sigma\rangle, B) = 1$ by the coprime argument of A.15, so write $X = h^{-1}\sigma(h)$, lift h to $\widetilde{h}$ and take $\widetilde{h}^{-1}\sigma(\widetilde{h})$. This is a lift satisfying the exact involution relation. A lift of the local section in the first row can be made to have value one on c without changing its projection: its value at c is a cocycle in the pro-three kernel, and the same coprime argument conjugates it to one. The two rows can therefore be lifted independently and combined using (A.26.3).

A.19 supplies the first-row lift to N for every metabelian section. For a class-two section B.7– B.8 and A.19 say that this first row lifts exactly when its local class belongs to $\mathscr{L}_q$. This proves the asserted surjectivity and (A.26.1). If the target is unramified at q, all its full and metabelian lifts are unramified there by their zero-inertia results. The same inertia-kernel argument as in A.24 therefore identifies the fiber with A.20.

Now fix a ramified target in the image and a normalized lift. Its local lifting class is unique by the injectivity of the ramified cusp packet in class two. As in A.24, its local geometric centralizer is trivial both in the target and in B. Thus local lifts of the exact target with value one at c form one free conjugacy orbit under J_B^c. Use that conjugacy to fix the entire local first row to the chosen lift. No residual geometric conjugation remains, because the fixed local section has trivial centralizer. The remaining global fiber is consequently the literal set of second-row lifts of the target value at σ.

Write a second-row lift uniquely as jX_0, $j \in J_B$. Its involution equation is

$$jX_0\sigma(j)\sigma(X_0) = 1 \quad \Longleftrightarrow \quad j\Theta(j) = 1.$$

The identity $X_0\sigma(X_0) = 1$ proves that $\Theta^2 = 1$. Coprime H^1 for this action on J_B gives every solution as $h^{-1}\Theta(h)$. Two choices of h give the same solution precisely when they differ by left multiplication by J_B^{Θ}. This proves (A.26.2), including its coset orientation. The same argument with target N/N'' and kernel N'' gives the last assertion. □

Statement A.27 (Proved). The normalized integral Frobenius at an inert prime and the explicit Heisenberg test at seventeen. *Retain the free pro-three basis $u, v = cuc$ of P_3 from A.10, and write $[u, v] = uvu^{-1}v^{-1}$. For an inert rational prime $q \neq 3$, conjugate its local order-two complement in $P_3 \rtimes \langle c\rangle$ to $\langle c\rangle$. Let $\varphi_q \in P_3$ be the resulting arithmetic Frobenius of the prime of K_0 of norm q^2. Its image in the integral class-two quotient is exactly*

$$\varphi_q \bmod \gamma_3 P_3 = u^{a_q} v^{a_q} [u, v]^{-a_q^2/2}, \qquad a_q = \frac{\log(-q)}{\log 7} \in \mathbb{Z}_3. \tag{A.27.1}$$

The image in (A.27.1) is independent of the conjugator used to identify the local order-two complement with $\langle c\rangle$.

Put $\theta = \zeta_9$, $\beta^3 = \lambda$, and

$$u_i = 1 - \zeta^i\theta \quad (0 \le i \le 2), \qquad \omega = u_1 u_2^2, \qquad L = K_0(\theta, \beta, \omega^{1/3}). \tag{A.27.2}$$

Then L/K_0 is the full exponent-three class-two quotient of P_3: its Galois group has order 27 and presentation

$$\langle \sigma, \tau, \rho \mid \sigma^3 = \tau^3 = \rho^3 = 1, \quad \rho \text{ central}, \quad [\sigma, \tau] = \rho \rangle. \tag{A.27.3}$$

The field is Galois over $\mathbb{Q}$, and complex conjugation acts by

$$c\sigma c = \tau\sigma, \qquad c\tau c = \tau^{-1}, \qquad c\rho c = \rho^{-1}. \tag{A.27.4}$$

For every prime $q \equiv 8 \pmod 9$, the Frobenius of the prime of K_0 above q is trivial in (A.27.3). In particular this holds at 17, although $v_3(a_{17}) = 1$ in the integral formula (A.27.1). For example,

$$a_{17} \equiv 78 \pmod{81}, \qquad -a_{17}^2/2 \equiv 36 \pmod{81}. \tag{A.27.5}$$

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379); the first inert prime outside A.25 in Tasks A.6 and B.4.

Proof. The image of the local unramified group at q in the relative quotient over K_0 is procyclic with quotient C_2 and pro-three kernel. Its order-two subgroup is the prime-to-three component of this procyclic group. All complements to P_3 in $P_3 \rtimes C_2$ are P_3-conjugate, by the coprime argument of A.15. After the specified conjugation, φ_q commutes with c.

There is an exact integral logarithm description of the free class-two pro-three group on u, v: its underlying lattice has basis $U, V_0, [U, V_0]$, and multiplication is $X * Y = X + Y + \frac{1}{2}[X, Y]$. Here $U = \log u$ and $V_0 = \log v$. The fraction $1/2$ belongs to $\mathbb{Z}_3$; multiplication and the inverse maps are integral polynomial maps, so this identifies the pro-three group and the lattice with their compact topologies. The involution interchanges U, V_0 and negates $[U, V_0]$. Thus its fixed subgroup is exactly $\{\exp(a(U + V_0)) : a \in \mathbb{Z}_3\}$. The cyclotomic character sends both u, v to 7 and sends φ_q to q^2. Since the logarithm is injective on $1 + 3\mathbb{Z}_3$, it follows that $2a \log 7 = 2 \log(-q)$, giving $a = a_q$. The product formula in this lattice yields (A.27.1). Two allowed conjugators differ by an element of P_3^c; its image belongs to the displayed procyclic fixed subgroup and therefore commutes with the image of φ_q. This proves the asserted independence, with its exact ambiguity identified.

For the field in (A.27.2), direct multiplication gives

$$u_0 u_1 u_2 = \lambda, \qquad \frac{\sigma(\omega)}{\omega} = \frac{u_0^3}{\lambda}, \qquad c(\omega) = -\omega, \tag{A.27.6}$$

where initially $\sigma(\theta) = \zeta\theta$. The classes of ζ and λ are independent in $K_0^\times / K_0^{\times 3}$: the valuation of λ at the prime (λ) is one, whereas ζ is a unit and is not a cube in K_0. Consequently $K_0(\theta, \beta)/K_0$ has degree nine and λ is not a cube in $K_0(\theta)$. The kernel of

$$K_0(\theta)^\times / K_0(\theta)^{\times 3} \longrightarrow K_0(\theta, \beta)^\times / K_0(\theta, \beta)^{\times 3}$$

is the line generated by λ. To see this directly, a cube root in the cyclic cubic extension has an eigenvector under its Galois group and hence is a scalar multiple of $1, \beta$, or β^2. If the class of ω belonged to that line, the class of $\sigma(\omega)/\omega$ would be trivial. This contradicts (A.27.6). Thus adjoining $\gamma = \omega^{1/3}$ has degree three.

Choose the cube roots β, γ so that actual complex conjugation satisfies $c(\beta) = -\theta^2\beta$ and $c(\gamma) = -\gamma$. Each choice is possible by multiplying an initial cube root by a power of ζ; the ratios before and after that multiplication differ by the corresponding power of ζ. The following explicit automorphisms are then well-defined:

	θ	β	γ
σ	$\zeta\theta$	β	$u_0\gamma/\beta$
τ	θ	$\zeta\beta$	γ
ρ	θ	β	$\zeta\gamma$
c	θ^{-1}	$-\theta^2\beta$	$-\gamma$

(A.27.7)

Their powers and products, using (A.27.6), prove (A.27.3)–(A.27.4). All the radicands are units away from three, and the degrees are powers of three. The Kummer polynomials and their derivatives therefore show that the extension is unramified at every finite prime away

from three. It is consequently a quotient of P_3. The free exponent-three class-two group on two generators has the normal forms $\sigma^a\tau^b\rho^z$, with $a,b,z\in\mathbb{F}_3$, and order 27. The displayed quotient has this order, so the quotient map identifies its kernel exactly with the closed subgroup $P_3^3\gamma_3P_3$.

For a rational prime $q\equiv -1 \pmod 9$, the rational Frobenius in $\mathrm{Gal}(L/\mathbb{Q})$ acts on θ as $\theta\mapsto\theta^{-1}$. It consequently has the form gc with $g=\tau^b\rho^z$. Formula (A.27.4) gives $(gc)^2=gc(g)=1$. Its square is the Frobenius over K_0, proving its triviality in (A.27.3). Equivalently, at every residue choice of θ in $\mathbb{F}_{q^2}$, (A.27.6) gives $\omega^q=-\omega$, whence

$$\omega^{(q^2-1)/3}=(-1)^{(q+1)/3}=1.$$

The exponent on the right is even because the odd prime q is -1 modulo nine.

The program `code/a_frobenius17.py` checks (A.27.6) in the exact ring $\mathbb{Z}[\theta]=\mathbb{Z}[T]/(T^6+T^3+1)$, all 729 products under (A.27.4), and all three choices of θ at each of the 29 primes $q\equiv 8 \pmod 9$ through 1000. At seventeen one choice gives $\theta=9+11\zeta$ and $\omega=3+6\zeta$, with cube roots $2+\zeta$, $16+\zeta$, and $16+15\zeta$. The same program computes a_{17} by unique successive lifting of $7^a=-17$, without a floating-point logarithm. It gives (A.27.5) and the output through modulus 3^{12}. Finally $v_3(\log(-17))=v_3(-18)=2$ and $v_3(\log 7)=1$, proving $v_3(a_{17})=1$. □

Statement A.28 (Proved). The exact equivariant one-relator presentation at every inert prime. *Let $q\neq 3$ be any inert rational prime, and let $P_q=G_{K_0,\{3,q\}}(3)$ and $\Gamma_q=P_q\rtimes\langle c\rangle$. After the local-complement identification used in A.27, choose a tame generator t and an arithmetic Frobenius ϕ at the prime over q, with*

$$c(t)=t^{-1},\qquad c(\phi)=\phi,\qquad \phi t\phi^{-1}=t^{q^2}. \tag{A.28.1}$$

Choose a lift $x\in P_q$ of $u\in P_3$ and put $y=cxc$. The lift can and will be chosen with $\kappa_q(x)=0$ modulo three. Let $F=F_3(x,y,t)$ be the free pro-three group with involution $c(x)=y$, $c(y)=x$, $c(t)=t^{-1}$. Its natural equivariant map to P_q is surjective. There is a c-fixed pro-three word $W\in F$ mapping to the actual local Frobenius ϕ, and for every such choice the natural map gives the exact presentations

$$\begin{aligned} P_q &\simeq F/\overline{\langle\!\langle WtW^{-1}t^{-q^2}\rangle\!\rangle},\\ \Gamma_q &\simeq \bigl(F/\overline{\langle\!\langle WtW^{-1}t^{-q^2}\rangle\!\rangle}\bigr)\rtimes\langle c\rangle. \end{aligned} \tag{A.28.2}$$

These are continuous pro-three presentations, with the indicated finite operator group in the second line. In particular

$$P_q^{\mathrm{ab}}\simeq\mathbb{Z}_3^2\oplus\mathbb{Z}_3/(q^2-1),\qquad \dim_{\mathbb{F}_3}H^2(P_q,\mathbb{F}_3)=1. \tag{A.28.3}$$

At $q=17$, W lies in $\Phi(F)$, and the relator in (A.28.2) lies in the third Zassenhaus subgroup of F. Its image on killing t is the actual Frobenius of A.27; in particular its integral class-two image is given by (A.27.1) and (A.27.5).

For any of the geometric quotients B under consideration, (A.28.2) gives the following exact section parameterization. Normalize the value at c to one; assign the values $a,c(a),b$ at x,y,t. If $\Omega_W(a,b)$ denotes the value at W of this continuous free-group cocycle, the remaining equations are precisely

$$c(b)=b^{-1},\qquad \Omega_W(a,b)F_{q^2}(b)\Omega_W(a,b)^{-1}=b^{q^2}. \tag{A.28.4}$$

Classes are the orbits under B^c acting by

$$(a,b)\longmapsto(h^{-1}a\,x(h),\ h^{-1}bh). \tag{A.28.5}$$

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379); Tasks A.6 and B.4 beyond the prime families A.16 and A.25.

Proof. The Kummer group for K_0 with ramification allowed at three and q has the basis ζ, λ, q. The first two classes are pulled back from P_3, and the last has value one on t. Adjusting an initial lift of u by a power of t gives the specified value $\kappa_q(x) = 0$; because $c(\kappa_q) = -\kappa_q$, the value at y is also zero. Thus x, y, t form a basis of the Frattini quotient, and the continuous map $F \to P_q$ is onto. The involution is equivariant by its definition on these three generators.

The local generators (A.28.1) can be chosen in the relative tame local group: the unramified $\mathbb{Z}_3$ generator and its order-two component commute, and the latter acts on three-primary inertia as the Teichmüller scalar -1. Coprime complement conjugacy identifies this order-two element with the given global c. The local pro-three group maps injectively into P_q. For an explicit realization, the extension

$$K_0(\mu_{3^\infty}, q^{1/3^\infty})/K_0 \tag{A.28.6}$$

is pro-three and unramified outside three and q. At q its cyclotomic part contains every unramified extension of three-power degree: $q^2 \in 1 + 3\mathbb{Z}_3$ has infinite order, so the local cyclotomic degrees are unbounded. The uniformizer roots supply every tame three-power ramification degree. The standard tame description then identifies its local compositum with the maximal pro-three extension of $K_{0,q}$. Consequently the local decomposition map into P_q has trivial kernel.

Since ϕ is c-fixed and $F \to P_q$ is an equivariant pro-three surjection, the coprime H^1 argument of A.15 proves $F^c \to P_q^c$ is onto. This supplies $W \in F^c$ as stated. Put $r = WtW^{-1}t^{-q^2}$ and $G_W = F/(r)$, where (r) denotes the closed normal subgroup it generates. As

$$c(r) = Wt^{-1}W^{-1}t^{q^2} = t^{-q^2}r^{-1}t^{q^2},$$

this subgroup is stable under c. There is a surjective equivariant map $G_W \to P_q$, inducing an isomorphism on $H^1(-, \mathbb{F}_3)$. Here $r \in \Phi(F)$ since $q^2 \equiv 1 \pmod 3$.

The degree-two comparison is the exact one proved in A.14: full-to-maximal-pro-three inflation is injective, and the full global group has

$$H^2(G_{K_0,\{3,q\}}, \mathbb{F}_3) \simeq \{(z_3, z_q) \in (\tfrac{1}{3}\mathbb{Z}/\mathbb{Z})^2 : z_3 + z_q = 0\}. \tag{A.28.7}$$

The Kummer, Picard, and Brauer comparisons are precisely Milne II.2.9, II.2.1, and II.2.2(a) [Mil06], applied to the localization of the same principal ideal domain. Thus restriction $H^2(P_q, \mathbb{F}_3) \to H^2(D_{q^2}, \mathbb{F}_3)$ is injective. The assignments $\phi \mapsto W$, $t \mapsto t$ define a continuous map $D_{q^2} \to G_W$, whose composite to P_q is the actual local map. It follows that inflation $H^2(P_q, \mathbb{F}_3) \to H^2(G_W, \mathbb{F}_3)$ is injective. The five-term kernel argument proved in A.14 now shows that $G_W \to P_q$ is an isomorphism. This proves (A.28.2), including the stated operator action.

Abelianizing its single relator gives exactly $(1 - q^2)t$ and no relation on x, y, proving the first formula of (A.28.3). This abelianization has nonzero finite torsion; P_q is therefore not free pro-three. The minimal relation criterion used in A.10 gives $H^2(P_q, \mathbb{F}_3) \neq 0$, while its injection into $H^2(D_{q^2}, \mathbb{F}_3)$ bounds the dimension by one. The latter local group has dimension one, as follows from the Hochschild–Serre sequence for $\mathbb{Z}_3(1) \rtimes \mathbb{Z}_3$ with trivial $\mathbb{F}_3$ coefficients. This proves the second formula.

At seventeen the Frobenius values on ζ, λ are zero by (A.25.4). The chosen local Frobenius fixes the uniformizer roots, so its value on q is also zero modulo three. Conjugating the local embedding does not change these degree-one values. Therefore $W \in \Phi(F)$. If $D_n(F)$ denotes the Zassenhaus filtration, then $\Phi(F) = D_2(F)$, so $[W, t] \in D_3(F)$; also $t^{q^2-1} = t^{288} \in D_9(F)$.

The relator thus belongs to $D_3(F)$. The map killing t identifies the quotient with the unramified group P_3 and sends W to the normalized Frobenius already computed in A.27.

Finally a normalized cocycle on $F \rtimes C_2$ is exactly the arbitrary value a on x, its conjugate on y, and the value b on t satisfying $c(b) = b^{-1}$. The continuous value at the pro-three word W is defined by the free pro-three universal property; this defines Ω_W without replacing W by a truncated word. Arithmetic inertia at $q \neq 3$ acts trivially on the original geometric group, while W acts as its actual Frobenius F_{q^2}. Evaluating r gives exactly the second equation (A.28.4), and conjugating a normalized cocycle gives (A.28.5). These computations prove both directions of the parameterization. The word W in this assertion retains its possible dependence on t; (A.27.1) computes its stated quotient, not that full dependence. Solving (A.28.4) in the full geometric group for every prescribed metabelian target at seventeen remains the specific unresolved step recorded in the ledger. □

Statement A.29 (Proved). Local cuspidality at every finite derived level and its global one-prime consequences. *Write $N^{(0)} = N$ and $N^{(m+1)} = \overline{[N^{(m)}, N^{(m)}]}$, and put $Q_m = N/N^{(m)}$ for $m \geq 2$. For every finite extension $k/\mathbb{Q}_q$, $q \neq 3$, every continuous section with geometric quotient Q_m is either unramified or lies in exactly one actual cusp packet. Its inertia is trivial exactly when its abelian inertia is zero. Every such local section lifts to N, and the map from full to Q_m sections is bijective on the ramified classes. Thus the exact local description in A.19 holds with Q_m in place of N/N'', including the actual central Frobenius coordinates of its image in class two.*

For every $q \in \Sigma \cup \Xi$ and every $m \geq 2$, the global map

$$\mathrm{Sec}_{\{3,q,\infty\}}(N) \longrightarrow \mathrm{Sec}_{\{3,q,\infty\}}(Q_m) \tag{A.29.1}$$

is onto. For ramified targets its fiber is the right-coset set $N^{(m)}/(N^{(m)})^c$ when $q \in \Sigma$, and the left-coset set $(N^{(m)})^{\Theta}\backslash N^{(m)}$ when $q \in \Xi$, with the chosen-lift involution Θ of A.26. Unramified fibers are the exact unramified fibers of A.20 for the map $N \to Q_m$.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379); the successive derived approximations in Tasks A.6 and B.4.

Proof. We first identify the precise peripheral intersection statement used below. Each boundary generator c_d extends to a free pro-three basis of N: in the standard surface presentation, eliminate a different boundary generator. Yamaguchi's Proposition 1.1.6 [Yam23] gives the centralizer for every nonzero integer power. For $a = 3^s u$ with $u \in \mathbb{Z}_3^\times$, continuous exponentiation gives $c_d^{3^s} = (c_d^a)^{u^{-1}}$ and $c_d^a = (c_d^{3^s})^u$. The two elements generate the same closed procyclic subgroup and hence have the same centralizer. We therefore obtain

$$C_{Q_j}(c_d^a) = \langle c_d \rangle \quad (j \geq 2,\ a \in \mathbb{Z}_3 \setminus \{0\}). \tag{A.29.2}$$

If $g c_d^a g^{-1} = c_e^b$ with $a, b \neq 0$, abelianization gives $a e_d = b e_e$. The distinct boundary lines are disjoint away from zero, and each e_d is primitive. It follows that $d = e$ and $a = b$. Formula (A.29.2) then forces $g \in \langle c_d \rangle$. This proves the required intersection and normalizer claims through explicit maps to V and the stated centralizer theorem.

We induct on m, starting with A.19. Let a local section in Q_m have tame value t, and denote its image in Q_{m-1} by $\bar{t}$. By induction $\bar{t}$ is either one or conjugate to c_d^a, $a \neq 0$. In the latter case choose such a conjugator and lift it through $Q_m \to Q_{m-1}$; this changes the representative by an allowed geometric conjugation and makes $\bar{t} = c_d^a$ exactly.

Choose a decreasing cofinal sequence of characteristic open normal subgroups M_n of Q_{m-1}, and let $H_n \subset N$ be the inverse image of $\langle \bar{t} \rangle M_n$. Then

$$N^{(m-1)} \subset H_n, \qquad N^{(m)} \subset H_n', \qquad H_\infty := \bigcap_n H_n = \begin{cases} N^{(m-1)}, & \bar{t} = 1, \\ N^{(m-1)}\langle c_d^a \rangle, & \bar{t} = c_d^a. \end{cases} \tag{A.29.3}$$

These H_n need not be normal in N; the finite covers and their boundary points below are described by cosets and double cosets, respectively. The characteristic choice and the section's Frobenius relation make $H_n/N^{(m)}$ stable under the section. Pulling back $(H_n/N^{(m)})s(G_k)$ from $Q_m \rtimes G_k$ gives an actual arithmetic cover with geometric group H_n. It contains original arithmetic inertia because $t \in H_n/N^{(m)}$. Thus this cover has the good tame compactification used in A.17–A.19. The inclusion $N^{(m)} \subset H_n'$ identifies its actual geometric abelianization with that of $H_n/N^{(m)}$. In particular the chosen lift of t has the exact Frobenius eigenvalue $\#\mathbb{F}_k$ in H_n^{ab}, rather than only an eigenvalue in a further quotient. The compactification weight argument places it in the integral boundary submodule at every n.

As proved in A.19 by stabilization in every finite quotient and compact lifting of cosets,

$$H_\infty^{\mathrm{ab}} \simeq \varprojlim_n H_n^{\mathrm{ab}}. \tag{A.29.4}$$

We compute the inverse limit of the boundary modules with their actual transition maps. The boundary points above e are $H_n\backslash N/I_e$; the loop at $H_n g I_e$ is the appropriate conjugate of $c_e^{j_n(g,e)}$, where

$$j_n(g,e) = [I_e : I_e \cap g^{-1}H_n g]. \tag{A.29.5}$$

Transition maps sum over inverse images and multiply each term by the ratio of these ramification indices. The diagonal relation has transition map equal to the covering degree. The index $[N : H_n]$ is unbounded, because the group Q_{m-1} cannot have an open procyclic subgroup: its quotient V has rank ten. The diagonal tower consequently has zero limit, and its first derived limit is zero by the compact equation argument in A.19. The inverse limit of boundary modules is therefore the inverse limit of the weighted free boundary modules in (A.29.5).

If $\bar{t} = 1$, every intersection $I_e \cap g^{-1}H_\infty g$ is trivial, since each cusp group injects into Q_{m-1} by its primitive abelian image. Every ramification index tends to infinity. This growth is uniform on each clopen set of the compact double coset space: otherwise nested nonempty closed sets with bounded indices would give a point with nontrivial limiting intersection. The weighted transition maps are thus divisible by arbitrary powers of three at each fixed finite stage. Their inverse limit is zero. By (A.29.4), t is zero in $H_\infty^{\mathrm{ab}} = N^{(m-1)}/N^{(m)}$, proving $t = 1$.

Now let $\bar{t} = c_d^a$ and write $a = 3^s\varepsilon$ with $\varepsilon \in \mathbb{Z}_3^\times$. A bounded index in (A.29.5) is equivalent to a nontrivial intersection of $gI_e g^{-1}$ with $\langle c_d^a \rangle$ in Q_{m-1}. By (A.29.2), this forces $e = d$ and the image of g to belong to I_d. There is therefore exactly one surviving boundary double coset in $H_\infty\backslash N/I_d$. Its ramification index stabilizes at 3^s. Its 3^s sheets under $I_d/\langle c_d^a \rangle$ constitute one boundary point, not 3^s independent boundary generators. On every clopen set away from this point the same compactness argument gives uniform unbounded ramification. The inverse boundary module is consequently the single copy of $\mathbb{Z}_3(1)$ generated by the class of $c_d^{3^s}$. Projecting H_∞ to $\langle c_d^a \rangle$ shows that the class of t is $\varepsilon[c_d^{3^s}] = [c_d^a]$ in (A.29.4).

Put $A_m = N^{(m-1)}/N^{(m)}$ and let T denote conjugation by c_d^a. The image of H_∞ in Q_m is $A_m \rtimes \langle c_d^a \rangle$. Its closed commutator subgroup is exactly $(T-1)A_m$: the image is closed because A_m is compact. The equality of abelianizations just proved gives $t(c_d^a)^{-1} \in (T-1)A_m$, which is equivalent to $t = h^{-1}c_d^a h$ for some $h \in A_m$. This proves actual cusp conjugacy. After making this conjugation, compare the Frobenius value with that of the actual tangential cusp section. Their ratio centralizes c_d^a, hence lies in I_d by (A.29.2). The complete local section therefore lies in the actual cusp packet. Its parameters and uniqueness are exactly the Kummer parameters of A.19 and B.8, and its full lift is the corresponding cusp section. In the unramified case an arbitrary Frobenius value lifts to N, as proved in A.19. This completes the induction and the local statement.

For the global consequences, the arithmetic presentations A.15–A.16 and A.25 are independent of the geometric quotient. Replace the local metabelian lifting input in A.24 and A.26 by the local result just proved. The local ramified lift is still unique and has trivial full local geometric centralizer; the independent involution datum in A.26 still lifts by coprime H^1. The identical explicit maps then prove (A.29.1) and its stated fibers with kernel $N^{(m)}$. For an unramified target, the local zero-inertia result forces all full lifts to be unramified at q, so the comparison with A.20 is exact. □

Statement A.30 (Proved). The mixed Heisenberg test at seventeen and the exact first relation degree. *Use the presentation at $q = 17$ in A.28. For its Frobenius word and defining relator, the stronger exact conclusions are*

$$W \in D_3(F) \setminus D_4(F), \qquad r = WtW^{-1}t^{-289} \in D_4(F) \setminus D_5(F). \tag{A.30.1}$$

Here $D_n(F)$ is the Zassenhaus filtration, equivalently the filtration by degree in the mod-three Magnus expansion of the free pro-three group $F = F_3(x, y, t)$. If x, y, t have Magnus variables X, Y, T, the degree-four part of $r-1$ has coefficient 2 on both X^3T and Y^3T. These coefficients are independent of the other degree-three coefficients of $W - 1$.

The arithmetic input detecting the only possible degree-two component of W is the following explicit Heisenberg extension. Let

$$\alpha^3 = 3, \qquad \beta^3 = 17, \qquad u_0 = 2 + \alpha^2, \qquad u_i = \sigma^i(u_0), \quad \sigma(\alpha) = \zeta\alpha,$$

and set

$$\omega = u_1u_2^2, \qquad L_{3,17} = K_0(\alpha, \beta, \omega^{1/3}). \tag{A.30.2}$$

This field is Galois over K_0, has the group (A.27.3), and is unramified outside three and seventeen. At the prime with $\alpha \equiv 7 \pmod{17}$, the Frobenius fixing the uniformizer roots has trivial image in the center of this Heisenberg quotient.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379); the next arithmetic relation at the first remaining inert prime in Tasks A.6 and B.4.

Proof. The norm calculation is exact:

$$u_0u_1u_2 = 2^3 + \alpha^6 = 17, \qquad \frac{\sigma(\omega)}{\omega} = \frac{u_0^3}{17}. \tag{A.30.3}$$

The classes of 3 and 17 in $K_0^\times/K_0^{\times 3}$ are independent, by their valuations at the primes above three and seventeen. The degree and automorphism argument of A.27 applies with these two radicands and the norm witness u_0. In particular the last cubic extension in (A.30.2) is nontrivial: otherwise the class of ω over $K_0(\alpha)$ would lie on the line generated by 17, contradicting (A.30.3). For $\gamma^3 = \omega$, the explicit automorphisms are

	α	β	γ
σ	$\zeta\alpha$	β	$u_0\gamma/\beta$
τ	α	$\zeta\beta$	γ
ρ	α	β	$\zeta\gamma$

(A.30.4)

and $[\sigma, \tau] = \rho$. Choose positive real cube roots α, β in the fixed complex embedding. Complex conjugation interchanges u_1, u_2 and satisfies

$$c(\omega)\omega = (17/u_0)^3, \qquad c(\gamma) = 17/(u_0\gamma).$$

The second equality follows from $|\gamma|^2 = 17/u_0 > 0$, for any choice of the cube root γ. Thus the field is Galois over $\mathbb{Q}$, with

$$c\sigma c = \sigma^{-1}, \qquad c\tau c = \tau^{-1}, \qquad c\rho c = \rho. \tag{A.30.5}$$

The radicands are units away from three and seventeen, so the Kummer polynomials are unramified at every other finite prime.

At the chosen prime over seventeen, the residue field of K_0 is $\mathbb{F}_{17^2}$ and $\alpha = 7$ is a cube root of 3. In the basis $1, \zeta$, direct finite-field multiplication gives

$$(u_0, u_1, u_2) = (0,\ 4 + 2\zeta,\ 2 + 15\zeta), \qquad \omega = 7 + 10\zeta, \qquad \omega^{96} = 1. \tag{A.30.6}$$

Here ω is a unit, so its cubic residue is exactly the Frobenius value on γ. The specified local Frobenius fixes α and the uniformizer root β, and (A.30.6) shows that it fixes γ as well. Its Heisenberg image is therefore the identity. In particular this value is unchanged by conjugating the local embedding to align its order-two complement with c.

We now compare this arithmetic calculation with the actual free presentation. By A.28, $W \in D_2(F)$ and $c(W) = W$. The quotient $D_2(F)/D_3(F)$ has basis

$$[x, y], \quad [x, t], \quad [y, t] \quad \text{over } \mathbb{F}_3.$$

Indeed commutators generate this quotient, cubes already have Magnus degree three, and these three degree-two commutator polynomials are linearly independent. The involution negates the first basis vector and sends the last two to each other's negatives. Its fixed subspace is exactly the line generated by $[x, t] - [y, t]$.

Under the quotient (A.30.2), the Frattini images of x, y, t on the two Kummer coordinates $(3, 17)$ are

$$x \longmapsto (1, 0), \qquad y \longmapsto (2, 0), \qquad t \longmapsto (0, 1). \tag{A.30.7}$$

For the first coordinate this uses $\kappa_3 = -\kappa_\zeta - \kappa_\lambda$ and the two Frobenius rows of A.10. The second uses the lift choice in A.28. Consequently the fixed wedge generator maps to $(1 - 2)\rho = -\rho \neq 0$. The trivial Frobenius value in (A.30.6) forces the class of W on that line to vanish. This proves $W \in D_3(F)$.

In F^{ab}, its coordinates on x, y are a_{17}, a_{17} by A.27, and its coordinate on t is zero because it is fixed by the involution which inverts t. We have $a_{17} \equiv 6 \pmod 9$. Setting $Y = T = 0$ in its Magnus expansion therefore gives $(1 + X)^{a_{17}}$. The coefficient of X^3 is

$$\binom{a_{17}}{3} \equiv a_{17}/3 \equiv 2 \pmod 3.$$

The same holds for Y^3. Thus $W \notin D_4(F)$. If W_3 is its homogeneous degree-three term, the first possible term of $[W, t] - 1$ is $W_3T - TW_3$. The coefficient of X^3T in this expression is exactly 2, and likewise for Y^3T; no other monomial of W_3 can contribute to either coefficient. The factor $t^{1-289} = t^{-288}$ has first possible Magnus degree nine, since $288 = 9 \cdot 32$. It follows that r has exact degree four, proving (A.30.1).

The exact ring identities, all three residue choices, and the nonzero Magnus coefficients with arbitrary remaining cubic terms are reproduced by `code/a_frobenius17_mixed.py` and its output. This calculation determines the first relation degree; the remaining coefficients of W and the full lifting equation (A.28.4) continue to be retained in the unresolved global problem. □

Statement A.31 (Proved). The nonzero cubic Frobenius term at seventeen and its failure to be a cube. *Let $L_0 = K_0(\alpha)$ with $\alpha^3 = 3$, and let $H = G_{L_0,\{3\}}(3) \subset P_3$. Put*

$$\pi = \frac{\lambda}{\alpha}, \qquad \eta = \alpha^2 - 2, \qquad \delta = 1 + \zeta^2\alpha + \zeta\pi. \tag{A.31.1}$$

The four Kummer classes of ζ, π, η, δ form an $\mathbb{F}_3$-basis of $H^1(H, \mathbb{F}_3)$. At every prime of L_0 above seventeen, the Frobenius row on this basis is

$$(0, 0, 0, 1). \tag{A.31.2}$$

Choose $m \in P_3^c$ whose abelian coordinates in the basis u, v are $(1, 1)$, and put $n = vu^{-1}$. Then m, n are a free pro-three basis, $c(m) = m$, $c(n) = n^{-1}$, and the normalized Frobenius of A.27 satisfies the exact congruence

$$\varphi_{17} \equiv m^6[n, [n, m]] \pmod{D_4(P_3)}. \tag{A.31.3}$$

Moreover φ_{17} is not a cube of any element of P_3.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379); the explicit nonprimitive Frobenius in Tasks A.6 and B.4.

Proof. The subgroup H is exactly $\ker(\kappa_3 : P_3 \to \mathbb{F}_3)$. Indeed every finite Galois three-extension of L_0 unramified outside three has a Galois closure over K_0 with three-group Galois group: take the compositum of its three conjugates, whose kernel over L_0 embeds in a product of three-groups. This proves the stated comparison with the open subgroup of P_3. The subgroup is free of rank four by the pro-three Schreier formula. Thus its first cohomology has dimension four.

The element π is a unit away from three. For $\sigma(\alpha) = \zeta\alpha$, direct multiplication gives

$$N_{L_0/K_0}(\eta) = 1, \qquad \delta^3 = \zeta\eta^2\sigma(\eta), \qquad N_{L_0/K_0}(\delta) = \zeta. \tag{A.31.4}$$

Consequently η and δ are units at every finite prime. The following four Frobenius evaluations prove independence of their Kummer classes:

q	$\bar{\zeta}$	$\bar{\alpha}$	κ_ζ	κ_π	κ_η	κ_δ
2	ζ	1	1	2	0	2
2	ζ	ζ	1	1	2	2
2	ζ	ζ^2	1	0	1	0
61	13	4	2	0	1	0

(A.31.5)

The first three residue fields are $\mathbb{F}_4$ with $\zeta^2 + \zeta + 1 = 0$; the last is $\mathbb{F}_{61}$. In each row, raise the unit residue to $(\#\mathbb{F} - 1)/3$ and express the answer as a power of the displayed residue of ζ. The integer determinant of the four-by-four matrix in (A.31.5) is 2, hence is nonzero modulo three. The dimension calculation then proves the basis assertion.

At seventeen choose the residue field $\mathbb{F}_{17^2}$ and first take $\alpha = 7$. The four residues are

$$\zeta, \qquad 5 + 12\zeta, \qquad 13, \qquad 16 + 3\zeta.$$

Their 96th powers give $(1, 1, 1, \zeta)$. The other two choices $\alpha = 7\zeta, 7\zeta^2$ give the same four Kummer values. This proves (A.31.2), including the effect of any conjugation permuting the three primes above seventeen.

Coprime H^1 for the kernel of $P_3 \to P_3^{\mathrm{ab}}$ makes $P_3^c \to (P_3^{\mathrm{ab}})^c$ onto. This gives m with the specified abelian vector. Together with $n = vu^{-1}$ its Frattini vectors have determinant 2, so m, n are a free basis. The actions of c are exactly those stated, and

$$\kappa_3(m) = 0, \quad \kappa_3(n) = 1, \quad \kappa_\zeta(m) = 1 \quad \text{in } \mathbb{F}_3, \tag{A.31.6}$$

by the two rows of A.10. Thus conjugation by n acts on L_0 as σ. One completely specified choice of m uses only continuous pro-three square roots. Put $d_0 = uc(uv)^{-1/2}$ and $h_0 = (d_0c)^{1/2} \in P_3$, and take $m = h_0^{-1}(uv)h_0$. Indeed $(uc)^2 = uv$, so d_0 is an involution commuting with uv. The identity $c(d_0c)c = (d_0c)^{-1}$ gives $h_0ch_0^{-1} = d_0$ and hence $c(m) = m$. Its abelian vector is still $(1, 1)$. Square roots here are the unique continuous powers $g^{1/2}$ in the closed procyclic group generated by g; this defines every word and comparison in the formula.

We give the finite-group comparison which permits use of (A.31.2) in degree three. A Schreier basis of H is $n^3, m, nmn^{-1}, n^2mn^{-2}$. Modulo $\Phi(H)$ these form a basis of an elementary abelian group of order 3^4. Conjugation by n fixes n^3 and cyclically permutes the other three basis vectors. Hence $P_3/\Phi(H)$ has nilpotence class at most three, exponent dividing nine, and commutator subgroup of exponent three. The defining product formula

$$D_4(P_3) = P_3^9\, \gamma_2(P_3)^3\, \gamma_4(P_3)$$

therefore proves $D_4(P_3) \subset \Phi(H)$. Every Kummer character in (A.31.5), in particular $\ell = \kappa_\delta$, annihilates this subgroup.

The vector space $D_3(P_3)/D_4(P_3)$ has basis

$$m^3, \quad n^3, \quad [m,[m,n]], \quad [n,[n,m]]. \tag{A.31.7}$$

Collection gives these generators; their Magnus polynomials are respectively M^3, N^3, $M^2N + MNM + NM^2$, and $N^2M + NMN + MN^2$, which are independent. On this basis c has signs $+, -, -, +$. The Frobenius lies in D_3 by A.27 and is fixed by c, so its class has form $Am^3 + B[n,[n,m]]$. Its abelian coordinates in the m, n basis are $(a_{17}, 0)$, whence $A = a_{17}/3 = 2$ modulo three.

For $h \in H$, Kummer theory identifies $\ell(nhn^{-1})$ with $\kappa_{\sigma^{-1}\delta}(h)$. Since $-2 = 1$ in $\mathbb{F}_3$, we obtain

$$\begin{aligned} \ell([n,[n,m]]) &= ((\sigma^{-1} - 1)^2\kappa_\delta)(m) \\ &= \kappa_{N_{L_0/K_0}\delta}(m) = \kappa_\zeta(m) = 1. \end{aligned} \tag{A.31.8}$$

Also $\ell(m^3) = 0$. Formula (A.31.2) gives $\ell(\varphi_{17}) = 1$, so $B = 1$. This proves the full degree-three congruence (A.31.3), rather than only its cyclotomic component.

Finally, a cube root of φ_{17} in P_3 would have abelian vector $(a_{17}/3, a_{17}/3)$ in the u, v basis. Its κ_3 value is zero, so it would belong to H. The character κ_δ annihilates every cube in H, contradicting (A.31.2). This proves the last assertion.

The exact unit identities and root are verified by `a_frobenius17_unit_root.py` in `code/`; the residue matrix and all three Frobenius rows at seventeen are verified by `a_frobenius17_sunits.py` in the same directory. Their outputs are included. An initial attempted list $\zeta, \pi, \eta, \sigma(\eta)$ had rank three, not four; the exact cubic root δ in (A.31.4) supplies the missing unit. No basis assertion relies on that rejected list. □

Statement A.32 (Proved). The exact marked global image and ramified fibers on $\Sigma \cup \Xi$. *Put $R = N/\gamma_3(I_{\mathrm{mark}})$, $A = N/\gamma_3 N$, and $J_R = \ker(R \to A)$. For every $q \in \Sigma \cup \Xi$, the three global class-two images coincide and are exactly*

$$\begin{aligned} \operatorname{im}\big(\mathrm{Sec}_{S_q}(N) \to \mathrm{Sec}_{S_q}(A)\big) &= \operatorname{im}\big(\mathrm{Sec}_{S_q}(N/N'') \to \mathrm{Sec}_{S_q}(A)\big) \\ &= \operatorname{im}\big(\mathrm{Sec}_{S_q}(R) \to \mathrm{Sec}_{S_q}(A)\big) = \mathrm{res}_q^{-1}(\mathscr{L}_q), \end{aligned} \tag{A.32.1}$$

where $S_q = \{3, q, \infty\}$ and $\mathscr{L}_q$ is the actual local unramified and cusp-packet set of A.24.

Fix a ramified class-two target in this image and let $d \in D(\mathbb{Q}_q)$ be its unique local cusp. Define its local marked lifting-class set by

$$\mathcal{L}_{q,d} = \begin{cases} \{*\}, & d \neq Q_c, \\ \mathcal{C}_{\mathbb{Q}_q,Q_c}, & d = Q_c, \end{cases} \tag{A.32.2}$$

using exactly the reference and quotient of B.12–B.13. Restriction gives a surjection from the global marked fiber onto $\mathcal{L}_{q,d}$. For each element of this local class set, choose a normalized global marked lift above it. The inverse image of that element is precisely

$$\begin{cases} J_R/J_R^c, & q \in \Sigma, \\ J_R^\Theta \backslash J_R, & q \in \Xi, \end{cases} \tag{A.32.3}$$

with right and left cosets, respectively. In the second case $\Theta = \mathrm{Ad}(X_0) \circ \sigma$ is the actual involution specified by the chosen lift in A.26. Thus (A.32.2)–(A.32.3) describe the entire fiber without identifying different local marked classes or asserting a canonical product decomposition. Targets unramified at q have exactly the unramified fiber of A.20 for $R \to A$.

The exact image of $\mathrm{Sec}_{S_q}(N) \to \mathrm{Sec}_{S_q}(R)$ consists of the classes unramified at q, the unmarked ramified cusp classes, and the marked ramified classes whose local parameter in $\mathcal{C}_{\mathbb{Q}_q,Q_c}$ is zero. Its nonempty ramified fibers are given by (A.32.3) with J_R replaced by $\gamma_3(I_{\mathrm{mark}})$ and the corresponding chosen full lift. In particular there are global marked sections supported at $\{3, q\}$ which have no full lift, although their class-two image has a full lift. These examples still have no full lift after any finite number-field extension.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379); the marked higher image and fibers in Task A.6.

Proof. The local marked classification B.12–B.14 proves that its exact class-two image is $\mathscr{L}_q$. The local fiber over an unmarked ramified cusp is a singleton; the local fiber over the marked cusp is the full group $\mathcal{C}_{\mathbb{Q}_q,Q_c}$. Each class in the latter group is an actual local section, with the Frobenius representative $c_{Q_c}^r z F_0$ and inertia $c_{Q_c}^a$ of (B.12.3). Its class-two image retains a, r and kills z through the actual morphism $\gamma_2 I_{\mathrm{mark}}/\gamma_3 I_{\mathrm{mark}} \to A$.

For $q \in \Sigma$, a normalized global section is its arbitrary local pair modulo R^c, by A.15–A.16. Any prescribed local marked lifting class therefore globalizes. To make its projection the exact local pair of the given global class-two target, lift the required local geometric conjugator through $R \to A$. There is no further arithmetic relation. For $q \in \Xi$, first normalize the local c-value to one using coprime H^1 in the pro-three kernel. The independent involution value in the second row of (A.26.3) lifts the prescribed class-two value by the same coprime argument. It combines with the selected local row to give the required global marked section. This proves surjectivity onto (A.32.2) and the marked equality in (A.32.1). The other two equalities are A.24 and A.26.

We check the centralizers used to pass from local to global conjugation. The class-two local section has trivial geometric centralizer: its image in V would be Frobenius-fixed, hence zero by the weight calculation, and the remaining element in $\bigwedge^2 V$ is zero for the same reason. At an unmarked ramified cusp in R, its projection to the filled full group has trivial centralizer by B.7, and the kernel has no inertia-fixed element in either M or Z by the free label actions in B.14. Hence its centralizer in R is zero. At the marked cusp the centralizer of the inertia element is $\langle c_{Q_c} \rangle Z$ by (B.12.6). A full-section centralizer in this group has zero cusp coordinate because multiplication by $q - 1$ is injective on $\mathbb{Z}_3$; its central coordinate is also zero, because $q^2 \bigwedge^2 \Phi - 1$ is injective on Z. This last injectivity follows on every finite

free permutation quotient from its eigenvalues, and then in the inverse limit. It is unaffected by the central Frobenius parameter z. Thus every ramified local marked section used here has trivial geometric centralizer.

Fix a local marked lifting class. Exact local lifts of the target pair form one free conjugacy orbit under J_R. Indeed their conjugator projects to a centralizer of the target, which has just been proved trivial. For Σ, the remaining global conjugators are J_R^c. With the cocycle convention of A.24, their action on the local orbit parameter is right multiplication, giving J_R/J_R^c. For Ξ, normalized local rows form a free orbit under J_R^c. Fix one such row by this conjugation; no geometric conjugator remains. The remaining data are exactly the lifts of the independent involution value. As in (A.26.2), writing them as jX_0 gives $j\Theta(j) = 1$ and the bijection $J_R^{\Theta}\backslash J_R \to \{j : j\Theta(j) = 1\}$. This proves (A.32.3) with its coset orientations.

The same presentation arguments apply to $N \to R$. B.7 and B.12–B.14 show that a local marked class has a full lift exactly for the parameters listed in the statement. The lifting class is unique when ramified. Thus the local condition is necessary, the arithmetic presentations make it sufficient, and the same centralizer and coset argument gives the asserted fibers. If a class-two or marked target is unramified at q, the zero-abelian-inertia results of B.7 and B.12 force every higher lift here to be unramified there as well. The comparison with A.20 is consequently exact.

For existence of the last examples, start with any nonzero marked tame exponent and choose a nonzero element of $\mathcal{C}_{\mathbb{Q}_q,Q_c}$; B.13 gives such elements, including an explicit central commutator detected in a finite label quotient. The preceding construction gives the global marked section, while its class-two image belongs to (A.32.1). The group $\mathcal{C}_{k,Q_c}$ is the unramified first cohomology of the actual central coefficient module Z in B.12. On this module the cusp inertia acts trivially. Its tame first-cohomology component has Frobenius $Q\bigwedge^2\Phi$, with no invariant vector, so it is the whole $H^1_{\mathrm{cts}}(G_k, Z)$. For any finite local extension, restriction followed by corestriction on this continuous cohomology is multiplication by its degree; this is the usual finite coset-sum chain map and is valid for compact coefficients. B.13 proves that the source group is torsion-free. Its restriction is therefore injective. A nonzero central parameter remains nonzero at every place of every finite number-field extension above q, and the local full image still has central parameter zero by B.12. This proves persistence of the failure of full lifting. □

Statement A.33 (Proved). A class-three lift for every finite-support locally cuspidal class-two section. *Let S' be any finite set of rational primes containing three, with the real place included. Put $A = N/\gamma_3 N$ and $M_3 = \gamma_3 N/\gamma_4 N$. Every section of A over $G_{\mathbb{Q},S'}$ whose restriction at each $q \in S' \setminus \{3\}$ belongs to the actual local image $\mathscr{L}_q$ of B.8 has a section lift to $N/\gamma_4 N$ over the same arithmetic group. Its lifting fiber is a torsor under*

$$H^1(G_{\mathbb{Q},S'}, M_3) = H^1(G_{\mathbb{Q},\{3,\infty\}}, M_3) = \mathbb{Z}_3^{166} \oplus (\mathbb{Z}/3)^7 \oplus (\mathbb{Z}/9)^{56} \oplus (\mathbb{Z}/27)^{25} \oplus (\mathbb{Z}/81)^3. \quad \text{(A.33.1)}$$

These assertions concern the actual common class-three quotient of the full and metabelian groups.

There is also an exact obstruction to choosing this global lift with prescribed actual cuspidal local class-three restrictions. Let $J \subset S' \setminus \{3\}$ be the primes where the class-two target is ramified. Choose any global class-three lift s_3. At each $q \in J$ the unique actual cuspidal class-three lifting class determines a difference $\Delta_q(s_3) \in M_3/(F_q - 1)M_3$. The element

$$[(\Delta_q(s_3))_{q\in J}] \in \operatorname{coker}\left(H^1(G_{\mathbb{Q},\{3,\infty\}}, M_3) \xrightarrow{\mathrm{res}} \bigoplus_{q\in J} M_3/(F_q - 1)M_3 \right) \quad \text{(A.33.2)}$$

is independent of s_3. It vanishes exactly when a global class-three lift can have all the indicated cuspidal restrictions. Every full or metabelian global lift of the class-two target forces (A.33.2) to vanish.

Proof. The action on the central kernel M_3 is the actual action induced from V; conjugation by the class-two section does not change it. Its finite quotients have three-group image over K_0. The composition-series argument in A.1 therefore gives

$$H^i(G_{\mathbb{Q},\{3,\infty\}}, M_3/3^n) = 0 \quad (i \geq 2).$$

For $X = \operatorname{Spec} \mathbb{Z}[1/3]$, apply the actual localization triangle (A.1.2) with $M_3/3^n$ and remove the finite set $S' \setminus \{3\}$. Its degree-two term injects into the sum of the residue groups $H^1(\mathbb{F}_q, M_3(-1)/3^n)$. These are precisely the local groups $H^2(G_{\mathbb{Q}_q}, M_3/3^n)$, by the tame inertia calculation in A.1. Thus the actual global lifting obstruction, restricted to each local arithmetic group, is its corresponding local lifting obstruction. An actual local full lift exists by membership in $\mathscr{L}_q$, so all these restrictions are zero.

The finite-S' degree-one groups with finite coefficients are finite. Their first derived limit consequently vanishes. Taking inverse limits therefore gives an injection

$$H^2_{\mathrm{cts}}(G_{\mathbb{Q},S'}, M_3) \longrightarrow \bigoplus_{q \in S' \setminus \{3\}} H^2_{\mathrm{cts}}(G_{\mathbb{Q}_q}, M_3), \tag{A.33.3}$$

and the continuous obstruction is zero. This proves existence of a continuous global class-three lift, without a passage from unrelated finite-level choices.

Every rational Frobenius eigenvalue on $M_3(-1)$ has complex absolute value strictly greater than one. Indeed M_3 is the degree-three multiplication kernel of A.22, and every eigenvalue is a product of three eigenvalues on V; their absolute values are at least $q^{3/2}$ before the twist. Hence $H^0(\mathbb{F}_q, M_3(-1)) = 0$. Degree-one localization and the same finite-S' inverse-limit argument identify the two first-cohomology groups in (A.33.1). The Smith calculation of A.22 proves the displayed decomposition. The target section has trivial geometric centralizer: its image in V is Frobenius-fixed and therefore zero, and its remaining image in $\bigwedge^2 V$ is zero for the same reason. The ordinary central-extension lifting action is consequently a torsor on section classes, with no further stabilizer quotient.

Locally the same weight calculation kills the tame term in $H^1(G_{\mathbb{Q}_q}, M_3)$, so this group is exactly $M_3/(F_q - 1)M_3$. A ramified class-two cusp packet fixes its actual full cuspidal lifting class, by B.7–B.8, and therefore fixes its class-three image. The difference from s_3 is consequently an element of this first-cohomology group. Replacing s_3 by a global central twist adds the restriction of that twist to every local difference. This proves the independence and the asserted equivalence in (A.33.2). A.19 forces every ramified local metabelian restriction to be the actual cusp packet, and B.7 gives the same conclusion for the full group. Their class-three images therefore make every local difference zero, proving the final necessity assertion. □

Statement A.34 (Proved). The restriction map at seventeen and a class-three lift that stops at class four. *Use the actual arithmetic generators m, n, c of A.31. On the degree-three lattice M_3, the restriction map*

$$H^1(G_{\mathbb{Q},\{3,\infty\}}, M_3) \longrightarrow H^1(G_{\mathbb{Q}_{17}}, M_3) = M_3/(F_{17} - 1)M_3 \tag{A.34.1}$$

has image of dimension 15 *after reduction in the local target modulo three. The target modulo three has dimension* 164*, and the cokernel of (A.34.1) modulo three therefore has dimension* 149*.*

Fix free geometric generators $x_0, \dots, x_9$ whose abelianizations are the integral basis of A.7, with the first eight entries the non-O cusp generators. In M_3 put $e = [[x_8, x_0], x_9]$. The exact normalized cocycle

$$\beta(c) = 0, \qquad \beta(m) = (1 + c)e, \qquad \beta(n) = 0 \tag{A.34.2}$$

is unramified outside three. Start with the actual full O-cuspidal section whose tangent parameter is 17, project it to $N/\gamma_4 N$, and twist by β. The resulting global section s_3 has the same class-two image as that full cuspidal section, but s_3 has no lift to $N/\gamma_5 N$. In particular it has no full or metabelian lift, and no lift to the marked quotient with marked cusp Q_c.

Thus a class-two target with a full lift can have a class-three lifting class with a nonzero next obstruction. The theorem does not assert non-lifting of that class-two target: its original full cuspidal lift is specified above.

Proof. All the following matrices act on the reduction $\overline{M}_3$. Write U, V_0, C for the actual matrices of u, v, c in A.22, and put

$$M = UV_0, \qquad L = V_0U^{-1}, \qquad A_0 = M - 1, \qquad B_0 = L - 1, \qquad \Pi_+ = (1+C)/2. \tag{A.34.3}$$

The action over K_0 is abelian by the explicit isogeny and CM comparisons in A.7. Thus these really are the actions of m, n, independently of the chosen c-fixed lift m. The exact matrices give

$$A_0^3 = B_0^3 = A_0^2B_0^2 = 0, \quad A_0B_0 = B_0A_0, \quad CMC = M, \quad CLC = L^{-1}. \tag{A.34.4}$$

In particular the fourth power of the augmentation ideal of this coefficient action is zero modulo three. Normalized cocycles are prescribed freely by

$$a \in \ker(C - 1), \qquad b \in \ker(LC + 1) \tag{A.34.5}$$

as their values at m, n. These subspaces have dimensions $164, 166$. They are reductions of the corresponding integral eigensublattices, since the two operators C and LC are involutions and two is invertible. The exact free presentation with $c(m) = m$ and $c(n) = n^{-1}$ proves that every integral such pair is an actual normalized cocycle.

For any cocycle, evaluation on $D_5(P_3)$ is zero modulo three. Evaluation on $\gamma_5(P_3)$ contains four augmentation operators and is zero by (A.34.4). The coefficient action kills all commutators, so evaluation on cubes of commutators is three times a value and is zero. Evaluation on ninth powers is also zero: the action of every element has order dividing three, and its nine-term norm is three times its three-term norm. These are the factors generating $D_5(P_3)$. Consequently (A.31.3) leaves only the three possible degree-four Hall terms

$$K_1 = [m, [m, [m, n]]], \quad K_2 = [n, [m, [m, n]]], \quad K_3 = [n, [n, [m, n]]].$$

Their cocycle values, and the value of the specified degree-three term, are respectively

$$\begin{array}{c|l} m^6[n,[n,m]] & (2A_0^2 + B_0^2)a - A_0B_0b \\ K_1 & -A_0^2B_0a + A_0^3b \\ K_2 & -A_0B_0^2a + A_0^2B_0b \\ K_3 & -B_0^3a + A_0B_0^2b. \end{array} \tag{A.34.6}$$

These formulas follow by repeated use of $f(gh) = f(g) + gf(h)$; commuting coefficient actions give $f([g, h]) = (1 - h)f(g) + (g - 1)f(h)$.

The action of the normalized rational Frobenius at seventeen is C modulo three. Indeed its square is φ_{17}, and (A.27.1) gives the latter action as $M^{a_{17}}$ with $a_{17} \in 3\mathbb{Z}_3$, whereas $M^3 = 1$

modulo three. The rational Frobenius is exactly $c\varphi_{17}^{1/2}$ in its decomposition group. The local target modulo three is therefore $\overline{M}_3/(C-1)\overline{M}_3 \simeq \overline{M}_3^C$. Its Frobenius value is half the Π_+ projection of the value at φ_{17}.

The program `a_frobenius17_eval_m3.py` in the `code` directory applies the four operators in (A.34.6) to the two eigenspaces (A.34.5), using the full integral basis and matrices in `a_degree3_matrices.json`. All operations here are exact in $\mathbb{F}_3$. After projection by Π_+, the three degree-four operators have ranks $0, 1, 0$. The specified degree-three operator has rank 15. Adding the middle operator with coefficient 0, 1, or 2 gives the same column space, of rank 15 in every case. The saved image basis and the elimination routine certify these comparisons, so unknown higher coefficients of the actual Frobenius cannot change the image. Coboundaries restrict to zero in the local quotient; (A.34.5) supplies all integral normalized cocycles. This proves the rank and cokernel assertions for the actual map (A.34.1).

For (A.34.2), the same calculation has zero contribution from every unknown degree-four term. In the multidegree basis of A.22, e has zero-based index 87, namely $(\{0, 8, 9\}, 8)$. Let z be the reduction modulo three of its local Frobenius value. It is C-fixed. The program records all its 36 nonzero coordinates and forms the actual Lie bracket $[z, e_O]$ in the degree-four free Lie algebra by its injection into the free associative algebra. Here $e_O = -\sum_{j=0}^{7} e_j$ in the fixed abelian basis. For reproducibility, a single nonzero coefficient is

$$[z, e_O]_{X_0X_0X_0X_2} = 1 \quad \text{in } \mathbb{F}_3. \tag{A.34.7}$$

The complete polynomial has 1098 nonzero words. It is saved in `a_frobenius17_eval_m3.json`. The code converts each multidegree difference to its integral Hall bracket $[[x_i, x_j], x_k]$, expands the brackets, and checks directly with the actual matrix on V that

$$C[z, e_O] = -[z, e_O]. \tag{A.34.8}$$

Thus this is a nonzero element of the actual degree-four free Lie lattice modulo three, with its stated sign.

Locally choose the uniformizer Kummer roots for 17 so that the reference cusp section has inertia c_O and Frobenius F_0. The twist (A.34.2) changes no inertia, since it is unramified there, and changes Frobenius to zF_0 modulo $\gamma_4 N$. In a putative lift to $N/\gamma_5 N$, its inertia must be $c_O k$ for some $k \in L_4 := \gamma_4 N/\gamma_5 N$. An arbitrary choice of the lifted Frobenius differs only by a central L_4 value, which has no effect on its conjugation of inertia. The tame relation therefore gives the necessary equation

$$(F_{17} - 17)k = -17[z, e_O] \quad \text{in } L_4. \tag{A.34.9}$$

Modulo three its left side is $(C+1)k$, in the plus eigenspace. By (A.34.7)–(A.34.8) its right side is a nonzero minus eigenvector. This is impossible because two is a unit modulo three. Conjugating a prospective lifting class would not avoid the equation: lift its class-three conjugator through the surjection $N/\gamma_5 N \to N/\gamma_4 N$ to make its projection the prescribed representative, and then apply the same calculation.

This proves the local obstruction and therefore the failure of a global class-four lift. A.19 would make any local metabelian lift of this ramified class an actual cusp section and hence give a full local lift, contradicting (A.34.9). B.14 gives the identical consequence for a marked lift because O is an unmarked cusp. These exact quotient comparisons prove the remaining non-lifting assertions. The original O-cusp section remains a full lift of the shared class-two image, as asserted. □

Statement A.35 (Proved). A fifteen-dimensional toric family with zero first mod-three gluing class. *Let $B \in (\bigwedge^2 W)^{c=-1}$. This eigensublattice has rank fifteen. The*

exact prescriptions

$$\gamma_B(c) = 0, \qquad \gamma_B(m) = 0, \qquad \gamma_B(n) = B \tag{A.35.1}$$

define an unramified-outside-three class-two central cocycle. Twist the class-two image of the global O-cuspidal section with tangent parameter 17 *by γ_B. Its restriction at seventeen is the same actual class-two cusp section, including its Frobenius coordinate. This target has a global class-three lift with unchanged local inertia. For the exact gluing class (A.33.2), its image in the cokernel of (A.34.1) modulo three is zero. No assertion of integral vanishing of (A.33.2) is made here.*

Proof. Over K_0 the toric lattice is scalar cyclotomic. Thus m acts on $\bigwedge^2 W$ by 49^2, whereas n acts as one. Its complex-conjugation signs are $(13, 15)$, from the signs $(3, 5)$ on W. The free presentation of A.31 proves (A.35.1). At the normalized local group, φ_{17} is fixed by c. The cocycle value at this element lies both in $\mathbb{Z}_3 B$ and in the plus eigenspace of c, so is zero. Its value on tame inertia is zero because the cocycle is unramified. Its value on rational Frobenius $c\varphi_{17}^{1/2}$ is also zero. This proves the exact local class-two assertion.

Here is an actual global class-three lift. In A.28 first choose x with $\kappa_{17}(x) = 0$ exactly: right multiplication by a unique power of the tame generator achieves this, and the correction is in the Frattini subgroup when the initial value is zero modulo three. The conjugate y, and the resulting generators m, n, then also have zero Kummer value. Keep the tame section value c_O. Choose $h \in \gamma_2 N/\gamma_4 N$ which is a product of toric commutators with degree-two image $-B/2$. Give m section value one and n section value $h^{-1}(nc)(h)$ relative to the reference cusp section. The latter value obeys the exact involution relation because $(nc)^2 = 1$. Its degree-two image is B. The fixed word W in A.28 has zero degree-two difference from the reference, by the same c-fixed argument used for φ_{17}. Its remaining difference in class three is central, and therefore does not change its conjugation of c_O. The actual defining relation $WtW^{-1} = t^{17^2}$ is satisfied. The universal property of that proved presentation gives the claimed global class-three lift.

We calculate its Frobenius difference modulo three, retaining the actual path origins. For every cusp d, let z_d be its framed Abel–Jacobi point and $h_d = 3z_d$ the torus point of A.9. Choose compatible division points $z_{d,j}$ and toric division points $h_{d,j}$. The compatible torsion points $3z_{d,j} - h_{d,j}$ form $v_d \in V$, and their actual proper image in $T/3$ is $d - O$. Thus the path cocycle has the exact identity

$$3b_{g,d} = \kappa(h_d)(g) + gv_d - v_{gd}. \tag{A.35.2}$$

The toric component of v_d is constrained by the actual torsion point $z_d - h_{d,1}$; it has not been replaced by an arbitrary lift. Altering the chosen z-division points changes v_d by $3V$ and changes b by the corresponding actual coboundary. We may therefore choose its proper component to be the specified integral cusp label $t_d = (a_d, b_d)$ of A.7. For g over K_0 the proper component of (A.35.2) is $(T_g - 1)t_d/3$.

Put

$$H_m = 2I_2, \qquad H_n = (T_n - I_2)/3 = \frac{1}{7}\begin{pmatrix} -4 & -1 \\ 1 & -5 \end{pmatrix}. \tag{A.35.3}$$

These are the proper components of the path operators for m, n, since $T_m = 7I_2$. The upper-right blocks in the actual integral action on V are

$$\Phi_m = 14\Psi, \qquad \Phi_n = -\Psi H_n. \tag{A.35.4}$$

Modulo three all the diagonal K_0 actions on V are one. On a toric commutator $[e_d, e_e]$, its change in degree three is the derivation with cusp values $-[H_g t_d, e_d]$, together with toric

terms depending on the full framed origins. The subsequent augmentation kills every toric term; on a term with one proper T factor it replaces that factor by Φ_g.

More explicitly, use the abelian coefficient group $\gamma_2 N/\gamma_4 N$, with the action twisted by the reference cusp section. In the submodule generated by toric commutators and their images, three successive augmentation operators are zero modulo three. A first augmentation produces degree-three terms with at most one T factor; the second produces only W terms; the third is zero. The c-normalization used above changes the chosen n-value only by degree-three terms with at most one T factor, so these changes also vanish after two augmentations. The tame augmentation sends a toric commutator to its bracket with e_O, a pure W term, and its product in either order with every other augmentation is zero here.

For noncommuting coefficient operators $A_0 = m - 1$ and $B_0 = n - 1$, the leading crossed-derivation value at $[n, [n, m]]$, when the m-value is zero and the n-value is B, is

$$(A_0 B_0 - 2B_0 A_0)B. \tag{A.35.5}$$

This follows by differentiating brackets: $D[X, Y] = XD(Y) - YD(X)$, including the action $[X, Y]$ on the source value. All the higher group corrections contain three augmentations and are zero. The term m^6 contributes zero. A.30 proves $W \in D_3(F_3(m, n, t))$ for the actual three-generator word, so there are no degree-two tame-letter terms. Its degree-three terms involving the tame letter have two augmentations and vanish by the identities just proved; a restricted tame cube has zero tame source value. The unknown terms in D_4 have three augmentations and are also zero. Thus (A.31.3) computes the whole Frobenius difference in this calculation.

For each cusp, the proper-path contribution to (A.35.5) is, up to the common cusp-bracket sign,

$$\Phi_m H_n - 2\Phi_n H_m = 14\Psi H_n + 4\Psi H_n = 18\Psi H_n \equiv 0 \pmod 3. \tag{A.35.6}$$

The framed toric terms were killed before this calculation, so no unknown origin term remains. The exact program `a_frobenius17_toric_twists.py` in the `code` directory expands all twenty-eight toric commutators in the free associative algebra, using (A.35.3)–(A.35.4), and verifies (A.35.6). Twenty-seven have nonzero individual summands, but their stated difference is always zero. Hence the chosen class-three lift has zero local Frobenius difference in $M_3/3$. Inertia was unchanged by construction. Its class in the local quotient and therefore the gluing cokernel modulo three is zero, which proves the statement. □

Statement A.38 (Proved). An actual Frobenius basis with completely specified finite class-two action. *Let $F_2 = \langle x, y\rangle$ be the geometric pro-three group of $\mathbb{P}^1 - \{0, 1, \infty\}$, based at the unit tangent at zero, and let*

$$N = \ker(F_2 \longrightarrow (\mathbb{Z}/3)^2), \qquad K = N^3 \gamma_3 N. \tag{A.38.1}$$

Here N^3 is the closed subgroup generated by cubes. There is an actual Frobenius u at the prime $(7, \zeta - 2)$ of K_0 such that $u, v = cuc$ are a free basis of P_3, their integral action on V is exactly A.7, and their action on N/K is the following fully specified finite action. Complex conjugation is induced by $x \mapsto x^{-1}, y \mapsto y^{-1}$. The restriction of Frobenius u to N/K equals the restriction of the following automorphism of F_2/K:

$$x \longmapsto x^7, \qquad y \longmapsto f_0^{-1} y^7 f_0. \tag{A.38.2}$$

In the ordered Schreier basis

$$\begin{gathered}(yxy^{-1}x^{-1},\ y^2xy^{-2}x^{-1},\ y^3,\ xyxy^{-1}x^{-2},\ xy^2xy^{-2}x^{-2},\ xy^3x^{-1},\\ x^3,\ x^2yxy^{-1},\ x^2y^2xy^{-2},\ x^2y^3x^{-2}),\end{gathered} \tag{A.38.3}$$

write N/K as $\mathbb{F}_3^{10} \oplus \bigwedge^2 \mathbb{F}_3^{10}$ with product $(a,z)(b,w) = (a+b, z+w+\frac{1}{2}a\wedge b)$. The word-class representative f_0 defining this finite automorphism has coordinates

$$a_f = (0,0,1,1,2,0,2,0,1,2), \tag{A.38.4}$$

and, in lexicographic order $i<j$, the forty-five central coordinates are

$$\begin{aligned} z_f = (&2,1,2,1,1,2,0,1,2,\ 1,1,0,0,0,2,2,0,1,\\ &1,1,1,1,1,0,1,1,0,0,1,2,0,0,1,0,1,1,\\ &0,0,1,2,0,2,2,0,0). \end{aligned} \tag{A.38.5}$$

Thus (A.38.2)–(A.38.5) specify every coordinate of the images of the ten geometric generators. They specify a representative for the induced finite action; the actual Ihara word can differ by the kernel of its map to that action. The arithmetic basis is obtained from any initial Frobenius basis by at most four explicitly described conjugations below. The underlying Galois action and tangential origin are retained; all cocycles and all arithmetic Frobenius words are transported by the resulting actual source automorphism.

Proof. We first specify the geometric comparison. The map from the Fermat curve is

$$t = -z^3/x^3, \qquad (-z/x)^3 = t, \qquad (-y/x)^3 = 1-t. \tag{A.38.6}$$

It is the connected degree-nine cover classified by $F_2 \to (\mathbb{Z}/3)^2$. At O, the specified parameter is $s = z/x$, so $t = -s^3$. The change from the tangent -1 to the unit tangent is the based pullback $t^{1/3^n} \mapsto -s^{3/3^n}$, using the compatible rational odd roots of -1. It has zero Kummer cocycle. This retains the given tangential arithmetic splitting. The Schreier transversal $x^a y^b$, $0 \le a,b < 3$, gives exactly (A.38.3). It also gives the actual conjugation actions of x and y on N/K; using these actions and the stated class-two product is a finite word-collection algorithm for (A.38.2).

For the chosen tangential paths the actual Ihara action is $x \mapsto x^7$, $y \mapsto f^{-1}y^7 f$, with $f \in F_2'$. Its symmetry, hexagon and pentagon are the actual Galois relations [Sch05, §§I.3, I.6]; we use the hexagon order

$$f(x,y)x^3 f(z,x)z^3 f(y,z)y^3 = 1, \qquad z = (xy)^{-1}. \tag{A.38.7}$$

The comparison with A.7 is the integral unimodular matrix saved by `a_ihara_lattice.py`. It is obtained from the actual order-nine deck translations, and intertwines both complex conjugation and the integral Frobenius homology matrix. Consequently the finite equations below constrain the actual arithmetic action in the specified geometric basis.

We give the finite elimination with its source-lifting conditions. The exact collector for (A.38.3) works in F_2/K, of order 3^{57}. Its intersection with the lower central and Zassenhaus filtrations is computed by taking literal group commutators, cubes, and their invariant spans. Through degree seven the kernel of the comparison from the free source has one degree-six generator, $[x,y]^3$, and four degree-seven generators. The literal words for all five are included in the certificate and are individually checked to map to one in F_2/K. No source direction below degree six is lost. The exact commutator calculation is separately verified by the full class-two multiplication law, retaining its quadratic conjugation terms.

The homology equation, $f \in F_2'$, symmetry, and (A.38.7) leave three possible homology rows and an affine central system over each row. Requiring a lift of these same equations over $\mathbb{Z}/9$ retains all next-digit variables. We then impose the full group pentagon through source degree seven, including the five actual source-kernel words just specified. For precision, the

pentagon is evaluated in the fourth-strand free group of $PB_4 = F_3 \rtimes PB_3$. The Artin maps are

$$r_1 : (X, Y, Z) \mapsto (Y, Y^{-1}XY, Z), \qquad r_2 : (X, Y, Z) \mapsto (X, Z, Z^{-1}YZ). \tag{A.38.8}$$

Put $A_{12} = r_1^2$, $A_{23} = r_2^2$, $A_{13} = r_2 r_1^2 r_2^{-1}$, and $A_{14} = \mathrm{Inn}(X)$, $A_{24} = \mathrm{Inn}(Y)$, $A_{34} = \mathrm{Inn}(Z)$. These are the actual pure Artin maps [CFR12, (2.1)–(2.3)]; direct free-word reduction verifies every relation and $A_{12}A_{13}A_{23} = \mathrm{Inn}((XYZ)^{-1})$. The five factors have pairs

$$\begin{aligned} &(A_{12}, A_{23}A_{24}), \quad (A_{13}A_{23}, A_{34}), \quad (A_{23}, A_{34}), \\ &(A_{12}A_{13}, A_{24}A_{34}), \quad (A_{12}, A_{23}). \end{aligned} \tag{A.38.9}$$

If they are denoted $F_0, \ldots, F_4$, the residual is $F_0F_1F_4^{-1}F_3^{-1}F_2^{-1}$. Forgetting the fourth strand sends this residual to one, so it is an element of the actual free fiber and its displayed action is inner.

All evaluations use the denominator-free Magnus substitution $X_i \mapsto 1 + X_i$ in the associative algebra over $\mathbb{F}_3$, truncated after degree eight. If $\Delta_A = \Phi_A - 1$, then the source word $1 + \sum_w c_w w$ acts as $1 + \sum_w c_w \Delta_{w_1} \cdots \Delta_{w_k}$. Each Δ_A raises augmentation degree. Terms of source degree at least eight therefore do not affect the target generators through degree eight. Conversely, the first nonzero degree-d term h_d of the fiber residual gives $[h_d, X_i]$ in degree $d + 1$ of its inner action. A positive-degree tensor commuting with all three generators is zero. Hence the computation detects precisely all source residuals through degree seven. This uses the actual free fiber and the displayed maps throughout.

The degree-two source coefficient must also be retained integrally. Write $f = [x, y]^{f_2}$ modulo $\gamma_3 F_2$. The integral Heisenberg collector applied to the hexagon with $m = (\chi - 1)/2$ gives

$$3f_2 = \frac{m(m+1)}{2}. \tag{A.38.10}$$

This identity is in the torsion-free group $\mathbb{Z}_3$; at $\chi = 7$ it gives $f_2 = 2$ before any reduction. Its reduction modulo nine is well-defined on the source used above, since the Heisenberg group over $\mathbb{Z}/9$ kills $D_8(F_2)$. The five source-kernel words have degree-two coefficients $(3, 0, 0, 0, 0)$, so this final scalar row is imposed on the combined source word, including those kernel corrections.

Here is the complete finite elimination result. "Invisible" means the kernel of the explicitly computed map to the source metabelian quotient, followed by the intrinsic Johnson map.

equations imposed	$\operatorname{rank} J$	$\operatorname{rank}(J, H)$	$\operatorname{rank} J\vert_{\ker H}$
mod-nine symmetry and hexagon	3	5	2
also full degree-seven pentagon	3	4	1
also integral f_2 mod 9	2	3	0

(A.38.11)

The source metabelian map in this table is the literal Fox map: write $f = [x, y]^h$ in F_2'/F_2'', with $X = x - 1$, $Y = y - 1$. In the finite comparison its coefficient ring is

$$\mathbb{F}_3[X, Y]/(X^6, X^3Y^3, Y^6, X^5Y^2, X^2Y^5). \tag{A.38.12}$$

It is computed by $D_y(f) = Xh$. Thus every map in the last column of (A.38.11) is an actual quotient map of the same based free group.

The row reductions give a more precise conclusion. Let J_0 be the 45×10 central matrix obtained from (A.38.4)–(A.38.5). There are two specified matrices L_1, L_2 such that every actual Frobenius in the initial basis has

$$J = J_0 + aL_1 + bL_2, \qquad a = h_{03} - 2, \quad b = h_{05} \quad \text{in } \mathbb{F}_3. \tag{A.38.13}$$

Here h_{ij} is the coefficient of X^iY^j in (A.38.12). The matrices L_1, L_2, of respective ranks seven and one, are saved with all 450 entries each. The nine matrices in (A.38.13) are distinct. Formula (A.38.13) is at this point a necessary arithmetic constraint; the realization follows below.

For reproducibility, the complete pentagon matrix has 29,520 coefficient rows. Its six candidate columns and five source-kernel columns have rank seven, and the five kernel columns have rank five. Each of the three branches has coefficient and augmented rank seven on its 24 retained parameters. The additional scalar row (A.38.10) has rank one in each branch. The supplied particular solutions and full nullspaces verify every coefficient row. The original and degree-pruned pentagon evaluations give identical matrices; the independent integral Heisenberg audit verifies (A.38.10) and every filtered solution against all 29,520 rows. The files in the `code` directory with prefixes `a_ihara_pentagon`, `a_ihara_source_scalar`, `a_ihara_final_parameters`, `de_fermat_pentagon`, and `de_fermat_scalar_audit` retain the exact words, matrices, row reductions, and executed outputs. These finite identities establish (A.38.11)–(A.38.13), including the affine origin.

We now realize the stated value by an arithmetic morphism. For a current actual basis $u, v = cuc$, two operations are

g	$u' = gug^{-1}, \quad v' = cu'c$	(a', b')
v	$vuv^{-1}, \quad c(vuv^{-1})c$	$(a-1, b+a-1)$
$[u,v]$	$[u,v]u[u,v]^{-1}, \quad c([u,v]u[u,v]^{-1})c$	$(a, b+1)$.

(A.38.14)

These are direct identities of the full 55×55 matrices $\left(\begin{smallmatrix} U & 0 \\ J & \bigwedge^2 U \end{smallmatrix}\right)$ and the fixed complex-conjugation matrix. The program `a_ihara_frobenius_conjugacy.py` verifies all nine inputs of both operations. Applying the first operation a times, using the representatives $a \in \{0, 1, 2\}$, changes the pair to $(0, b + a(a-1)/2)$. Applying the second operation the negative of that last second coordinate times gives $(0, 0)$. At most four operations are needed. Every step conjugates an actual Frobenius, so the final element is an actual Frobenius at the same prime of K_0.

The assignments $u \mapsto u'$, $v \mapsto cu'c$ preserve the two independent Frattini classes and commute with c. They therefore define a surjective endomorphism of the free pro-three group P_3. A finitely generated profinite group is Hopfian: its finitely many open normal subgroups of each fixed index are permuted by inverse image under a surjection, forcing its kernel into every open normal subgroup. Thus the endomorphism is an actual c-equivariant automorphism ψ.

The full integral homology action is also retained. The splitting after pullback by [3] in A.7 is unique, hence G_{K_0}-equivariant. Its isogeny $T_D \times E \to J_{E,D}$ induces the rational equivariant identification with $W_{\mathbb{Q}_3} \oplus T_{\mathbb{Q}_3}$. The embedding of the chosen integral graph coordinates into this diagonal representation is

$$\iota(w, t) = (w + \Psi t/3, t). \tag{A.38.15}$$

The actual isogeny on Tate modules is $j(w, t) = (w, 3t)$ in the rational diagonal coordinates, or $\iota^{-1} j(w, t) = (w - \Psi t, 3t)$ in graph coordinates. Thus the equivariant comparison retains its multiplication by three on the proper factor. On this rational direct sum every $g \in G_{K_0}$ acts by $(\chi(g) I_8, T_g)$, where T_g is multiplication in the commutative algebra $\mathbb{Q}_3[\zeta]$. These actions commute. The integral lattice embeds in this rational representation, so their integral actions commute as well. Arithmetic conjugation consequently preserves the exact A.7 matrix.

Finally we give the transport of every parameter used later. For the original action ρ, the new action is $\rho \circ \psi$. The exact semidirect-product isomorphism is $(N/K) \rtimes_{\rho\circ\psi} \Gamma \to (N/K) \rtimes_{\rho} \Gamma$, $(h, g) \mapsto (h, \psi(g))$, where $\psi(c) = c$; its inverse uses ψ^{-1}. A cocycle is pulled back by $z \mapsto z \circ \psi$.

In particular,

$$z(gug^{-1}) = z(g)\,\rho(g)(z(u))\,\rho(gug^{-1})(z(g)^{-1}), \tag{A.38.16}$$

with the additive version $z(g)+\rho(g)z(u)-\rho(gug^{-1})z(g)$ for an abelian module. This formula retains the reference section and its origin. In the free presentation of A.28, make the same substitutions on x, y and fix t. They again define a c-equivariant IA automorphism, and the actual new Frobenius word is

$$W_{q,\mathrm{new}} = \psi^{-1}(W_{q,\mathrm{old}}). \tag{A.38.17}$$

Indeed the old quotient map composed with ψ sends this word to the same local Frobenius and sends the transformed relator to $W_{q,\mathrm{old}} t W_{q,\mathrm{old}}^{-1} t^{-q^2}$. The definitions of m, n by the pro-three square-root operations commute with this actual automorphism. An IA automorphism of a free pro-three group acts identically on its associated restricted Lie algebra. Since A.30 proves $W_{17} \in D_3$, the initial terms in A.31 and A.37 survive unchanged modulo D_4.

For completeness this also preserves the ordinary weight-three coefficients modulo three used to identify those terms. The degree-one and degree-two Hall coefficients of W_{17} are divisible by three. Under a basis-conjugating automorphism, the changes to a degree-one power with exponent $3a$ have weight-two and weight-three coefficients divisible by $3a$ or $\binom{3a}{2}$; both are divisible by three. A weight-two power with exponent $3b$ changes by weight-three multiples of $3b$. The automorphism is the identity on the third lower central graded quotient. Hence the stated ordinary coefficients are unchanged as well. All higher terms, and the integral lower coordinates when needed, are transported by (A.38.17). This proves the realization and every specified comparison. □

Statement A.39 (Proved). Global exclusions at seventeen inside the exact local cusp packet. *Use the actual arithmetic basis of A.38, with m, n and the normalized local Frobenius $\varphi = \varphi_{17}$ transported as there. Put $M_2 = \bigwedge^2 V$ and define the integral normalized cochain module and its actual evaluation map by*

$$\begin{aligned} D &= M_2^{c=1} \oplus M_2^{nc=-1}, \\ E_\varphi : D &\longrightarrow M_2^{c=1}, \quad (a,b) \longmapsto \gamma_{a,b}(\varphi), \\ R &= \ker E_\varphi. \end{aligned} \tag{A.39.1}$$

Here $\gamma_{a,b}$ is the unique continuous cocycle on $P_3 \rtimes \langle c\rangle$ with values $(a,b,0)$ at (m,n,c). The lattice D has rank 45, and R is a saturated lattice of rank 24.

For $r \in R$, twist the class-two image of the global O-cuspidal section with tangent parameter 17 by γ_r. This defines an injectively parameterized family of global class-two section classes, unramified outside $\{3, 17, \infty\}$. Every member has the same exact local O-cusp section at seventeen, including its Frobenius coordinate, and every member has a global class-three lift.

There is an $\mathbb{F}_3$-linear obstruction on $R/3R$ of rank at least three which vanishes on every member lifting to any of

$$N, \qquad N/N'', \qquad N/\gamma_3(I_{Q_c}). \tag{A.39.2}$$

Consequently, from any integral basis $r_1, \ldots, r_{24}$ of the kernel in (A.39.1), at least three of the resulting twenty-four class-two section classes have none of the lifts in (A.39.2). This is an unconditional finite-family existence certificate. It does not select an unevaluated individual basis vector as a counterexample. No assertion of full local liftability at three is included. An individually specified member of R with nonzero obstruction, including all integral generator coordinates and a nine-term detecting functional, is given in B.20.

Proof. We first prove the integral assertions, before using any finite computation. The relations $c(m) = m$, $c(n) = n^{-1}$ give precisely the two conditions in (A.39.1). Since m, n

are a free pro-three basis, they are sufficient. The signs of c on V are $(4,6)$, so its signs on M_2 are $(21,24)$. The involution nc is conjugate to c by a pro-three square root of n, and therefore has the same signs. Its integral eigenspaces are direct summands, since two is a unit in $\mathbb{Z}_3$. Thus D is free of rank 45.

Since φ is c-fixed, its cocycle value belongs to $M_2^{c=1}$. For h in that invariant lattice, the global coboundary $g \mapsto (g-1)h$ belongs to D and evaluates to $(F_\varphi - 1)h$. The rational eigenvalues of F_φ on M_2 are

$$17^4, \qquad -17^3, \qquad 17^2. \tag{A.39.3}$$

Indeed on the rational equivariant splitting of A.7 the toric factor has eigenvalue 17^2 and the proper factor has scalar eigenvalue -17. None of (A.39.3) is one. The map $F_\varphi - 1$ is consequently invertible over $\mathbb{Q}_3$ on the 21-dimensional c-fixed target. Evaluation has rational rank 21. Its kernel R has rank 24, and $D/R \simeq \operatorname{im} E_\varphi$ embeds in a free $\mathbb{Z}_3$-module. Thus R is saturated and

$$R/3R \hookrightarrow D/3D \tag{A.39.4}$$

has dimension 24.

For $r \in R$ the unramified-outside-three cocycle has zero value at c and φ and zero tame inertia value. Its restriction to the local group at seventeen is therefore zero. This proves the exact local class-two assertion. The family is injective on global section classes as well. A geometric conjugator between two such class-two sections has invariant image in V, which is zero by Frobenius weights. It therefore lies in M_2. Normalization at c makes it c-fixed. If the difference of two members of R were its coboundary, evaluation at φ would give $(F_\varphi - 1)h = 0$, hence $h = 0$ by (A.39.3).

Here is a definite integral class-three lift of each member. Write

$$L = \gamma_2 N/\gamma_4 N, \qquad 0 \longrightarrow M_3 \longrightarrow L \longrightarrow M_2 \longrightarrow 0. \tag{A.39.5}$$

This is an exact sequence of free $\mathbb{Z}_3$-modules. Lift the two values of r equivariantly for the involutions c and nc, using their averaging idempotents. In the free presentation of A.28 take these as the L-valued cochain values at m, n, keep the reference tame section value c_O, and keep the value at c equal to one. The resulting crossed cochain β has $\beta(t) = 0$. Its value at the actual word W projects to $E_\varphi(r) = 0$, so $\beta(W) \in M_3$. That value is central in $N/\gamma_4 N$ and commutes with the unchanged tame section. The relation $WtW^{-1} = t^{289}$ is therefore satisfied exactly. Equivalently, its crossed value is $(1 - T^{289})\beta(W) = 0$, where T is the reference tame action on L and is the identity on M_3. The proved one-relator presentation supplies the global class-three lift.

We describe the finite comparison computing its remaining local obstruction. It uses the actual action of A.38 and does not require degree-three images of the geometric generators. Let X_i be their logarithms in the rational class-three nilpotent Lie algebra. The logarithm maps L onto the integral lattice with degree-two basis $[X_i, X_j]$ and degree-three Hall basis $[[X_i, X_j], X_k]$, with the usual Jacobi identifications. To check integrality, the sole correction needed is

$$\log[x_i, x_j] = [X_i, X_j] + \tfrac{1}{2}[X_i + X_j, [X_i, X_j]], \tag{A.39.6}$$

and two is invertible in $\mathbb{Z}_3$. Thus this is an actual lattice comparison. If a class-two automorphism has homology matrix A and central columns J_i, its action on $\overline{L} = L/3L$ is

$$\widetilde{G} = \begin{pmatrix} \bigwedge^2 A & 0 \\ \Theta(A,J) & \operatorname{Lie}_3(A) \end{pmatrix}, \quad \Theta(A,J)_{ij} = [J_i, AX_j] + [AX_i, J_j]. \tag{A.39.7}$$

Unknown degree-three terms in the images of X_i enter these brackets only in degree four and hence disappear in (A.39.5). Formula (A.39.7) is therefore determined by A.38.

For the O reference section, e_O is the seventh Schreier generator of A.38 and

$$T(z,w) = (z, w + [e_O, z]). \tag{A.39.8}$$

The source generators m, n have zero 17-Kummer value by A.35 and its transported construction. Their actions on L are consequently the original arithmetic actions. The program `a_ihara_gamma2_action.py` constructs (A.39.7), verifies composition and every involution and tame relation, and compares the entire degree-three block with the independent integral lattice of A.13. For the augmentation ideal generated by $m-1, n-1, t-1$ on $\overline{L}$, the dimensions of successive images are

$$375,\ 249,\ 132,\ 47,\ 4,\ 0. \tag{A.39.9}$$

In particular its fifth power is zero. A crossed cochain therefore kills D_6 of the arithmetic free group: the derivation of the sixth augmentation power lies in the fifth power applied to the module.

Every source term that can affect the answer is retained. The group D_3/D_6 is abelian of exponent three. On it the map

$$\mathrm{av}(w) = (w\, c(w))^2 \tag{A.39.10}$$

is exactly the c-fixed averaging idempotent. A.31 and A.37 give the known cubic part $m^6[n,[n,m]][t,[t,m]]^2$. The two unknown cubic terms have Lyndon words $012, 021$ in the order (m,n,t). All eighteen degree-four and all forty-eight degree-five Lyndon terms are then retained, with (A.39.10) applied to each. These exhaust the source freedom: the only restricted-power terms in these degrees are the degree-three generator cubes, and their coefficients are the already fixed abelian Frobenius coordinates. The source comparisons of A.38 ensure that these are the actual transported coefficients of W.

The local mod-three M_3 target is its c-fixed space of dimension 164. Divide by the actual rank-fifteen global restriction image in A.34; denote the resulting linear projection by Q, with 149-dimensional target. The factor two from norm Frobenius to rational Frobenius is a unit and does not affect any of the following ranks. The canonical integral lifts in (A.39.5) give an ambient linear map on $D/3D$. Evaluating every term just listed and applying Q shows that only the following three unknown terms survive:

$$[m,[n,t]], \qquad [[m,t],n], \qquad [m,[[m,n],n]]. \tag{A.39.11}$$

Every degree-five contribution vanishes. We retain the three actual coefficients as arbitrary members of $\mathbb{F}_3$; the rank bound below holds for all twenty-seven triples.

All local conjugators and global class-three corrections are also retained. The image of a local conjugator in V is invariant, hence zero by weights. Its degree-two part belongs to $M_2^{c=1}$ after normalization at c. Choose any of its 21-dimensional equivariant lifts h to $\overline{L}$. Allow the enlarged finite comparison in which

$$\beta(t) = (T-1)h, \qquad Q\big(\beta(W) - (F_\varphi - 1)h\big) \tag{A.39.12}$$

is evaluated, while the values at m, n are the chosen cochain lifts. This includes every reduction of a local comparison coming from a putative lift. Degree-three changes of h are already killed by the local coboundary quotient. Allowing every degree-two h only enlarges the set of finite corrections, so a nonzero obstruction after this allowance still excludes an integral lift.

Write the map (A.39.12), for its actual coefficient triple ϵ, as $G_\epsilon r + H_\epsilon h$. The complete finite calculation is

$\epsilon \in \mathbb{F}_3^3$	$\operatorname{rank} G_\epsilon$	$\operatorname{rank} H_\epsilon$	$\operatorname{rank}(G_\epsilon, H_\epsilon)$	$\operatorname{rank}(G_\epsilon \bmod \operatorname{im} H_\epsilon)$
each of the twenty-seven triples	30	6	30	24.

(A.39.13)

Here G_ϵ has 45 columns and H_ϵ has 21 columns. The images of all H_ϵ are the same six-dimensional subspace, so the last column can be computed in one fixed 143-dimensional target. For a reproducible specification of all their entries, take the ascending pivot columns of $1+c$ and $1-nc$ on $M_2/3$ for the two source bases, lift them by the same operators on $L/3$, and evaluate the words above using

$$\begin{aligned} \beta(ab) &= \beta(a) + a\beta(b), \\ \beta([a,b]) &= (1 - aba^{-1})\beta(a) + (a - [a,b])\beta(b). \end{aligned} \tag{A.39.14}$$

Use the literal Jacobi reductions in the basis of A.13 and the projection Q just defined. These formulas and A.38 specify every entry without an arithmetic oracle. The program `a_frobenius17_mixed_gluing.py`, run with `-local-conjugator`, saves the full matrices, all twenty-seven coefficient cases, and their exact elimination ranks. Its certificate also saves each of the sixty-eight candidate word maps before discarding the zero ones. In particular (A.39.13) retains the two mixed inertia terms and the quartic term, rather than assigning their values. The independent program `bc_gluing_audit.py` instead uses truncated noncommutative Magnus polynomials and the derivation formula $\beta((g_1-1)\cdots(g_d-1)) = (g_1-1)\cdots(g_{d-1}-1)\beta(g_d)$. It reproduces every entry of all sixty-eight word maps, the common six-dimensional image, and all twenty-seven rank computations.

Any two global class-three lifts of the fixed class-two target differ by an M_3 cocycle. Such a cocycle is unramified outside three: at every other finite prime the possible tame value lies in $H^0(\mathbb{F}_q, M_3(-1))$, which is zero by weights. Its effect is therefore in the actual global restriction image already removed by Q; this is also the localization comparison in A.33. Thus vanishing of $G_\epsilon r$ modulo $\operatorname{im} H_\epsilon$ is a necessary condition for a global class-three lift whose restriction at seventeen is the actual cusp section.

For the actual, unassigned triple ϵ, (A.39.13) says that the kernel of this ambient map on the 45-dimensional space $D/3D$ has dimension 21. The embedded subspace $R/3R$ in (A.39.4) has dimension 24. Its restriction consequently has rank at least $24-21=3$. This is the asserted obstruction. It is obtained from actual integral cochains in R; no mod-three local condition has been substituted for the exact kernel in (A.39.1).

Finally every lift in (A.39.2) would supply the forbidden locally cuspidal class-three lift. The full group maps to $N/\gamma_4 N$. The metabelian quotient does also, because $N'' \subset \gamma_4 N$, and its local section is the actual O cusp packet by A.19. For the marked quotient, the image of I_{Q_c} in V is the single line generated by e_{Q_c}. Hence $[I_{Q_c}, I_{Q_c}] \subset \gamma_3 N$ and $\gamma_3(I_{Q_c}) \subset \gamma_4 N$, proving the actual map to the same class-three quotient. The marked cusp is Q_c, whereas the ramified cusp here is O; B.14 identifies its local marked section with the unique actual O cusp section. The local tame exponent is one, and its unramified cusp parameter is zero because $17-1$ is a three-adic unit. Thus all three types of lift are excluded whenever the obstruction is nonzero. A basis of R spans its reduction, so an obstruction of rank at least three is nonzero on at least three basis members. This proves the finite-family assertion and the stated global exclusions. □

Statement A.40 (Proved). The excluded family also fails at the next nilpotent stage. *Every member of the family in A.39 with nonzero stated obstruction has no global lift to $N/\gamma_5 N$. Thus the twenty-four-member finite family there contains at least three class-two sections which lift globally to $N/\gamma_4 N$ and have no global lift to $N/\gamma_5 N$. The explicitly evaluated section in B.20 is one such member.*

Proof. First conjugate a proposed lift, through the actual surjection to $N/\gamma_3 N$, so that its class-two projection is the specified cocycle. The coprime involution argument in the

kernel then normalizes its value at c to one. Every global class-three lift differs from the lift constructed in A.39 by an M_3 cocycle. A.39 proves that this difference is unramified outside three, even when additional ramification was initially allowed. Hence its actual tame inertia at seventeen is still exactly c_O in $N/\gamma_4 N$.

Write F_0 for the reference rational Frobenius and F' for the Frobenius of the proposed lift. In class three their difference is an element $z \in M_3$. The relative local group has the exact relation

$$\mathrm{Frob}_{17} = c\,\varphi_{17}^{1/2}. \tag{A.40.1}$$

Its involution commutes with φ_{17}, so normalization at c makes z c-fixed. The integral A.7 matrix gives $\varphi_{17} = 1$ on $V/3$. Thus its square root is also one on every graded quotient modulo three, and the cocycle norm operator $1 + \rho(\varphi_{17}^{1/2})$ is $2I$ modulo three. The rational-Frobenius difference z is consequently one half of the norm-Frobenius difference modulo three. Nonzero obstruction in A.39 implies

$$\bar{z} \neq 0, \qquad C\bar{z} = \bar{z} \quad \text{in } M_3/3M_3, \tag{A.40.2}$$

after every permitted global class-three correction.

Let $M_4 = \gamma_4 N/\gamma_5 N$. In class four write the inertia and Frobenius of the hypothetical lift in the exact order

$$\tau' = c_O a_4, \qquad F' = z b_4 F_0, \qquad a_4, b_4 \in M_4. \tag{A.40.3}$$

A choice of lift of z from M_3 only changes b_4. The reference cusp relation is $F_0 c_O F_0^{-1} = c_O^{17}$. The group M_4 is geometrically central at this stage, and collecting the relation $F'\tau' F'^{-1} = (\tau')^{17}$ gives

$$(F_{17} - 17)a_4 = -17[z, e_O] \quad \text{in } M_4. \tag{A.40.4}$$

Indeed conjugating c_O^{17} by the chosen lift of z contributes $17[z, e_O]$; b_4 commutes with inertia, while the two contributions from a_4 are $F_{17}a_4$ and $17a_4$. This derives the sign in the stated ordering and retains the actual reference relation.

On $M_4/3M_4$ the action of rational Frobenius is C by (A.40.1). Since $17 \equiv -1 \pmod 3$, equation (A.40.4) becomes

$$(C + 1)\bar{a}_4 = [\bar{z}, e_O]. \tag{A.40.5}$$

The left side is c-invariant. The right side is c-anti-invariant, because $C\bar{z} = \bar{z}$ and $Ce_O = -e_O$. It remains to check its nonvanishing in the actual ordinary lower-central graded quotient.

The ordinary free Lie algebra on $V/3$ embeds in its tensor algebra: the Hall–Lyndon expansions have distinct unit leading words, giving an integral comparison before reduction. The element e_O is a primitive basis letter. A homogeneous tensor P of degree three commuting with that letter must be a scalar multiple of e_O^3. To see this directly, the equality $e_O P = P e_O$ forces every word of P to start with e_O; cancel that initial letter and repeat. The ordinary degree-three free Lie subspace has zero component in the multidegree consisting of that letter alone, since every bracket on one letter vanishes. Thus

$$[\bar{z}, e_O] = 0 \text{ in } M_4/3M_4 \quad \Longrightarrow \quad \bar{z} = 0. \tag{A.40.6}$$

By (A.40.2) the right side of (A.40.5) is nonzero. The plus and minus eigenspaces of an involution over $\mathbb{F}_3$ intersect trivially, so (A.40.5) is impossible. This excludes every global class-four lift of that class-two target, while A.39 already supplies its global class-three lift. □

Statement A.41 (Proved). The exact completed Fox image and fibers on the cusp family. *For the family R in A.39 put*

$$Q = N/N'', \qquad M = N'/N'', \qquad \Lambda = \mathbb{Z}_3[[V]], \qquad \mathfrak{a} = \ker(\Lambda \to \mathbb{Z}_3), \qquad J = \mathfrak{a}M. \tag{A.41.1}$$

Give M the arithmetic action twisted by the actual reference O section with tangent parameter 17. *Thus m, n, c have their original actions, and tame inertia acts by conjugation by c_O. Define*

$$\begin{aligned} D_M &= M^{c=1} \oplus M^{nc=-1}, \qquad D_J = J^{c=1} \oplus J^{nc=-1}, \\ E_W : D_M &\longrightarrow M^{c=1}, \qquad E_W(a,b) = \beta_{a,b}(W), \end{aligned} \tag{A.41.2}$$

where $\beta_{a,b}$ is the continuous crossed cochain on the free arithmetic source with values $(a, b, 0, 0)$ at (m, n, t, c). These are continuous linear maps of compact $\mathbb{Z}_3$-modules. The natural quotient $M \to M_2$ gives $\pi : D_M \twoheadrightarrow D$, with kernel D_J, and the complete metabelian image on this family is exactly

$$R_{\mathrm{met}} = \pi(\ker E_W) \subset R. \tag{A.41.3}$$

Equivalently, it is the kernel of the explicit continuous map

$$\begin{aligned} \mathfrak{o}_M : R &\longrightarrow J^{c=1}/E_W(D_J), \\ r &\longmapsto [\, E_W(\widetilde{r}\,)\,], \end{aligned} \tag{A.41.4}$$

where $\pi(\widetilde{r}) = r$. Every nonempty fiber of $\mathrm{Sec}(Q) \to \mathrm{Sec}(N/\gamma_3 N)$ over this family is an affine copy of

$$\ker(E_W : D_J \longrightarrow J^{c=1}). \tag{A.41.5}$$

There is no remaining conjugacy quotient after the actual local normalization in the proof. Formulas (A.41.2)–(A.41.5) retain the completed arithmetic Fox operator. They do not assign a numerical value to an unevaluated coefficient of W or of its action.

Proof. We construct the completed module and its maps. Let $x_1, \dots, x_{10}$ be the actual geometric Schreier basis in A.38, and denote their images in V by T_i. The completed Fox map gives the exact sequence

$$0 \longrightarrow M \longrightarrow \Lambda^{10} \xrightarrow{(T_1-1,\dots,T_{10}-1)} \mathfrak{a} \longrightarrow 0. \tag{A.41.6}$$

One direct proof uses the actual finite abelian covers with geometric subgroups $H_k = N' N^{3^k}$. Their based cellular chain maps send the ith lifted edge to $T_i - 1$. The kernel is H_k^{ab}. Inverse limits of these compact chain groups are exact, and the descending-subgroup comparison proved in A.19 identifies $\varprojlim H_k^{\mathrm{ab}}$ with N'/N''. This proves (A.41.6) and its based comparison. The action of geometric conjugation by x_i on M is multiplication by T_i. Consequently

$$J = \mathfrak{a}M = \gamma_3 N/N'', \qquad M/J = N'/\gamma_3 N = \bigwedge^2 V = M_2. \tag{A.41.7}$$

These equalities identify the quotient map used in the statement, including its integral lattice.

The algebra Λ is $\mathbb{Z}_3[[T_1 - 1, \dots, T_{10} - 1]]$, an integral domain. In (A.41.6) the reference tame action T is multiplication by the actual deck element T_O, the image of c_O. Therefore

$$1 - T^{289} : M \longrightarrow M \quad \text{is injective.} \tag{A.41.8}$$

Indeed it is multiplication by the nonzero power series $1 - T_O^{289}$ on a submodule of the torsion-free module Λ^{10}. This also proves that the tame fixed subgroup of M is zero.

The arithmetic action on M is semilinear over its actual action on Λ: $g(\lambda z) = g(\lambda) g(z)$. Its matrices are the Fox matrices of the actual based geometric automorphisms. Thus the continuous group-ring homomorphism from the free arithmetic source to $\mathrm{End}_{\mathbb{Z}_3}(M)$ and the completed Fox derivatives specify (A.41.2) explicitly as

$$E_W(a,b) = \rho\left(\frac{\partial W}{\partial m}\right) a + \rho\left(\frac{\partial W}{\partial n}\right) b. \tag{A.41.9}$$

The derivative with respect to t has zero cochain value. Here ρ is the reference twisted action just defined; neither its semilinear part nor the inertia dependence of W is removed. Since the word W is c-fixed and the cochain is normalized at c, its value is c-fixed. This proves the target in (A.41.2).

Averaging for the order-two actions lifts both eigenspaces through the surjection $M \to M_2$. Hence $\pi : D_M \to D$ is onto with kernel D_J. Reduction of (A.41.9) is exactly E_φ, because the tame action on M_2 is trivial and killing t sends W to the same actual φ. For $r \in R$, $E_W(\widetilde{r})$ therefore lies in $J^{c=1}$. Two choices of $\widetilde{r}$ differ by D_J, proving (A.41.4) is well-defined. The domains are compact and all maps are continuous, so $E_W(D_J)$ is closed. Choosing equivariant lifts of a finite basis of D makes (A.41.4) an explicitly continuous linear map.

We next prove that these kernels classify the actual section classes. A cocycle β with the values in (A.41.2), relative to the full reference cusp section, is a cocycle on the free arithmetic source. Its value on the relator is

$$\beta(WtW^{-1}t^{-289}) = (1 - T^{289})\beta(W). \tag{A.41.10}$$

By (A.41.8) it descends to the proved arithmetic presentation exactly when $E_W(a, b) = 0$. Its values at t, c, W then vanish, so its local section at seventeen is the literal reference section. Its class-two image lies in R by the preceding comparison. This constructs actual global metabelian sections from every member of $\ker E_W$.

Conversely, take any global metabelian section whose class-two class is the target indexed by $r \in R$. Lift a global class-two conjugator through $Q \to N/\gamma_3 N$ so that its projection equals the exact target cocycle; then normalize the value at c in the kernel. At every prime outside three and seventeen its abelian inertia is zero. A.19 implies that its entire metabelian inertia there is zero. Thus it factors through the arithmetic group of A.28 even if arbitrary additional ramification was initially allowed.

At seventeen A.19 places it in the actual O packet. The tame exponent is one, and the unramified cusp parameter vanishes since $17 - 1$ is a three-adic unit. Hence a local geometric conjugator identifies it with the reference cusp section. Its image in $N/\gamma_3 N$ centralizes that local section. This centralizer is trivial: its image in V is Frobenius-invariant and therefore zero; its remaining image in M_2 is likewise zero by (A.39.3). Consequently the conjugator lies in J. The normalization at c makes it c-fixed. Conjugating globally by this same element preserves the exact class-two target and makes the whole local section equal to the reference.

The resulting difference from the reference is an M-valued cocycle, because its abelian projection is the same. Its values at t, c, W are zero, so it gives a member of $\ker E_W$ reducing to r. The normalization is unique: the ratio of two such conjugators is tame-invariant in M, and (A.41.8) makes that ratio zero. More generally, a global conjugator between two normalized representatives has invariant image in V, hence lies in M, and its local inertia commutation again forces it to be zero. This proves the claimed bijection with section classes, including the absence of a remaining quotient.

The image formula (A.41.3), the obstruction formula (A.41.4), and the affine fiber (A.41.5) now follow from the exact sequence of these actual linear maps. In particular R_{met} is a closed $\mathbb{Z}_3$-submodule of the finite free lattice R. Its restriction has the explicit nonzero obstruction of A.39; its integral image has not been replaced by that first finite obstruction.

There is also an exact finite-quotient interpretation. Use the cofinal, arithmetic-stable quotients

$$M_{a,b} = M/(3^a M + \mathfrak{a}^b M), \qquad a, b \geq 1. \tag{A.41.11}$$

They are finite, since (A.41.6) is a finitely generated module over the noetherian ring Λ. The formal-series topology and a continuous surjection $\Lambda^r \to M$ show that these quotients are cofinal: a surjective homomorphism of compact Hausdorff groups is open, and the images

of the powers of $(3, \mathfrak{a})$ form the corresponding neighborhood basis. The identical argument applies to the finitely generated module J. For each pair, take the actual finite images of the arithmetic action and of the completed Fox derivatives in (A.41.9). The finite system asks for the two equivariant cochain values reducing to r mod 3^a and satisfying $E_W = 0$ in $M_{a,b}$. Membership in (A.41.3) is equivalent to solvability of every one of these finite linear systems. To prove the reverse direction, lift their solution conditions to closed subsets of the compact module D_M. Every finite collection is satisfied at a common refinement (a, b), so these subsets have the finite intersection property. Their intersection gives an actual solution in D_M. This establishes the integral criterion without an assumption that the transition maps on solution sets are onto. The finite inputs are the actual images of W and of the action; no calculation of them is inferred merely from writing the operator. □

Statement A.42 (Proved). A complete affine Fox criterion for every one-prime inert metabelian target. *Let $q \neq 3$ be inert in K_0, and let s be any class-two section unramified outside $\{3, q, \infty\}$. Use the actual presentation of A.28, written in the free basis m, n, t with $c(m) = m$, $c(n) = n^{-1}$, $c(t) = t^{-1}$, and put $Q_q = q^2$. Retain its actual c-fixed Frobenius word W. Normalize $s(c) = 1$, and choose an equivariant lift $\widetilde{s}$ of its three generator values to $Q = N/N''$ on this free source. Such a lift always exists before imposing the single relator.*

Let ρ_α be the action on $J = \gamma_3 N/N''$ obtained by conjugation by $\widetilde{s}$, where α is the prescribed abelian section. Set

$$U = \rho_\alpha(m), \quad B = \rho_\alpha(n), \quad T = \rho_\alpha(t), \quad C = \rho_\alpha(c), \quad \Phi = \rho_\alpha(W). \tag{A.42.1}$$

This action depends only on α and descends to the actual arithmetic group. In particular $\Phi T \Phi^{-1} = T^{Q_q}$. Define

$$\begin{aligned} \mathcal{D}_\alpha &= J^{C=1} \oplus J^{BC=-1} \oplus J^{TC=-1}, \\ \mathcal{J}_\alpha &= J^{T^{Q_q} C=-1}, \\ \mathcal{E}_W(a,b,d) &= \rho_\alpha\!\left(\frac{\partial W}{\partial m}\right) a + \rho_\alpha\!\left(\frac{\partial W}{\partial n}\right) b + \rho_\alpha\!\left(\frac{\partial W}{\partial t}\right) d, \\ \mathcal{L}_s(a,b,d) &= (1 - T^{Q_q}) \mathcal{E}_W(a,b,d) + \left(\Phi - \sum_{i=0}^{Q_q - 1} T^i\right) d, \\ e_s &= \widetilde{s}(W t W^{-1} t^{-Q_q}) \in \mathcal{J}_\alpha. \end{aligned} \tag{A.42.2}$$

The full global metabelian image consists exactly of the targets for which

$$[e_s] = 0 \quad \text{in} \quad \mathcal{J}_\alpha / \mathcal{L}_s(\mathcal{D}_\alpha). \tag{A.42.3}$$

For every target in that image, its fiber is an affine torsor under the explicitly specified compact group

$$\frac{\ker \mathcal{L}_s}{\{((U-1)h, (B-1)h, (T-1)h) : h \in J^{C=1}\}}. \tag{A.42.4}$$

All source actions, the affine error, and the finite images of W in these formulas are the actual based ones. The criterion is an exact completed linear system, including its affine origin and every conjugation. Its numerical evaluation for arbitrary input is not asserted.

Proof. Use the completed module and actual quotient maps of (A.41.6)–(A.41.7). The kernel of $Q \to A = N/\gamma_3 N$ is exactly the abelian compact module J. Every section can be normalized at c by the coprime involution argument. A generator value at m is fixed by c,

and it lifts through $Q^c \to A^c$. A generator value at n is a cocycle for the involution nc; it lifts by coprime complement conjugacy. The same assertion holds for the involution tc and the tame generator. These three choices are independent in the free source with operators. They give the stated equivariant $\widetilde{s}$ before any relator equation is required.

Two such lifts of a generator differ by $J \subset M$. Conjugation by M is trivial on the abelian module M, and any two generator values with the same image in V differ by M. Thus the induced action on M, and on J, depends only on the abelian generator value. This proves its dependence only on α. The relator of A.28 maps under $\widetilde{s}$ to an element of J, since the class-two target is an actual section. This element acts trivially on J. Thus ρ_α kills the relator and descends to the actual arithmetic group. Its involutions satisfy $C^2 = (BC)^2 = (TC)^2 = 1$, and its local relation is exactly $\Phi T \Phi^{-1} = T^{Q_q}$.

All other equivariant free-source lifts with the same exact class-two target are uniquely of the form

$$\widetilde{s}'(g) = \beta(g)\widetilde{s}(g), \tag{A.42.5}$$

where β is a J-valued crossed cochain for ρ_α, normalized at c. The source involution relations give $C\beta(m) = \beta(m)$, $BC\beta(n) = -\beta(n)$, and $TC\beta(t) = -\beta(t)$. Hence their generator values are precisely the domain $\mathcal{D}_\alpha$ in (A.42.2). The completed Fox formula gives $\beta(W) = \mathcal{E}_W(a, b, d)$, with no discarded tame term.

Write $r = WtW^{-1}t^{-Q_q}$. Direct crossed-cochain collection gives

$$\beta(r) = (1 - T^{Q_q})\beta(W) + \left(\Phi - \sum_{i=0}^{Q_q - 1} T^i\right)\beta(t). \tag{A.42.6}$$

For example the first three factors give $(1-\Phi T\Phi^{-1})\beta(W)+\Phi\beta(t)$; the last factor subtracts the displayed geometric sum. This proves the operator $\mathcal{L}_s$. The source identity $c(r) = t^{-Q_q} r^{-1} t^{Q_q}$ implies

$$Ce_s = -T^{-Q_q} e_s. \tag{A.42.7}$$

It gives the identical relation for $\beta(r)$, proving the target $\mathcal{J}_\alpha$ of the operator.

The error of (A.42.5) on the relator is the sum $e_s + \mathcal{L}_s(a, b, d)$ in the actual abelian kernel J. The proved presentation therefore shows that it descends to a global metabelian section exactly when this sum is zero. This proves (A.42.3). The image of $\mathcal{L}_s$ is closed, because its domain is compact. A different equivariant free lift changes e_s by $\mathcal{L}_s$ of the corresponding cochain and leaves ρ_α unchanged. Thus it translates the affine solution space by that same cochain and preserves the obstruction and the resulting sections. This gives the explicit comparison between all choices of affine origin.

It remains to identify the quotient on the solution space. The geometric centralizer of the class-two target in A is trivial. Its image in V would be Galois-invariant, hence zero by Frobenius weights; its remaining central image in M_2 is likewise zero. Thus any geometric conjugator between two lifts with the same exact target lies in J. To preserve normalization at c it must lie in $J^{C=1}$. Conjugation by h^{-1} changes (A.42.5) by the coboundary

$$((U - 1)h, (B - 1)h, (T - 1)h). \tag{A.42.8}$$

Its image under $\mathcal{L}_s$ is $(\rho_\alpha(r) - 1)h = 0$. These are all remaining conjugations, proving (A.42.4). Their image is compact and closed, so this is a Hausdorff compact quotient.

The criterion also applies to lifts initially allowed ramification at other primes. The abelian target is unramified there, and A.19 makes the entire local metabelian section unramified. Such a lift consequently factors through the same arithmetic group. To change the chosen class-two representative, lift its geometric conjugator through $Q \to A$ and conjugate every free lift and every actual action by that element. On M this is the explicit deck multiplication by its image

in V. For normalized representatives the conjugator in A is c-fixed; choose its lift through the actual surjection $Q^c \to A^c$, preserving the normalization literally. It intertwines the Fox operators, their errors, and their solution quotients. Thus (A.42.3)–(A.42.4) characterize section classes and not just the chosen representatives.

For finite evaluation, reduce these actual maps on a cofinal system of arithmetic-stable finite quotients of J, for example $J/(3^a J + \mathfrak{a}^b J)$. The generator actions and the completed Fox derivatives reduce to finite operators, and e_s reduces to its actual finite relator error. Vanishing of (A.42.3) is equivalent to solvability of all these finite affine linear systems. The proof is the compact finite-intersection argument in A.41, applied to the closed solution conditions in $\mathcal{D}_\alpha$. It retains compatibility across every precision and every augmentation degree. These finite inputs must be evaluated from the actual arithmetic word and action; the operator description does not supply numerical coefficients that have not been computed.

On the special family R, the reference cusp section is available. The unique local conjugation of A.41 sets the tame correction to zero and the Frobenius correction to zero. The injective operator $1 - T^{289}$ then reduces (A.42.6) exactly to $E_W = 0$, and the same conjugation removes the quotient (A.42.8). This is the explicit map from the general affine criterion to the image and fibers in A.41. □

Statement A.43 (Proved). Finite arithmetic image and fiber tests with arbitrary ramification support. *Let s be an exact continuous class-two section, normalized by $s(c) = 1$, and let α be its abelian projection. The following construction applies to each of*

$$G = N, \qquad G = N/N'', \qquad G = N/\gamma_3(I_{Q_c}). \tag{A.43.1}$$

It gives an exact image criterion and every fiber using explicit finite arithmetic fields, finite groups, and finite cocycle sets. The target may have infinite ramification support.

Put $H_0 = N$, $H_{n+1} = \Phi(H_n) = H_n^3[H_n, H_n]$, with closures understood. The finite tests below are indexed by $n \geq 1$; set

$$\begin{aligned} P_n &= N/H_n, \qquad A_n = N/(H_n\gamma_3 N), \\ G_n &= P_n, \quad P_n/P_n'', \quad P_n/\gamma_3(I_n), \end{aligned} \tag{A.43.2}$$

in the three cases, where I_n is the normal closure of the actual image of c_{Q_c} in P_n. Write s_n for the exact target cocycle in A_n, and let

$$S_n = \{3, \infty\} \cup \{q \neq 3 : \alpha \bmod 3^n \text{ is ramified at } q\}. \tag{A.43.3}$$

These are explicitly determined finite increasing sets at each precision. All finite geometric groups and their based arithmetic actions are constructed by the iterated elementary abelian covers in the proof.

Let F_n be the finite Galois field splitting the action on G_n. Let K_n be its compositum with the finite Galois field cut out by the target homomorphism $g \mapsto (s_n(g), \rho_{A_n}(g))$. If s_n is ramified outside S_n, the solution set below is empty immediately. Otherwise put

$$d_n = |G_n|[F_n : \mathbb{Q}], \qquad B_n = \prod_{p \in S_n \setminus \{\infty\}} p^{d_n(1+\lfloor \log_p d_n \rfloor)}. \tag{A.43.4}$$

Set $\Omega_0 = \mathbb{Q}$. Construct Ω_n as the compositum of Ω_{n-1} and all Galois extensions $E/\mathbb{Q}$ containing K_n, unramified outside S_n, and of degree at most d_n. The degree and discriminant bound B_n make this a terminating finite field enumeration. Put $\Delta_n = \mathrm{Gal}(\Omega_n/\mathbb{Q})$.

Define $X_n(s, G)$ by finite enumeration of all functions $z : \Delta_n \to G_n$ satisfying

$$\begin{aligned} z(gh) &= z(g)\, g(z(h)), \qquad z(c) = 1, \\ \pi_n z &= s_n, \end{aligned} \tag{A.43.5}$$

and the following finite additional conditions: for every $k \le n$, its projection to G_k is trivial on the inertia groups at every prime of $S_n \setminus S_k$. The target and coefficient actions in (A.43.5) are inflated through their specified finite fields. A failed immediate test makes all subsequent solution sets empty.

There are explicitly induced transition maps between these finite sets, and

$$[s] \in \operatorname{im}\bigl(\operatorname{Sec}(G) \to \operatorname{Sec}(N/\gamma_3 N)\bigr) \quad \Longleftrightarrow \quad X_n(s,G) \neq \varnothing \text{ for every } n. \tag{A.43.6}$$

Let $J_n = \ker(G_n \to A_n)$, with its actual c-action. The entire fiber over $[s]$ is

$$\varprojlim_n (X_n(s,G)/J_n^c), \tag{A.43.7}$$

where $h \in J_n^c$ acts by $z(g) \mapsto h^{-1}z(g)g(h)$. Thus the finite quotients in the fiber use the kernels J_n, with no replacement by a finite-level centralizer of the target.

Each stage is a terminating arithmetic construction from the finite target data at that precision. Formula (A.43.6) is an infinite sequence of finite tests; no finite stopping rule for every positive instance is asserted.

Proof. First we construct the finite geometric inputs. The subgroups H_n are characteristic, open, and cofinal in N. Indeed the iterated Frattini series of every finite three-group terminates, so every finite quotient of N kills some H_n. Induction also gives $x^{3^n} \in H_n$ for every $x \in N$. The image of H_n in V is exactly $3^n V$: cubes multiply its image by three and commutators have zero image. Thus $A_n^{\mathrm{ab}} = V/3^n V$, and S_n is computed from the same finite target table as s_n. Consequently every group in (A.43.2) has exponent dividing 3^n. Closedness of the geometric kernels and compactness give

$$G = \varprojlim G_n, \qquad N/\gamma_3 N = \varprojlim A_n. \tag{A.43.8}$$

Images of closed commutator subgroups under a surjection are the corresponding closed commutator subgroups. This proves the displayed formulas P_n/P_n'' and $P_n/\gamma_3(I_n)$ for the actual quotient maps. The marked normal subgroup is arithmetic-stable since Q_c is rational.

There is an algebraic construction of every P_n with its based arithmetic action. At the stage corresponding to H_n, write the actual open cover as $Y = C \setminus D$, where C is smooth proper and D is its reduced boundary. Choose an auxiliary interior point $b \in Y(\overline{\mathbb{Q}})$, construct the generalized Jacobian $J_{C,D}$, and use

$$j_b : Y \longrightarrow J_{C,D}, \qquad P \longmapsto [\mathcal{O}_C(P-b), \text{its boundary frames}].$$

The meromorphic section associated with $P - b$ is nonzero on D and specifies these frames. Pull back the finite étale map $[3] : J_{C,D} \to J_{C,D}$ along j_b. The induced map on geometric abelian three-adic homology is an isomorphism: on the proper quotient this is the usual Abel–Jacobi isomorphism, and on the boundary kernel it sends each inertia generator to the corresponding toric generator, with the same diagonal relation. The two exact sequences and this map prove the assertion integrally. For the proper comparison and generalized cover construction see [Mil08, Chapter III, Propositions 9.1 and 9.7]. It follows that this pullback is connected and has kernel $H_n^3[H_n, H_n] = H_{n+1}$.

All the constructions just used are finite algebraic operations: smooth normalization, divisor and framed-line bundle charts, the Jacobian group law, multiplication by three, and finite fiber products. They supply equations for the maximal elementary abelian three-cover without requiring an algorithm for relations among an infinite group of divisor classes. Choosing its lift of the prescribed tangential Puiseux point retains the based cover. In particular no divisor $P - d$ with $d \in D$ has been used in the generalized Jacobian. Changing

the auxiliary point to b' changes j_b by the translation $[b-b']$. If $3a=[b-b']$, the actual map between the two pullbacks is $(P,z)\mapsto(P,z+a)$. The prescribed Puiseux lift fixes its remaining three-torsion deck ambiguity.

The arithmetic descent is specified by an actual subgroup. Under the given tangential splitting, the stage-n cover corresponds to $H_n\rtimes G_{\mathbb{Q}}$ inside $N\rtimes G_{\mathbb{Q}}$; in the full arithmetic fundamental group take the inverse image of this subgroup. Its next pointed cover uses $\Phi(H_n)\rtimes G_{\mathbb{Q}}$ in the same way. The geometric isomorphism with the [3]-pullback is the one matching its prescribed tangential lift. Thus the splitting specifies the genuine framed descent, including the conjugations in its Galois maps. Computing a finite normal compositum of the equations and these based maps realizes that descent algebraically. This is also the iterated Kummer-cover construction in C.16. Only finitely many Puiseux coefficients are needed to distinguish the finitely many branches; Galois permutation of those coefficients and the finite equations computes the based action on P_n. Finite group quotient operations then compute G_n, A_n, their cusp images, and the field F_n. Thus the coefficient action is constructed from the curve.

We next prove the uniform ramification bound that makes the arithmetic sets finite even for infinite support. At every $q\neq 3$, each section of any group in (A.43.1) has either trivial tame inertia or inertia conjugate to

$$c_d^{a_q},\qquad a_q\in\mathbb{Z}_3, \tag{A.43.9}$$

at a rational local cusp. For N this is B.7–B.8, and for N/N'' it is A.19. For the marked quotient the two cases are explicitly B.12 and B.14: at the marked cusp B.12 gives exactly the inertia $c_d^{a_q}$, including when there is an additional central Frobenius parameter; at an unmarked cusp B.14 gives the actual cusp section. The final zero-inertia assertion of B.12 proves that zero abelian inertia makes the entire marked inertia trivial.

Every cusp image e_d is primitive in V. For the eight non-origin cusps this follows from the toric basis, and e_O is minus their sum. Thus $a_qe_d=0$ in $V/3^n$ implies $3^n\mid a_q$. The exponent bound on G_n then kills (A.43.9). Wild inertia at $q\neq 3$ is pro-q and has trivial image in a three-group. Hence every actual section above the target reduces to an unramified section outside S_n at level G_n. This proves the bound using only the abelian target at that precision. It also proves the immediate rejection when the finite class-two target itself ramifies outside S_n.

For precision, the sets in (A.43.3) are the ramification sets of actual finite cocycles. At $q\neq 3$ the coefficient action is unramified, so their inertia restrictions and their vanishing are independent of a coboundary change. The good compactification of the given curve makes every F_n unramified outside three. The fields K_n are finite and have degree at most $|A_n|[F_n:\mathbb{Q}]\le d_n$: use the target homomorphism with the action quotient $\mathrm{Gal}(F_n/\mathbb{Q})$ in its second component. After the immediate check they too are unramified outside S_n.

We give the field bound in (A.43.4). For a number field $E/\mathbb{Q}$ of degree at most d_n, unramified outside S_n, pass in each completion first to its maximal unramified subextension. For its remaining ramification index e, the derivative of an Eisenstein polynomial for a uniformizer has terms with pairwise distinct valuations modulo e. The different exponent is therefore at most $e-1+ev_p(e)$. Multiplying by residue degrees and summing gives

$$v_p\operatorname{disc}(E)\le d_n(1+\lfloor\log_p d_n\rfloor). \tag{A.43.10}$$

Taking the product proves the bound B_n. Apply the finite polynomial enumeration of C.14 and its checked Hermite bound, now at every prime of S_n: enumerate monic integral polynomials of degree at most d_n whose roots have the generous bound $2(B_n+1)$, factor them, and perform the exact degree, Galois, containment and ramification tests. This

lists every required field with its embeddings in the fixed algebraic closure. Taking their finite compositum with Ω_{n-1} constructs Ω_n. Decomposition and inertia subgroups are then computed by finite number-field operations. No feasibility estimate for this deliberately large enumeration is used.

Every continuous G_n-valued cocycle z unramified outside S_n defines the actual finite homomorphism

$$g \longmapsto (z(g), \rho_{G_n}(g)) \quad \text{into} \quad G_n \rtimes \mathrm{Gal}(F_n/\mathbb{Q}). \tag{A.43.11}$$

Its kernel cuts out a Galois extension of degree at most d_n. If its projection is s_n, this field contains K_n. It is one of the fields used in Ω_n, so every candidate cocycle factors through Δ_n. Conversely, every function satisfying (A.43.5) inflates to an actual global cocycle with the stated ramification. This proves that the finite enumeration misses no candidate. Normalization at complex conjugation and all local inertia checks are finite group checks on these same fields.

The additional lower-level ramification conditions are needed for the transition maps. For $z \in X_n$, its projection to G_k is unramified outside S_k. The homomorphism (A.43.11) at level k therefore cuts out one of the fields included in Ω_k. It follows that the projected cocycle factors through Δ_k. It satisfies all the conditions defining X_k, so projection gives the asserted transition map. This also describes its finite computation: deflate the projected function through the known restriction $\Delta_n \twoheadrightarrow \Delta_k$. The transition maps need not be onto.

Suppose first that a class above $[s]$ exists. Lift a geometric conjugator through $G \to N/\gamma_3 N$ to make its projection the exact chosen target cocycle. Its value at c is then in the pro-three kernel of this map. Coprime complement conjugacy in that kernel normalizes it to one while preserving the exact target. Its projections satisfy every ramification bound proved above, hence give a compatible point of $\varprojlim X_n$.

Conversely, nonempty finite sets X_n with the specified transition maps have a nonempty inverse limit: in the compact product of the finite sets, every finite list of compatibility equations is satisfied by projection from a common higher level. A compatible sequence of the inflated cocycles defines a continuous cocycle into $G = \varprojlim G_n$, because the inverse-limit topology tests continuity in each finite quotient. Its cocycle equation, normalization at c, and projection to the exact target follow at every level. It is the required global section. This proves (A.43.6), including the case of an infinite union of the sets S_n.

We finally prove the fiber formula rather than replacing its conjugation groups by finite approximations without comparison. Put $J = \ker(G \to N/\gamma_3 N)$. Its image in G_n is exactly J_n: if a lift of an element of J_n has class-two image in the image of H_n, multiply it by the inverse of an element of the image of H_n with that same class-two image. The resulting lift lies in J. Thus $J \twoheadrightarrow J_n$, and the same argument gives $J_n \twoheadrightarrow J_k$ for $n \geq k$. The kernels are pro-three and c-stable. Coprime complement conjugacy gives the actual surjections

$$J^c \twoheadrightarrow J_n^c, \qquad J_n^c \twoheadrightarrow J_k^c. \tag{A.43.12}$$

The global geometric centralizer of the target in $N/\gamma_3 N$ is trivial. Projection to V and then to M_2 reduces this assertion to their zero invariant groups, proved by Frobenius weights. Hence the global conjugators between two lifts of the exact target lie in J. The normalizations at c restrict them exactly to J^c. The fiber is therefore $(\varprojlim X_n)/J^c$.

This quotient is canonically the right side of (A.43.7). For surjectivity, choose representatives of a compatible sequence of finite orbits recursively. A conjugator needed to match representatives at level n lifts to level $n+1$ by (A.43.12), so they can be chosen compatible. For injectivity, if two compatible sequences are conjugate at every level, the corresponding sets of conjugators in the compact group J^c are nonempty, closed, and nested, again by (A.43.12). Their intersection gives one global conjugator. This proves (A.43.7) and retains every fiber identification.

At each precision the target input used by the procedure is a finite Galois presentation of its projected cocycle and its table. All subsequent field and finite-group operations terminate. The infinite assertion in (A.43.6) remains an infinite assertion, in agreement with the decision obstruction in A.12. The construction provides finite certificates of exclusion when a stage is empty and exact compatible finite descriptions of all nonempty fibers. □

Statement A.44 (Proved). Inherited comparison and every marked coefficient layer. *Let F be the free pro-three geometric group of $\mathbb{P}^1 \setminus \{0,1,\infty\}$, with the based generators x, y of A.38. The actual Fermat covering and its specified tangential comparison identify*

$$N = \ker\bigl(F \longrightarrow (\mathbb{Z}/3)^2\bigr), \qquad c_{Q_c} = y^3. \tag{A.44.1}$$

Put $J_y = \overline{\langle\langle y \rangle\rangle}_F$ and $I = I_{Q_c}$. The inherited quotients and their actual comparison maps are

$$\begin{aligned} N/N'' &\twoheadrightarrow H_{\mathrm{met}} := \mathrm{im}(N \to F/F''), \\ &\quad \ker = (N \cap F'')/N'', \\ N/\gamma_3 I &\twoheadrightarrow H_{\mathrm{depth}} := \mathrm{im}(N \to F/\gamma_3 J_y), \\ &\quad \ker = (N \cap \gamma_3 J_y)/\gamma_3 I. \end{aligned} \tag{A.44.2}$$

These are continuous genuine $G_{\mathbb{Q}}$-equivariant surjections, with the origins of A.38. The second is precisely the pro-three instance of LM7 (E25.1). At the prounipotent level over $\mathbb{Q}_3$ the image in each specified quotient of the base group is the whole base quotient. On abelian Lie algebras the map from the intrinsic group has kernel of dimension eight.

Set $Q = N/\gamma_3 I$ and $M_r^{\mathrm{mark}} = \gamma_r Q/\gamma_{r+1} Q$. There is an actual quotient of graded $\mathbb{Z}_3$-Lie algebras

$$\mathrm{gr}_\gamma Q = \mathcal{L}(V)/\gamma_3 \mathfrak{j}, \qquad \mathfrak{j} = (e_{Q_c})_{\mathcal{L}(V)}. \tag{A.44.3}$$

Every graded piece is finite free. Choose the free factor B of rank nine obtained by omitting the generator y^3 from the ten literal Schreier generators of A.38, and put $V_B = B^{\mathrm{ab}}$. As a choice of integral bases,

$$\begin{aligned} M_r^{\mathrm{mark}} \simeq \mathcal{L}_r(V_B) \;\oplus\; & V_B^{\otimes(r-1)} e_{Q_c} \\ & \oplus \Bigl(\bigwedge\nolimits^2 (\mathrm{T}(V_B) e_{Q_c})\Bigr)_r . \end{aligned} \tag{A.44.4}$$

Here a tensor word of length a followed by e_{Q_c} has ordinary degree $a+1$. The arithmetic action is the one induced by (A.44.3) from the actual action on V; (A.44.4) does not impose a block diagonal arithmetic action. For $r \geq 2$ its rank is

$$\frac{1}{r} \sum_{d \mid r} \mu(d) 9^{r/d} + 9^{r-1} + \frac{(r-1)9^{r-2} - \mathbf{1}_{2\mid r}\, 9^{r/2-1}}{2}. \tag{A.44.5}$$

In particular the ranks in degrees one through four are $10, 45, 330, 2466$.

For $r \geq 3$, put $A_{r,q} = M_r^{\mathrm{mark}}(-1)$ and $D_{r,q} = \mathrm{Frob}_q - 1$ on that lattice. Every $D_{r,q}$ is injective, and, with the restricted families of F.1, the complete coefficient formulas are

$$\begin{aligned} H^2_{\mathrm{cts}}(G_{\mathbb{Q}}, M_r^{\mathrm{mark}}) &= \mathcal{R}(A_r)/D_r\mathcal{R}(A_r), \\ \varprojlim{}^1{}_n H^1(G_{\mathbb{Q}}, M_r^{\mathrm{mark}}/3^n) &= \frac{\mathcal{R}(A_r) \cap \prod_{q \neq 3} D_{r,q} A_{r,q}}{D_r \mathcal{R}(A_r)} \neq 0, \\ H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, M_r^{\mathrm{mark}}) &= M_r^{\mathrm{mark}}/(u-1)(M_r^{\mathrm{mark}})^c. \end{aligned} \tag{A.44.6}$$

The quotient functor (A.44.3), applied to the matrices of A.7, gives all action matrices and hence an integral Smith calculation of the last group at every degree. Degrees one and two

are V and $\bigwedge^2 V$; their derived limits are those of F.5 and are also nonzero. Together with A.21–A.22, these formulas specify the cofinal finite-rank towers and the derived-limit term of every layer required in Task A.6. Their exact images and all fibers are the finite arithmetic sets of A.43.

Proof. The morphism of curves is $t = -z^3/x^3$. The two cube roots are $-z/x$ and $-y/x$, so its deck quotient is the indicated $(\mathbb{Z}/3)^2$. The based pullback $t^{1/3^n} \mapsto -s^{3/3^n}$ of A.38 identifies the arithmetic splittings, not just the outer actions. The Schreier loop y^3 is the marked cusp loop, with the same chosen path. Thus $I \subset J_y$, and $N'' \subset F''$. Passing to these closed subgroups proves the two kernels in (A.44.2). Compactness makes their images closed, so the displayed quotient maps are continuous surjections. Their arithmetic equivariance follows from this based covering map. They therefore also induce the specified pushforward maps on section classes.

For the rational prounipotent assertion, use the continuous morphism from the $\mathbb{Q}_3$-prounipotent completion of N to that of F. Its image contains x^3 and y^3. In every characteristic-zero nilpotent quotient, their logarithms are $3\log x$ and $3\log y$, so the image Lie algebra contains both $\log x$ and $\log y$. More explicitly, the continuous free-Lie homomorphism

$$\log x \longmapsto \tfrac{1}{3}\log_N(x^3), \qquad \log y \longmapsto \tfrac{1}{3}\log_N(y^3)$$

is a section of this morphism on the completed Lie algebras. It maps the second derived ideal into the second derived ideal. It also maps the ideal generated by $\log y$ into the intrinsic ideal generated by $\log_N(c_{Q_c})$, since $c_{Q_c} = y^3$. It therefore descends through both specified quotients and proves their inverse-limit surjectivity directly. Their depth and derived ideals have zero image in the two-dimensional abelianization. The source has ten-dimensional abelianization, and the displayed logarithms prove surjectivity there, giving kernel dimension eight. This argument concerns the specified prounipotent image of the covering map; it does not substitute completion for either integral kernel in (A.44.2).

We now prove the integral layer statement. The actual Schreier basis gives $N = B *_3 \langle c_{Q_c} \rangle$. The kernel I of its retraction to B is the free pro-three group on the continuous family $bc_{Q_c}b^{-1}$ indexed by $b \in B$. One can verify this directly: the free pro-three group on that profinite family, with B acting by left translation, forms a semidirect product with B. Sending its named generators to $bc_{Q_c}b^{-1}$ and sending B identically to B defines a homomorphism to N. The inverse sends c_{Q_c} to the generator indexed by 1 and sends the nine generators of B to themselves. The two compositions are the identity on their topological generating families.

Let

$$\begin{aligned} \Lambda_B &= \mathbb{Z}_3[[B]], \quad \mathfrak{a}_B = \ker(\Lambda_B \to \mathbb{Z}_3), \\ M &= \Lambda_B e, \qquad Z = \widehat{\bigwedge}^2_{\mathbb{Z}_3} M. \end{aligned} \tag{A.44.7}$$

The last completion is over finite coefficient and finite label quotients. The free class-two quotient of I is the group on $M \oplus Z$ with law

$$(m, z)(m', z') = (m + m', z + z' + \tfrac{1}{2} m \wedge m'). \tag{A.44.8}$$

Its named cusp conjugate is $(be, 0)$, and B acts by left multiplication on M and its induced exterior-square action on Z. The coefficient $1/2$ is an integral three-adic unit. The class-two collection formula first proves this model for finite generating sets and then, through finite partitions of B, for the completed family. The two universal maps just constructed give the actual continuous isomorphism

$$Q \simeq B \ltimes (M \oplus Z), \qquad \bar{I} := I/\gamma_3 I \simeq M \oplus Z. \tag{A.44.9}$$

No arithmetic stability of the chosen free factor B is required for this geometric isomorphism.

For $r \geq 1$ define

$$M_r^* = \mathfrak{a}_B^{r-1} M,$$
$$Z_r^* = \overline{\sum_{\substack{a,b\geq 0 \\ a+b\geq r-2}} \mathfrak{a}_B^a M \wedge \mathfrak{a}_B^b M}. \tag{A.44.10}$$

and set $G_r = \gamma_r B \ltimes (M_r^* \oplus Z_r^*)$. We prove $G_r = \gamma_r Q$, including the closures. The integral Magnus map identifies Λ_B with the noncommutative formal-series ring on nine variables, and maps $\gamma_r B$ into $1 + \mathfrak{a}_B^r$. Thus $\gamma_r B$ raises the augmentation degree on M by at least r, and raises total degree on a wedge by at least r. Formula (A.44.8) gives $[G_r, G_s] \subset G_{r+s}$. Since $G_1 = Q$, this proves $\gamma_r Q \subset G_r$.

For the reverse inclusion, the projection of a commutator with $b \in B$ to M is multiplication by $b - 1$. Iterating such commutators on the cusp conjugates shows that

$$\gamma_{a+1} Q \cap \bar{I} \longrightarrow \mathfrak{a}_B^a M \quad \text{is onto.} \tag{A.44.11}$$

Indeed these commutators give every monomial in the nine augmentation generators times every be; their closed linear span is the indicated ideal. The image is closed because the source is compact. Bracketing a lift of $m \in \mathfrak{a}_B^a M$ with a lift of $m' \in \mathfrak{a}_B^b M$ gives exactly $(0, m \wedge m')$ by (A.44.8). Hence $Z_r^* \subset \gamma_r Q$. A lift in (A.44.11) of any $m \in M_r^*$ has central component in Z_r^* by the already proved inclusion. Remove that component to obtain $(m, 0)$ in $\gamma_r Q$. Finally the embedded $\gamma_r B$ belongs to $\gamma_r Q$. These facts prove equality with G_r.

Magnus monomials now give $\mathrm{gr}_{\mathfrak{a}_B} M = \mathrm{T}(V_B)e$. On Z their unordered distinct pairs give an integral basis at every total degree. The completed product introduces no relations: at any fixed degree there are only finitely many such words and pairs, and the coefficient projections are the actual continuous inverse-limit maps. This proves (A.44.4) and finite freeness.

For completeness the comparison with the free Lie quotient is explicit. The free Lie algebra $\mathcal{L}(V_B)$ acts on $\mathcal{L}(\mathrm{T}(V_B)e)$ by extending left concatenation as derivations. There are inverse Lie homomorphisms

$$\mathcal{L}(V_B \oplus \mathbb{Z}_3 e) \;\rightleftarrows\; \mathcal{L}(V_B) \ltimes \mathcal{L}(\mathrm{T}(V_B)e). \tag{A.44.12}$$

One sends V_B, e to the named generators. The other sends $v_1 \cdots v_a e$ to $\mathrm{ad}(v_1) \cdots \mathrm{ad}(v_a)e$ and is the identity on V_B. The universal property of the free Lie algebra extends these maps, and their composition is the identity on V_B, e and all their iterated actions. Quotienting the second free Lie factor by its third lower central term leaves $\mathrm{T}(V_B)e \oplus \bigwedge^2(\mathrm{T}(V_B)e)$. Under the actual maps in (A.44.11), this is exactly the graded group just computed. It proves (A.44.3) integrally. Equivariance uses the actual action on V and the arithmetic-stable cusp line $\mathbb{Z}_3 e_{Q_c}$, so it also proves the stated prescription for arithmetic matrices.

The first summand of (A.44.4) has the Witt rank in (A.44.5), and its second has rank 9^{r-1}. For the third, there are $(r-1)9^{r-2}$ ordered pairs of tensor words of total ordinary degree r. An equal pair exists only for even r, with $9^{r/2-1}$ choices. Subtract these pairs and divide by two to count unordered distinct pairs. This proves (A.44.5). The program `a_marked_layers.py` in `code/` independently enumerates all ordinary Lyndon words on ten letters through degree five, retains marked-letter counts zero, one and two, and checks all three summands and the ranks $45, 330, 2466$. Its exact output and JSON certificate are saved.

Finally, over K_0 the finite coefficient representations of these graded modules have three-group image, since they are functorial quotients of tensor powers of V. The composition-series argument of A.1 gives $H^i(G_{\mathbb{Q},S}, M_r^{\mathrm{mark}}/3^n) = 0$ for $i \geq 2$ and $S = \{3, \infty\}$. The normalized free-source calculation of A.10 gives the complex $[(M_r^{\mathrm{mark}})^c \xrightarrow{u-1} M_r^{\mathrm{mark}}]$. In degree $r \geq 3$, all Frobenius weights after twisting by -1 have absolute value at least $q^{r/2-1} > 1$; hence $D_{r,q}$

is injective. F.1–F.3 therefore give every formula in (A.44.6). The nonzero free summand $\mathcal{L}_r(V_B)$ makes each coefficient lattice nonzero. Chebotarev at splitting primes of the reduction of $M_r^{\mathrm{mark}}(-1)$ modulo 3^n makes all entries of $D_{r,q}$ divisible by arbitrarily large powers of three. F.3 then proves the strict nonvanishing of its derived limit. This also proves the same nonvanishing after every finite number-field extension, as in F.5.

The finite-rank quotients are the actual $N/(\gamma_3 I\,\gamma_r N)$ of A.21, and their coefficient kernels are precisely the free lattices just computed. The metabelian tower uses the actual quotients $N/(N''\gamma_r N)$ and the lattices of A.22. A.21 proves their cofinality and the comparison on section classes. At every layer the residue families in (A.22.2) or (A.44.6) retain the integrality and restricted-convergence requirements of F.1. The finite sets of A.43 additionally retain all inherited ramification conditions; their compact inverse limit gives the exact global image and fiber without omitting any of these layer obstructions. □

1.8 Cofinal higher quotients and the first further coefficient

All commutator subgroups in this subsection are closed, and all group quotients are pro-three. Put

$$R_{\mathrm{met}} = N'', \quad R_{\mathrm{mark}} = \gamma_3(I_{\mathrm{mark}}), \quad Q_R = N/R, \quad Q_{R,r} = N/(R\gamma_r N) \quad (r \geq 3).$$

We also use $R = 1$ for the full group. The notation Sec_S means cocycles inflated from $G_{\mathbb{Q},S}$, $S = \{3, \infty\}$, with its actual action. The manuscript comparison is *La Longue Marche*, §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Statement A.20 (Proved). Every unramified class-two section lifts to the full group. *For each of the three choices of R, the map*

$$\mathrm{Sec}_S(Q_R) \longrightarrow \mathrm{Sec}_S(N/\gamma_3 N)$$

is surjective. Its fibers have the following exact description. Use u, c of A.10 and let B be a pro-three quotient of N. Writing B^c for the fixed subgroup of genuine complex conjugation, we have a bijection

$$\mathrm{Sec}_S(B) = B/\{x \sim h^{-1}x\,u(h) : h \in B^c\}. \tag{A.20.1}$$

For $p : B \twoheadrightarrow N/\gamma_3 N$, put $J = \ker p$. The fiber over the class represented by x in (A.20.1) is

$$p^{-1}(x)/\{y \sim h^{-1}y\,u(h) : h \in J^c\}. \tag{A.20.2}$$

These orbit sets specify all identifications between the lifts.

Proof. The based action on N is unramified outside three. Over K_0 its image on $N/\Phi(N) = V/3$ is a three-group: the toric and elliptic graded pieces are trivial and the extension is unipotent. The kernel of $\mathrm{Aut}(N) \to \mathrm{Aut}(V/3)$ is pro-three. Indeed its image on a characteristic finite three-group quotient acts trivially on the Frattini quotient, and successive lower exponent-three factors show that this automorphism group has three-power order. Thus the whole action over K_0 factors through P_3 of A.10. For a finite three-group-valued cocycle, its graph together with this action also has three-group image. The cocycle therefore factors through the same relative quotient $\Gamma = P_3 \rtimes \langle c \rangle$. Taking characteristic inverse limits proves this for B.

We have $H^1(C_2, B) = 1$. For finite three-groups, conjugacy of complements of order two follows inductively by averaging in a central elementary abelian subgroup. For B, the solution sets are nonempty closed subsets of the compact group, so the finite intersection property gives the same result. Normalize a cocycle by $z(c) = 1$. Since u, cuc freely generate

P_3, the value $z(u)$ is arbitrary and determines z uniquely. The conjugations preserving this normalization are precisely B^c, and their action is (A.20.1).

For a surjection with this involution, $B^c \to (B/J)^c$ is surjective: the defect of any lift of a fixed element is a C_2-cocycle in J, which the same argument removes. Lifting x now proves surjectivity on section classes, and the fixed-subgroup assertion moves every representative of the fiber above x itself. Its remaining conjugators project to the stabilizer of x under $(N/\gamma_3 N)^c$. That stabilizer centralizes the whole Γ-section; it is trivial by C.1 applied to the inflated arithmetic section. Thus exactly J^c remains, giving (A.20.2). □

Statement A.21 (Proved). Cofinality and the exact comparison of section limits. *For every number field L and each specified R, the natural maps are an isomorphism of pro-three groups and a bijection of pointed sets, respectively:*

$$Q_R \xrightarrow{\sim} \varprojlim_{r\geq 3} Q_{R,r}, \qquad H^1_{\mathrm{cts}}(G_L, Q_R) \xrightarrow{\sim} \varprojlim_{r\geq 3} H^1_{\mathrm{cts}}(G_L, Q_{R,r}). \tag{A.21.1}$$

Each $Q_{R,r}$ has a finite central series of finitely generated $\mathbb{Z}_3$-module factors, and every finite quotient of Q_R factors through one of them. They are cofinal finite-rank quotients for this section problem.

Proof. A finite three-group is nilpotent: the class equation gives a nontrivial center, and induction gives a central series. Consequently every finite quotient kills some $\gamma_r Q_R$. This proves cofinality and that their intersection is trivial. Compactness gives the first isomorphism; a finite collection of compatible coordinates lifts to Q_R, proving surjectivity. The lower-central factors of N are generated over $\mathbb{Z}_3$ by the finitely many iterated commutators of fixed degree in its ten generators. Their quotients give the finite-rank assertion.

For compatible cohomology classes, choose representatives successively. The image of the next cocycle is conjugate to the preceding one; lift the conjugator through the surjective group map and modify the new cocycle. This gives exactly compatible continuous representatives. Their inverse limit is continuous for the inverse limit topology and is a cocycle. For injectivity, the sets of elements of Q_R conjugating two cocycles at level r are nonempty closed nested subsets of a compact group. Their intersection supplies one conjugator at every level. This proves the second assertion of (A.21.1). □

Statement A.22 (Proved). Metabelian coefficient lattices and the next lifting fiber. *For $r \geq 2$ put $M_r = \gamma_r(N/N'')/\gamma_{r+1}(N/N'')$. There is a natural $G_{\mathbb{Q}}$-equivariant isomorphism*

$$M_r \simeq \ker\bigl(V \otimes \mathrm{Sym}^{r-1} V \longrightarrow \mathrm{Sym}^r V\bigr), \qquad \operatorname{rank} M_r = (r-1)\binom{r+8}{r}, \tag{A.22.1}$$

where the arrow is multiplication. For $r \geq 3$, write $A_{r,q} = M_r(-1)$ and $D_{r,q} = \mathrm{Frob}_q - 1$. Then

$$\begin{aligned} H^2_{\mathrm{cts}}(G_{\mathbb{Q}}, M_r) &= \mathcal{R}(A_r)/D_r\mathcal{R}(A_r), \\ \varprojlim{}^1 H^1(G_{\mathbb{Q}}, M_r/3^n) &= \frac{\mathcal{R}(A_r) \cap \prod_{q\neq 3} D_{r,q}A_{r,q}}{D_r\mathcal{R}(A_r)} \neq 0, \\ H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, M_r) &= M_r/(u-1)M_r^c. \end{aligned} \tag{A.22.2}$$

The matrices on M_r are the kernel functor (A.22.1) applied to the explicit matrices on V in A.7, giving a finite integral prescription for every first-cohomology Smith factor.

*The next coefficient of the full, metabelian, and marked depth-two towers is the same rank-*330 *lattice* M_3*, and*

$$H^1(G_{\mathbb{Q}}, M_3) \simeq \mathbb{Z}_3^{166} \oplus (\mathbb{Z}/3)^7 \oplus (\mathbb{Z}/9)^{56} \oplus (\mathbb{Z}/27)^{25} \oplus (\mathbb{Z}/81)^3. \tag{A.22.3}$$

Every nonempty lifting fiber from class two to class three is a torsor under (A.22.3).

Proof. We verify the integral lattice in (A.22.1). In a metabelian graded Lie ring the adjoint generators commute on the derived ideal. Antisymmetry and Jacobi collect every degree-r bracket into left-normed brackets

$$[x_i, x_j, x_{k_3}, \dots, x_{k_r}], \qquad i > j, \quad j \le k_3 \le \cdots \le k_r. \tag{A.22.4}$$

To prove independence integrally, use $\Lambda = \mathbb{Z}_3[[X_1, \dots, X_{10}]]$ and the upper triangular matrices with diagonal $(1 + X_i, 1)$ and upper-right entry $e_i \in \Lambda^{10}$. Their closed generated group is pro-three in the $(3, X_1, \dots, X_{10})$-adic topology, so the free pro-three universal property defines a continuous homomorphism from N. Commutators are translations, so it kills N''. The translation of $[x_i, x_j]$ is $X_i e_j - X_j e_i$; each further right bracket multiplies its leading term by $-X_k$.

In a fixed multidegree of total degree r, let J be the variable support. The multiplication kernel consists of coefficient vectors indexed by J with sum zero. The brackets (A.22.4) in that multidegree map, up to their common sign, to the standard differences with distinguished index $j = \min J$ and other index $i \in J \setminus \{j\}$. They are an integral basis. Collection proves spanning and these matrices prove independence, hence the absence of three-torsion and the asserted isomorphism. The bracket/multiplication construction is natural in V, proving Galois equivariance. Subtracting the symmetric-power ranks gives the displayed rank formula.

Over K_0 these finite coefficient representations have three-group image. The composition-series proof of A.1 gives $H^i(G_S, M_r/3^n) = 0$ for $i \ge 2$. Normalized cocycles of A.10 give the finite-S complex $[M_r^c \xrightarrow{u-1} M_r]$. For $r \ge 3$, the weights give injective $D_{r,q}$, as in F.5, so the integral degree-one residue invariants vanish. Localization and F.1–F.3 prove (A.22.2), including the nonzero derived limit.

The metabelian relation starts in degree four. The marked relation also does: modulo $\gamma_3 N$, conjugates of the marked cusp generator differ from it by central elements, so its normal closure is abelian there. Thus $[I_{\mathrm{mark}}, I_{\mathrm{mark}}] \subset \gamma_3 N$ and $\gamma_3(I_{\mathrm{mark}}) \subset \gamma_4 N$. This proves the common degree-three coefficient.

The exact program `code/a_degree3_lattice.py` constructs the 330-dimensional multiplication kernel by these multidegree differences, verifies the induced matrices, and gives involution eigenspace ranks $164, 166$. The 330 by 164 matrix $(u-1)|M_3^c$ has Smith valuations

$$0^{73}, \quad 1^7, \quad 2^{56}, \quad 3^{25}, \quad 4^3.$$

All 164 pivots are obtained modulo 3^{10}; their valuations are at most four. The recorded invertible row and column operations therefore certify the exact three-adic Smith factors, giving (A.22.3). C.1 makes the target geometric centralizer trivial, so no additional stabilizer identifies the central lifting torsor. □

Statement A.23 (Proved). The complete higher images and fibers in Task A.6. *For each of* $G = N/N''$ *and* $G = N/\gamma_3(I_{\mathrm{mark}})$*, let* $X_n(s, G)$*,* G_n*,* A_n *and* J_n *be the explicitly constructed finite arithmetic data of A.43 for an exact class-two target* s*. Then the image consists exactly of the classes satisfying*

$$X_n(s, G) \neq \varnothing \quad \textit{for every } n,$$

and its entire fiber is

$$\varprojlim_{n} X_n(s,G)/J_n^c.$$

Every finite test terminates and uses the actual based arithmetic action and finite target table. The inherited ramification constraints give its transition maps, including for infinitely ramified targets. The lower-central quotients of A.21 are cofinal for the same section problem. Their metabelian coefficient lattices and derived limits are A.22; their marked coefficient lattices and derived limits are A.44. The exact maps to the inherited quotients, with both kernels, are (A.44.2).

Proof. The two geometric kernels are closed and arithmetic-stable. The quotient maps in (A.43.2) are their actual finite images, and (A.43.8) identifies their inverse limits with the specified groups. Normalization at complex conjugation is possible inside each pro-three lifting kernel. A.43 proves the uniform ramification bound from the abelian target and enumerates every possible finite graph field within its explicit degree and discriminant bounds. The resulting finite sets therefore give both directions of the image criterion, with continuous compatible cocycles.

For the fiber, A.43 proves that the image of the actual geometric kernel is J_n, and that its c-fixed subgroup surjects onto J_n^c. The global class-two centralizer is trivial. Its compact conjugator comparison consequently gives precisely the displayed inverse limit of orbits. No extra finite-level centralizer is divided out.

A.21 proves both the group and section-class comparisons with the stated finite-rank lower-central towers. A.22 and A.44 identify the actual coefficient lattices and actions; their residue formulas use the image on restricted integral families and retain its nonzero first derived limit at every positive higher layer. Thus convergence has not been replaced by independent local solvability. Formula (A.44.2) comes from the same based Fermat covering and gives the requested inherited comparisons. These statements together settle every image, fiber, cofinality and coefficient-layer requirement of Task A.6. The finite arithmetic stages form an infinite criterion; no total stopping algorithm is used. □

1.9 Explicit nine-torsion covers at the first auxiliary Frobenius

Statement A.36 (Proved). Miller covers and their two exact norm comparisons at seven. *The Fermat curve in the task has the following specified isomorphism with the minimal Weierstrass curve $E_m : y_m^2 + y_m = x_m^3 - 7$:*

$$x_m = \frac{-3z}{x+y}, \quad y_m = \frac{4x-5y}{x+y}, \qquad [x:y:z] = [y_m+5 : 4-y_m : -3x_m]. \tag{A.36.1}$$

It carries O to the Weierstrass origin and carries P, Q_c to $(3,-5),(3,4)$, respectively. All of $E_m[9]$ is defined over the unramified degree-three extension of $\mathbb{Q}_7$. Its reduction over $k = \mathbb{F}_7[\theta]/(\theta^3-2)$ has exactly 81 points. One point above P under multiplication by three is

$$Q = (2+\theta^2,\ 1+\theta+6\theta^2), \qquad \varphi Q - Q = (6,2), \tag{A.36.2}$$

where φ raises the residue coordinates to their seventh powers. The unique nine-torsion lifts give the corresponding identities over the unramified local extension.

For each of the nine points Q above either $d = P$ or $d = Q_c$, let $v_R = x_m - x_m(R)$ and let $\ell_{R,S}$ be the line through R, S, tangent at R when $R = S$, normalized to have y_m coefficient one. Then the actual function

$$a_Q = \frac{\ell_{Q,Q}\ell_{2Q,Q}}{v_{2Q}v_d}, \qquad \operatorname{div}(a_Q) = 3(Q) - (d) - 2(O), \tag{A.36.3}$$

defines a connected cubic cover $w^3 = a_Q$ which is finite étale over U. Its geometric character is an explicit Kummer class with nonzero cusp valuations at d, O. With $\iota(R) = -R$, the following function identities hold:

$$a_Q\, \iota^* a_Q = \frac{(x_m - x_m(Q))^3}{x_m - x_m(d)}, \qquad \prod_{i=0}^{2} a_{\varphi^i Q} = g_Q^3. \tag{A.36.4}$$

In the second identity, put $Q_i = \varphi^i Q$, $S = Q_0 + Q_1$, and

$$g_Q = \frac{\ell_{Q_0,Q_1}\ell_{S,Q_2}}{v_S v_d}. \tag{A.36.5}$$

This function is defined over $\mathbb{Q}_7$, and its reduction is defined over $\mathbb{F}_7$.

For the exact Fermat parameter $s = z/x$ and its unit tangent, the regularized constants are

$$(s^2 a_Q)(O) = 9, \qquad (s^2(x_m - 3))(O) = (s^2(x_m - 3\zeta^2))(O) = 9. \tag{A.36.6}$$

In the large model $Y^2 = X^3 - 432$, $X = 4x_m$, $Y = 8y_m + 4$, the corresponding normalized Miller function is $4a_Q$, with regularized constant 36. For the residue embedding $\zeta \mapsto 2 \in \mathbb{F}_7$, the cubic residue of 9 is $4 = \zeta^2$, whereas that of 36 is 1. These are arithmetic cover data; no central Galois action on an intrinsic Heisenberg quotient is inferred from them.

Proof. Substitution in (A.36.1) gives the identity

$$(x + y)^3(y_m^2 + y_m - x_m^3 + 7) = 27(x^3 + y^3 + z^3).$$

The displayed inverse gives the reverse identity. The rational maps extend across their omitted points because both projective curves are smooth. Their composites on the displayed dense affine opens are the identity, proving the claimed isomorphism and its stated cusp images. The discriminant of the minimal equation is -3^9, so it has good reduction at seven. Multiplication by nine is étale over $\mathbb{Z}_7$, and reduction identifies its nine-torsion over an unramified extension with the nine-torsion of the residue curve.

The exact program `a_miller_f343_root.py` constructs k, lists all points of $y_m^2 + y_m = x_m^3$, and applies the stated Weierstrass group law. There are 324 points, 81 killed by nine, and nine killed by three. Every three-torsion point has coordinates in $\mathbb{F}_7$. It lists all nine preimages of each specified cusp and verifies (A.36.2) and each Frobenius orbit. The geometric nine-torsion has order 81 because nine is invertible in characteristic seven; the enumerated points therefore exhaust it. Finite étale lifting proves the local assertion, rather than merely an assertion about an unrelated finite-field group.

The divisors of the normalized line and vertical functions are

$$\begin{aligned} \operatorname{div}(\ell_{R,S}) &= (R) + (S) + (-R - S) - 3(O), \\ \operatorname{div}(v_R) &= (R) + (-R) - 2(O). \end{aligned}$$

Their substitution proves (A.36.3). At every point of U the valuation is a multiple of three, so the normalized cubic cover is unramified there. Characteristic zero then makes it étale. Its valuation -1 at d proves that the function is not a cube and that the cover is connected. This gives an actual finite cover whose Kummer character is defined by the displayed function.

Writing $t = -x_m/y_m$ at the origin gives $x_m = t^{-2} + O(t)$, $y_m = -t^{-3} + O(1)$. Each normalized line has leading term $-t^{-3}$ and each vertical function has leading term t^{-2}. Consequently a_Q and g_Q have leading term t^{-2}. The divisors of the two sides of the first identity in (A.36.4) agree, and their leading coefficients at O are both one. Their ratio is therefore the constant one on the smooth proper curve, proving that identity.

For the second identity, write $D_Q = \varphi Q - Q$. The finite computation and finite étale comparison give $D_Q \in E_m[3](\mathbb{Q}_7)$. Thus $Q_i = Q + iD_Q$, and $Q_0 + Q_1 + Q_2 = 3Q + 3D_Q = d$. The line-divisor calculation in (A.36.5) gives

$$\operatorname{div}(g_Q) = (Q_0) + (Q_1) + (Q_2) - (d) - 2(O).$$

This divisor is fixed by φ. Its normalized leading coefficient is one, so $\varphi(g_Q) = g_Q$. The function descends to $\mathbb{Q}_7$. Comparing divisors and the same leading coefficient proves the cube identity. The finite-field program also multiplies the function numerators in the exact ring $k[x_m, y_m]/(y_m^2 + y_m - x_m^3)$ and verifies both identities by cross-multiplication, for all 18 points.

Finally on the Fermat chart $x = 1$, $y = -(1 + s^3)^{1/3}$ gives $x_m = 9s^{-2} + O(s)$ and $y_m = -27s^{-3} + O(1)$. Hence $s = 3t + O(t^4)$, which proves (A.36.6). The large-model line and vertical functions are eight and four times the corresponding minimal-model functions, so their two-line quotient is $4a_Q$. Reduction of the retained constants gives $9 \equiv 2$, $36 \equiv 1 \pmod 7$; the cubic residue map on $\mathbb{F}_7^\times$ is $u \mapsto u^2$. This proves the stated tangent phases without dropping the origin term. The program retains the characteristic-zero coordinate identity, tangent expansions, complete field coordinates, line coefficients, and all exact outputs. □

Statement A.37 (Proved). An order-eighty-one arithmetic quotient and a tame-letter Frobenius coefficient at seventeen. *Put $K_0 = \mathbb{Q}(\zeta)$, $\zeta^2 + \zeta + 1 = 0$, and $L_{17} = K_0(\beta)$ with $\beta^3 = 17$. Let $\sigma(\beta) = \zeta\beta$ and set*

$$\delta = \frac{-7 - \zeta - 3\beta - \beta^2}{3}, \qquad \delta_i = \sigma^i(\delta), \qquad r_i^3 = \delta_i \quad (i \in \mathbb{Z}/3). \tag{A.37.1}$$

The extension

$$M_{17} = L_{17}(r_0, r_1, r_2)/K_0 \tag{A.37.2}$$

is Galois, unramified outside three and seventeen, with group

$$G = (\mathbb{F}_3)^3 \rtimes \langle s \rangle, \qquad s^3 = 1, \tag{A.37.3}$$

where $s(r_i) = r_{i+1}$ and $s(\beta) = \zeta\beta$. For $a = (a_0, a_1, a_2)$ write $\tau_a(r_i) = \zeta^{a_i} r_i$, $\tau_a(\beta) = \beta$. Then $s\tau_a s^{-1} = \tau_{Sa}$, where $(Sa)_i = a_{i-1}$. The group has order 81, lower-central orders $81, 9, 3, 1$, exponent nine, and trivial fourth three-Zassenhaus subgroup.

Use the arithmetic generators m, n, t of A.28 and A.31, with the exact Kummer normalization of A.35. Their modulo-three characters are

	m	n	t
κ_ζ	1	0	0
κ_{17}	0	0	1

. (A.37.4)

The restriction map to (A.37.2), composed with the actual free pro-three presentation, is a continuous epimorphism

$$F_3(m, n, t) \longrightarrow G_{K_0,\{3,17\}}(3) \longrightarrow G. \tag{A.37.5}$$

Let $W \in D_3(F_3(m, n, t))$ be the actual arithmetic Frobenius word of A.30. Under (A.37.5),

$$W \longmapsto \tau_{(2,2,2)}, \qquad [t, [t, m]] \longmapsto \tau_{(1,1,1)}. \tag{A.37.6}$$

In the restricted degree-three Lie algebra $D_3(F_3)/D_4(F_3)$, this map annihilates the three generator cubes and every ordinary degree-three component except the multidegree $(1, 0, 2)$

component generated by $[t,[t,m]]$. Consequently the coefficient of this bracket in the full W is exactly $2 \in \mathbb{F}_3$. This records a coefficient involving tame inertia; the map killing t in A.31 does not retain it.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379); Tasks A.6 and B.4.

Proof. For $a,b,c \in K_0$ direct multiplication of the three conjugates gives

$$N_{L_{17}/K_0}(a+b\beta+c\beta^2) = a^3+17b^3+289c^3-51abc. \tag{A.37.7}$$

At $(a,b,c)=(-7-\zeta,-3,-1)$ the right side is 27ζ. Thus

$$\delta_0\delta_1\delta_2 = \zeta. \tag{A.37.8}$$

Each δ_i is integral away from three. Since its norm is a unit, it is a unit at every finite prime away from three. Kummer extensions by its cube root are unramified at such primes, because three is invertible there and the radical is a unit. The extension L_{17}/K_0 is ramified only at three and seventeen. This proves the asserted ramification bound once the degree and Galois group are established.

The three classes of δ_i are independent in $L_{17}^{\times}/L_{17}^{\times 3}$. Indeed the map

$$\mathbb{F}_3[\langle\sigma\rangle] \longrightarrow L_{17}^{\times}/L_{17}^{\times 3}, \qquad \sigma^i \longmapsto [\delta_i] \tag{A.37.9}$$

is a module homomorphism. Every nonzero submodule of the regular module $\mathbb{F}_3[u]/(u-1)^3$ contains its socle, generated by $1+u+u^2$. By (A.37.8) that element maps to $[\zeta]$. The latter is nonzero: adjoining a cube root of ζ gives $K_0(\mu_9)$, a cubic extension unramified at seventeen, whereas L_{17}/K_0 is Eisenstein at the unique prime over seventeen. These two cubic extensions therefore have different ramification indices and cannot coincide. Thus the kernel in (A.37.9) is zero.

Kummer theory now gives $\mathrm{Gal}(M_{17}/L_{17}) = (\mathbb{F}_3)^3$ with the displayed diagonal actions. Independence also proves that the defining radical algebra is a field of degree 27 over L_{17}. Cyclic permutation of its generators defines the specified extension s of σ, with $s^3=1$. This proves the semidirect presentation (A.37.3). Its commutators are

$$\gamma_2 G = (S-1)(\mathbb{F}_3)^3, \quad \gamma_3 G = (S-1)^2(\mathbb{F}_3)^3 = \langle \tau_{(1,1,1)} \rangle, \quad \gamma_4 G = 1. \tag{A.37.10}$$

The three dimensions are $2,1,0$. Every cube in G is diagonal central, while $G^9=1$ and $(\gamma_2 G)^3 = 1$. The defining formula $D_j(G) = \prod_{i3^a \geq j} (\gamma_i G)^{3^a}$ therefore gives $D_4(G)=1$.

We next compute the actual local Frobenius. The residue field of K_0 at seventeen is $k = \mathbb{F}_{289} = \mathbb{F}_{17}[\zeta]$. At its unique extension to L_{17} the radical β reduces to zero, so every δ_i has the same residue

$$d_0 = (-7-\zeta)/3 = 9+11\zeta, \qquad d_0^3 = \zeta, \qquad d_0^{96} = \zeta^2. \tag{A.37.11}$$

In particular d_0 is not a cube in k. The polynomial $R^3 - d_0$ defines the unramified residue extension of degree three. Its roots obey

$$R^{289} = d_0^{96} R = \zeta^2 R. \tag{A.37.12}$$

An arithmetic Frobenius fixing the chosen uniformizer roots fixes β and acts by $r_i \mapsto \zeta^2 r_i$ for all three radicals. To see that the residue calculation determines this exact automorphism, the ratio of any two possible images of r_i is in μ_3, whose reduction is injective at seventeen. This

proves the first formula in (A.37.6). It is independent of conjugating the chosen decomposition subgroup because this image is central.

Finally put $v = r_0 r_1 r_2$, so $v^3 = \zeta$. For an element $\tau_a s^k$, the two characters $\kappa_\zeta, \kappa_{17}$ are respectively $a_0 + a_1 + a_2$ and k. Thus the actual images of the normalized generators have the form

$$m \mapsto \tau_a, \qquad n \mapsto \tau_b, \qquad t \mapsto \tau_c s, \qquad \sum a_i = 1, \quad \sum b_i = \sum c_i = 0. \tag{A.37.13}$$

We have made no choice of their individual coordinates. All three images have cube one. Conjugation by $\tau_c s$ on the abelian kernel is S, and

$$(S-1)^2 a = (1 + S + S^2)a = (1, 1, 1), \qquad (S-1)^2 b = 0. \tag{A.37.14}$$

Every bracket with two entries from the abelian kernel vanishes. The only surviving ordinary degree-three bracket is therefore $[t, [t, m]]$; its image is the second formula in (A.37.6). The standard free restricted degree-three basis consists of the ordinary degree-three free Lie part and the three generator cubes. The preceding calculations identify the entire map on that basis, so the Frobenius image proves that its indicated coefficient is two. This argument uses the full three-generator word and the actual arithmetic restriction (A.37.5).

For comparison with the lower Heisenberg test, the fixed field of the diagonal subgroup in (A.37.10) is

$$L_{17}(v, w), \qquad v^3 = \zeta, \quad w = r_0^2 r_1, \quad w^3 = \delta_0^2 \delta_1. \tag{A.37.15}$$

The two character rows $(1, 1, 1)$ and $(2, 1, 0)$ are independent, so this field has degree nine over L_{17}, hence degree 27 over K_0. The action satisfies $s(w) = (v/\delta_0)w$. Its weighted radical has residue $d_0^3 = \zeta$, which already has the cube root d_0 in k. The Frobenius image computed above becomes the identity in this quotient, exactly as follows from its diagonal kernel. This specifies both the lower quotient map and the coefficient it forgets.

The exact program `a_wreath17_root.py` verifies (A.37.7), the residue field and its actual degree-three radical extension, all 81 group elements, the lower-central orders, and every one of the 729 triples in (A.37.13). The preliminary norm search and its exact algebraic certification are retained in `a_norm_zeta_17.py`. □

2 B. Geometric classes and centralizers

2.1 Geometric classes and the cyclotomic centralizer

We retain the tangent at O with parameter z/x and tangent one. All inertia groups in this section are their images in the geometric pro-3 group N. We write κ for continuous pro-3 Kummer classes. A cuspidal section means a section contained in the decomposition group of a lifted cusp, with geometric conjugacy taken in the group actually displayed. The manuscript comparison is §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Statement B.1 (Proved). The three abelian cusp packets, with their framed origins. *Put $J = J_{E,D}$ and use the frame coordinates of LM7 (E24.4). Let z_P be the framed line bundle $\mathcal{O}(P - O)$ with frames $(x/z)^{-1}$ at P, z/x at O, and the canonical section at the other cusps. Define z_Q in the same way from $Q_c - O$, using y/z at Q_c. Put $\alpha_P = \kappa(z_P)$, $\alpha_Q = \kappa(z_Q)$ and $\alpha_O = 0$. The three packets in $H^1(G_\mathbb{Q}, V)$ are*

$$\mathscr{P}_d^V = \alpha_d + e_d\, \mathcal{R}_\mathbb{P}(\mathbb{Z}_3), \qquad d = O, P, Q_c. \tag{B.1}$$

Here $e_d\, \mathcal{R}_\mathbb{P}(\mathbb{Z}_3)$ denotes the image of the continuous homomorphism on cohomology induced by the $G_\mathbb{Q}$-equivariant inclusion $\mathbb{Z}_3(1) \to V$, $1 \mapsto e_d$.

The torus-point relation specifying the second framed origin is

$$z_P + z_Q = (3^{-1}, 3^{-1}, -\lambda^{-1}, -\lambda^{-1}, \zeta\lambda^{-2}). \tag{B.2}$$

In the presentation (E24.7), eliminate the first λ-generator and retain z'. Write g_2, g_3 for the remaining two λ-generators and τ_3 for ζ in the third quadratic factor. After pro-3 completion the origins have toric coordinates zero, divisible-cohomology coordinate zero, and remaining coordinates

$$\alpha_P = \kappa(z'), \qquad \alpha_Q = \kappa(2z' + g_2 + g_3 - \tau_3). \tag{B.3}$$

The inertia maps in the original frame coordinates are

$$\begin{aligned} i_O(u) &= (u^{-1}, u^{-1}, u^{-1}, u^{-1}, u^{-1}), \\ i_P(u) &= (u, 1, 1, 1, 1), \qquad i_Q(u) = (1, u, 1, 1, 1). \end{aligned} \tag{B.4}$$

Their images under the two coordinate-unit characters are respectively $(-\kappa u, -\kappa u)$, $(\kappa u, 0)$ and $(0, \kappa u)$. The three sets in (B.1) are pairwise disjoint, and each displayed parameter map is injective.

Proof. The tangential Albanese class at d, relative to the tangent at O, is represented in the generalized Picard group by $\mathcal{O}(d - O)$ with its local parameter frames. This description can be obtained directly from the divisor construction: replace d and O by points with local parameters s_d and s_O, take the line bundle of their difference, and divide its local frames by the leading powers s_d^{-1} and s_O. The resulting fibers at the reduced boundary are the displayed frames. Applying multiplication by 3^n identifies the finite abelian path torsor with the torsor of 3^n-division points of this framed line bundle. The inverse limit therefore gives its Kummer class in $T_3 J = V$. Changing the nonzero tangent at d multiplies its local root system by the root system of a scalar u and changes the abelian cocycle by $e_d\kappa(u)$. The same description applies to an arbitrary continuous section of the cusp decomposition extension, because that extension is $\mathbb{Z}_3(1) \rtimes G_{\mathbb{Q}}$ and its section classes are $H^1(G_{\mathbb{Q}}, \mathbb{Z}_3(1))$.

We verify the origin calculation. The rational function

$$h = \frac{x + y + z}{x + y}$$

has divisor $P + Q_c - 2O$. Its leading values are -1 after division by x/z at P, -1 after division by y/z at Q_c, and -3 after multiplication by $(z/x)^2$ at O. On the representatives $[0 : -\zeta : 1]$, $[-\zeta : 0 : 1]$ and $[1 : -\zeta : 0]$ its values, normalized by -3, are $-\lambda^{-1}, -\lambda^{-1}, \zeta\lambda^{-2}$. These are exactly (B.2). The expansions follow, for example, from $y/z = -(1 + (x/z)^3)^{1/3}$ near P. The relation $3 = -\zeta^2\lambda^2$ verifies the three quadratic values without choosing complex approximations.

For comparison, multiplication by $(x + z)/(x + y)$ trivializes $\mathcal{O}(3Q_c - 3O)$ and gives

$$3z_Q = (3^{-1}, -3^{-2}, -\lambda^{-2}, -\lambda^{-1}, \zeta\lambda^{-3}). \tag{B.5}$$

Together with (E24.6), this checks the cube of (B.2) in all five coordinates. The program `code/bc_checks.py` verifies all these identities over the exact field $\mathbb{Q}[\zeta]$.

Signs disappear in pro-3 Kummer theory: -1 has compatible rational 3^n-th roots. Relation (E24.7) then reads $3z' = -g_1 - 2g_2 - 3g_3 + 2\tau_3$. The quadratic coordinates in (B.2) are $-g_1 - g_2 - 2g_3 + \tau_3$. Subtracting z' and eliminating g_1 gives $2z' + g_2 + g_3 - \tau_3$, proving (B.3). The last factor of (E24.1) is zero because these are Kummer classes of actual rational generalized-Jacobian points; we use the splitting there with its rational-point completion as kernel.

Normalizing the frame at O to one gives (B.4). Applying (E24.4) gives the three asserted toric images; the quadratic norm of a rational scalar is its square. These images prove injectivity of each parameter map. Filling all cusps gives the continuous $G_{\mathbb{Q}}$-equivariant quotient $V \twoheadrightarrow T_3E$. It sends the packets to the point classes of O, P, Q_c, respectively. Their pairwise differences have exact order three. Reduction at two embeds rational odd torsion into $E(\mathbb{F}_2)$, which has order three, as checked by the program. Consequently none of these nonzero differences is three-divisible in $E(\mathbb{Q})$, and its modulo-three Kummer class is nonzero. This proves disjointness. □

Statement B.7 (Proved). Every nontrivial tame inertia in a full section is cuspidal. *Let $q \neq 3$ be a rational prime, put $k = \mathbb{Q}_q$, and use the actual pro-three fundamental group N of the marked Fermat cubic with the integral unit tangent at O as origin. For a continuous section $s : G_k \to N \rtimes G_k$, its restriction to inertia is either trivial or has image in one cuspidal inertia subgroup. In the latter case the whole section is contained in that cusp's decomposition group. The cusp is k-rational and the lifted cusp is unique. Moreover, a full section is unramified at q exactly when its image in V is unramified there.*

Proof. The marked curve and the tangent have good reduction at q. Thus the original based action of inertia on N is trivial, by prime-to-q specialization. Its cocycle restriction is a homomorphism. Wild inertia, being pro-q, maps to one; the only tame factor which can map to N is $\mathbb{Z}_3(1)$. Let $t = s(\tau_q)$ denote the geometric value on a generator of that factor, and assume $t \neq 1$. If $F = s(\phi_q)$ is the full Frobenius lift, the section relation is the exact group equality

$$FtF^{-1} = t^q. \tag{B.7.1}$$

Take a characteristic open normal subgroup M of N and put $H = M\overline{\langle t \rangle}$. Such M form a cofinal family, since N is finitely generated. Equation (B.7.1) and the characteristic property of M show that H is stable under $s(G_k)$. The open arithmetic subgroup $H\, s(G_k)$ is therefore the subgroup of the pro-three arithmetic fundamental group defining a finite etale cover $Y_H \to U_k$. Its geometric group is H, by the exact finite-three-group cover comparison proved in B.4.

We justify the reduction property needed for weights. The subgroup $H\, s(G_k)$ contains the original arithmetic inertia: its geometric cocycle values are powers of t, which lie in H. Original inertia centralizes N, so it acts trivially on every coset in N/H. The cover consequently descends to the smooth marked integral model by prime-to-q specialization. Its compactification is smooth and proper over the valuation ring. Locally at a boundary point this last assertion is the tame normalization $z^e = x$ after an unramified extension, with e prime to q; away from the boundary it is etale. These charts are exactly the tame-cover specialization comparison, also obtained from the logarithmic comparison in Stix, Theorem 3.4.6 [Sti02].

Let X_H be this compactification. The boundary sequence (B.8), with $\mathbb{Z}_3$ coefficients, is equivariant for the actual arithmetic Frobenius F of this descended cover. The image of t in $T_3J_{X_H}$ satisfies $Fv = qv$ by (B.7.1). All Frobenius eigenvalues on this Tate module have complex absolute value $\sqrt{q}$, by Deligne, Theorem (1.6) [Del74], applied in degree one to the smooth proper special fiber. The Tate module of its Jacobian has these same eigenvalues for arithmetic Frobenius, by the Abel–Jacobi comparison and the dual Weil pairing. Thus $v = 0$. The image of t in H^{ab} is in the boundary submodule.

The cyclic finite three-group H/M is generated by the image of t. Its homomorphism from H^{ab} consequently maps the boundary submodule onto all of H/M. At least one individual cusp inertia maps onto this cyclic group: otherwise every such image would be in its unique maximal subgroup. It follows that t fixes a cusp in the boundary set of the normal cover

associated to M. The inverse images of these finite fixed-cusp sets in $\coprod_d N/I_d$ are nonempty closed subsets with the finite-intersection property. Compactness gives a lifted cusp fixed by t at every level, so t belongs to its actual inertia group $I_{\widetilde{d}}$.

Distinct cusp inertia groups have trivial intersection by the boundary and cover argument in B.4. Hence this lifted cusp is unique. Equation (B.7.1) shows that F normalizes $I_{\widetilde{d}}$, because its image has the nontrivial element t^q in common with that group. Its normalizer is the cusp decomposition group, again by the explicit stabilizer argument in B.4. The inertia image and F topologically generate the image of the local section; therefore that whole image lies in the decomposition group. Its projection is all G_k, forcing $d \in D(k)$.

Finally the inclusion $I_{\widetilde{d}} \to V$ sends its generator to the primitive nonzero boundary vector e_d. It is injective. Thus $t \neq 1$ has nonzero abelian image, while a trivial full inertia image certainly has trivial abelian image. This proves the last assertion. □

Statement B.8 (Proved). The exact local image of ramified full sections in class two. *With $k = \mathbb{Q}_q$, $q \neq 3$, let $A = N/\gamma_3 N$. For each $d \in D(k)$ choose the coordinate unit tangent used in C.10, its actual section $s_{d,1}$, and a peripheral generator c_d based at O. The ramified part of the image of full sections in* $\mathrm{Sec}_k(A)$ *is precisely the disjoint union*

$$\coprod_{d \in D(k)} \left\{ \left[g \longmapsto c_d^{\kappa_q(u)(g)} s_{d,1}(g) \right] : u \in k^{\times \wedge_3},\ v_q(u) \neq 0 \right\}. \tag{B.8.1}$$

Here the expression is multiplication in the arithmetic group, followed by its actual class-two quotient. Its parameter space is explicitly

$$k^{\times \wedge_3} = \mathbb{Z}_3 \kappa(q) \oplus \mu_{3^{v_3(q-1)}}(k). \tag{B.8.2}$$

There are nine packets for $q \equiv 1 \pmod 3$ and three for $q \equiv 2 \pmod 3$. In the central cuspidal lattice $I/[N,I] = \mathbb{Z}_3(1)[D]$ the tame vector of a packet is $v_q(u)c_d$, with exactly this one cusp coordinate nonzero. In particular the mixed-inertia counterexamples in A.5 and A.6 cannot lift to full N.

Proof. The decomposition extension for a rational cusp is $\mathbb{Z}_3(1) \rtimes G_k$ using its unit tangent. Its sections modulo inertia conjugacy are the Kummer group (B.8.2). The decomposition inclusion is a continuous homomorphism of actual arithmetic groups; after quotienting the geometric group it gives exactly (B.8.1). The Kummer cocycle equation and $s_{d,1}(g) c_d s_{d,1}(g)^{-1} = c_d^{\chi_3(g)}$ verify directly that the displayed product is a section. The unit-tangent section is unramified, and the tame Kummer exponent is $v_q(u)$, giving the stated ramification condition. Every displayed class has its full cuspidal lift. Conversely B.7 puts every ramified full section in one of these decomposition extensions.

To check that the displayed parametrization remains injective in class two, first pass to a finite unramified extension splitting all cusps. The boundary inclusion $e_d : \mathbb{Z}_3(1) \to W$ is then split, and $H^1(k,W) \to H^1(k,V)$ is injective because $T_3 E^{G_k} = 0$. Restriction on the Kummer group is injective by the same invariant argument as C.10. Thus two parameters at the same cusp cannot coincide. Different cusps map in the proper elliptic quotient to the Kummer classes of different three-torsion points. Reduction at $q \neq 3$ injects three-primary torsion into the finite group $E(\mathbb{F}_q)$; its nonzero elements are not divisible by every power of three. The formal group is pro-q, on which multiplication by three is invertible. Hence a nonzero difference of two cusp points has nonzero integral local Kummer class, proving disjointness.

The decomposition of $k^\times$ into a uniformizer, the roots of unity of order $q-1$, and principal units gives (B.8.2), since the principal units are pro-q. The two possible cusp counts follow

from the actual field of the six nonrational cusps, $\mathbb{Q}(\zeta_3)$. Projection to $I/[N,I]$ sends the peripheral generator to its indicated basis element, proving the tame-vector formula. The vectors in A.5 and A.6 involve $c_O + c_P$ and therefore do not have this form. □

Statement B.9 (Proved). All inherited class-two cusp origins. *Use exactly the diagonal coordinates of LM7 E.29 for H, based at the lifted cusp P over $t = 0$, and put*

$$f(g) = \frac{\chi_3(g)^2 - 1}{24} \in \mathbb{Z}_3(2). \tag{B.9.1}$$

The three rational cusp packets in $\mathrm{Sec}(H)$ *have the following continuous cocycle representatives:*

	$u(g)$	$v(g)$	$w(g)$
P	$\kappa(a)(g)$	0	0
Q_c	0	$\kappa(a)(g)$	$f(g)$
O	$-\kappa(a)(g)$	$-\kappa(a)(g)$	$f(g)/2$

$$a \in \mathbb{Q}^{\times \wedge 3}. \tag{B.9.2}$$

Each row is injectively parametrized and the rows are pairwise disjoint. Above inherited abelian class $(0,0)$ *the three unit tangents give, respectively, the central labels* $0, 2, 1 \in \mathbb{Z}/3$ *relative to* $g \mapsto (\chi_3(g)^2 - 1)/3$. *The actual arithmetic map from the intrinsic group based at* O *to this* H*-extension is*

$$(n, g) \longmapsto (q_O(n)(0, 0, f(g)/2), g), \tag{B.9.3}$$

after the geometric path choices in the proof. Thus (B.9.2) retains the change of section origin between the O *and* P *base points.*

Proof. Write $t = -u_0^3$, $1 - t = -v_0^3$ for the cubic cover of the tripod, with $u_0 = x/z$, $v_0 = y/z$. The standard path from the unit tangent at zero to the unit tangent at one lifts from P to Q_c: along the real interval its cube roots are $u_0 = -t^{1/3}$ and $v_0 = -(1-t)^{1/3}$. It reaches exactly the displayed rational cusps. At either end the lifted local parameter has tangent minus one relative to u_0 or v_0. Changing that sign has zero pro-three Kummer class, with the explicit compatible rational roots $(-1)^{1/3^n} = -1$. The local decomposition-section comparison therefore changes no section class.

Let X, Y be the logarithms of the standard tripod loops and $C = [X, Y]$. The H lattice is $\mathbb{Z}_3(3X) \oplus \mathbb{Z}_3(3Y) \oplus \mathbb{Z}_3 C$. Wickelgren's evaluation of the standard path cocycle in the subsection "Evaluating the 2-nilpotent quotient of $\mathfrak{f}$", equation $f(\sigma) = (\chi(\sigma)^2 - 1)/24$ [Wic12], says that the section at Q_c has central cocycle fC relative to P. The sign follows from the actual action convention $g(y) = \mathfrak{f}_g^{-1} y^{\chi(g)} \mathfrak{f}_g$: the lift $\mathfrak{f}_g s_P(g)$ conjugates y to its $\chi(g)$-th power. Thus the P, Q_c origins and their peripheral generators are $0, fC$ and $3X, 3Y$.

Consider the rational coordinate interchange $(x, y, z) \mapsto (y, x, z)$ on the cubic. It covers $t \mapsto 1 - t$, interchanges P, Q_c, and fixes O. Transporting the fundamental group along the same standard path gives an actual automorphism of the arithmetic H-extension. Its geometric map swaps the two abelian coordinates and negates C, and its change of arithmetic origin is fC. There are no additional linear central terms in its geometric map: because fC is central, the geometric map commutes with the diagonal $G_{\mathbb{Q}}$-action; a term from weight one to weight two would satisfy $(\chi - \chi^2)c = 0$ and therefore is zero in the torsion-free lattice. Consequently the exact automorphism is

$$((u, v, w), g) \longmapsto ((v, u, -w + f(g)), g). \tag{B.9.4}$$

Its formula also checks directly that it is a continuous group homomorphism and that its square is the identity. It preserves H because the covering subgroup is the kernel of the two abelian coordinates modulo three.

At O, use the local parameter $s = z/x$. The regularized leading terms of u_0, v_0 are $1, -1$; thus the inherited abelian class of the unit tangent is zero. Conjugating by an element of H makes its abelian cocycle identically zero. Its remaining class is some $h \in H^1(G_{\mathbb{Q}}, \mathbb{Z}_3(2))$. The coordinate interchange takes s to z/y, whose leading coefficient at O is -1. The rational root comparison above makes these two pro-three tangent sections conjugate. Since (B.9.4) acts on central classes by $h \mapsto -h + [f]$, we obtain $2h = [f]$. Multiplication by two is invertible on this group, so $h = [f/2]$. A central geometric conjugation then chooses the representative $f/2$.

We also check the inertia generator in this gauge. Its abelian coordinates are $(-1, -1)$, since both u_0 and v_0 have a simple pole in s. Write its central coordinate as w_0. The section just chosen is central, so its cyclotomic normalization of the inertia generator and the diagonal action give $\chi(g)^2 w_0 = \chi(g) w_0$ for every g. The lattice is torsion-free and the character is nontrivial, hence $w_0 = 0$. This determines the actual inertia image and avoids an unsupported choice of a conjugate of $(xy)^{-3}$. Twisting each of the three sections by its Kummer parameter now gives (B.9.2), using the displayed group law of E.29. For each row the two abelian entries are proportional, so that its BCH cross term is zero. Equation (B.9.3) is the arithmetic map with exactly the calculated image of the intrinsic reference section; centrality of that image makes its geometric part q_O equivariant for the stated diagonal action.

Finally $H^1(G_{\mathbb{Q}}, \mathbb{Z}_3(2)) = \mathbb{Z}/3$ with generator $(\chi_3^2 - 1)/3$ by E.29m. The coefficients $1/8$ and $1/16$ reduce to 2 and 1 modulo three. The three abelian packet lines meet only at zero, and their central classes there are distinct. This proves all injectivity and disjointness assertions. The exact rational-arithmetic program `code/bc_h_packets.py` verifies the BCH automorphism, the three cocycle identities and the three distinct central labels; its output is included alongside the code. □

Statement B.10 (Proved). The full global image at the specified prime seven. *Let $A = N/\gamma_3 N$, and use the normalized pair coordinates (a, b) and the A^c-action of A.15 for sections supported at $\{3, 7, \infty\}$. Such a class has a full N-lift exactly in the following cases:*

$$b = 1, \quad \text{or its restriction to } G_{\mathbb{Q}_7} \text{ belongs to one of the nine ramified packets (B.8.1).} \tag{B.10.1}$$

Equivalently these are precisely the classes which lift to every lower-central quotient. A full lift can be chosen unramified outside $\{3, 7, \infty\}$.

Here is an exact packet parametrization in the pair coordinates. Fix the embedding $\zeta_3 \equiv 2 \pmod 7$, a Frobenius ϕ fixing compatible roots of 7, and the actual unit-tangent values $g_d = s_{d,1}(\phi) \in A$. Use the same based inertia element $c_d \in A$ in each case. The ramified part of the global image is

$$\left\{ [(c_d^{2j} g_d, c_d^v) \cdot h] : d \in D(\mathbb{Q}_7), \quad v \in \mathbb{Z}_3 \setminus \{0\}, \quad j \in \{0, 1, 2\}, \quad h \in A/A^c \right\}. \tag{B.10.2}$$

Here $(a, b) \cdot h = (h^{-1} a F(h), h^{-1} b h)$, so A/A^c means right cosets, and the displayed parameters are injective after these explicit choices. The symbols g_d retain the actual tangential Frobenius origins; they are not replaced by arbitrary solutions of a graded normalization equation.

Proof. The exact isomorphism of arithmetic groups in A.14 and the normalized-cocycle construction in A.15 identify every global class with the asserted pair. If $b = 1$, lift a arbitrarily to $\widetilde{a} \in N$ and take $\widetilde{b} = 1$. The full tame equation holds, and A.15 extends this pair to the required global section. If $b \neq 1$ and a full section exists, B.7 forces its local restriction to be cuspidal. Conversely a local cuspidal class in (B.8.1) has its actual full local lift. Conjugate this lift by a lift in N of the geometric element comparing its class-two image to the specified pair (a, b). Its generator values then project to that exact pair. A.15

extends them, with their conjugate values on the second local factor, to a full normalized global section. This proves both directions of (B.10.1), including the ramification support. The nilpotent assertion follows either from A.15's compactness proof or from B.11 below.

For (B.10.2), the completed local Kummer group is $\mathbb{Z}_3\kappa(7) \oplus (\mathbb{Z}/3)\kappa(\zeta_3)$. Choose compatible roots of ζ_3. Its unramified cocycle is $(\chi_3 - 1)/3$, which has value two at ϕ and value zero on tame inertia. The uniformizer cocycle has values zero and one there. The full decomposition-group formula of B.8 therefore gives exactly the pair $(c_d^{2j} g_d, c_d^v)$. Taking $j + 3$ instead changes the pair by conjugation with c_d, so the three chosen representatives cover precisely the unit parameter group.

Different d, v, j have different local section classes by B.8. Within each local class the geometric centralizer in A is trivial: its abelian image would be fixed by Frobenius on V, and its remaining central image would be fixed on $\bigwedge^2 V$; the respective eigenvalue absolute values are $\sqrt{7}, 7$ and $7, 7^{3/2}, 49$, all greater than one. Thus the full A-conjugacy orbit of that local pair is freely parametrized by $h \in A$. The normalized global conjugacies are exactly A^c by A.15, acting on the right. This proves the stated right-coset parametrization and its injectivity. □

Statement B.11 (Proved). No new ramification above class two and compactness for a finite set of primes. *Let S be any finite set of rational places containing 3 and infinity, and let s_2 be a class-two section unramified outside S. Every lift of s_2 to $N/\gamma_r N$, $r \geq 3$, is unramified outside the same set S. Moreover the following assertions are equivalent:*

$$\begin{gathered} s_2 \text{ lifts to } N; \\ s_2 \text{ lifts to } N/\gamma_r N \text{ for every } r \geq 3. \end{gathered} \tag{B.11.1}$$

The separately given nilpotent lifts need not be compatible. The conclusion supplies a compatible continuous full lift, also unramified outside S.

Proof. For each r, iterated commutators give a continuous $G_{\mathbb{Q}}$-equivariant surjection

$$V^{\otimes r} \longrightarrow \gamma_r N/\gamma_{r+1} N. \tag{B.11.2}$$

The target is a finite free $\mathbb{Z}_3$-module. This integral assertion follows by Hall collection in the free pro-three group: basic commutators of weight r span the quotient, and their Magnus expansions have the linearly independent leading Hall Lie monomials. Collection at finite exponent quotients, followed by compactness, gives the same spanning assertion over $\mathbb{Z}_3$. The leading monomials prove independence and exclude torsion. Thus (B.11.2) is the actual integral graded comparison being used.

For $q \neq 3$, Frobenius eigenvalues on V have weights one and two: on the elliptic factor their absolute values are $\sqrt{q}$ and on the boundary factor they are q. The eigenvalues on (B.11.2) are therefore among products of r such eigenvalues. For $r \geq 3$ their absolute values are at least $q^{r/2} > q$. This eigenvalue conclusion is valid for a quotient of the tensor representation and does not assume semisimplicity of the extension defining V.

Suppose a lift from class $r-1$ to class r acquires inertia at a prime outside S. Its geometric inertia values lie in the central kernel $\gamma_r N/\gamma_{r+1} N$ and, as in B.7, factor through the tame three-part. The exact tame relation requires its value t to satisfy $F_q t = qt$; conjugation by the section's Frobenius value acts trivially on this central kernel. For $r \geq 3$ the preceding eigenvalue calculation and torsion-freeness give $t = 0$. Induction starting at the fixed class-two section proves that every higher nilpotent lift has the asserted support.

Let $K_0 = \mathbb{Q}(\zeta_3)$ and let S' be its places over S. The relative pro-three quotient Γ_S of $G_{\mathbb{Q},S}$ has kernel $G_{K_0,S'}(3)$ and quotient of order two. That pro-three kernel is finitely generated:

its $\mathbb{F}_3$-characters form the finite Kummer group of S'-units and the finite three-primary ideal class group. The finite-dimensional Frattini quotient generates a pro-three group, by the same finite-quotient maximal-subgroup argument used in A.14. Thus Γ_S is finitely generated. The actual geometric action factors through it, as proved in A.15 from the three-group action on the Frattini quotient of N. Every cocycle under consideration also factors through Γ_S: over K_0 its graph is a homomorphism to a pro-three semidirect product.

Fix generators $g_1, \dots, g_m$ of Γ_S and a cocycle representative of s_2 on it. Consider the compact space of tuples in N^m whose projections are its fixed generator values. Choose a free profinite presentation on these generators, with its closed subgroup of relators. For each $r \geq 3$, impose all those relators on the lifted tuples in $(N/\gamma_r N) \rtimes \Gamma_S$. These are closed conditions: at any finite quotient each profinite word is a continuous function of the generator values. A lift at level r makes the resulting closed subset nonempty, after conjugating its projection to the fixed representative; the required geometric conjugator lifts because $N/\gamma_r N \to N/\gamma_3 N$ is surjective. The closed sets are nested. Compactness supplies one tuple satisfying every level of every relator.

The intersection $\bigcap_r \gamma_r N$ is trivial: a nontrivial element survives in a finite three-group, and that group is nilpotent. Thus the tuple satisfies all relators in $N \rtimes \Gamma_S$ itself and defines a continuous section. Composing with $G_{\mathbb{Q},S} \to \Gamma_S$ gives the desired full cocycle with the prescribed class-two projection. This proves the nontrivial implication in (B.11.1); the other is induced by the continuous quotient maps. □

Statement B.2 (Refuted). The inherited class-two image need not be a point or cusp image. *The exclusion proposed in Lead 1 is false. There are sections of $N \rtimes G_{\mathbb{Q}}$ whose image in* $\mathrm{Sec}(H)$ *has toric coordinates $(\kappa(3), 2\kappa(3))$. Every such image lies outside the three rational cusp packets, and $U(\mathbb{Q})$ has no point producing it. It therefore supplies the asserted counterexample already with a full geometric pro-3 lift.*

Proof. We use exactly the onto map in LM7 E.34: the coordinate-unit homomorphism on the unramified group $H^1(G_{\mathbb{Q},\{3,\infty\}}, V)$ has image $\mathbb{Z}_3\kappa(3)^2$, and every class in this source lifts to the full N. Its projectivity step is checked as follows. Over $\mathbb{Q}(\zeta)$, the maximal pro-3 Galois group unramified away from three is free on two generators: Kummer theory gives the two classes $\zeta, 1-\zeta$; the class group of $\mathbb{Z}[\zeta, 1/3]$ is zero, and the three-primary Brauer group supported at its single omitted finite place is zero by the sum of local invariants. Thus degree-two cohomology with $\mathbb{F}_3$ vanishes. The free-pro-3 criterion follows by solving finite central embedding problems. Complex conjugation has order two, so averaging gives the same degree-two vanishing for the relative semidirect product. The geometric action and every abelian cocycle factor through this relative group: their restrictions over $\mathbb{Q}(\zeta)$ take values in pro-3 groups. Successive finite geometric embedding problems therefore have solutions; finite generation of the relative group and compactness make them compatible. This is the full-group lifting step used from E.34. The direct-summand coordinate-unit map is the explicit algebraic section $(a, b) \mapsto (a, b, 1, 1, 1)$ of (E24.4), so its onto property requires no further lifting assertion. These arguments also agree with the free-Galois-group mechanism in Hoshi, Theorem 3.5 [Hos10]; that theorem concerns the specified proper Jacobian sections, while the preceding relative-group argument is the needed one for all ten coordinates here.

By (B.4), the cusp projections are $(c, 0)$, $(0, c)$ and $(-c, -c)$, with $c \in H^1(G_{\mathbb{Q}}, \mathbb{Z}_3(1))$. The valuation at three sends the proposed vector to $(1, 2)$, which lies on none of the three lines, already over $\mathbb{F}_3$. The code verifies this finite separation. Conjugation in H does not alter the abelian cohomology class, so no choice of central lift can change this conclusion. LM7's computation $E(\mathbb{Q}) = \{O, P, Q_c\}$ gives $U(\mathbb{Q}) = \varnothing$.

The drafter's suggested vector $\kappa(3)(e_O + e_P)$ does have an intrinsic abelian image outside the packets (B.1): its elliptic image is zero and its toric image is not diagonal. Its inherited toric image is $(0, -\kappa(3))$, on the Q_c line. That particular toric calculation therefore gives no exclusion in $\mathrm{Sec}(H)$. The vector used above supplies the required exclusion there. □

Statement B.3 (Proved). The cusp packets in the cuspidally central quotient. *In $N^{\mathrm{cc}} = C \times_T V$, let the generator of the central $\mathbb{Z}_3(1)$ of C be the elliptic commutator $[a, b]$. Choose the reference C-section above the elliptic class zero to be the tangent at O; above $\kappa(P - O)$ and $\kappa(Q_c - O)$ choose the images of the respective unit tangents. Use these references for every abelian class in the corresponding three elliptic fibers. With these explicit choices for the torsors denoted by $s(\alpha, a)$, the cusp packets are*

d	α	a
O	$e_O\kappa(u)$	$-\kappa(u)$
P	$\alpha_P + e_P\kappa(u)$	0
Q_c	$\alpha_Q + e_{Q_c}\kappa(u)$	0

(B.6)

Here u ranges over the full pro-3 completion of $\mathbb{Q}^\times$. This parametrization is injective on each row, and the rows have disjoint images.

Proof. The morphism $U \hookrightarrow E \setminus \{O\}$ fills the cusps P, Q_c, so their inertia maps to the identity in C. At O, the once-punctured elliptic presentation gives $c_O = -[a, b]$ in the central lattice of C. Consequently a cusp twist $\kappa(u)$ changes the C-coordinate by $-\kappa(u)$ at O and by zero at P, Q_c. The V-coordinate is exactly (B.1). The map to V and its already proved packet disjointness and parameter injectivity give both remaining assertions. □

Statement B.4 (Proved). Lead 4 for the full geometric pro-three group. *For every continuous section $s : G_{\mathbb{Q}} \to N \rtimes G_{\mathbb{Q}}$ put $\Delta = \ker(\chi : G_{\mathbb{Q}} \to \widehat{\mathbb{Z}}^\times)$. Then*

$$C_N(s(\Delta)) = \begin{cases} I_{\widetilde{d}}, & s \text{ is cuspidal at the lifted cusp } \widetilde{d}, \\ 1, & s \text{ is noncuspidal.} \end{cases} \tag{B.7}$$

In the first case the lifted cusp is unique and its image is one of O, P, Q_c. Thus both implications in Lead 4 hold for the actual pro-3 group, with its genuine section action.

Proof. We give the pro-3 argument. No fixed element is lifted from a quotient of a full profinite fixed group.

For an open subgroup $H \subset N$, the corresponding finite geometric cover has maximal pro-3 fundamental group H. To check this comparison, choose an open normal subgroup $M \subset H$ of N; N/M is a finite 3-group and H/M is subnormal in it. Pulling back a subnormal chain expresses the cover as a tower of covers of 3-power degree with normal successive groups. A finite 3-group quotient of a subgroup at one stage becomes a quotient of a finite 3-group after taking the finitely many conjugates and then the extension by the stage's 3-group. Induction proves that the topology induced from the original maximal pro-3 quotient is the full pro-3 topology on the subgroup. This proves the comparison. For its compactification X_H and geometric boundary D_H, the oriented surface presentation gives the exact sequence

$$\mathbb{Z}_3(1) \xrightarrow{\delta} \mathbb{Z}_3(1)[D_H] \longrightarrow H^{\mathrm{ab}} \longrightarrow T_3 J_{X_H} \longrightarrow 0. \tag{B.8}$$

Every arrow is a continuous homomorphism, and the last two arrows are induced respectively by inertia and by filling the cusps. Its only boundary relation is the diagonal sum.

Each lifted inertia group is $\mathbb{Z}_3(1)$. Two distinct lifted cusps have inertia groups meeting trivially. Indeed, separate the cusps in a finite normal pro-3 cover, refine until there are at least three boundary points, and take a positive 3-power of a purported common element so that it lies in the cover group. It remains nonzero in $\mathbb{Z}_3(1)$ and (B.8) puts it on two different boundary lines; those lines have zero intersection. The refinement exists because an inertia group has infinite index: an open procyclic subgroup would give a hyperbolic cover whose abelianization has rank one, contradicting its surface presentation. Normal covers therefore have unboundedly many points above a cusp. The same intersection argument shows that the geometric normalizer of nontrivial inertia is that inertia group, and its arithmetic normalizer is the corresponding cusp decomposition group: a normalizer fixes the lifted cusp, and stabilizers are precisely those groups. These geometric assertions also agree with Nakamura's pro-class statements, Lemmas (2.1.2) and (2.1.4) [Nak94].

Let $z \in C_N(s(\Delta))$. The action through s on the finitely generated pro-3 group N has an open stabilizer for each open subgroup H: a characteristic open subgroup inside H reduces the assertion to an action on a finite group. Consequently H is realized by a cover over some number field L, and $s|_{G_L}$ lifts to that cover when H contains z. The image of z in (B.8) is fixed by $G_{L(\mu_\infty)}$. Ribet's Appendix Theorem 1 [Rib81] says that an abelian variety over a number field has finite torsion over its full cyclotomic extension. Its consequence here is exact: a nonzero invariant vector of $T_3 J_{X_H}$ would give cyclotomic rational torsion points of unbounded 3-power order. Thus the image of z in the last term of (B.8) is zero.

Apply this to $H = M\langle z\rangle$, for every open normal $M \subset N$. The image of z in H^{ab} belongs to the boundary subgroup. The quotient H/M is cyclic of 3-power order and generated by the image of z. Its boundary images generate it, and a cyclic 3-group generated by subgroups equals one of those subgroups. Thus z fixes a cusp in the cover associated to M. The finite sets of such fixed cusps are nonempty at every normal level. A common refinement supplies every finite set of compatibility requirements; compactness gives a lifted cusp $\widetilde{d}$ fixed by z. Hence $z \in I_{\widetilde{d}}$.

We verify that all nonzero elements of the centralizer belong to one inertia group. More generally let a closed subgroup $B \subset N$ consist entirely of inertia elements. The set of cusps with $B \cap I_c \neq 1$ is B-stable. If it were a single orbit while every $B \cap I_c$ were proper, B would be a union of conjugates of one proper closed subgroup. A finite quotient in which this subgroup remains proper contradicts this: in the finite coset action the average number of fixed cosets is one and the identity fixes more than one, so some element fixes none. We can therefore choose two distinct B-orbits of cusps unless B is contained in one inertia group. Compactness separates these orbits at a finite normal level M; the open subgroup BM has two distinct corresponding cusps. When it has only two cusps, its compactification has positive genus. Every element of B maps trivially to the compactification group. A degree-three finite unramified cover of that compactification consequently gives a further open subgroup containing B with six cusps. We have thus arranged at least three cusps while keeping the specified two distinct.

Take nonzero z, z' in the two inertia intersections. Their boundary images are ae_1, be_2, with $a, b \neq 0$. Both zz' and z^2z' have nonzero boundary image. With four or more cusps, neither image can be on a single boundary line. With three cusps, lying on one line forces respectively $a = b$ and $2a = b$, a contradiction. This contradicts the defining property of B and proves the desired single-inertia assertion.

For $C = C_N(s(\Delta)) \neq 1$, we have proved $C \subset I_{\widetilde{d}}$. Each $s(\sigma)$, $\sigma \in \Delta$, fixes a nonzero element there; the inertia-intersection result forces it to fix $\widetilde{d}$. Its inertia action is $\chi_3(\sigma) = 1$, so $I_{\widetilde{d}} \subset C$. Thus equality holds. Normality of Δ makes C stable under all of $s(G_{\mathbb{Q}})$. The arithmetic normalizer identity puts the entire section in the decomposition group of $\widetilde{d}$. Its base cusp is rational, and uniqueness follows from inertia intersections. Conversely a cuspidal

section fixes its inertia over Δ; the same argument gives equality in (B.7). □

Statement B.5 (Proved). The cuspidal members of the full unramified family. *Let $\mathscr{F}_S$ be the image of $H^1(G_{\mathbb{Q},\{3,\infty\}}, N)$ in $H^1(G_{\mathbb{Q}}, N)$, using the integral tangent at O. Its cuspidal subset is the disjoint union of the three packets obtained by twisting the unit tangents at O, P, Q_c by $t\kappa(3)$, $t \in \mathbb{Z}_3$. Every other full section class in $\mathscr{F}_S$ has centralizer 1 over Δ. The centralizer of each cuspidal member is its complete pro-3 inertia group.*

Proof. The good marked model over $\mathbb{Z}[1/3]$ and its integral nonzero tangents give actions and tangential cocycles unramified outside three. This is the based smooth-specialization step used in E.34; the comparison is made on covers with their local parameter root fiber functors, so it applies to the genuine action. For a twist $u \in \mathbb{Q}^{\times,\wedge_3}$, restriction to inertia at $q \neq 3$ has abelian value $v_q(u)e_d$. The vector e_d is primitive and nonzero in V, so a section in $\mathscr{F}_S$ forces $v_q(u) = 0$. Unique factorization in the completed Kummer group then gives $u = 3^t$, with $t \in \mathbb{Z}_3$. Conversely these twists are unramified outside three. Statement B.4 identifies every cuspidal full section with exactly one of these packets and computes all remaining centralizers. The abelianized images are the three affine lines in B.1 with the parameter restricted to $t\kappa(3)$.

This assertion concerns full section classes. The surjection from this family to its abelianized classes in E.34 need not have one-point fibers. An abelian class outside the three displayed lines has exclusively noncuspidal full lifts. We have not deduced cuspidality of every lift above a point on one of those lines. □

Statement B.12 (Proved). The marked local classification when the filled-curve projection is unramified. *Let $k/\mathbb{Q}_q$ be finite, $q \neq 3$, with residue cardinality Q, and let $d = Q_c$ be the marked cusp. Write*

$$\mathscr{Q} = N/\gamma_3 I_{\mathrm{mark}}, \qquad B = N/I_{\mathrm{mark}}, \qquad \mathscr{I} = I_{\mathrm{mark}}/\gamma_3 I_{\mathrm{mark}}, \qquad Z = \gamma_2 I_{\mathrm{mark}}/\gamma_3 I_{\mathrm{mark}}. \tag{B.12.1}$$

All groups carry the actual arithmetic action, and the quotient B is the full geometric pro-three group of the curve obtained by filling d. Choose tame three-inertia τ, arithmetic Frobenius ϕ, the peripheral generator c at d, and the unit tangential reference $s_{d,1}$ compatibly. Put $F_0 = s_{d,1}(\phi)$. The induced based action Φ of F_0 on B is the action of the point d on the filled curve. There are actual equivariant identifications

$$M := \mathscr{I}^{\mathrm{ab}} = \mathbb{Z}_3[[B]](1), \qquad Z = \widehat{\bigwedge}^2 M, \qquad (F_0)_M = Q\Phi, \quad (F_0)_Z = Q^2 \widehat{\bigwedge}^2 \Phi. \tag{B.12.2}$$

Here the completed modules are inverse limits over finite continuous partitions of the profinite label set B; the label 1 is the chosen lift of the marked cusp.

Every local section of $\mathscr{Q}$ whose projection to B is unramified and whose tame inertia is nontrivial has, up to geometric conjugation, exactly the following form:

$$s(\tau) = c^a, \qquad s(\phi) = c^r \exp(z) F_0, \qquad \begin{cases} a \in \mathbb{Z}_3 \setminus \{0\}, \\ r \in \mathbb{Z}_3/(Q-1)\mathbb{Z}_3, \\ [z] \in \mathcal{C}_{k,d} := Z/(Q^2 \widehat{\bigwedge}^2 \Phi - 1)Z. \end{cases} \tag{B.12.3}$$

In particular its B-section is the point section at d. The quotient in (B.12.3) uses the actual image; that image is closed because Z is compact. Every displayed parameter occurs, and two displayed sections are conjugate exactly when all three parameters agree.

Under $\mathscr{Q} \to N/\gamma_3 N$, their image is exactly the ramified local cusp packet at d. For each member of that packet the entire local fiber is the group $\mathcal{C}_{k,d}$, acting simply transitively after the displayed reference is fixed. Finally a local marked section whose abelian tame inertia in V is zero is itself unramified.

Proof. The filling map gives a geometric free product $N = B *_3 \langle c \rangle$ after a choice of free lift of B. Its normal kernel is the free pro-three group on the profinite set of lifted marked points, a torsor under B. The two inverse semidirect-product maps used in F.25, now with $b \mapsto bcb^{-1}$, prove this assertion for the actual filling kernel. The maximal class-two quotient consequently has abelianization $\mathbb{Z}_3[[B]](1)$ and center its completed exterior square. Finite partitions and the finite-rank Hall basis prove these assertions and their inverse-limit compatibility. Galois permutes the labeled inertia generators, up to conjugacy and cyclotomic power. This proves (B.12.2); at the distinguished label the unit tangent also gives the exact equality $F_0 c F_0^{-1} = c^Q$.

We record the fixed-point and measure facts needed below. For any unramified B-section, every positive power of its Frobenius automorphism has fixed subgroup one in B. On B^{ab} the eigenvalues have complex absolute values Q or $\sqrt{Q}$; the same assertion for the degree-r lower-central quotient follows from its equivariant Hall injection into $(B^{\mathrm{ab}})^{\otimes r}$. Inner twists do not change the action on this graded quotient. Thus no such power has eigenvalue one in any positive degree. The intersection of the closed lower central groups is one, as finite three-group quotients are nilpotent. This proves the fixed-subgroup assertion for the full B. It follows that an affine Frobenius action on a B-torsor has at most one point fixed by any given positive power. A finite orbit must therefore be a single fixed point, and there is at most one such point in total.

The closure of that affine action is virtually pro-three. Indeed the translation subgroup is pro-three and the kernel of the action on the Frattini quotient of the finitely generated free pro-three group is pro-three. An integer power removes the finite prime-to-three part. For an invariant $\mathbb{Z}_3$-measure on such a torsor, all mass outside its possible single periodic point is zero. The same assertion holds with $\mathbb{F}_3$ coefficients. To see this precisely, stabilize a clopen set avoiding the periodic point by an open subgroup of the pro-three closure. Every orbit there is infinite. For any n, compactness supplies a finite invariant partition on which every orbit has size divisible by 3^n. Pushforward sums make the measure of the clopen set divisible by 3^n; over $\mathbb{F}_3$ one step already makes it zero. This is the integral measure argument of F.25, applied to the actual permutation action.

Now let the projection of s to B be unramified. Wild inertia and the tame factors other than three map trivially to the geometric pro-three group. Write $t = s(\tau) \in \mathscr{I}$ and $F = s(\phi)$, so $FtF^{-1} = t^Q$. On M the action of F is $Q\Psi$ for the affine Frobenius on the marked-point path torsor. The image m of t therefore satisfies $\Psi m = m$. The measure argument makes $m = a\delta_x$ for an actual fixed label x, or makes it zero when there is no such label.

When $m = 0$, the element t is in Z. The fixed relation says $(Q\widehat{\bigwedge}^2\Psi - 1)t = 0$ after dividing by the unit Q. This operator is injective. On every finite label quotient the permutation has finite order, so all its eigenvalues are roots of unity; none becomes one after multiplication by the rational integer $Q > 1$. Tensoring with $\mathbb{Q}_3$ therefore proves injectivity on that finite free lattice. The inverse-limit comparison in (B.12.2) then proves injectivity on Z. Hence $t = 1$.

For nontrivial t we consequently have $a \neq 0$. The fixed path x is exactly an isomorphism between the projected section and the point section at the filled point d: on inertia both actions are trivial by good marked reduction, and on the unramified quotient it is the displayed Frobenius fixed-path condition. Conjugate by a lift of that path in $\mathscr{Q}$. The label becomes 1, the B-projection is the specified point section, and $F = hF_0$ with $h \in \mathscr{I}$.

Use the integral class-two logarithm, with its sole denominator two, and set $e = \delta_1$. Write

$$t = (ae, z), \qquad b = h_{\mathrm{ab}}.$$

The equality $F_0 c F_0^{-1} = c^Q$ removes any central origin term on the line $\mathbb{Z}_3 e$. Class-two

collection now gives the exact equation

$$(Q\widehat{\bigwedge}^2\Phi - 1)z = a\, e \wedge b. \tag{B.12.4}$$

In fact $F_0 t = (Qae, Q^2(\bigwedge^2 \Phi)z)$ and conjugating by h adds $aQ\, b \wedge e$; comparison with $t^Q = (Qae, Qz)$ gives (B.12.4).

Augmentation and the fixed label give the equivariant splitting $M = \mathbb{Z}_3 e \oplus M^0$. Projection of (B.12.4) to $\widehat{\bigwedge}^2 M^0$ is zero, by the same finite-permutation eigenvalue argument. Thus $z = e \wedge v$ for a unique $v \in M^0$, and

$$(Q\Phi - 1)v = ab^0, \qquad b^0 = b - \mathrm{aug}(b)e. \tag{B.12.5}$$

The operator $Q\Phi - 1$ on M^0 is injective modulo three. If $Q = 1 \bmod 3$, its kernel consists of Φ-invariant $\mathbb{F}_3$-measures; if $Q = -1 \bmod 3$, every element of its kernel is fixed by Φ^2. Either way the measure argument supports it only at 1, and augmentation zero removes that last possibility. Reduction of the completed module commutes with the finite label inverse limits: compatible lifts exist by compactness, or by the surjective transition maps of the finite free modules. Writing $a = 3^s u$ with u a unit, repeated reduction of (B.12.5) now gives $v \in 3^s M^0 = aM^0$. For a lift h_1 of v/a, the explicit conjugation $t \mapsto h_1 t h_1^{-1}$ changes t to c^a. This is an integral divisibility argument; no division by a nonunit has been assumed.

An element of $\mathscr{Q}$ centralizing c^a must have trivial B-image, because the action of B on the label torsor is free and $a\delta_b = a\delta_1$ forces $b = 1$. Inside $\mathscr{I}$, the commutator equation is $ae \wedge h_{\mathrm{ab}} = 0$. Torsion-freeness and the augmentation splitting give $h_{\mathrm{ab}} \in \mathbb{Z}_3 e$. Consequently the exact centralizer is

$$C_{\mathscr{Q}}(c^a) = \langle c \rangle Z. \tag{B.12.6}$$

The relation with F_0 puts FF_0^{-1} in this centralizer, giving precisely $c^r \exp(z)F_0$. Conjugating by $c^u \exp(z')$ changes the two Frobenius parameters by $(Q-1)u$ and $(Q^2 \bigwedge^2 \Phi - 1)z'$. There is no cross term, since Z is central in $\mathscr{I}$ and c commutes with it. Abelianization fixes a, and (B.12.6) proves that these are all possible identifications. Conversely every displayed pair satisfies the tame relation. Continuous powering gives the homomorphism on the procyclic unramified quotient, and the tame semidirect-product presentation gives an actual continuous local section. This proves (B.12.3).

All conjugates of the marked generator have the same image e_d in V. Their pairwise commutators therefore vanish in $\gamma_2 N/\gamma_3 N$. Thus the actual quotient homomorphism sends $Z = \gamma_2 I_{\mathrm{mark}}/\gamma_3 I_{\mathrm{mark}}$ to one in $N/\gamma_3 N$. Formula (B.12.3) has exactly the stated class-two cusp image. Conversely, any marked section above a ramified class-two cusp at d has zero abelian inertia in the filled group. The full local theorem B.7 for that filled curve makes its B-projection unramified, so the preceding classification includes its entire fiber. The same argument starts from any marked section with zero inertia in V: its B-projection is unramified, while $ae_d = 0$ forces $a = 0$. The already proved zero-m case then makes $t = 1$. □

Statement B.13 (Proved). The central marked local fiber is a countable product of three-adic integers. *The group $\mathcal{C}_{k,d}$ in (B.12.3) is a nonzero, torsion-free, compact metrizable $\mathbb{Z}_3$-module. After a choice of divisible basis of its discrete dual, there is a continuous $\mathbb{Z}_3$-linear isomorphism*

$$\mathcal{C}_{k,d} \simeq \prod_{n \geq 1} \mathbb{Z}_3. \tag{B.13.1}$$

The coordinates in (B.13.1) depend on that stated choice; the quotient and the section action in (B.12.3) are intrinsic to the displayed reference. In particular there are actual local marked sections with the same ramified class-two cusp image which are not cuspidal and have no full-N lift.

Proof. Let $D = Q^2 \bigwedge^2 \Phi - 1$. The completed exterior square identifies with the antisymmetric direct summand of $\mathbb{Z}_3[[B \times B]]$ by the idempotent $(1 - \mathrm{swap})/2$. Modulo three, D is $\bigwedge^2 \Phi - 1$. The only periodic point of Φ on B is 1, so the only periodic point on $B \times B$ is $(1, 1)$. The invariant-measure argument of B.12 makes every invariant $\mathbb{F}_3$-measure a multiple of $\delta_{(1,1)}$. Its antisymmetric part is zero. Thus D is injective modulo three.

If $3z = Dv$, reduction modulo three gives $v = 3v_1$, and cancellation in the torsion-free module Z gives $z = Dv_1$. This proves that the cokernel has no three-torsion, and hence no torsion at all. It is compact and metrizable because B is a finitely generated pro-three group, its finite characteristic quotients form a countable cofinal family, and all the coefficient quotients are finite. The image DZ is compact and closed.

We prove infinite rank with actual finite quotient maps. Put $V_B = B^{\mathrm{ab}}$, of rank nine, and $\Omega_n = V_B/3^n V_B$. The homomorphism $B \to \Omega_n$ induces a continuous surjection

$$Z/3Z \longrightarrow \bigwedge^2 \mathbb{F}_3[\Omega_n].$$

It sends D to $\bigwedge^2 \Phi_n - 1$, and therefore induces a surjection from $\mathcal{C}_{k,d}/3$ onto the coinvariants of that finite permutation module. If m is the order of Φ on $V_B/3$, its order on Ω_n divides $m3^{n-1}$: expand the powers of the integral matrix $\Phi^m = 1 + 3A$. The exterior basis vectors $\delta_0 \wedge \delta_v$, $v \neq 0$, form an invariant direct summand. No orbit reverses its orientation, because Φ_n fixes 0 and does not send a nonzero vector to 0. Their coinvariants have one independent vector for each orbit, and consequently dimension at least

$$\frac{3^{9n} - 1}{m3^{n-1}}.$$

This tends to infinity. A concrete nonzero test already at $n = 1$ takes a rational unmarked cusp vector v and sums the coefficients on $\delta_0 \wedge \delta_v$ and, when distinct, on $\delta_0 \wedge \delta_{-v}$. Frobenius multiplies that cusp vector by Q, so the sum kills D modulo three and takes the value one on an actual central commutator $[c, bcb^{-1}]$ whose label b maps to v.

For an explicit description of the isomorphism in (B.13.1), let $\mathcal{C}^\vee$ be the group of continuous characters of $\mathcal{C}_{k,d}$ into $\mathbb{Q}_3/\mathbb{Z}_3$. It is countable and three-primary. Injectivity of multiplication by three on $\mathcal{C}_{k,d}$ makes multiplication by three surjective on $\mathcal{C}^\vee$, so the latter is divisible. This dual assertion can be checked without a continuity assumption left implicit: a finite-valued character on the closed subgroup $3\mathcal{C}_{k,d}$ factors through its image in a finite quotient of $\mathcal{C}_{k,d}$, and extends there because $\mathbb{Q}_3/\mathbb{Z}_3$ is divisible. Pulling that finite character back supplies the continuous extension. Choose a basis of its subgroup killed by three, and compatible successive division chains for these basis elements. Independence and exhaustion follow by induction on the order of a character; this gives an isomorphism $\mathcal{C}^\vee \simeq \bigoplus_{n \geq 1} \mathbb{Q}_3/\mathbb{Z}_3$. The index set is countably infinite by the unbounded finite dimensions just proved. Evaluation on those chosen division chains is the continuous homomorphism in (B.13.1). Duality, or separation of points by finite characters followed by compactness, proves that it is an isomorphism.

The cuspidal class with the parameters a, r has $[z] = 0$. By B.12 a nonzero $[z]$ cannot be removed by any geometric conjugation, so it is not cuspidal. A full local lift would be cuspidal by B.7, and would retain that property under the actual quotient to $\mathscr{Q}$. Thus these sections have no full lift. □

Statement B.14 (Proved). The complete local marked image and the unmarked ramified fibers. *Retain the local data and actual quotients of B.12. If the projection of a section of $\mathscr{Q}$ to B is ramified, that section is itself cuspidal at a unique unmarked cusp $d' \in D(k) \setminus \{d\}$. For a fixed ramified B-section the lift to $\mathscr{Q}$, up to geometric conjugation, is unique and is the image of its actual full cusp section.*

Consequently the complete local image of $\mathrm{Sec}_k(\mathscr{Q})$ *in* $\mathrm{Sec}_k(N/\gamma_3 N)$ *equals the local full image: all unramified classes and the disjoint ramified cusp packets of B.8. The fiber over a ramified unmarked cusp class is a singleton; the fiber over a ramified marked cusp class is exactly the group* $\mathcal{C}_{k,d}$ *of B.12–B.13.*

Proof. By the full local theorem B.7 applied to the filled curve, a ramified projected section is cuspidal at an unmarked rational cusp d'. Filling d is an isomorphism on the inertia and tangent data at d'. Thus this projected section has an actual full cusp lift, which we choose as reference. It remains to prove that every lift to $\mathscr{Q}$ is conjugate to that reference.

Let X be the profinite torsor of paths on the filled curve from this reference at d' to the filled point d. In these coordinates $M = \mathbb{Z}_3[[X]](1)$. The geometric inertia image of the reference is $J_0 = \langle c_{d'}^a \rangle \simeq \mathbb{Z}_3$, with $a \neq 0$. It acts freely on X by translation, and diagonally on $X \times X$. Both are pro-three actions without finite orbits. The integral measure lemma therefore gives $M^{J_0} = 0$ and $(\mathbb{Z}_3[[X \times X]])^{J_0} = 0$. Taking the antisymmetric direct summand gives $Z^{J_0} = 0$ as well.

We justify continuous cohomology and orbit coinvariants with these compact coefficients. For a procyclic three-group generated by T acting continuously on a compact pro-three module A, evaluation at T gives

$$H^1_{\mathrm{cts}}(J_0, A) = A/(T-1)A. \tag{B.14.1}$$

Every value $x \in A$ defines a continuous cocycle: its norm sum at T^{3^n} tends to zero in every finite coefficient quotient, since on that quotient the action has three-power order and the repetitions are eventually divisible by its exponent. These finite cocycles are compatible and give the stated continuous cocycle. Coboundaries are exactly $(T-1)A$; this image is compact and closed. In the permutation module case, pushforward of measures gives an actual isomorphism

$$M/(T-1)M \simeq \mathbb{Z}_3[[J_0\backslash X]](1). \tag{B.14.2}$$

At each finite continuous invariant partition, permutation coinvariants are the free module on its orbits. Inverse limits preserve this description: the finite systems of correcting elements have compact solution sets, so the kernel of pushforward is the actual closed image $(T-1)M$. Equivalently the universal continuous homomorphism from the completed free module to an orbit-constant module has exactly this kernel. The same construction for $X \times X$ commutes with the idempotent $(1 - \mathrm{swap})/2$, and hence computes $Z/(T-1)Z$ as the corresponding antisymmetric direct summand of the orbit module.

Let Ψ denote the actual Frobenius permutation on $J_0\backslash X$. The induced Frobenius action on (B.14.1) contains an inverse factor Q: conjugation sends T to T^Q, so the pullback of a cocycle evaluated at the generator is multiplied by Q^{-1} in coinvariants. On M the coefficient action has factor Q. Thus the action on (B.14.2) is precisely Ψ, with no remaining Tate factor.

There are no finite Ψ-orbits. We construct the map proving this assertion. Filling all remaining punctures maps the full pro-three group of the filled curve onto T_3E. On path torsors it gives a continuous equivariant map from X to the proper path torsor from d' to d. The group J_0 maps to one in T_3E, because it is peripheral inertia. The map consequently descends to $J_0\backslash X$. A point there periodic under Ψ^n would give a fixed point of the proper path torsor under Frobenius to the nth power. Original arithmetic inertia already acts trivially on this torsor by good reduction. Such a fixed point would make the completed Kummer class of $d - d'$ zero over the finite unramified degree-n extension of k.

That difference is a nonzero three-torsion point of E. At every finite extension of a local field of residue characteristic different from three, the reduction kernel is pro-residue-characteristic and the three-primary completion of $E(k')$ is its finite three-primary torsion

subgroup. In particular its nonzero three-torsion point remains nonzero in that completion. The inverse-limit Kummer injection then makes its class nonzero in $H^1(k', T_3E)$. Being divisible once, or several times, in a larger finite group would not make this completed class zero. This proves the required absence of periodic points.

The Frobenius action on the compact quotient set is virtually pro-three, as in B.12. Its invariant integral measure group is therefore zero. Equations (B.14.1)–(B.14.2) show

$$H^1_{\mathrm{cts}}(G_k, M) = 0. \tag{B.14.3}$$

For precision, the unramified quotient causes no additional term: its possible first cohomology term has coefficient $M^{J_0} = 0$. One can obtain the same conclusion without invoking a spectral sequence. If the inertia cocycle is a coboundary, remove it. The tame relation then makes the Frobenius value J_0-invariant, hence zero. Wild and non-three tame inertia have no nonzero homomorphism to these compact pro-three modules.

For Z the inverse-Q factor on (B.14.1) removes one of its two coefficient Tate factors. Thus Frobenius acts on $Z/(T-1)Z$ by Q times the permutation of the diagonal-J_0 orbit set in $X \times X$, restricted to the antisymmetric summand. This group has no invariant vector. On each finite invariant partition, a cycle of length h would force $(Q^h - 1)v = 0$ in a free $\mathbb{Z}_3$-module, which implies $v = 0$. The inverse-limit orbit-module comparison then gives the same vanishing integrally. There is no extra orbit reversal: if a translation in J_0 sends (x, y) to (y, x), its square fixes x; freeness and the absence of order-two elements in J_0 imply $x = y$. Together with $Z^{J_0} = 0$, this proves

$$H^1_{\mathrm{cts}}(G_k, Z) = 0. \tag{B.14.4}$$

The difference between two lifts of the fixed B-section is a continuous $\mathscr{I}$-cocycle, for the reference action just used. By (B.14.3) its image in M is a coboundary. Lift a correcting element from M to $\mathscr{I}$ and conjugate the section. Its cocycle now has values in the central subgroup Z. Equation (B.14.4) removes this last cocycle by a central conjugation. This proves uniqueness of the actual cusp lift in the ramified projected case.

If a class-two section is unramified, choose any lift of its geometric Frobenius value to $\mathscr{Q}$. Continuous powering defines an unramified lift on the procyclic unramified Galois quotient. The ramified cases are exactly B.12 and the case just proved, giving the full image assertion. For a ramified unmarked cusp class, its projected B-section is the unique cusp class determined by its class-two image: the nonzero inertia line identifies the cusp, and the completed local Kummer parameter is already injective in the abelian cusp packet. The uniqueness above therefore gives the singleton fiber. The marked fiber is precisely B.12–B.13. The class-two tame identity, the correcting-conjugation sign, and the finite orbit-sum certificates for the nonzero marked central fiber are checked in `code/bc_marked_checks.py` and its output. □

Statement B.15 (Proved). An explicit cleaned Heisenberg detector at seven. *Put $k = \mathbb{F}_7(\theta)$, $\theta^3 = 2$, and write $[a, b, c] = a + b\theta + c\theta^2$. On $E_k : y^2 + y = x^3$, let $\zeta = 2$ and*

$$\begin{aligned} P &= (3, 2), \qquad P^- = (3, 4), \\ Q &= ([4, 1, 1], [2, 0, 2]), \\ S &= ([0, 6, 5], [1, 4, 3]), \\ T_1 &= ([3, 3, 3], [6, 2, 2]), \qquad T_2 = -T_1. \end{aligned} \tag{B.15.1}$$

The following functions are explicit:

$$\begin{aligned} a &= \frac{x^2+[1,6,5]x+[0,1,4]+[4,4,4]y}{x-3}, \quad b = x-3, \\ f_1 &= \frac{x^2+[2,3,5]x+[6,4,6]+[6,1,6]y}{x-[3,3,3]}, \\ f_2 &= \frac{x^2+[2,3,5]x+[0,3,0]+[1,6,1]y}{x-[3,3,3]}, \\ m &= \frac{x^2+[2,5,2]x+[6,2,2]+[2,4,5]y}{x-[3,3,3]}, \quad f = \frac{m}{f_1 f_2^2 (x-[3,3,3])}. \end{aligned} \tag{B.15.2}$$

Define $p(v)=\sum_{i=0}^{5} p_i v^i$ *and* $h(v)=\sum_{i=0}^{3} h_i v^i$ *by*

i	0	1	2	3	4	5
p_i	$[3,0,0]$	$[0,6,6]$	$[6,1,5]$	$[4,0,2]$	$[4,1,2]$	$[1,5,4]$
h_i	$[6,1,4]$	$[6,0,1]$	$[2,0,0]$	$[1,5,4]$		

(B.15.3)

In $k(E)(v)$ *with* $v^3=b$*, put*

$$u(v)=\frac{p(v)+y}{vh(v)}, \qquad r = f\,u(v)u(\zeta v)^2. \tag{B.15.4}$$

Then

$$v^3=b, \qquad w^3=a, \qquad z^3=r \tag{B.15.5}$$

defines a connected Galois cover of degree 27*, with Heisenberg group of exponent three. It is finite étale over* $U_k = E_k \setminus E_k[3]$*. The actual deck generators are*

	v	w	z
σ	ζv	w	$u(\zeta^2 v)z/w$
τ	v	ζw	z
ρ	v	w	ζz,

$$[\sigma,\tau]=\rho. \tag{B.15.6}$$

For the Fermat tangent parameter s *at* O*, the first two radicals retain the leading coefficient* 9*, whose reduction is* 2*. The cleaned central radical satisfies* $r \sim s^4$*. Thus the based Puiseux choices are*

$$v \sim \theta s^{-2/3}, \qquad w \sim \theta s^{-2/3}, \qquad z \sim s^{4/3}. \tag{B.15.7}$$

There is also an explicit Frobenius transport on the compositum of the three coefficient-conjugate covers. Let a superscript $[j]$ *mean raising every coefficient to* 7^j*, with* j *read modulo three, and use the same choices (B.15.7) for* w_j, z_j*. The arithmetic Frobenius at the framed tangent is*

$$\begin{aligned} \Phi(v) &= \zeta^2 v, \qquad \Phi(w_j) = \zeta^2 w_{j+1}, \\ \Phi(z_j) &= \frac{\zeta w_{j+1} z_{j+1}}{u^{[j+1]}(\zeta v)}, \qquad \Phi^3 = 1. \end{aligned} \tag{B.15.15}$$

Proof. All coefficients in this proof are computed in the displayed finite field. The group law gives $3Q=P$ and $3S=T_2$. The function a is the normalized Miller function

$$\operatorname{div}(a) = 3(Q) - (P) - 2(O). \tag{B.15.8}$$

For example, multiplying the tangent line at Q and the line through $2Q, Q$, and dividing by the vertical lines at $2Q, P$, gives (B.15.2); the common factor $x - x(2Q)$ cancels in both

numerator components. This is an equality of rational functions with the retained tangent leading coefficient, rather than an identification of divisor classes.

The full norm identity is

$$N_{k(E)(v)/k(E)}u = a. \tag{B.15.9}$$

Here is a direct finite calculation of it. Write

$$R(B) = [6,1,0] + [0,3,1]B + [0,2,2]B^2 + [5,4,0]B^3.$$

Expansion using $y^2 + y = x^3$, $v^3 = x - 3$ gives

$$N(p(v) + y) = a(x-3)R(x-3), \qquad N(h(v)) = R(x-3). \tag{B.15.10}$$

Since $N(v) = x - 3$, (B.15.9) follows without a scalar ambiguity. The norm calculation can also be checked by the cubic identity $N(A + Bv + Cv^2) = A^3 + bB^3 + b^2C^3 - 3bABC$.

We next exhibit the removal of every auxiliary ramification point. The cubic h is irreducible over k. Let α be its root and let

$$R_i = (\alpha^3 + 3, -p(\zeta^i\alpha)), \qquad i = 1, 2. \tag{B.15.11}$$

These are points of $E(k(\alpha))$; the candidate with $i = 0$ does not lie on E. Their Frobenius orbits have degree three. Denote the associated closed-point divisors by $\mathcal{R}_1, \mathcal{R}_2$. Exact addition of the three conjugates gives $\mathrm{Tr}(R_i) = T_i$. The two successive line constructions for $R_i + R_i^{343} + R_i^{343^2} = T_i$ give precisely f_i:

$$\mathrm{div}(f_i) = \mathcal{R}_i - (T_i) - 2(O). \tag{B.15.12}$$

The coefficients descend to k after cancellation of the intermediate vertical factor; they are the coefficients displayed in (B.15.2). Likewise m is the normalized Miller function with divisor $3(S) - (T_2) - 2(O)$. Since $T_1 = -T_2$, the vertical function has divisor $(T_1) + (T_2) - 2(O)$. Therefore

$$\mathrm{div}(f) = 3(S) + 6(O) - \mathcal{R}_1 - 2\mathcal{R}_2. \tag{B.15.13}$$

At $\mathcal{R}_i$, the valuations modulo three of $u(v)u(\zeta v)^2$ are i, on all three sheets. For example at $v = \alpha$ the denominator contributes -1, and the numerator contributes 1, 2, or 0 according to whether its zero occurs at $\alpha, \zeta\alpha$, or $\zeta^2\alpha$. The two actual points have indices $1, 2$, giving residues $1, 2$. Formula (B.15.13) cancels them. At the three lifts of Q the numerator orders are $0, 0, 3$; the central valuations are therefore multiples of three. The extra zero S in (B.15.13) also has order three. Equations (B.15.8)–(B.15.10) list all other zeros and poles: they lie over P, P^-, O. The first two cubic extensions are unramified at all interior points, and the last radical now has valuation divisible by three there. Since the residue characteristic is seven, the normalized cover is étale over U_k.

For completeness, the field degrees and deck morphisms can be verified directly. The classes of a, b in $k(E)^\times/k(E)^{\times 3}$ are independent: the valuation at P^- first forces the coefficient of b to vanish, and the valuation at P then forces the coefficient of a to vanish. Hence $[k(E)(v,w) : k(E)] = 9$. Writing $\sigma(v) = \zeta v$, (B.15.9) gives

$$\frac{\sigma(r)}{r} = \frac{u(\zeta^2 v)^3}{a}. \tag{B.15.14}$$

If r became a cube in $k(E)(v,w)$, the Kummer kernel would give $r = a^j c^3$ in $k(E)(v)$ for some j. Applying σ and (B.15.14) would make $a = (u(\zeta^2 v)c/\sigma(c))^3$ in $k(E)(v)$, contradicting the independence just proved. The last degree is therefore three. Equations (B.15.14) and

(B.15.9) prove that (B.15.6) preserves all defining equations, that $\sigma^3 = 1$, and that $[\sigma, \tau] = \rho$. These are actual automorphisms of the degree-27 field.

Finally use $t = -x/y \sim s/3$ at O. The functions a, b, f_1, f_2, m and the vertical factor all have leading term $9s^{-2}$. Thus $u(v) \sim v$ and $f \sim s^6/3^6$. The original central radical has leading term $\zeta^2\, 9s^{-2}$, so after the twist its coefficient is $\zeta^2\, 9/3^6 = 1$ in k and its exponent is four. This proves (B.15.7). In particular no replacement of the minimal-model tangent constant by 36 is made.

To prove the last assertion, embed the fields in the Puiseux field using (B.15.7). Coefficientwise seventh power, fixing $s^{1/3}$, acts on $k((s^{1/3}))$ with order three, and fixes the Laurent expansions of x, y. It sends the chosen leading coefficient θ to $\theta^7 = \zeta^2\theta$. Moreover (B.15.9) gives the exact identity

$$\frac{r^{[j]}(\zeta^2 v)}{r^{[j]}(v)} = \frac{a^{[j]}}{u^{[j]}(\zeta v)^3}.$$

Thus the proposed image of z_j in (B.15.15) has the correct cube. Its leading coefficient is one, because $w_{j+1}/u^{[j+1]}(\zeta v)$ has leading coefficient ζ^{-1}. The uniqueness of the chosen cube root with this leading coefficient proves the formula. It respects every relation among the conjugate covers, since it is the restriction of the actual coefficient automorphism of the ambient Puiseux field. Iterating three times gives $w_j^3/N(u^{[j]}) = 1$ in the multiplier of z_j, verifying $\Phi^3 = 1$ directly as well.

The exact polynomial identities, the finite-field extensions, the three-conjugate additions, and the displayed coefficients are independently replayed by `code/bc_miller_norm_rr.py` and `code/bc_miller_heisenberg.py`, with their saved JSON and text outputs. This explicit based cover is a detector for the remaining central-action calculation; its construction and the conjugate-cover morphism (B.15.15) do not by themselves assign its central monodromy values on an uncomputed fixed Schreier basis. □

Statement B.16 (Proved). The integral depth-one map through the cyclic-nine cover. *Let F be the free pro-three group on the standard tripod inertia generators x, y, with $[x, y] = xyx^{-1}y^{-1}$. For $f \in F'$ write its metabelian image as $[x, y]^{h(X,Y)}$, where $X = x - 1$, $Y = y - 1$ act by conjugation. There is an actual continuous metabelian quotient in which*

$$\begin{aligned} x &\longmapsto (1, 0), \qquad y \longmapsto (0, 1), \\ (m, a)(n, b) &= (m + t^a n, a + b), \\ m, n &\in \mathbb{Z}_3[[t - 1]], \qquad a, b \in \mathbb{Z}_3. \end{aligned} \tag{B.16.1}$$

In this quotient the image of f is $(m_f, 0)$, with

$$m_f(t) = (1 - t)h(0, t - 1). \tag{B.16.2}$$

Let $H_9 = \ker(F \to \mathbb{Z}/9)$, where $x \mapsto 0$, $y \mapsto 1$. Its actual cover is

$$z^9 = 1 - t_{\text{base}}, \qquad Y_9 = \mathbb{P}^1 \setminus \{0, \infty, \mu_9\}. \tag{B.16.3}$$

In $H_9^{\text{ab}}/3$, write the image of f as $\sum_{j=0}^{8} c_j\, y^j x y^{-j}$; its y^9 coordinate is zero. Then

$$\begin{pmatrix} h_{0,3} \\ h_{0,5} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 2 & 2 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \\ c_6 \\ c_7 \\ c_8 \end{pmatrix} \quad \text{in } \mathbb{F}_3. \tag{B.16.4}$$

For the standard path β from $\overrightarrow{01}$ to $\overrightarrow{10}$ and the convention $f_g = \beta(g\beta)^{-1}$, these coordinates are explicit framed Kummer coordinates on (B.16.3). The relevant endpoint-to-start ratios are

$$r_0 = 9, \qquad r_j = \frac{-\zeta_9^j}{1-\zeta_9^j} \quad (1 \le j \le 8), \qquad \prod_{j=0}^{8} r_j = 1. \tag{B.16.5}$$

Choose their compatible roots by continuing the units $z - \zeta_9^j$ along the lifted path from $z = 1$ to $z = 0$, using the actual start velocity $-1/9$ for $j = 0$. If

$$\frac{g(r_{\chi(g)^{-1}j}^{1/3^n})}{r_j^{1/3^n}} = \zeta_{3^n}^{\kappa_j(g)}, \tag{B.16.6}$$

where the index is taken modulo nine, then $c_j = -\kappa_j(g) \pmod 3$. In particular (B.16.4)–(B.16.6) evaluate the two coefficients by finite cyclic-cover Kummer data for the full cyclotomic action; no restriction to $\ker\chi$ is imposed.

Proof. The group in (B.16.1) is a compact pro-three group. The universal property of F gives the stated continuous homomorphism. Its target is metabelian, and conjugation by x acts trivially on its additive normal subgroup, while conjugation by y multiplies that subgroup by t. Direct multiplication sends $[x, y]$ to $(1 - t, 0)$. Therefore the induced commutator-module map sends X to zero and Y to $t - 1$, proving (B.16.2).

The lift of a positive loop about $t_{\mathrm{base}} = 1$ in (B.16.3) sends z to $\zeta_9 z$. Consequently the Schreier generators $y^j x y^{-j}$ are the actual positive inertia loops at $z = \zeta_9^j$, and y^9 is the positive inertia loop at $z = 0$. The functions $z - \zeta_9^k$ and z have valuations respectively δ_{jk} and zero on the first nine generators, and respectively zero and one on the last generator. Their Kummer characters thus give the actual dual coordinate maps on $H_9^{\mathrm{ab}}/3$. The Schreier basis has ten members, so these coordinate maps are an isomorphism with $\mathbb{F}_3^{10}$. Since $f \in F'$, its total y exponent is zero; hence its y^9 coordinate is zero.

Reducing (B.16.2) modulo three and $t^9 - 1$ gives

$$\sum_{j=0}^{8} c_j (1+Y)^j = -Y h(0, Y) \pmod{Y^9}, \tag{B.16.7}$$

because $(1 + Y)^9 - 1 = Y^9$ in characteristic three. The coefficients of Y^4, Y^6 give $h_{0,3} = -\sum_j \binom{j}{4} c_j$ and $h_{0,5} = -\sum_j \binom{j}{6} c_j$, which are exactly (B.16.4). Only these indicated coefficients are recovered: the Y^8 coefficient of h is lost under multiplication by Y modulo Y^9.

The path β lifts to the real interval from $z = 1$ to $z = 0$. At the start, $z-1 = -t_{\mathrm{base}}/9+\cdots$, and at the end $z - 1 = -1$. Their framed ratio is therefore nine. For $j \neq 0$ both endpoint values are ordinary nonzero values, giving (B.16.5). The product identity follows from $\prod_{j=1}^{8}(1 - \zeta_9^j) = 9$ and $\prod_{j=1}^{8}(-\zeta_9^j) = 1$. The continued roots obey the product identity as well: their arguments occur in opposite pairs, and the $j = 0$ root is the positive real root of nine.

To check the sign in (B.16.6), choose one root of the unit at the start and continue it along β to the end. The path $g\beta$ transports the Galois conjugate start root to the Galois conjugate end root, with the unit index permuted by $\chi(g)$. Following β and then $(g\beta)^{-1}$ multiplies the start root by the inverse of the endpoint-to-start Kummer multiplier. Thus the positive inertia coordinate of f_g is $-\kappa_j(g)$, as asserted. This argument constructs the maps for every g; it does not extend a formula stated only on $\ker\chi$ by an unspecified normalization.

The program `code/bc_depth_one.py` checks the actual Schreier coordinates against the X coefficient of the abelianized Fox derivative $D_y(f) = Xh(X, Y)$ on explicit commutator words. It also verifies the exact cyclotomic-field identities (B.16.5), their degree-nine to

degree-three transition products, the Frobenius-seven index permutation, and the two rows in (B.16.4). The saved output has status `PASS`. □

Statement B.17 (Proved). The actual norm-289 depth-one Frobenius coordinates. *Let g_{17} be an arithmetic Frobenius of norm 289 at the prime above seventeen of $K_0 = \mathbb{Q}(\zeta_3)$, viewed through the specified tangential arithmetic action. Use the path and commutator conventions of B.16, and write the metabelian Ihara word as $f_{g_{17}} = [x, y]^{h(X,Y)}$. Then*

$$h(0, Y) \equiv -Y^6 - Y^7 \pmod{(3, Y^8)}, \qquad h_{0,3} = h_{0,5} = 0 \quad \text{in } \mathbb{F}_3. \tag{B.17.1}$$

An explicit residue-field embedding is

$$\begin{aligned} k &= \mathbb{F}_{17}[\theta]/(\theta^2 - 3), \\ \overline{\zeta}_9 = 7 + 4\theta, \qquad \overline{\zeta}_3 &= \overline{\zeta}_9^3 = 8 + 15\theta. \end{aligned} \tag{B.17.2}$$

For the nine framed ratios of B.16 the actual local Kummer coordinates and geometric cusp coordinates are

$$\begin{aligned} (\kappa_j(g_{17}))_{j=0}^8 &= (0, 1, 2, 0, 1, 2, 0, 1, 2), \\ (c_j)_{j=0}^8 &= (0, 2, 1, 0, 2, 1, 0, 2, 1). \end{aligned} \tag{B.17.3}$$

The entries are in $\mathbb{F}_3$. These values hold for each prime of $\mathbb{Q}(\zeta_9)$ above seventeen, with its compatible embedding of $\zeta_3 = \zeta_9^3$.

Proof. The integer three is a nonsquare modulo seventeen, and direct multiplication in (B.17.2) gives $\overline{\zeta}_9^9 = 1$, $\overline{\zeta}_9^3 \neq 1$, and $\overline{\zeta}_9^{17} = \overline{\zeta}_9^{-1}$. Choose the embedding of $\mathbb{Q}(\zeta_9)$ into $\overline{\mathbb{Q}}_{17}$ that reduces to (B.17.2). Its completion is the unramified quadratic extension of $\mathbb{Q}_{17}$, which is also the completion of K_0. Every r_j in (B.16.5) is a unit there. The polynomial $Z^3 - r_j$ is finite etale on the local unit locus: its derivative is a unit at every root. Its roots therefore lie in finite unramified extensions.

Let ρ_j be the cube root selected by the actual path of B.16. Reduction is injective on μ_3. Arithmetic Frobenius acts on its residue as the 289th power, so the exact local comparison is

$$\overline{\frac{g_{17}(\rho_j)}{\rho_j}} = \overline{\rho}_j^{288} = \overline{r}_j^{96} = \overline{\zeta}_3^{\kappa_j(g_{17})}. \tag{B.17.4}$$

Here g_{17} fixes μ_9, so there is no index permutation in (B.16.6). It also fixes μ_3; changing a cube root by an element of μ_3 does not change (B.17.4). Thus this calculation computes the prescribed based cocycle itself modulo three. Changing the representative by local inertia also leaves it unchanged because these unit extensions are unramified.

We can evaluate every symbol without choosing any radical extension. For $1 \leq j \leq 8$, the seventeen-power Frobenius sends $\overline{r}_j$ to $\overline{r}_{-j}$, with indices modulo nine, and

$$\overline{r}_j \overline{r}_{-j} \in \mathbb{F}_{17}^{\times}, \qquad \frac{\overline{r}_j}{\overline{r}_{-j}} = -\overline{\zeta}_9^j. \tag{B.17.5}$$

Every element of $\mathbb{F}_{17}^{\times}$ has trivial 96th power. Since $96 \equiv 6 \pmod 9$, (B.17.5) gives

$$\kappa_j + \kappa_{-j} = 0, \qquad \kappa_j - \kappa_{-j} = 2j \quad \text{in } \mathbb{F}_3.$$

Thus $\kappa_j = j \pmod 3$. The ratio $r_0 = 9$ also has trivial symbol, which proves the first vector in (B.17.3). The actual sign comparison of B.16 gives $c_j = -\kappa_j$ and proves the second. The argument uses only primitivity of the chosen ninth root and its cube, so it applies to every embedding above seventeen.

For completeness, the entire specialization through this precision follows from the same nine symbols. In $\mathbb{F}_3[t]/(t^9-1)$,

$$\sum_{j=0}^{8} c_j t^j = -t\frac{d}{dt}\left(\sum_{j=0}^{8} t^j\right) = t(t-1)^7.$$

With $t = 1+Y$, equation (B.16.7) becomes $-Yh(0,Y) = Y^7 + Y^8 \pmod{Y^9}$. Division by Y at the stated precision proves (B.17.1). No coefficient of degree eight or higher is assigned by this division.

The standalone program `code/bc_depth_one_frob17.py` executes all field operations in (B.17.2), evaluates the nine symbols by 96th powers, verifies (B.17.5), reconstructs the cyclic group-ring polynomial and the two rows of B.16, and repeats the computation for all six primitive ninth roots in k. The saved JSON and text outputs have status `PASS`. □

Statement B.18 (Proved). The full finite arithmetic image and the exact Frobenius action at seventeen. *Use the actual source basis of A.38 and the c-adapted basis m, n of A.31. Let*

$$\rho : P_3 \longrightarrow \mathrm{Aut}(N/(N^3\gamma_3 N))$$

be the actual arithmetic action. Its image G has order 243, nilpotency class three, and exponent nine. In the $10+45$ coordinates of A.38, its linear image has order nine, and its identity-linear kernel has dimension three over $\mathbb{F}_3$. The exact dimensions are

	$\gamma_2(G)$	$\gamma_3(G)$	$\gamma_4(G)$
$\dim_{\mathbb{F}_3}$	2	1	0

	$D_2(G)$	$D_3(G)$	$D_4(G)$
$\dim_{\mathbb{F}_3}$	3	2	0.

(B.18.1)

In particular $D_4(P_3)$ acts trivially on this specified geometric quotient. Put $M = \rho(m)$ and $N_* = \rho(n)$. Then the entire actual Frobenius action at seventeen is

$$\rho(\varphi_{17}) = M^6[N_*, [N_*, M]] = M^6 \neq 1. \tag{B.18.2}$$

It has order three. Its Johnson block has rank two and annihilates all nine cusp vectors. The full matrix is retained in `bc_ihara_finite_image_certificate.npz`.

Proof. An automorphism in these coordinates has matrix

$$\begin{pmatrix} A & 0 \\ J & \bigwedge^2 A \end{pmatrix}.$$

Its identity-linear subgroup is the additive group $\mathrm{Hom}(\mathbb{F}_3^{10}, \bigwedge^2 \mathbb{F}_3^{10})$. For such a matrix X, conjugation is the exact map

$$X \longmapsto (\bigwedge^2 A) X A^{-1}. \tag{B.18.3}$$

The script below reconstructs $\rho(u)$ and $\rho(c)$ from the A.38 word and its literal Schreier map, forms $\rho(v) = \rho(c)\rho(u)\rho(c)$, and applies the actual square-root source maps defining m, n. These reconstructed matrices agree with the independently saved arithmetic certificate.

The two linear matrices of M, N_* commute, each has order three, and their nine products are distinct. Define Z_0, Z_1, Z_2 to be the additive Johnson blocks, respectively, of

$$M^3, \qquad [M, N_*], \qquad [M, [M, N_*]].$$

Exact substitution gives three linearly independent vectors and the conjugation table

$$\begin{array}{c|ccc} & Z_0 & Z_1 & Z_2 \\ \hline \mathrm{Ad}(M) & Z_0 & Z_1+Z_2 & Z_2 \\ \mathrm{Ad}(N_*) & Z_0 & Z_1 & Z_2. \end{array} \tag{B.18.4}$$

It also gives $N_*^3 = 1$. The subgroup generated by the three Z_i is therefore normal, is elementary abelian of order 27, and is contained in the actual image. After quotienting by it, the images of M, N_* commute and have order three. The linear image has order nine, so the quotient is exactly $(\mathbb{Z}/3)^2$. This proves $|G| = 243$, and shows that the displayed subgroup is the whole identity-linear kernel.

The normal closure of $[M, N_*]$ is exactly $\langle Z_1, Z_2\rangle$ by (B.18.4). Its next commutator is $\langle Z_2\rangle$, which is central. Thus the lower-central assertions in (B.18.1) follow as actual group statements. Every cube lies in the identity-linear kernel, and that kernel has exponent three; consequently every element has ninth power one. Since M^3 is nontrivial, the exponent is nine. The cubes span $\langle Z_0, Z_2\rangle$, as verified on the 243 distinct normal forms. This also proves the two asserted Zassenhaus dimensions.

The defining formula for this filtration gives

$$D_4(P_3) = \overline{\gamma_4(P_3)(\gamma_2 P_3)^3 P_3^9}.$$

Each factor has trivial image by the preceding calculations. Equation (A.31.3), transported by the actual automorphism in A.38, therefore gives the equality in (B.18.2), not only a leading-term comparison. Table (B.18.4) gives $[N_*, [N_*, M]] = 1$. The matrix M^6 has order three, and its 45×10 Johnson block has rank two. Multiplication by the nine actual cusp columns of the Schreier certificate gives zero in every case.

The standalone `code/bc_ihara_finite_image.py` computes the full normal commutator module by the exact maps (B.18.3), checks (B.18.4), and independently enumerates the subgroup generated by M, N_* using literal matrix multiplication. Both computations give the same 243 elements. It verifies all ninth powers, the cube span, the Frobenius word, its rank and cusp values, and saves every indicated basis and matrix. Its JSON and text outputs have status `PASS`. □

Statement B.19 (Proved). A finite precision determining the integral local kernel. *Let $M_2 = \bigwedge^2 V$, and let*

$$\mathcal{D} = M_2^c \oplus \ker(1 + \rho(n)c : M_2 \longrightarrow M_2)$$

be the rank-45 lattice of normalized source cocycles used in A.39. Define the actual Frobenius evaluation

$$E_\varphi : \mathcal{D} \longrightarrow M_2^c, \qquad E_\varphi(\gamma) = \gamma(\varphi_{17}), \qquad R = \ker E_\varphi.$$

The integral operator $B = (F_\varphi - 1)|M_2^c$ has three-adic Smith exponents

$$\underbrace{1,\dots,1}_{6},\quad \underbrace{2,\dots,2}_{7},\quad \underbrace{3,3}_{2},\quad \underbrace{4,\dots,4}_{6}. \tag{B.19.1}$$

Identify $R/3R$ with its image in $\mathcal{D}/3\mathcal{D}$. Then its exact finite-precision description is

$$\frac{R}{3R} = \mathrm{im}\left(\ker\left(\mathcal{D}/243\mathcal{D} \xrightarrow{E_\varphi} M_2^c/243M_2^c\right) \longrightarrow \mathcal{D}/3\mathcal{D}\right). \tag{B.19.2}$$

More explicitly, for every integral $d \in \mathcal{D}$ with $E_\varphi(d) \in 243M_2^c$, put

$$h = B^{-1}E_\varphi(d) \in 3M_2^c, \qquad \gamma = d - \delta h, \qquad (\delta h)(g) = g(h) - h. \tag{B.19.3}$$

Then $\gamma \in R$ and $\gamma \equiv d \pmod{3\mathcal{D}}$. The matrix B and the bound (B.19.1) are computed in the attached certificate. Evaluating the separate arithmetic matrix $E_\varphi \bmod 243$ remains the finite calculation needed to list the resulting 24-dimensional subspace individually.

Proof. At seventeen, cubing permutes $\mathbb{F}_{17}$. Consequently the reduction $y^2 + y = x^3 - 7$ has one x for each y, and its point count, including infinity, is eighteen. Its Frobenius polynomial is $X^2 + 17$, so the norm-289 Frobenius on T is exactly -17. All cusps are rational over K_0, and its action on W is multiplication by 289. The actual graph-lattice comparison (A.7.4) therefore gives

$$F_\varphi|V = \begin{pmatrix} 289 I_8 & 102\Psi \\ 0 & -17 I_2 \end{pmatrix}. \tag{B.19.4}$$

We take its full exterior square and restrict to the c-fixed lattice. The latter has rank 21. The projection to $\bigwedge^2 T$ is zero on it, because c acts there by -1; hence it is also the fixed lattice inside $\mathcal{K}$.

The program constructs a basis from 21 independent columns modulo three of $1 + c$ on M_2. This is an actual $\mathbb{Z}_3$ basis of the fixed lattice: $\frac{1}{2}(1+c)$ is an integral idempotent, and the selected columns span its reduction modulo three, so Nakayama's lemma gives the full lattice. An invertible minor of three-adic unit determinant gives the coordinate restriction of (B.19.4). Both the rectangular matrix in M_2 and its 21×21 coordinate matrix have the exponents (B.19.1). An independent elimination modulo 3^8 gives the same list. The determinant check is

$$\det B = 83520^{13}\, 4914^8, \qquad v_3(\det B) = 50.$$

Indeed the rational fixed subspace has thirteen toric wedges, of eigenvalue 289^2, and eight mixed wedges, of eigenvalue $-17 \cdot 289$. The inverse matrix verifies that $81B^{-1}$ is integral over $\mathbb{Z}_3$, while $27B^{-1}$ is not. This proves the stated bound with its actual integral comparison.

For $h \in M_2^c$, the coboundary δh is normalized at c and belongs to $\mathcal{D}$. Its evaluation is exactly

$$E_\varphi(\delta h) = Bh.$$

Thus $\operatorname{im} E_\varphi$ contains $B(M_2^c)$, which contains $81M_2^c$ by (B.19.1). The formula (B.19.3) is therefore integral and has $h \in 3M_2^c$ whenever $E_\varphi(d)$ is divisible by 243. It gives $E_\varphi(\gamma) = 0$ and leaves the reduction of d modulo three unchanged. Conversely every element of R reduces to an element of the displayed finite kernel. Since the target of E_φ is torsion-free, $R \cap 3\mathcal{D} = 3R$. This proves both the claimed identification and (B.19.2).

The exact reconstruction, point count, two Smith computations and inverse-denominator check are executed in `code/bc_frobenius17_precision.py`. Its JSON and text outputs have status `PASS`, and include the fixed-lattice basis and the full integral matrix B. □

Statement B.20 (Proved). An individually specified class-two target with no class-four lift. *Use the actual arithmetic basis m, n, c of A.38 and A.31. Write the ten integral A.7 basis vectors as*

$$\begin{aligned}(v_0, \dots, v_7) &= (e_P, e_{Q_c}, e_A, e_{\bar{A}}, e_B, e_{\bar{B}}, e_C, e_{\bar{C}}),\\ v_8 = (\Psi(1,0)/3, (1,0)), &\qquad v_9 = (\Psi(0,1)/3, (0,1)) \quad \text{in } L_\Psi \subset W_{\mathbb{Q}_3} \oplus T_{\mathbb{Q}_3}.\end{aligned} \tag{B.20.1}$$

Let C_2 be the actual action of c on $M_2 = \bigwedge^2 V$. Define the integral vectors

$$b_0 = 3v_8 \wedge v_9 + \sum_{i=0}^{7} \big(\Psi_{i,2} v_i \wedge v_8 - \Psi_{i,1} v_i \wedge v_9\big), \qquad b = (1 - C_2) b_0. \tag{B.20.2}$$

Here the columns of Ψ are numbered one and two. In the basis $v_i \wedge v_j$, $i < j$, all nonzero coordinates of b are the following; every other coordinate is zero:

(i,j)	b_{ij}	(i,j)	b_{ij}	(i,j)	b_{ij}
$(0,8)$	-2	$(1,8)$	2	$(2,6)$	-1
$(2,7)$	1	$(2,8)$	2	$(2,9)$	2
$(3,6)$	1	$(3,7)$	-1	$(3,9)$	-2
$(4,6)$	1	$(4,7)$	-1	$(4,8)$	-2
$(4,9)$	-2	$(5,6)$	-1	$(5,7)$	1
$(5,9)$	2	$(6,8)$	2	$(6,9)$	-2
$(7,8)$	-2	$(7,9)$	2	$(8,9)$	$6.$

(B.20.3)

The prescriptions

$$\gamma(m) = 0, \qquad \gamma(n) = b, \qquad \gamma(c) = 0 \tag{B.20.4}$$

define an actual continuous M_2-cocycle unramified outside three. Let $s_{O,17}$ be the actual O-cuspidal section with tangent parameter seventeen. In $A = N/\gamma_3 N$, put

$$s_b(g) = \gamma(g) s_{O,17}(g). \tag{B.20.5}$$

This is an actual class-two section, with abelian class $\kappa_{17} e_O$. Its restriction at seventeen equals the specified local cuspidal section exactly. It has a global lift to $N/\gamma_4 N$, but no global lift to $N/\gamma_5 N$. Consequently it has no full, metabelian, or marked-depth lift.

The projection of (B.20.4) to $\bigwedge^2 T$ has the exact generator values $0, 6$. If $a = \kappa_3(n) \in \mathbb{Z}_3^\times$ in this actual source basis, this cocycle is $(6/a)\kappa_3$. Only $a \equiv 1 \pmod 3$ is used; its integral value has not been replaced by one.

Proof. We first prove the integral local condition. On V, the actions of m, n are respectively $G_u G_v$ and $G_v G_u^{-1}$ from (A.7.4)–(A.7.5). They commute, and the first commutes with c. Set

$$L_0 = \ker(n - 1 : M_2 \to M_2) \cap \ker(c + 1 : M_2 \to M_2).$$

This is a closed saturated lattice stable under m, n, c and their inverses. Direct substitution in the exact graph matrices gives

$$n(b_0) = b_0, \qquad n(b) = b, \qquad c(b) = -b. \tag{B.20.6}$$

For the first equality, the rational comparison is also explicit: the vector 3 times the wedge of the two pure proper vectors has the mixed coordinates in (B.20.2), proper coefficient three, and toric coefficient $\frac{1}{3}\Psi_1 \wedge \Psi_2$. Subtracting this last toric vector gives exactly b_0. Both pure proper determinant and toric wedges are fixed by n, so the equality transfers through the actual graph embedding. No reduction of a rational splitting modulo three is taken.

The full lattice is

$$L_0 = (\bigwedge^2 W)^{c=-1} \oplus \mathbb{Z}_3 b, \qquad \operatorname{rank} L_0 = 16. \tag{B.20.7}$$

Here is an integral verification. The toric anti-invariant lattice has rank fifteen. Choose the lexicographically first independent columns modulo three of $1 - c$ on its 28 wedges. These are an actual integral basis by the order-two projector and Nakayama's lemma. Adjoining b gives sixteen columns independent modulo three. The stacked rational matrices $n - 1, c + 1$ have rank 29, so these columns span their sixteen-dimensional common rational kernel. Their unit minor makes their integral span primitive in M_2, proving (B.20.7). The code also computes

the exact matrix of m on this basis and its inverse; both are integral over $\mathbb{Z}_3$. This verifies the stated stability on the actual lattice.

Since m, n freely generate P_3, their prescribed values give a continuous crossed homomorphism. The identities $c(m) = m$, $c(n) = n^{-1}$ and (B.20.6) prove its exact c-equivariance, so it extends with value zero at c. Every value lies in L_0. The actual norm-289 Frobenius φ_{17} is fixed by c, hence

$$\gamma(\varphi_{17}) = c\gamma(\varphi_{17}) = -\gamma(\varphi_{17}) = 0. \tag{B.20.8}$$

The final equality uses that two is invertible. Thus (B.20.4) lies in the exact integral lattice R of A.39, not merely in a finite approximation to it. Its inflation has zero tame value at seventeen and zero value at the normalized local involution as well. These three values prove the exact local equality in (B.20.5).

We give a finite functional detecting the ensuing gluing obstruction. Let $X_0, \dots, X_9$ be the ordered Schreier generators in (A.38.3), and use the integral degree-three basis

$$H_{\mu,i} = [[X_i, X_{\min \mu}], X_j],$$

where μ is a nondecreasing multiset of size three, $i > \min \mu$ occurs in it, and j is the remaining index after removing i and one occurrence of $\min \mu$. Write $q_{\mu,i}$ for the corresponding coordinate functional. Over $\mathbb{F}_3$, define

$$\begin{aligned} \ell = 2\big(&q_{001,1} + q_{002,2} + q_{003,3} + q_{004,4} \\ &+ q_{088,8} + q_{188,8} + q_{288,8} + q_{388,8} + q_{488,8}\big). \end{aligned} \tag{B.20.9}$$

All these indices refer to the Schreier basis, not to an unidentified change of proper lifts. The program uses the exact unimodular $A.7$-to-Schreier matrix P and the actual map $\bigwedge^2 P$ on b. It verifies that this map intertwines the full m, n, c actions on M_2 before evaluating the source cochain.

With the independent Magnus evaluator of `bc_gluing_audit.py`, the functional has the following exact values. It kills the entire global M_3 restriction image, all twenty-one retained local-conjugator parameters, and all sixty-eight unknown-word contributions on the cocycle (B.20.4). Its value on the known Frobenius contribution is

$$\ell(\Delta) = 1. \tag{B.20.10}$$

Only three of those unknown maps are nonzero on the general source, and all three vanish on the whole lattice L_0 modulo three. The known map vanishes on its fifteen-dimensional toric summand and is nonzero on b. In the fixed 143-dimensional quotient of the independent audit, its vector has 87 nonzero entries. All twenty-seven remaining arithmetic coefficient possibilities give exactly this same vector.

These are evaluations of the actual global gluing map of A.39. Changing the integral class-three lift adds a global M_3 correction, which (B.20.9) kills. The retained source truncation and all local comparison parameters are those independently verified there, so (B.20.10) applies to the actual arithmetic Frobenius. A.33 gives a global class-three lift; A.40 proves that this nonzero gluing class prevents every class-four lift. The actual quotient maps and local cusp comparisons in A.39–A.40 give the full, metabelian and marked non-lifting conclusions on section classes.

Finally the proper projection of b is six by (B.20.3). The positive-root Kummer cocycle κ_3 has value zero at c, and hence at the c-fixed generator m. Its value at n is a unit by (A.31.6). Equality on the actual free source generators proves the asserted identity with $(6/a)\kappa_3$.

The standalone `code/bc_explicit17.py` executes the integral lattice, saturation, action and basis comparisons, then every stated functional and gluing evaluation. It saves all 45

coordinates, the exact rank-16 basis, the transition matrix, the full nonzero vector and the nine-term functional. Its JSON, NPZ and text certificates have status `PASS`. □

Statement B.6 (Proved). The complete global one-prime image and its fibers. *Fix a rational prime $q \neq 3$, put $A = N/\gamma_3 N$, and let s be a continuous class-two section unramified outside $S = \{3, q, \infty\}$. Choose its exact cocycle representative with $s(c) = 1$, and write α for its projection to V. Define*

$$\begin{aligned} &H_0 = N, \qquad H_{n+1} = H_n^3[H_n, H_n], \\ &P_n = N/H_n, \qquad A_n = P_n/\gamma_3 P_n, \qquad J_n = \gamma_3 P_n, \\ &S_n = \{3, \infty\} \cup \{q : \alpha|_{I_q} \bmod 3^n \neq 0\}. \end{aligned} \tag{B.6.1}$$

All subgroups are closed. The exact finite target s_n is the image of s in A_n.

Use the explicitly constructed finite based action on P_n and its splitting field F_n from A.43. Let K_n be the compositum of F_n with the field of the target graph (s_n, ρ_{A_n}). If s_n is ramified outside S_n, the test fails. Otherwise form the finite Galois field Ω_n as the compositum of Ω_{n-1} and all Galois extensions $E/\mathbb{Q}$ containing K_n, unramified outside S_n, of degree at most $|P_n|[F_n : \mathbb{Q}]$, starting with $\Omega_0 = \mathbb{Q}$. The discriminant and polynomial bounds in (A.43.4)–(A.43.10) give a terminating finite enumeration of these fields.

Let X_n be the finite set of functions $z : \mathrm{Gal}(\Omega_n/\mathbb{Q}) \to P_n$ satisfying the actual cocycle equation, $z(c) = 1$, and $z \bmod J_n = s_n$ literally. For every $k \leq n$, require its projection to P_k to be trivial on inertia at the primes of $S_n \setminus S_k$. These conditions give the actual transition maps of A.43. Then

$$\begin{aligned} [s] \in \mathrm{im}(\mathrm{Sec}(N) \longrightarrow \mathrm{Sec}(A)) &\iff X_n \neq \varnothing \quad \text{for all } n \\ &\iff [s] \text{ lifts to } N/\gamma_r N \quad \text{for all } r \geq 3. \end{aligned} \tag{B.6.2}$$

Every such full or nilpotent lift is unramified outside S. The complete fiber on section classes is

$$\varprojlim_n X_n/J_n^c, \tag{B.6.3}$$

where $h \in J_n^c$ acts by $z(g) \mapsto h^{-1} z(g) g(h)$. Thus (B.6.2) is an exact sequence of terminating finite arithmetic tests, and (B.6.3) retains all compatible conjugacy data. A bounded stopping rule for every positive instance is not asserted.

The necessary local condition is precisely the unramified class or a member of the actual cusp packets (B.8.1). It is sufficient at seven by B.10, and on the prime sets Σ and Ξ by A.16 and A.25. It is insufficient at seventeen. The integral section of B.20, with abelian class $\kappa(17)e_O$ and central generator values $\gamma(m) = 0$, $\gamma(n) = b$, has exactly the local O-cuspidal class-two restriction. It has a global $N/\gamma_4 N$ lift and no $N/\gamma_5 N$ lift, hence no full, metabelian, or marked-depth lift.

Proof. For the full group, the three quotient maps in (A.43.2) specialize exactly to (B.6.1). The image of H_n in V is $3^n V$, by induction on the Frattini operation. Hence $A_n^{\mathrm{ab}} = V/3^n V$, and the finite target table determines the displayed set S_n. The actual quotient map sends $\gamma_3 N$ onto $\gamma_3 P_n$, which proves the asserted kernel J_n. This identifies every group and action in the finite test with its A.43 counterpart.

The inherited ramification conditions ensure that projection to P_k factors through Ω_k; they are retained even when the prime q first enters the sets S_n at a later precision. When $\alpha(I_q) = 0$, it never enters, so an inertia value visible only in the central class-two target is rejected as soon as it is detected in some A_n. This is consistent with B.7's detection of full inertia in V. A.43 now gives the first equivalence in (B.6.2) and the exact fiber (B.6.3).

Its comparison uses the surjections $\ker(N \to A)^c \twoheadrightarrow J_n^c$ and the trivial global geometric centralizer of the target in A. It therefore identifies global section classes, without replacing the conjugation kernel by a finite-level centralizer. B.11 gives the second equivalence and the common ramification bound, including for separately chosen nilpotent lifts.

For the obstruction coordinates requested in Task B.4, entry into A is the residue, tame-relation and convergence criterion of Part A. The next central coefficient is the actual free lattice $M_3 = \gamma_3 N/\gamma_4 N$. Its residue complex at q is

$$[\, M_3(-1) \xrightarrow{F_q - 1} M_3(-1)\,], \qquad F_q t = qt \text{ for a tame inertia value } t \in M_3. \tag{B.6.4}$$

Here F_q in the complex acts on the twisted module and the second F_q acts on M_3; the equations are the same under the chosen tame generator. A.33 computes the continuous obstruction using this localization map. For the fixed finite set S, the finite-coefficient first-cohomology groups are finite, so their first derived limit is zero. Actual local cuspidal lifts kill the local degree-two obstructions and give a global class-three lift. Its remaining obstruction to having the prescribed local cuspidal classes is exactly the restriction cokernel (A.33.2). A.39 evaluates a nonzero family in that cokernel; A.40 proves the class-four exclusion; B.20 specifies an individual member with all integral coordinates and an explicit nine-term finite functional. These facts prove the stated failure of local sufficiency.

The finite tests (B.6.2) retain every subsequent geometric layer and global relation. They thus give the full one-prime image at seventeen and at every other prime, including the classes for which the local condition alone does not decide lifting. B.1, B.3 and B.9 separately give all three global cusp packets and their inherited central origins, as required in Task B.1. □

3 C. Restriction and germs

3.1 Restriction, descent, and persistent obstruction germs

Write $A = N/\gamma_3 N$ and $\mathrm{Sec}_L = H^1_{\mathrm{cts}}(G_L, A)$ for a number field $L \subset \overline{\mathbb{Q}}$. The action is the restriction of the action based at the fixed tangent at O. Our maps below use that actual action, and all equivalences of sections are by conjugation in A. This is the germ construction requested in CK C9, cote 149, §3, manuscript p. 33, scan 66, applied to the stated class-two quotient; the Fermat comparison is §51.16, folios 733v–736v, scans 172–178.

Statement C.1 (Proved). No arithmetic centralizer in the class-two geometric group. *For every number field L and every continuous section $s : G_L \to A \rtimes G_L$,*

$$C_A(s(G_L)) = 1. \tag{C.1}$$

The assertion uses the whole G_L, in contrast to the subgroup $\ker\chi$ in B.4; the comparison map between the two assertions is the inclusion of subgroups, which reverses inclusions of their centralizers.

Proof. Let h be in the displayed centralizer. Its image in $V = A^{\mathrm{ab}}$ is fixed by G_L, since inner conjugation is trivial on abelianization. Choose a finite prime v of L above two, with residue size $q = 2^f$. The eigenvalues of arithmetic Frobenius on $W \otimes \mathbb{Q}_3$ have complex absolute value q: they are q times roots of unity from the finite cusp permutation action. On $T_3E \otimes \mathbb{Q}_3$ the two eigenvalues have absolute value $q^{1/2}$. Here the necessary elliptic calculation is explicit: $E(\mathbb{F}_2)$ has three points by `bc_checks.py`; the Frobenius F_2 has determinant two by the Weil pairing, and $\det(1 - F_2) = \deg(1 - F_2) = 3$ because $1 - F_2$ is separable with precisely those three kernel points. Thus its characteristic polynomial is $X^2 + 2$. On the base change at v the Frobenius is F_2^f, which proves the asserted absolute values directly. The

characteristic polynomial of the extension V is the product of those on W and T, so one is not an eigenvalue. Consequently $V^{G_L} = 0$, and h lies in the central group $\gamma_2 N/\gamma_3 N = \bigwedge^2 V$. Inner conjugation acts trivially there. Its Frobenius eigenvalues, as seen from the filtration induced by $0 \to W \to V \to T \to 0$, have absolute values $q^2, q^{3/2}, q$. Again none is one. The lattice is torsion-free, so its invariant subgroup is zero. Hence $h = 1$. □

Statement C.2 (Proved). All restriction maps are injective and finite Galois descent is exact. *For number fields $L \subset L'$, restriction is the injective map of pointed sets*

$$\mathrm{res}_{L'/L} : \mathrm{Sec}_L \longrightarrow \mathrm{Sec}_{L'}, \qquad [z] \longmapsto [z|_{G_{L'}}]. \tag{C.2}$$

For L'/L finite Galois it identifies its source with precisely the $\mathrm{Gal}(L'/L)$-fixed classes in its target. Writing ρ for the based action on A, the action by $g \in G_L$ is represented by

$$(g \cdot z)(h) = \rho(g) z(g^{-1}hg), \quad h \in G_{L'}. \tag{C.3}$$

Thus the kernel of (C.2), as a pointed-set map, is its base point, and its image in the invariant set is the entire set.

Proof. Restriction preserves the continuous cocycle equation and geometric conjugation, proving that (C.2) is defined. For (C.3), conjugate the section graph by $(1, g)$ in $A \rtimes G_L$. This gives exactly the formula and proves continuity and the cocycle equation. When $g \in G_{L'}$, the cocycle identity gives

$$\rho(g) z(g^{-1}hg) = z(g)^{-1} z(h) \rho(h) z(g),$$

so the action is conjugation by $z(g)^{-1}$ and is trivial on section classes. Therefore it factors through the specified finite Galois group.

For injectivity, conjugate two sections so that their restrictions to $G_{L'}$ agree. Choose a finite extension M/L, Galois over L, with $M \supset L'$. The restrictions agree on the normal subgroup G_M. For each $g \in G_L$, the element $t(g)s(g)^{-1} \in A$ centralizes $s(G_M)$: both sections induce, by conjugation, the same map $s(h) \mapsto s(ghg^{-1})$ there. Statement C.1 gives $t(g)s(g)^{-1} = 1$, proving equality over G_L.

For the image assertion, let s represent a fixed class over L'. Consider its normalizer

$$B = N_{A \rtimes G_L}(s(G_{L'})).$$

It is a closed subgroup of a compact profinite group. Invariance of the class under (C.3) says precisely that each $g \in G_L$ is the projection of an element of B: use the conjugator in A comparing $g \cdot s$ with s. The kernel of $B \to G_L$ equals $C_A(s(G_{L'}))$. Indeed, an element projecting to one normalizes the graph only by preserving each of its points, since there is exactly one graph point over each $h \in G_{L'}$. This kernel is one by C.1. Hence $B \to G_L$ is a continuous bijective homomorphism between compact Hausdorff groups and has continuous inverse. That inverse is a section extending s, because its unique element over $h \in G_{L'}$ is $s(h)$. Injectivity already proved uniqueness up to geometric conjugacy. □

Statement C.3 (Proved). The action and fields of definition of germs. *The set*

$$\mathscr{G} = \varinjlim_{L \subset \overline{\mathbb{Q}}} \mathrm{Sec}_L \tag{C.4}$$

has a well-defined $G_{\mathbb{Q}}$-action obtained by (C.3) together with conjugation of the field of definition. Every transition map is injective. Each germ has an open stabilizer and a finite orbit. For every number field L,

$$\mathrm{Sec}_L \xrightarrow{\sim} \mathscr{G}^{G_L}. \tag{C.5}$$

In particular the image of $\mathrm{Sec}_{\mathbb{Q}}$ *is exactly the fixed-germ set. The fixed field of the stabilizer of a germ is its smallest field of definition.*

Proof. The formula transports a cocycle on G_L to a cocycle on G_{gL}, and respects restriction and conjugation. It therefore defines an action on the colimit. Restriction injectivity is C.2. A representative over L is fixed by G_L, since the same cocycle calculation used after (C.3) makes inner field conjugation trivial on its class. The stabilizer is consequently open.

Take a germ fixed by G_L, choose a field where it is represented, and enlarge it to a finite Galois extension M/L. For $g \in G_L$, the class and its translate are both defined over M. Their equality as germs means equality after restriction to a further finite field; injectivity in C.2 makes them equal already over M. The invariant class descends by C.2 to L. Uniqueness is again injectivity. This proves (C.5).

An open stabilizer $H \subset G_{\mathbb{Q}}$ is G_{L_0} for a number field L_0. Applying the same argument over L_0 proves definability there. Every other field of definition has its Galois group inside H, so it contains L_0. This proves the final assertion. □

Statement C.4 (Refuted). The R2 obstruction does not disappear as a germ. *Let* $c = 9\kappa(3) \cup \beta$ *be the nonzero R2 class specified by LM7 (E.32f), so that in* $H^2_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathbb{Z}_3(2)) = \mathcal{R}/d\mathcal{R}$ *it is represented by*

$$z_{q_j} = q_j - 1 \quad (j \geq 3), \qquad z_q = 0 \text{ at the other primes.} \tag{C.6}$$

For every finite number field L, $\mathrm{res}_{L/\mathbb{Q}}(c) \neq 0$. *In fact these restrictions all have infinite additive order. Consequently* c *gives a nonzero element of* $\varinjlim_L H^2_{\mathrm{cts}}(G_L, \mathbb{Z}_3(2))$. *The vanishing assertion in the first sentence of Task C.4 is false.*

Proof. For a nonzero integer m, the unique coordinatewise solution of $dy = mz$ has value m at every q_j and zero elsewhere, because multiplication by each nonzero $q - 1$ is injective in $\mathbb{Z}_3$. This family does not tend to zero 3-adically. Thus $mz \notin d\mathcal{R}$ and $mc \neq 0$ for every nonzero integer m.

We spell out the transfer used for this continuous cohomology. Set $G = G_{\mathbb{Q}}$, $H = G_L$ and $M = \mathbb{Z}_3(2)$. Use homogeneous continuous equivariant cochains, with alternating deletion differential. Choose representatives T for the finitely many left cosets $H\backslash G$, and write $r(g) = gt(g)^{-1} \in H$ for $g \in Ht(g)$. For an H-cochain c define the G-cochain

$$(\mathcal{T}c)(g_0, \ldots, g_n) = \sum_{t \in T} t^{-1} c(r(tg_0), \ldots, r(tg_n)). \tag{C.7a}$$

The coset permutation $tg = h(t,g)t'$ gives $t^{-1}h(t,g) = g(t')^{-1}$, proving equivariance; the formula commutes with the differential. All maps are continuous because the transversal is finite and each coset is open. Put $a_t(g) = t^{-1}r(tg)$. For a degree-n G-cochain f, the prism operator is

$$(Pf)(g_0, \ldots, g_{n-1}) = \sum_{t \in T} \sum_{i=0}^{n-1} (-1)^i f(a_t g_0, \ldots, a_t g_i, g_i, \ldots, g_{n-1}). \tag{C.7b}$$

The same coset permutation makes the sum equivariant. In $dP + Pd$ the internal mixed faces cancel in pairs; the two end faces are $[G : H]f$ and $-\mathcal{T}(\mathrm{res}\, f)$. Thus the formulas prove directly on continuous $\mathbb{Z}_3(2)$ cochains that

$$\mathrm{cor}_{L/\mathbb{Q}} \mathrm{res}_{L/\mathbb{Q}} = [L : \mathbb{Q}]. \tag{C.7}$$

Equivalently one can apply the same compatible formulas to every finite coefficient module; their cochain homotopies commute with reduction, so the assertion holds on the derived inverse limit as well as on finite-coefficient cohomology. This includes its $\varprojlim^1$ subgroup.

A zero restriction would make $[L:\mathbb{Q}]c = 0$, contrary to the first paragraph. A torsion restriction would, by the same calculation, make a nonzero integer multiple of c zero. Thus it has infinite order. An element maps to zero in a filtered colimit of groups exactly when it maps to zero at some later stage; no such stage exists here. □

Statement C.5 (Proved). Restriction is injective on the Tate-twist phantom subgroup. *For $\mathbb{Z}_3(2)$ the kernel of the finite-coefficient comparison map in R2 is torsion-free, and restriction from $\mathbb{Q}$ to any finite number field is injective on this kernel. This extends C.4 from its particular cup-product witness to every nonzero class in that kernel.*

Proof. The kernel is

$$\frac{\{z \in \mathcal{R} : z_q \in (q-1)\mathbb{Z}_3 \text{ for all } q\}}{d\mathcal{R}}.$$

Write $y_q = z_q/(q-1) \in \mathbb{Z}_3$. A nonzero integer multiple mz lies in $d\mathcal{R}$ exactly when $(my_q)_q$ is restricted. Multiplication by a fixed nonzero integer changes valuations by the fixed integer $v_3(m)$, so this condition is equivalent to $(y_q)_q$ being restricted. This proves torsion-freeness. Equation (C.7) then proves injectivity of restriction on this subgroup. No closure is taken in the denominator in this argument. □

Statement C.7 (Proved). The intrinsic phantom obstruction also survives every finite extension. *The nonzero obstruction in $H^2_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathcal{K})$ constructed in A.4 remains nonzero over every finite number field and in the filtered colimit over all number fields. More generally, restriction to every finite number field is injective on the phantom subgroup for $\mathcal{K}$ computed in A.2.*

Proof. In the residue presentation of A.2, every $D_q = \mathrm{Frob}_q - 1$ on $\mathcal{K}(-1)$ is injective. A locally zero class has a unique integral correction $y_q = D_q^{-1} z_q$ at each prime. A nonzero integer multiple of its class vanishes precisely when $(my_q)_q$ is restricted, which is equivalent to $(y_q)_q$ being restricted. Thus this phantom subgroup is torsion-free. The cochain transfer and prism formulas (C.7a)–(C.7b) apply verbatim with $M = \mathcal{K}$; they prove corestriction after restriction equals multiplication by the finite extension degree. Such multiplication is injective on the phantom subgroup. This proves the assertion and completes both persistence decisions asked for in Task C.4. □

Statement C.10 (Proved). The nine disjoint cusp germ packets and their Galois action. *Define the abelian germ parameter group*

$$\mathscr{K} = \varinjlim_{L \subset \overline{\mathbb{Q}}} H^1_{\mathrm{cts}}(G_L, \mathbb{Z}_3(1)) = \varinjlim_{L} \varprojlim_{n} L^\times / L^{\times 3^n}. \tag{C.10.1}$$

All transition maps are restriction. Choose at each cusp $d \in D(\overline{\mathbb{Q}})$ the unit tangent for the parameter x/z when $x = 0$, y/z when $y = 0$, and z/x when $z = 0$. These choices are $G_{\mathbb{Q}}$-equivariant. The map

$$\coprod_{d \in D(\overline{\mathbb{Q}})} \mathscr{K} \longrightarrow \mathscr{G}, \qquad (d, u) \longmapsto [s_{d,1} \text{ twisted by } u] \tag{C.10.2}$$

is injective, and its image is exactly the cuspidal germ set. There are nine packets. The $G_{\mathbb{Q}}$-action in these coordinates is

$$g(d,u) = (gd, gu), \tag{C.10.3}$$

where gu transports a Kummer class over L to its class over gL by the field isomorphism $g : L \to gL$. The parameter group has injective restriction maps, is divisible, and has torsion subgroup canonically μ_{3^∞}, with its usual Galois action. In particular, distinct torsion parameters and distinct cusp packets do not merge in the germ colimit.

Proof. The unit tangent gives a section of the decomposition extension at d. Over a number field L containing d, this extension is the semidirect product $\mathbb{Z}_3(1) \rtimes G_L$. Its continuous sections modulo inertia conjugacy are precisely $H^1(G_L, \mathbb{Z}_3(1))$. The decomposition homomorphism into $N \rtimes G_L$ and the continuous quotient $N \to A$ define the map in (C.10.2). Passing to a larger finite field restricts the same section, so the map is defined on (C.10.1). The fields containing d form a cofinal subsystem of the fields in that definition. Every cuspidal germ is, by definition, a section of one such decomposition extension, proving the image assertion once injectivity is checked.

First note that $H^0(G_L, \mathbb{Z}_3(1)) = 0$ for every finite number field L. If a one-cocycle becomes a coboundary over a finite extension, subtract that coboundary and enlarge to a finite Galois extension M/L so that the cocycle vanishes on G_M. For $g \in G_L$, its value is fixed by G_M, by applying the cocycle identity to $gG_Mg^{-1} = G_M$. The displayed invariant group is zero. The cocycle therefore vanishes on G_L, which proves injectivity of every restriction in (C.10.1).

Enlarge a common field of definition to contain all nine cusps. The boundary module is $W = \mathbb{Z}_3(1)[D]/\mathbb{Z}_3(1)\sum_{e\in D} e_e$. For a given d, the map $e_d : \mathbb{Z}_3(1) \to W$ is a split G_L-equivariant inclusion: omit any other cusp from the boundary basis. The map $H^1(G_L, W) \to H^1(G_L, V)$ is injective because $H^0(G_L, T_3E) = 0$, as proved in C.1. Thus $e_d : H^1(G_L, \mathbb{Z}_3(1)) \to H^1(G_L, V)$ is injective. The abelian image of the packet is exactly

$$\alpha_d + e_d H^1(G_L, \mathbb{Z}_3(1)), \tag{C.10.4}$$

where α_d is the Kummer class of the framed tangential divisor $d - O$. This follows from the same division-torsor comparison used in B.1, now over L. Consequently two parameters in one packet cannot become equal as abelian germs, and hence cannot become equal in $\mathscr{G}$.

Filling the boundary gives the continuous equivariant quotient $V \twoheadrightarrow T_3E$. It sends (C.10.4) to $\kappa_L(d - O)$. For $d \neq e$, the difference $d - e$ is a nonzero point of $E[3]$: the flex tangent relation is $3d \sim 3O$, and the nine distinct flexes exhaust $E[3]$. The integral Kummer class $\kappa_L(d - e)$ remains nonzero over every finite extension of L. Indeed the Mordell–Weil group of that extension is finitely generated. A nonzero three-primary torsion element in such a group is not divisible by 3^n for all n, so has nonzero image in its completion and in $H^1(T_3E)$ by the Kummer exact sequence. This argument uses the entire integral Kummer class; its modulo-three image alone can vanish after adjoining ninth division points. It proves that the nine packets remain pairwise disjoint in the colimit.

Our parameters and tangent value one are defined by the coordinate functions over $\mathbb{Q}$. Field conjugation therefore takes the chosen tangent at d to the chosen tangent at gd. Transporting the decomposition inclusion conjugates its geometric path, and geometric conjugacy is exactly the equivalence used in $\mathscr{G}$. Kummer functoriality proves (C.10.3), with no additional translation term.

For completeness, a finite-level Kummer class modulo three is killed by some finite extension: use the finite image of its affine cocycle representation. The long exact sequence of $0 \to \mathbb{Z}_3(1) \xrightarrow{3} \mathbb{Z}_3(1) \to \mu_3 \to 0$ then shows that the corresponding germ is divisible by three.

Multiplication by an integer prime to three is already invertible on each completed Kummer group. Thus $\mathscr{K}$ is divisible. At a fixed finite field, its completed Kummer group's torsion is precisely $\mu_{3^\infty}(L)$. To see this directly, use the divisor sequence from $L^\times$ to its principal fractional ideals. The latter group is free abelian and the sequence splits as abelian groups; the unit group is a finite roots-of-unity group plus a free abelian group by Dirichlet's theorem. The three-adic completion of a free abelian group is a subgroup of a product of copies of $\mathbb{Z}_3$, hence torsion-free. Only the three-primary roots of unity remain as torsion. Since all restrictions in (C.10.1) are injective, passing to the colimit introduces no further torsion and retains all roots of unity of three-power order. This proves the final assertions. □

Statement C.11 (Refuted). The pure-residue computation over the rationals does not extend verbatim to every number field. *Let*

$$L = \mathbb{Q}(\zeta_3, \sqrt{10}), \qquad S_L = \{v : v \mid 3\} \cup \{v : v \mid \infty\}.$$

For the actual rank-forty-four module $\mathcal{K}$,

$$H^2_{\mathrm{cts}}(G_{L,S_L}, \mathcal{K}) \neq 0. \tag{C.11.1}$$

Its inflation into $H^2_{\mathrm{cts}}(G_L, \mathcal{K})$ is injective, and every one of its classes has zero residue outside S_L. Consequently the pure-residue isomorphism (A.1.3) cannot be transported to this L by replacing the rational primes with its finite primes. The finite-S_L complex is a necessary part of Task C.1.

Proof. The polynomial $X^2 - 10$ has two simple roots modulo three and hence two roots in $\mathbb{Q}_3$ by Hensel's lemma. Therefore L has two places above three; each completion is $\mathbb{Q}_3(\zeta_3)$. The field has no real places. The Brauer invariant sequence gives

$$\mathrm{Br}(\mathcal{O}_L[1/3])[3] = \{(a, -a) : a \in \tfrac{1}{3}\mathbb{Z}/\mathbb{Z}\} \simeq \mathbb{F}_3. \tag{C.11.2}$$

Kummer theory and the arithmetic etale comparison surject $H^2(G_{L,S_L}, \mu_3)$ onto this group. These are the same comparison and invariant maps used in A.14: Milne, II.2.9, II.2.1 and Remark II.2.2(a) [Mil06]. Since $\mu_3 \subset L$, fixing $1 \mapsto \zeta_3$ identifies its coefficient with the trivial module $\mathbb{F}_3$. Thus $H^2(G_{L,S_L}, \mathbb{F}_3) \neq 0$.

The image on $\mathcal{K}/3$ is a finite three-group, because L contains $\mathbb{Q}(\zeta_3)$ and the action over that field has this property in A.1. A nonzero module for a finite three-group over $\mathbb{F}_3$ has a nonzero trivial quotient: its coinvariants are nonzero by nilpotence of the augmentation ideal. Choose a surjection $\mathcal{K}/3 \twoheadrightarrow \mathbb{F}_3$. The group G_{L,S_L} has three-cohomological dimension at most two. The long exact sequence therefore makes

$$H^2(G_{L,S_L}, \mathcal{K}/3) \longrightarrow H^2(G_{L,S_L}, \mathbb{F}_3)$$

surjective, so its source is nonzero.

The degree-two finite-coefficient groups are finite. Their inverse system is Mittag–Leffler, and the finite-coefficient groups in degree three vanish. The continuous-cohomology inverse-limit sequence then gives $H^3_{\mathrm{cts}}(G_{L,S_L}, \mathcal{K}) = 0$. Reduction modulo three in the long exact coefficient sequence yields a surjection

$$H^2_{\mathrm{cts}}(G_{L,S_L}, \mathcal{K})/3 \twoheadrightarrow H^2(G_{L,S_L}, \mathcal{K}/3).$$

This proves (C.11.1).

At every finite $v \nmid 3$ the eigenvalues on $\mathcal{K}(-1)$ have absolute values $\#k(v)$ or $\sqrt{\#k(v)}$, by the boundary/elliptic filtration. None is one, so $D_v = F_v - 1$ is injective on this lattice. The

localization triangle of F.1 over L thus has zero degree-zero residue group. Its long exact sequence injects the nonzero finite-S_L degree-two group into the global degree-two group, with image in the kernel of every displayed residue. This proves the precise failure of the proposed pure-residue replacement. □

Statement C.12 (Proved). The exact general number-field criterion with its retained arithmetic term. *For any number field L, put $S_L = \{v \mid 3, \infty\}$, $A_v = \mathcal{K}(-1)$ and $D_v = F_v - 1$ for $v \notin S_L$. Let $\mathcal{R}_L(A)$ denote the families in these integral lattices tending to zero three-adically. Then there are exact maps*

$$\begin{aligned} 0 \longrightarrow H^2_{\mathrm{cts}}(G_{L,S_L}, \mathcal{K}) &\xrightarrow{\inf} H^2_{\mathrm{cts}}(G_L, \mathcal{K}) \\ &\xrightarrow{\mathrm{res}} \mathcal{R}_L(A)/D\mathcal{R}_L(A) \longrightarrow 0. \end{aligned} \tag{C.12.1}$$

$$H^1_{\mathrm{cts}}(G_{L,S_L}, \mathcal{K}) \xrightarrow{\sim} H^1_{\mathrm{cts}}(G_L, \mathcal{K}). \tag{C.12.2}$$

The quotient in (C.12.1) uses the actual image of D. The first term is the inverse limit of the finite groups $H^2(G_{L,S_L}, \mathcal{K}/3^n)$.

For a section of N^{cc} over L, take its actual local path-origin residues r_v by the formula (A.8.3), with residue size $\#k(v)$ and the restricted G_L-action. Its intrinsic obstruction vanishes exactly when both of the following tests hold:

$$y_v = -D_v^{-1} r_v \in A_v \text{ for every } v \notin S_L, \qquad (y_v)_v \in \mathcal{R}_L(A); \tag{C.12.3}$$

$$\eta(s) = 0 \quad \text{in } H^2_{\mathrm{cts}}(G_{L,S_L}, \mathcal{K}). \tag{C.12.4}$$

Here, once (C.12.3) holds, $\eta(s)$ is the unique class whose inflation is the full continuous obstruction of s. Every nonempty class-two lift fiber is a torsor under $H^1_{\mathrm{cts}}(G_{L,S_L}, \mathcal{K})$, with no further geometric-stabilizer quotient. This statement retains the finite-S_L test; it does not claim an evaluation of that test for arbitrary L.

Proof. Apply the finite-coefficient localization triangle of F.1 and take its derived inverse limit. The residue complex is the actual two-term complex $[\mathcal{R}_L(A) \xrightarrow{D} \mathcal{R}_L(A)]$. Each D_v is injective by the weight calculation at the end of C.11, so its degree-zero cohomology is zero. The finite-S_L degree-one and degree-two finite-coefficient groups are finite, and hence their inverse systems are Mittag–Leffler. Three-cohomological dimension at most two gives $H^3_{\mathrm{cts}}(G_{L,S_L}, \mathcal{K}) = 0$ as in C.11. The indicated pieces of the localization long exact sequence are therefore precisely (C.12.1) and its degree-one isomorphism; the same finite-group Mittag–Leffler fact identifies the first term with its finite-coefficient inverse limit.

The local section equation used in A.8 is unchanged after restricting the arithmetic action to G_L: the inertia comparison cochains are the restrictions of the same based cusp paths, with the cusps now permuted by G_L. Applying that derivation over each completion L_v gives exactly r_v as the residue of the continuous obstruction class. Its residue class vanishes precisely under (C.12.3), because every D_v is injective and the sole integral correction is the displayed one. Exactness of (C.12.1) then gives the unique $\eta(s)$. Its vanishing is equivalent to vanishing of the original obstruction, proving necessity and sufficiency of the two tests.

Once a lift exists, two lifts of a fixed representative differ by a continuous $\mathcal{K}$-valued one-cocycle; central conjugacy changes it by a coboundary. A geometric stabilizer of the underlying N^{cc} section has fixed image in V, which is zero by C.1. Its remaining image is in $\mathbb{Z}_3(1)$ and is again zero. Thus no additional orbit identification occurs, and the fiber is the asserted full torsor. □

3.2 Algebraic evaluation of the point image

Statement C.16 (Proved). Finite algebraic formulas for every point's abelian and central coordinates. *Put $m = 3^n$. Two successive Kummer covers and a quotient of their finite deck group construct a based finite cover $X_n \to U$ with group*

$$P_n = N/(N^{3^n}\gamma_3 N). \tag{C.16.1}$$

Its multiplication, arithmetic action, transition maps, and fibers at algebraic points are computed by finite algebraic operations. The tangential base point is the specified $s = z/x$, tangent one.

Choose compatible lifts $y_1, \dots, y_{10}$ of a $\mathbb{Z}_3$-basis of V. For every $A \in U(L)$ the construction gives compatible finite cocycles

$$c_{A,n}(g) = (v_{A,n}(g), z_{A,n}(g)) \in (V/3^n) \oplus (\textstyle\bigwedge^2 V/3^n). \tag{C.16.2}$$

They are obtained by evaluating roots of rational functions at A and comparing the resulting fibers with the Puiseux fiber at O. Their inverse limit is a continuous cocycle whose class is the class-two point section of A. Thus (C.16.2) evaluates all forty-five central coordinates with their tangent origin.

Proof. We specify all the covers and comparison maps. The geometric constructions are over $\overline{\mathbb{Q}}$, and their finitely many coefficients lie in computable number fields.

For a smooth affine curve $Y = C \setminus B$, with C proper of genus g and B nonempty and finite, put

$$\mathcal{F}_m(Y) = \frac{\{f \in \overline{\mathbb{Q}}(C)^\times : \operatorname{ord}_x(f) \in m\mathbb{Z} \text{ for every } x \in Y\}}{\overline{\mathbb{Q}}(C)^{\times m}}. \tag{C.16.3}$$

Valuations give the exact sequence

$$0 \longrightarrow \operatorname{Jac}(C)[m] \longrightarrow \mathcal{F}_m(Y) \longrightarrow \ker\big((\mathbb{Z}/m)^B \xrightarrow{\sum} \mathbb{Z}/m\big) \longrightarrow 0. \tag{C.16.4}$$

The first map sends $[E]$ to the class of a function with divisor mE. An mth-power function makes E principal, proving injectivity. If every boundary valuation of f is also divisible by m, its full divisor is mE, proving exactness in the middle. For surjectivity, lift a boundary vector to a degree-zero divisor D on B. Choose E with $m[E] = [D]$ in the Jacobian and a function with divisor $D - mE$. Multiplication by m on the Jacobian is surjective over $\overline{\mathbb{Q}}$, so these choices exist. They give the required boundary valuations.

Choose a basis of $\operatorname{Jac}(C)[m]$ and the boundary vectors $b - b_0$. The preceding construction gives $r_Y = 2g + |B| - 1$ functions f_i forming a $\mathbb{Z}/m$-basis of $\mathcal{F}_m(Y)$. Normalize Y in

$$\overline{\mathbb{Q}}(C)(f_1^{1/m}, \dots, f_{r_Y}^{1/m}). \tag{C.16.5}$$

Their independence gives degree m^{r_Y}. At an interior point remove the m-divisible valuation from a Kummer equation; extracting a root of the remaining unit is etale in characteristic zero. Thus (C.16.5) is etale on Y. Conversely every cyclic etale cover of exponent dividing m comes from such a function, by function-field Kummer theory and this valuation test. This proves that (C.16.5) is exactly the maximal geometric abelian cover of exponent m.

All these are finite algebraic operations. Represent Jacobian classes by divisors, solve their finite m-division equations, and compute the principal functions by Riemann–Roch. The latter computation is given explicitly by Khuri-Makdisi's Theorem/Algorithm 5.5 [KM04]. Termination can also be certified without choosing a Jacobian implementation: enumerate

algebraic rational functions, test the valuation condition and their classes modulo mth powers, and retain independent functions until their generated subgroup has the known order m^{r_Y} from (C.16.4). Factoring $T^m - f$ over the algebraic function field decides the power tests. Equation (C.16.4) proves that this enumeration reaches the required order.

Apply (C.16.5) to U, and denote the result by $Y_m \to U$. Its intrinsic maximal pro-three group is $H_m = N^m[N, N]$. Indeed the normal closure over U of a finite three-group cover of Y_m again has three-group Galois group: intersect its finitely many conjugate subgroups, or use the associated three-group wreath product. This proves the comparison with the open subgroup of N. Apply (C.16.5) to Y_m, using its actual smooth compactification and full boundary. The composite cover of U has finite geometric group

$$E_m = N/(H_m^m[H_m, H_m]). \tag{C.16.6}$$

The denominator is characteristic in N, so this composite is geometrically Galois. Compute its finite deck group and take the fixed field of $E_m^m \gamma_3 E_m$; this defines X_n.

In $N/(N^m \gamma_3 N)$, the image of H_m is central and of exponent dividing m. Hence

$$H_m^m[H_m, H_m] \subset N^m \gamma_3 N. \tag{C.16.7}$$

The preimage of $E_m^m \gamma_3 E_m$ is therefore exactly $N^m \gamma_3 N$. The quotient maps induced by N give inverse isomorphisms $E_m/(E_m^m \gamma_3 E_m) \simeq N/(N^m \gamma_3 N)$. This proves the precise map in (C.16.1). Finite field automorphisms, fixed fields and the maps of their normalizations are computed by factoring finite algebraic equations.

Because m is odd, the Hall map

$$\begin{aligned} (V/m) \oplus (\textstyle\bigwedge^2 V/m) &\longrightarrow P_n, \\ (v, z) &\longmapsto \prod_i y_i^{v_i} \prod_{i<j} [y_i, y_j]^{\,z_{ij} - v_i v_j/2} \end{aligned} \tag{C.16.8}$$

is an explicit isomorphism, with the products in increasing order. Its source has multiplication

$$(v, z)(v', z') = (v + v', z + z' + \tfrac{1}{2} v \wedge v'). \tag{C.16.9}$$

Commuting successive generators verifies the map. Conversely this source is generated by the ten basis elements, has exponent m, and is of class two. The free-group map back is inverse on the generators. This proves both bijectivity and $|P_n| = m^{55}$. Abelian bases are read from the Kummer pairing with (C.16.3), and their lifts are found in the finite deck group. They can be chosen compatibly because $P_{n+1} \to P_n$ and the maps on their abelian kernels are surjective.

Embed the cover fields compatibly in the Puiseux field at O, using $s = z/x$ and fixed compatible symbols $s^{1/3^a}$. Galois acts on the algebraic coefficients and fixes these symbols. The characteristic subgroup in (C.16.1) is preserved. Equivalently there is a unique based isomorphism $\phi_{g,n} : {}^g X_n \to X_n$ carrying the conjugate chosen tangential fiber point to the chosen one. A deck transformation fixing a fiber point is the identity. Thus uniqueness gives $\phi_{gh,n} = \phi_{g,n}\, {}^g\phi_{h,n}$: these are actual arithmetic descent maps. They are computed from the finite function-field isomorphisms and the Puiseux branch condition. Compute coefficients until all finitely many branches are distinguished; distinct algebraic branches differ after finitely many terms. The coefficient action agrees with the path–Puiseux comparison of [HM19, §1.1, (1.1.1)–(1.1.3)], applied here to the actual finite covers of U.

For $A \in U(L)$ choose compatible points $p_{A,n}$ in $X_n(A)$ by the root equations (C.16.5) and the finite fixed-field maps. The functions may be chosen with valuation zero at every point above A: multiply them by mth powers with the required prescribed valuations, using

Riemann–Roch approximation. This leaves their classes in (C.16.3) unchanged. Alternatively, at a valuation mk, divide by a local parameter to the power mk and evaluate the resulting unit; its root is trivialized by that parameter to the power k. These are explicit isomorphisms of the Kummer equations. The based inclusions associated to $N^{3^{n+1}}\gamma_3 N \subset N^{3^n}\gamma_3 N$ supply the transition maps. Their maps on geometric fibers are surjective, allowing successive compatible choices.

Let deck transformations act on the right on fiber points, by composition with function-field automorphisms. Compute $c_{A,n}(g)$ from

$$\phi_{g,n}(g p_{A,n}) = p_{A,n}\, c_{A,n}(g). \tag{C.16.10}$$

All terms are algebraic points and maps already constructed. The descent identity gives

$$c_{A,n}(gh) = c_{A,n}(g)\rho_n(g)c_{A,n}(h). \tag{C.16.11}$$

For fixed n, the equations, deck maps, distinguishing Puiseux coefficients and fiber points lie in a finite Galois extension of L. Thus (C.16.10)–(C.16.11) are computed as functions on a finite Galois group. Formula (C.16.8) then extracts their coordinates.

Write the computed arithmetic action as

$$\rho_n(g)(v,z) = (R_g v, \textstyle\bigwedge^2 R_g\, z + \ell_g v). \tag{C.16.12}$$

Here ℓ_g is read from the ten images $\rho_n(g)y_i$ using (C.16.8). It is linear because the class-two addition $a+b = ab[a,b]^{-1/2}$ is recovered from the group law; automorphisms preserve this addition and the bracket. In particular the evaluated central cochain satisfies

$$\begin{aligned} z_{A,n}(gh) = z_{A,n}(g) + \textstyle\bigwedge^2 R_g\, z_{A,n}(h) + \ell_g(v_{A,n}(h)) \\ + \tfrac{1}{2} v_{A,n}(g) \wedge R_g v_{A,n}(h). \end{aligned} \tag{C.16.13}$$

This retains the off-diagonal arithmetic term as well as the tangential displacement.

Every finite quotient of $N/\gamma_3 N$ has finite three-power exponent, so it factors through some P_n. Therefore $N/\gamma_3 N \to \varprojlim_n P_n$ is an isomorphism: compactness proves surjectivity, and the kernel is tested in all finite quotients. The compatible finite cocycles thus give a continuous cocycle. The etale path comparison between the two fiber functors on each X_n is precisely (C.16.10); cofinality identifies its inverse limit with the point section. A different compatible fiber point changes the cocycle by the corresponding compatible geometric conjugation. This proves the asserted class. □

Statement C.17 (Proved). An evaluated nonzero central point coordinate. *Let $\alpha^3 = 2$, $\zeta = \zeta_3$, $\eta = \zeta_9$ with $\eta^3 = \zeta$, and $L = \mathbb{Q}(\zeta, \alpha)$. For $A = [1 : -\alpha : 1] \in U(L)$ an explicit quotient of P_1 of order 27 has zero abelian point coordinates and central coordinate*

$$-\kappa_L(\zeta) \in H^1(G_L, \mathbb{F}_3) \tag{C.17.1}$$

in the prescribed unit-tangent origin at O. This class is nonzero.

Proof. The functions $f = -x/z$ and $1 - f = (x+z)/z$ are units on U: the line $x + z = 0$ meets E only at Q_c, with multiplicity three. Pull back the cover

$$u^3 = f, \qquad v^3 = 1 - f, \qquad w^3 = A(u) := (1 - \zeta u)(1 - \zeta^2 u)^2. \tag{C.17.2}$$

It is the $l = 3, c = 1$ Heisenberg cover of [HM19, §2.1, (2.1.2), Theorem 2.1.4]. We verify its connected pullback and its deck maps. The valuations of $f, 1-f$ at P, O are $(1, 0)$ and

$(-1,-1)$, of determinant -1. Their Kummer classes are therefore independent and the first two equations have degree nine. Since

$$\frac{A(\zeta u)}{A(u)} = \frac{(1-u)^3}{1-u^3}, \tag{C.17.3}$$

the following are automorphisms:

	u	v	w
τ	ζu	v	$(1-u)w/v$
σ	u	ζv	w
z	u	v	$\zeta w.$

(C.17.4)

The identity $\prod_{j=0}^{2}(1-\zeta^j u) = 1-u^3$ gives $\tau^3 = 1$; also $\sigma^3 = z^3 = 1$ and $[\tau,\sigma] = z$. If $A(u) = b^3$ in the degree-nine field, its commuting automorphisms would satisfy $\tau(b)/b = \zeta^i(1-u)/v$ and $\sigma(b)/b = \zeta^j$ for some i, j. Their two orders of application would then differ by ζ, a contradiction. Thus the full degree is 27. Every zero and pole of these equations lies over a deleted cusp, so the normalization is etale on U. Its exponent-three class-two deck group is an actual quotient of P_1.

At A choose

$$u = -1, \qquad v = \alpha, \qquad w = -\eta; \tag{C.17.5}$$

indeed $A(-1) = -\zeta$. At O, put $r = s^{1/3}$. The branches are

$$u = -r^{-1}, \qquad v = r^{-1}(1+s)^{1/3}, \qquad w = \eta^2 r^{-1}(1+O(r)). \tag{C.17.6}$$

The last coefficient follows from $r^3 A(-r^{-1})|_{r=0} = \zeta^2$; the remaining unit series has coefficients in $\mathbb{Q}(\zeta)$. For $g \in G_L$ with $g\eta = \zeta^e \eta$, raw conjugation moves (C.17.5) by z^e and the tangent branch by z^{2e}. The based descent removes the latter displacement. Formula (C.16.10) therefore gives $z^{e-2e} = z^{-e}$. The raw arithmetic action commutes with (C.17.4), and its based correction is central, so the arithmetic action on the finite group over L is trivial. Geometric conjugation fixes this central cocycle.

Finally $X^3 - 2$ is Eisenstein at the prime over two in $\mathbb{Q}(\zeta)$. The distinct cyclic degree-three extension $\mathbb{Q}(\eta)/\mathbb{Q}(\zeta)$ is unramified there. Consequently $\eta \notin L$ and $\kappa_L(\zeta) \neq 0$. The exact polynomial identities, the twenty-seven deck elements and their multiplication, both fiber displacements, and the discriminants -108 and -3^9 are checked in `code/bc_point_coordinates.py` and its supplied output. □

Statement C.18 (Proved). The point-image set and its exact central membership criterion. *The point-image subset of the class-two germ set is*

$$\left\{ \left[\left(\varprojlim_n v_{A,n}, \varprojlim_n z_{A,n} \right) \right] : A \in U(\overline{\mathbb{Q}}) \right\}, \tag{C.18.1}$$

where each finite function is algebraically evaluated by (C.16.3)–(C.16.13). This parametrization is injective, disjoint from the nine packets (C.10.2), and equivariant for $A \mapsto gA$. Its stabilizer at A is $G_{\mathbb{Q}(A)}$.

Given a section and an algebraic point A, choose a common number field L of definition. Let d_n be the section cocycle in P_n and $c_{A,n}$ the evaluated cocycle above. Their germs agree exactly when every finite set

$$\mathcal{T}_n = \{h \in P_n : d_n(g) = h^{-1} c_{A,n}(g) \rho_n(g) h \text{ for all } g \in G_L\} \tag{C.18.2}$$

is nonempty. Each test is a computation on a finite Galois group. Combined with the injective abelian point map C.9, it gives a complete central membership criterion for (C.18.1).

Proof. C.16 proves (C.18.1), and C.9 proves injectivity, disjointness and the stated stabilizer. Conjugating the equations, divisors, roots and based comparison maps of C.16 carries the output at A to the output at gA, exactly as in (C.3), proving equivariance.

The equations in (C.18.2) factor through a finite Galois extension containing the fields of both finite cocycles and their actions. They are therefore finite tests. The inverse images of $\mathcal{T}_n$ in the compact group $N/\gamma_3 N$ are closed, nonempty and nested. Their intersection provides one conjugator valid at every precision, hence in $N/\gamma_3 N$. Conversely such a conjugator supplies every finite solution. Equality of germs is equality already over L by C.2. This proves (C.18.2) without assuming compatible choices of finite conjugators.

We spell out the transport to the other central coordinates. The Kummer pairing in (C.16.3) identifies P_n^{ab} with the actual $V/3^n$. For cusp vectors it is the valuation pairing; for proper vectors it is the torsion-divisor pairing in (C.16.4). Use the eight non-O cusp vectors and two lifts of an oriented Tate basis. Actual cusp generators are obtained by their Puiseux monodromy on the finite covers. Thus the basis comparison with (E36.4) takes place in the actual deck groups. For two compatible generator choices with the same abelian basis, write $y_i' = y_i \exp(t_i)$, with $t_i \in \bigwedge^2 V$. Substitution into (C.16.8) gives the exact map from coordinates in the primed generators to coordinates in the unprimed generators:

$$(v, z) \longmapsto (v, z + \sum_i v_i t_i). \tag{C.18.3}$$

An abelian basis change B gives B on v and $\bigwedge^2 B$ on central brackets, followed by the computed linear correction (C.18.3).

The based filling map q_O gives

$$q_O(v, z) = (\pi v, \omega z + \lambda(v)), \qquad \omega = \bigwedge^2 \pi, \tag{C.18.4}$$

where $\lambda(v_i)$ is read from the actual image $q_O(y_i)$ in the chosen basis of C. These images are computed by the finite maps of covers. One may take the two generators of C to be $q_O(y_1), q_O(y_2)$ and the other eight y_i to be peripheral; then $\lambda = 0$. The change to the involution-compatible basis of C.13 is (C.18.3), obtained by applying that actual involution to the generators. The orientation is $k = [a, b] = -\delta$, as in (E36.5), B.3 and A.8.

For the E36 parameter over a fixed abelian cocycle α, conjugate the computed cocycle by an actual geometric lift $\widetilde{h}$ of $h \in V$, using $c(g) \mapsto \widetilde{h}^{-1} c(g) \rho(g) \widetilde{h}$. Its abelian change is $v(g) \mapsto v(g) + (R_g - 1)h$. The vector giving the prescribed α is unique because $H^0(G_L, V) = 0$. Write the prescribed C-reference as $(\beta(g), w_\beta(g))$. After this explicit comparison the E36 parameter is the class of

$$\omega z_A(g) + \lambda(v_A(g)) - w_\beta(g) \quad \text{in the generator } k. \tag{C.18.5}$$

The two C-cocycles have identical abelian coordinate β; their group difference is precisely this central difference by (C.16.9). Over a prescribed class-two lift of the resulting N^{cc}-cocycle, the remaining fiber parameter is likewise the difference of the central cochains in $\mathcal{K} = \ker \omega$, after actual conjugation makes the N^{cc}-cocycles equal. This proves that the construction evaluates both the E36 parameter and the forty-four-dimensional fiber coordinate with their specified reference translations.

C.2–C.3 identify the germs fixed by G_L with Sec_L. In particular $\mathrm{Sec}_{\mathbb{Q}}$ is the entire $G_{\mathbb{Q}}$-fixed subset. Equations (C.10.2) and (C.18.1) supply both geometric image sets required in Task C.3. The criterion ranges over all n; it does not assert a bounded finite stopping rule for equality at all precisions. □

Statement C.6 (Proved). Complete class-two germ classification. *Task C.1 requests the full explicit residue complex, obstruction residues, convergence criterion, and fibers over*

every number field L. C.11 disproves a verbatim pure-residue extension of Part A; C.12 gives the exact replacement, including the finite-S_L arithmetic class and the lift torsor. C.13 evaluates the preceding cuspidally central obstruction by the local quadratic pairings at the places above three, with their global sum relation. C.14 computes the finite-S_L complexes by finite Galois cochain matrices and Artin–Verdier duality, and C.15 evaluates the retained class by explicit compact-support cup traces. Together with the restricted-family condition in C.12, these give the full number-field criterion and fibers requested in Task C.1. For Task C.2, C.1–C.3 give injective restriction, the actual Galois action and exact descent. Formula (C.5) identifies Sec_L *with all G_L-fixed germs. For Task C.3, C.8–C.9 prove injectivity of the point map and separation from cusp germs. C.10 gives the nine disjoint cusp parameter spaces with their torsion and Galois action. C.16 evaluates every point's abelian and central coordinates by finite algebraic operations at each precision; C.17 is an explicit nonzero central instance. C.18 gives the exact point-image set, its equivariance and its full central membership criterion. Its coordinate maps retain the E36 origin and the lift-fiber translation. Finally C.4–C.5 and C.7 prove that both specified phantom classes remain nonzero after every finite number-field extension and in the germ colimit, settling Task C.4.*

Proof. The cited statements provide the specified maps, arithmetic computations and geometric image sets. Their compatibility under finite extension is the cochain restriction of C.2 and the based fiber construction of C.16; hence they give one classification over the filtered system of number fields, as asserted. □

3.3 Separating the point germs by the nine tangent lines

The following calculation gives additional geometric information about Task C.3 directly in the abelian quotient. We work with the actual maps on fundamental groups induced by regular invertible functions on U. Their coefficient field is $K_0 = \mathbb{Q}(\zeta)$; enlarging a field of definition to contain K_0 is a restriction map of section sets, hence is permitted in the germ colimit.

Statement C.8 (Proved). An exact tangent-line separation identity. *Let ℓ_d be a row vector defining the flex tangent line at $d \in D$. The nine rows can be chosen as*

$$(1, \zeta^a, 0), \quad (0, 1, \zeta^a), \quad (\zeta^a, 0, 1) \qquad (a = 0, 1, 2).$$

Set $P_d = \ell_d^* \ell_d / 2$, a rank-one Hermitian projector on $\mathbb{C}^3$. The real-linear map

$$\mathrm{Herm}_3(\mathbb{C}) \longrightarrow \mathbb{R}^9, \qquad H \longmapsto \big(\mathrm{tr}(P_d H)\big)_d$$

is an isomorphism, and

$$\sum_d P_d = 3I_3, \qquad \mathrm{tr}(P_d P_e) = \begin{cases} 1 & d = e, \\ 1/4 & d \neq e. \end{cases} \tag{C.8.1}$$

The kernel after deleting coordinate d is the line $\mathbb{R}(4P_d - I_3)$. A positive semidefinite matrix with the eight undeleted measurements of vv^*, where $v \neq 0$ and $\ell_d(v) = 0$, is exactly vv^*.

Proof. Each row has norm squared two. The scalar product of two distinct rows has absolute value one: for rows with the same two nonzero positions this follows from $|1+\zeta| = |1+\zeta^2| = 1$; for different positions their supports meet in one coordinate. Summing the projectors cancels the off-diagonal terms and gives three on each diagonal entry. This proves (C.8.1). Their

Gram matrix is $(3I_9 + J_9)/4$, with eigenvalues 3 and 3/4, so the nine projectors are a real basis of the nine-dimensional Hermitian space. Moreover

$$\operatorname{tr}\bigl(P_e(4P_d - I_3)\bigr) = 3\delta_{de},$$

which proves the assertion about the deleted coordinate.

Write a matrix with the prescribed eight measurements as $vv^* + t(4P_d - I_3)$, $t \in \mathbb{R}$. The vector spanning $\operatorname{im} P_d$ is orthogonal to v, and its quadratic value is $3t$ times its norm squared. A nonzero vector orthogonal to both this line and v has quadratic value $-t$ times its norm squared. Positive semidefiniteness therefore gives both $t \geq 0$ and $t \leq 0$. Thus $t = 0$. The exact program `code/c_point_separation.py` verifies the rank, all 81 dual pairings, all nine tangent annihilations, and the eigenvalues $(3, -1, -1)$ over $\mathbb{Q}(\sqrt{-3})$. □

Statement C.13 (Proved). The initial cuspidally central obstruction over every number field. *Let $L \subset \overline{\mathbb{Q}}$ be a number field and let $\alpha \in H^1(G_L, V)$, with image $\beta \in H^1(G_L, T)$. Orient $\bigwedge^2 T \simeq \mathbb{Z}_3(1)$ by the commutator in the once-punctured elliptic curve, and write ω for this alternating pairing. For each $v \mid 3$ define*

$$q_v(\beta) = \tfrac{1}{2} \operatorname{inv}_v \omega_*(\beta_v \smile \beta_v) \in \mathbb{Z}_3. \tag{C.13.1}$$

Here $\operatorname{inv}_v : H^2_{\mathrm{cts}}(G_{L_v}, \mathbb{Z}_3(1)) \xrightarrow{\sim} \mathbb{Z}_3$ is the inverse limit of the normalized local Brauer invariant maps. Then

$$\alpha \in \operatorname{im}\bigl(\operatorname{Sec}_L(N^{\mathrm{cc}}) \to H^1(G_L, V)\bigr) \quad \Longleftrightarrow \quad q_v(\beta) = 0 \text{ for every } v \mid 3. \tag{C.13.2}$$

The tuple satisfies $\sum_{v|3} q_v(\beta) = 0$. Every nonempty fiber is a torsor under $H^1(G_L, \mathbb{Z}_3(1)) = \varprojlim_n L^\times/(L^\times)^{3^n}$, with no further stabilizer quotient. In particular, every abelian section lifts to N^{cc} over a number field having exactly one place above three, including $\mathbb{Q}$. This statement makes no assertion that every tuple subject to the displayed sum occurs as a self-pairing.

Manuscript comparison: §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379).

Proof. We first specify the genuine arithmetic action on the once-punctured elliptic class-two quotient C. The automorphism $[x : y : z] \mapsto [y : x : z]$ is elliptic inversion for origin O. It preserves $E \setminus \{O\}$ and has derivative -1 at O. Its comparison from the tangent -1 to the tangent 1 admits a $G_{\mathbb{Q}}$-fixed pro-three path: the compatible 3^nth roots of -1 can all be chosen to be -1. Composition with this path gives a genuine involution j of the based pro-three group, commuting with the arithmetic action. Applying the comparison twice gives the constant path, as the product of the two chosen roots is one at every level.

The integral class-two logarithm is available at three: its group law is the Baker–Campbell–Hausdorff law with the single denominator two. The central exact sequence of Lie lattices is

$$0 \longrightarrow \bigwedge^2 T \longrightarrow \log C \longrightarrow T \longrightarrow 0.$$

The involution acts as -1 on T and as 1 on the center. The operator $(1 - j)/2$ therefore gives a canonical lift of T to its negative eigenspace. An arithmetic automorphism commutes with j, so it preserves this lift. In these coordinates

$$(t, z)(t', z') = (t + t', z + z' + \tfrac{1}{2}\omega(t, t')), \qquad g(t, z) = (gt, \chi_3(g)z). \tag{C.13.3}$$

This proves the claimed diagonalization by an actual based morphism. In particular the central extension obstruction for a T-cocycle β is represented by the continuous two-cocycle

$$(g, h) \longmapsto \tfrac{1}{2}\omega(\beta(g), g\beta(h)). \tag{C.13.4}$$

At a finite place $v \nmid 3$, the curve has good reduction, so T is unramified. The two Frobenius eigenvalues have complex absolute value $\sqrt{Q_v}$. Thus both $T^{F_v=1}$ and $T(-1)^{F_v=1}$ vanish. The local inertia sequence gives

$$H^1(G_{L_v}, T) = T/(F_v - 1)T,$$

a finite group. This equality also follows at the integral level by the finite-coefficient inverse limit; the invariant towers are finite and have zero first derived limit. The self-pairing of any element of a finite group is torsion, whereas $H^2_{\mathrm{cts}}(G_{L_v}, \mathbb{Z}_3(1)) = \mathbb{Z}_3$ is torsion-free. Consequently (C.13.4) restricts to zero at every such place. Its restrictions at archimedean places vanish because three is odd.

For clarity, the required global injectivity retains the topology. At each finite level the Kummer sequence identifies $H^2(G_L, \mu_{3^n})$ with $\mathrm{Br}(L)[3^n]$. The global Brauer invariant exact sequence, followed by inverse limit, identifies its limit with

$$\left\{ (b_v)_v \in \mathcal{R}_{v\ mfinite}(\mathbb{Z}_3) : \sum_v b_v = 0 \right\}. \tag{C.13.5}$$

The sum converges because the family is restricted. Conversely, reduction of such a tuple modulo 3^n has finite support and zero sum, and hence gives the unique global Brauer class at that level; these classes are compatible. Statement F.5 gives $\varprojlim{}^1 H^1(G_L, \mu_{3^n}) = 0$, so Jannsen's sequence identifies the continuous integral second cohomology itself with (C.13.5). Equivalently one can apply the same Kummer inverse-limit argument to the completed multiplicative group. The local and global Brauer invariant theorems are used in their stated normalizations [Mil20b].

The only possibly nonzero entries of (C.13.4) are therefore precisely (C.13.1). Formula (C.13.5) proves both the sum relation and the vanishing equivalence. The arithmetic fiber product $N^{\mathrm{cc}} = C \times_T V$ transports this equivalence to (C.13.2). The central lifting fiber is acted on transitively by $H^1(G_L, \mathbb{Z}_3(1))$. Its potential stabilizer is the image of $H^0(G_L, V)$, which is zero: a good Frobenius outside three has no eigenvalue one on $V \otimes \mathbb{Q}_3$, since its boundary and elliptic eigenvalues have absolute values Q_v and $\sqrt{Q_v}$. As V is torsion-free, the integral invariants vanish as well. This proves the assertion about fibers. A tuple supported at one place and having sum zero is zero, which proves the last claim. □

Statement C.9 (Proved). Injective point germs and their separation from cusp germs. *The canonical map*

$$U(\overline{\mathbb{Q}}) \longrightarrow \varinjlim_{L} \mathrm{Sec}_L(N/\gamma_3 N)$$

is injective and $G_{\mathbb{Q}}$-equivariant. Its image is disjoint from all the cuspidal germs. Both assertions already hold after projecting to the germ colimit of $H^1(G_L, V)$. The smallest field of definition of a point germ is the field of definition of the point.

Proof. For distinct cusps d, e, the regular invertible function $f_{d,e} = \ell_d/\ell_e$ on U has divisor $3d - 3e$ on E. Its morphism $U \to \mathbb{G}_m$ gives a continuous map of geometric pro-three groups $N \to \mathbb{Z}_3(1)$, factoring through V. On the difference of the section classes of two points $A, B \in U(L)$, the induced cohomology map is exactly

$$\kappa_L(f_{d,e}(A)/f_{d,e}(B)). \tag{C.9.1}$$

This follows by pulling back the covers $z^{3^n} = t$ on $\mathbb{G}_m$; their fibers are the compatible division roots used to define the Kummer cocycle. Thus the comparison is a map of actual section torsors, with the tangential origin cancelling in the difference.

The kernel of $L^\times \to \varprojlim_n L^\times/L^{\times 3^n}$ is its group of roots of unity of order prime to three. Indeed every finite valuation of an infinitely three-divisible element is zero. Dirichlet's unit theorem then reduces the assertion to the finite roots-of-unity group. In particular every complex absolute value of a kernel element is one.

Equality of two abelian germs is equality after restriction to one finite number field containing A, B and K_0. By (C.9.1) the ratios of all these function values have absolute value one. Fix a complex embedding and normalize vector representatives a, b of A, B by $\ell_e(a) = \ell_e(b) = 1$. This is possible because the tangent at a flex meets E only at that cusp and $A, B \in U$. We obtain $|\ell_d(a)|^2 = |\ell_d(b)|^2$ for all nine d. C.8 gives $aa^* = bb^*$, hence the same projective point $A = B$.

Now compare a point $A \in U$ with a section in the packet of a cusp d. Choose $e \neq d$. For every $j \neq d$, the function $f_{j,e}$ has order zero at d and is a unit in its local ring. The morphism on the tangential packet is consequently the constant Kummer class of $f_{j,e}(d)$: extracting roots of a local unit gives that constant fiber, and the varying tangent parameter has exponent zero. Equality of abelian germs would therefore give, after normalization at ℓ_e,

$$|\ell_j(a)|^2 = |\ell_j(d)|^2 \quad (j \neq d).$$

The cusp vector satisfies $\ell_d(d) = 0$. The final assertion of C.8 applied to the positive matrix aa^* forces $A = d$, contradicting $A \in U$. This proves disjointness for every algebraic cusp and every completed Kummer parameter.

Equivariance follows from functoriality of the point section and restriction. Injectivity identifies the stabilizer of a point germ with the stabilizer of the point, which proves the assertion about its smallest field of definition. □

3.4 Finite arithmetic matrices over an arbitrary number field

Statement C.14 (Proved). An explicit finite Galois construction of the retained arithmetic groups. *Fix a number field $L \subset \overline{\mathbb{Q}}$ and put $S = \{v : v \mid 3\} \cup \{v : v \mid \infty\}$. For $n \geq 1$ write*

$$A_n = \mathcal{K}/3^n, \qquad A_n^D = \mathrm{Hom}(A_n, \mu_{3^n}), \qquad F_n = L(J_{E,D}[3^n], \mu_{3^n}), \qquad d_n = 3^{44n}[F_n : L]. \tag{C.14.1}$$

The homomorphisms in the dual have their usual contragredient Galois action. Let Ω_n be the compositum, inside the fixed algebraic closure, of all finite Galois extensions E/L containing F_n, unramified outside S, and satisfying $[E : L] \leq d_n$. Put $G_n = \mathrm{Gal}(\Omega_n/L)$. At each $v \mid 3$ choose a prolongation to Ω_n and write $D_{n,v} \leq G_n$ for its decomposition subgroup. These are finite fields and finite groups, obtained by the bounded enumeration in the proof.

For a finite G_n-module A, use the finite matrices

$$\begin{aligned} (\delta^0 b)(g) &= gb - b, \\ (\delta^1 z)(g,h) &= gz(h) - z(gh) + z(g). \end{aligned} \tag{C.14.2}$$

Define the following finite abelian groups by kernels and images of these matrices:

$$U_n = \ker\bigl(\delta^1 : \mathrm{Map}(G_n, A_n) \to \mathrm{Map}(G_n^2, A_n)\bigr)/\delta^0 A_n, \tag{C.14.3}$$

and

$$\begin{aligned} C_n = \{(z, (b_v)_{v\mid 3}) : \delta^1 z = 0,\ b_v \in A_n^D, \\ z(g) = gb_v - b_v \text{ for } g \in D_{n,v}\}/\{(\delta^0 b, (b)_{v\mid 3}) : b \in A_n^D\}. \end{aligned} \tag{C.14.4}$$

In (C.14.4), z ranges over all functions $G_n \to A_n^D$. There are canonical identifications

$$\begin{aligned} H^1(G_{L,S}, A_n) &\simeq U_n, \qquad & H^2(G_{L,S}, A_n) &\simeq C_n^\vee, \\ H^1_{\mathrm{cts}}(G_{L,S}, \mathcal{K}) &\simeq \varprojlim_n U_n, \qquad & H^2_{\mathrm{cts}}(G_{L,S}, \mathcal{K}) &\simeq \varprojlim_n C_n^\vee. \end{aligned} \tag{C.14.5}$$

Here $C_n^\vee = \mathrm{Hom}(C_n, \mathbb{Q}/\mathbb{Z})$; the transition maps are induced by coefficient reduction and its Cartier dual. Thus the arithmetic terms of C.12 are specified by finite arithmetic fields and finite matrices at every precision. The construction applies to every L; it does not assume that either displayed inverse limit is zero.

Proof. The reduction A_n is a submodule of $\bigwedge^2(V/3^n)$: the quotient $\bigwedge^2 T$ in the definition of $\mathcal{K}$ is torsion-free. The finite group scheme $J_{E,D}[3^n]$ therefore splits its action. Adjoining μ_{3^n} also splits the Cartier dual. Both coefficients are unramified outside S, because the original generalized Jacobian and its three-adic representation are unramified outside three. All the fields in (C.14.1) are algebraic data of the specified curve. For example, multiplication by 3^n on its framed divisor classes constructs $J_{E,D}[3^n]$; Riemann–Roch and factorization give its finite defining equations and their splitting field. No presentation of an unknown infinite Galois group is an input to the construction.

We give a bound for the field enumeration. For any integer $d \geq 1$, every extension E/L of degree at most d unramified outside S has

$$|\operatorname{disc}(E)| \leq B_L(d) := |\operatorname{disc}(L)|^d \prod_{v \mid 3} (Nv)^{d(1+e_v \lfloor \log_3 d \rfloor)}, \tag{C.14.6}$$

where e_v is the ramification index of $L_v/\mathbb{Q}_3$. Indeed, pass first to the maximal unramified subextension of a completion E_w/L_v. Its remaining totally ramified part has ramification index e and an Eisenstein polynomial P for a uniformizer π. The terms of $P'(\pi)$ have pairwise distinct valuations modulo e. Its valuation is therefore their minimum and is at most $e - 1 + ee_v v_3(e)$. Multiplying by the residue degree and summing over $w \mid v$ bounds the relative discriminant exponent by $d(1 + e_v \lfloor \log_3 d \rfloor)$. The tower formula for the discriminant proves (C.14.6).

Hermite's proof, in the explicit form of Milne, Theorems 8.42–8.43 [Mil20a], bounds the conjugates of a primitive integral element in terms of the absolute discriminant. Consequently one can list a finite collection of monic integral polynomials of degree at most $d[L : \mathbb{Q}]$ that contains a defining polynomial for every E in question. For a deliberately generous choice, a bound $2(B_L(d) + 1)$ for every conjugate suffices in that proof; the coefficient of degree $r - i$ is then bounded by $\binom{r}{i}[2(B_L(d) + 1)]^i$. Enumerate these polynomials, factor them, calculate rings of integers and discriminants, and retain precisely their embeddings over L that are Galois, contain F_n, have the required degree, and are unramified away from S. All these tests are finite exact number-field operations. Taking their compositum constructs Ω_n. The bound is used to prove termination; no feasibility estimate is asserted.

Let A be either A_n or A_n^D, and let $\rho : G_{L,S} \to \mathrm{Gal}(F_n/L)$ be the quotient. Every continuous crossed homomorphism $z : G_{L,S} \to A$ gives the continuous homomorphism

$$g \longmapsto (z(g), \rho(g)) : G_{L,S} \longrightarrow A \rtimes \mathrm{Gal}(F_n/L). \tag{C.14.7}$$

Its kernel cuts out a Galois extension containing F_n of degree at most $|A|[F_n : L] = d_n$, unramified outside S. It is one of the fields used in Ω_n. Thus every crossed homomorphism factors through G_n. Conversely every crossed homomorphism on G_n inflates to one on $G_{L,S}$, and the coboundaries on the two groups are exactly the same functions $g \mapsto gb - b$. This proves (C.14.3) and its analogue for A_n^D.

Use the compact-support complex

$$R\Gamma_c(\mathcal{O}_L[1/3], A_n^D) = \mathrm{Cone}\left(R\Gamma(G_{L,S}, A_n^D) \longrightarrow \bigoplus_{v\in S} R\Gamma(L_v, A_n^D) \right)[-1]. \tag{C.14.8}$$

The local complexes at infinite places here are the modified Tate complexes. They are acyclic for these odd-primary coefficients, including at a real place. Degree-one cycles in (C.14.8) are exactly a global one-cocycle together with the indicated local degree-zero trivializations. A change by a global zero-cochain is exactly the denominator of (C.14.4). Since the global cocycle factors through G_n, each local equation holds on G_{L_v} exactly when it holds on its image $D_{n,v}$. We have proved the explicit identification of C_n with that compact-support group.

Milne's Corollary II.3.3(a),(b) [Mil06], applied to the locally constant sheaf A_n^D, identifies its derived Cartier dual with A_n and gives the perfect pairing

$$H^2(G_{L,S}, A_n) \times C_n \longrightarrow \mathbb{Q}/\mathbb{Z}. \tag{C.14.9}$$

The Galois-to-etale comparison is the same II.2.9 used in C.11. This proves the second finite identification. Every finite group in degrees one and two is finite, so each coefficient tower is Mittag–Leffler. The continuous cohomology inverse-limit sequences therefore have zero first derived-limit term at finite S and give the last two identifications in (C.14.5). Coefficient naturality of (C.14.7)–(C.14.9) specifies all transition maps.

For a small arithmetic verification of the matrix convention, the program `c_arithmetic_matrices.py` in `code/` constructs the actual group of $\mathbb{Q}(\zeta_9, \sqrt[3]{3})/\mathbb{Q}$, of order 18. Over K_0 the two Kummer classes $\zeta, 3$ are independent and span the S-unit classes: $3 = -\zeta^2\lambda^2$. Thus this quotient captures all one-cocycles for $\mathbb{F}_3$ and μ_3 over $\mathbb{Q}$, including their coboundaries. Its decomposition group at three is the whole group. Indeed a Kummer line involving 3 has valuation nonzero modulo three over $K_{0,3}$; the remaining line ζ gives the ramified cyclotomic cubic extension. Hence no nontrivial cubic subextension is unramified there. In coordinates the group law is $(a,b,c)(a',b',c') = (a+a', b+(-1)^c b', c+c')$, with the first two coordinates modulo three and the last modulo two. The program verifies this law and both coefficient actions. The respective cocycle matrix ranks are 17 and 16, the coboundary ranks are 0 and 1, and both first-cohomology dimensions are one. Both compact-support dimensions are zero. A separately labelled duplicated-subgroup fixture checks the local-trivialization quotient; it is not assigned to a number field. These are checks of the finite matrices, with their complete output retained; the proof above establishes their general arithmetic use. □

Statement C.15 (Proved). The complete number-field class-two test with its finite arithmetic evaluations. *Let s be a continuous section of N^{cc} over L. Use the actual obstruction two-cocycle o and its origin-corrected residue family $r = (r_v)_{v\nmid 3,\infty}$ from A.8, restricted to G_L. Write $D_v = \mathrm{Frob}_v - 1$ on $\mathcal{K}(-1)$. First require*

$$y_v = -D_v^{-1} r_v \in \mathcal{K}(-1), \qquad (y_v)_v \in \mathcal{R}_L(\mathcal{K}(-1)). \tag{C.15.1}$$

The actual localization homotopy-fibre cycle (o, y) then supplies, at each n, a two-cocycle $\xi_n \in Z^2(G_{L,S}, A_n)$, by the comparison below. For every class represented by $(z, (b_v))$ in the explicit finite group C_n of C.14, choose a continuous two-cochain a with values in μ_{3^n} satisfying

$$da = \xi_n \cup z. \tag{C.15.2}$$

Such a cochain always exists. Define

$$\ell_n(s)(z,(b_v)) = \sum_{v|3} \mathrm{inv}_v\big((\mathrm{res}_v\,\xi_n)\cup b_v - \mathrm{res}_v\,a\big). \tag{C.15.3}$$

The expression in each local invariant is a two-cocycle. Formula (C.15.3) is independent of all the auxiliary cochain choices and is an element of $C_n^\vee$. There is an exact criterion

$$s \text{ lifts to } N/\gamma_3 N \iff \text{(C.15.1) holds and } \ell_n(s)=0 \text{ for every } n\ge 1. \tag{C.15.4}$$

Every nonempty fibre is a torsor under $\varprojlim_n U_n$, with U_n given by the finite matrices (C.14.3), and no further geometric-stabilizer quotient. Together with C.13 for the initial lift, C.14–C.15 give the number-field version requested in Task C.1: both the residue convergence and the arithmetic test are retained.

Proof. We specify the comparison that enters (C.15.2). Put $\mathcal{A}_v = \mathcal{K}(-1)$ and $R^\bullet = [\mathcal{R}_L(\mathcal{A}) \xrightarrow{D} \mathcal{R}_L(\mathcal{A})]$ in degrees zero and one. The localization triangle of F.1 provides a continuous cochain model and a quasi-isomorphism

$$R\Gamma(G_{L,S},\mathcal{K}) \longrightarrow \mathrm{Cone}(R\Gamma(G_L,\mathcal{K}) \longrightarrow R^\bullet[-1])\,[-1]. \tag{C.15.5}$$

Choose the cone sign so that a degree-two cycle is (o,y) with $do=0$ and $r+Dy=0$. This is the sign in (C.15.1). All restrictions, residue maps, and homotopies are the actual maps in the localization construction; in particular y is retained as part of the cycle. Reduction modulo 3^n in (C.15.5) gives its finite coefficient comparison. A representative ξ_n is any finite-coefficient two-cocycle on the left whose image is cohomologous to the reduction of (o,y) on the right. It exists because (C.15.5) is a quasi-isomorphism. Its cohomology class is uniquely determined. This prescription keeps the arithmetic class in the homotopy fibre, including when the finite-coefficient map from that class to global cohomology has a kernel.

There is a finite-data method for obtaining such representatives at any specified n. Continuous finite-valued cochains factor through finite Galois quotients. Enumerate finite S-ramified quotients and their normalized bar two-cocycles. For the comparison homotopies, enumerate finite global quotients with ramification allowed in a finite set T containing the ramification of o_n and the support of y_n; enumerate larger finite sets T as well. Use the tame resolution at the finite residue support. Thus only the arithmetic representative is required to factor through an S-ramified quotient; the global comparison homotopy retains its necessary ramification. Test the cochain identities in the finite multiplication tables. For the prescribed cycle a matching representative and homotopy occur, so this search terminates. The inverse in (C.15.5) is thereby a cochain comparison, not a choice of a splitting of the short exact sequence (C.12.1). At a finite coefficient level the residue support is finite: this is precisely where the restricted condition in (C.15.1) is used. As with C.14, this is an exact construction and no claim of practical running time is made.

The group $G_{L,S}$ has three-cohomological dimension at most two. Therefore $H^3(G_{L,S},\mu_{3^n})=0$, which proves existence in (C.15.2). Its right side is the standard evaluation cup product for $A_n\otimes A_n^D\to\mu_{3^n}$. Since $db_v=\mathrm{res}_v\,z$ and $d\xi_n=0$,

$$d\big((\mathrm{res}_v\,\xi_n)\cup b_v - \mathrm{res}_v\,a\big)=0.$$

We can obtain a by enumerating finite S-ramified quotients refining those through which ξ_n, z factor and solving the finite linear cochain equation. The vanishing just proved guarantees termination of this search as well.

In the compact-support cone (C.14.8) the product of ξ_n and $(z,(b_v))$ is represented by

$$\big(\xi_n\cup z,\,((\mathrm{res}_v\,\xi_n)\cup b_v)_v\big).$$

Subtracting the coboundary of $(a, 0)$ gives $(0, ((\mathrm{res}_v\, \xi_n) \cup b_v - \mathrm{res}_v\, a)_v)$. Normalize the compact-support trace by the sum of the local Brauer invariants on such a representative. This is the trace compatibility in Milne II.3, preceding Theorem 3.1 [Mil06]. The resulting trace is exactly (C.15.3). Each local invariant is that of a finite crossed-product algebra, so it is determined by finite extension, norm, and valuation data over L_v. The construction proves directly that (C.15.3) is the perfect pairing (C.14.9), applied to the class of ξ_n. It consequently descends to C_n and is independent of the representatives and of a. In particular, changing a by a global two-cocycle changes the sum by zero, by the global Brauer invariant relation.

At the integral level, injectivity of each D_v proves that (C.15.1) is exactly the vanishing of the residue obstruction, with its unique integral correction. Under that condition the class of (o, y) is the unique arithmetic class of C.12. By (C.14.5), this class vanishes exactly when all its finite components pair to zero with C_n. Those components are exactly (C.15.3). This proves both directions of (C.15.4). The finite-S first-cohomology inverse-limit identification and the stabilizer calculation of C.12 give the stated torsor and its precise conjugacy quotient. The global Milnor term is still the one computed in F.1–F.3; the additional finite arithmetic evaluations introduce no new first derived limit. Thus the criterion includes infinitely ramified classes, all torsion classes, every prime above three, and the modified real-place convention. □

4 D. Profinite leading terms

4.1 Sources, comparison maps, and the two remaining image problems

We use the generators and positive tangential normalization of the task. The source question is LM7 D.13, developing the comparison of groups in *La Longue Marche*, §51.15 (cote 140-4, scans 156–170). Only the supplied transcription, not the underlying scans, was available for this comparison. We make no assertion of a new reading of those scans.

Put $F = \langle x, y\rangle$, let $\widehat{F}$ be its profinite completion, and write F_ℓ for its maximal pro-ℓ quotient. The quotient $q_\ell : \widehat{F} \twoheadrightarrow F_\ell$ has characteristic kernel. It therefore induces a continuous homomorphism on outer automorphism groups. Let I^n denote the filtration obtained by acting trivially, up to inner automorphism, on $F_\ell/\Gamma_{n+1}F_\ell$. We filter each full profinite source by inverse images of I^n. The difference of an automorphism and the identity, read in $\Gamma_{n+1}/\Gamma_{n+2}$, is a derivation. Composition becomes addition in this quotient, and conjugating the automorphism changes that derivation by an inner derivation. Thus we obtain the specified maps

$$(I^nG/I^{n+1}G) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell \longrightarrow \mathrm{OutDer}_{\mathrm{sp}}(L_\ell)_n \longrightarrow \mathrm{OutDer}(L_\ell/L_\ell'')_n, \qquad L_\ell = \mathrm{Lie}_{\mathbb{Q}_\ell}(x, y).$$

Here G is one of the full profinite source groups, and the first tensor product means the rational span of its image in the integral graded automorphism module. Expanding each peripheral conjugacy equation gives the special-derivation equation. Expanding the commuting equations with the geometric S_3 gives invariance. These are necessary conditions on the images of the displayed maps.

Statement D.1 (Open). The weight-five image of the full profinite special group. *For $\ell \geq 5$, the dimension requested in Task D.1 has not been determined here. The target computed in R8 is*

$$(X + Y)\,\langle X^2, XY, Y^2\rangle_{\mathbb{Q}_\ell}[x, y].$$

Statement D.21 proves that this image has dimension either 1 *or* 3*: the actual geometric S_3 preserves it, and its invariant gamma line is already present by D.20. The remaining choice between* 1 *and* 3 *is not determined here.*

Statement D.2 (Open). The extra weight-nine direction for the full profinite gentle group. *For $\ell \geq 5$, we have neither constructed an element of $\widehat{\mathrm{GT}}_0$ with leading metabelian coordinate outside $\mathbb{Q}_\ell G_9$, nor proved that every such coordinate belongs to that line, where*

$$G_n = \frac{(X+Y)^n - X^n - Y^n}{XY}, \qquad p_{\mathrm{extra}} = X^2Y^2(X+Y)^3.$$

The requested choice between dimensions 1 and 2 remains open in this return.

Statement D.3 (Proved). The finite pentagon comparison in the weight-nine stratum. *At the compatible quotients $PB_n/(\gamma_{11}PB_n\, PB_n^{101})$, for $n = 3, 4$, the complete multiplier-one, weight-at-least-nine gentle shadow stratum has 101^{37} elements. Exactly 101^2 of these satisfy the complete pentagon. Their degree-nine leading spaces have dimensions four and one, respectively; their metabelian images have dimensions two and one. The latter image is the line generated by G_9. Statements D.7–D.8 give the finite comparison and an exact integer certificate. Its characteristic-zero comparison, together with the actual Galois lower bound in D.20, derives R9 in weight nine. The enumeration is of the entire stratum that can contribute a weight-nine leading term, with its degree-ten corrections; it is not an enumeration of other multipliers or of lower leading weights.*

4.2 What the linear and literature checks establish

Statement D.4 (Proved). Recomputed targets and the checked pentagon upper bound. *The original rational program reproduces R8's four rows in weights $1, \ldots, 10$. In weight nine its invariant metabelian image has basis $p_0 = p_{\mathrm{extra}}$ and*

$$p_1 = 2X^7 + 8X^6Y + 17X^5Y^2 + 23X^4Y^3 + 23X^3Y^4 + 17X^2Y^5 + 8XY^6 + 2Y^7, \qquad G_9 = \frac{15}{2}p_0 + \frac{9}{2}p_1.$$

The leading metabelian image of full $\widehat{\mathrm{GT}}$ is contained in $\mathbb{Q}_\ell G_n$ in positive odd weight n, and is zero in positive even weight. In particular p_{extra} is excluded in weight nine by this implication from the pentagon.

Proof. We reran the supplied programs, retaining their provenance under `code/de_source/`; this is a reproducibility check rather than an independently implemented rational Lie calculation. Their complete exact rational outputs and source witnesses are included. The rows are

n	1	2	3	4	5	6	7	8	9	10
special	0	0	1	0	3	0	6	4	13	12
metabelian image	0	0	1	0	3	0	5	0	7	0
S_3-invariants	0	0	1	0	1	0	2	1	4	2
their image	0	0	1	0	1	0	1	0	2	0.

The calculation solves $[f, y] = [g, x + y]$ in the free Lie algebra, then applies the canonical quotient $L \to L/L''$ to its solutions. This order of operations specifies the image being counted.

For the upper bound we use the opened primary text of Furusho, Corollary 0.3, Theorem 0.5, Proposition 5.1 and Lemma 5.2 [Fur11]. The leading homogeneous term of a profinite pentagon equation, after applying q_ℓ and the continuous rational unipotent completion, satisfies the infinitesimal pentagon: multiplication of factors of the same least positive degree becomes addition, and the standard braid insertion maps induce the corresponding insertion maps on the graded Lie algebras. Thus this leading term lies in $\mathfrak{grt}_1(\mathbb{Q}_\ell)$. Furusho's result

implies its gamma identity. We spell out the coordinate conversion. In the completed tensor algebra write $h = h_x x + h_y y$, and put $M(h) = (h_y y)^{\mathrm{ab}}$. Expansion gives

$$M([x,y]) = XY, \quad M([x,h]) = XM(h), \quad M([y,h]) = YM(h) \quad (h \in L').$$

Moreover $M([L',L']) = 0$, because the commutative images of both derived factors vanish. Hence $M(p[x,y]) = XYp$ on L'/L''. The homogeneous gamma identity reads $M(h) = a_n(X^n + Y^n - (X+Y)^n)$. Cancellation of XY gives the asserted line. The special equation implies $(X+Y) \mid p$: in L'/L'' it reads $-Yp = -(X+Y)q$, and $X+Y$ is relatively prime to Y. For even n, evaluating the numerator at $Y = -X$ makes $a_n = 0$. Weight one is zero directly. The rerun gamma verifier checks this Fox coordinate identity on all 224 Lyndon words in degrees two through ten and the displayed independent weight-nine vectors. □

The lower bound in R9 is established in D.20 from the opened primary Ihara formula, with its exact Fox coordinate and an extraction inside the actual compact Galois image. The initial Hain–Matsumoto audit reached Theorem 8.4, §8.4, Lemma 8.5 and Remark 8.6 [HM03]; that remark explicitly points to Ihara for nontriviality. The subsequent primary-source check opens Ihara's ICM lecture, §6.2, the theorem of §6.3, and (6.4.4), and supplies the missing direct comparison in D.20. Brown's motivic formula [Bro15, (1.3)] is not needed for that arithmetic proof. The source program's field `actual_Galois_and_GT_rational_rank` remains a literature-populated label; the arithmetic conclusion follows from the proof in D.20 rather than from that label.

Statement D.5 (Known). Cyclic-cover compatibility for every full profinite GT element. *Let $H_N = \ker(\widehat{F} \to \mathbb{Z}/N)$, where $x \mapsto 1$ and $y \mapsto 0$. The continuous homomorphism*

$$\delta_N : H_N \longrightarrow \widehat{F}, \quad x^N \mapsto x, \quad y \mapsto y, \quad x^a y x^{-a} \mapsto 1 \quad (0 < a < N)$$

is defined on a profinite free basis. For every $(\lambda, f) \in \widehat{\mathrm{GT}}$, with the convention $x \mapsto x^\lambda$, $y \mapsto f^{-1} y^\lambda f$, one has

$$\delta_N(f) = y^{c_N(f)} f, \qquad c_{MN}(f) = c_M(f) + c_N(f),$$

where $c_N(f)$ is the y coordinate of $\delta_N(f)$ in $\widehat{F}^{\mathrm{ab}}$. These are Enriquez's Theorem 8.1 and Remark 8.6 [Enr07].

Comparison with the covering map in LM7. For the negative Schreier generator $e_j = x^{-j} y x^j$, $0 < j < N$,

$$e_j = x^{-N}(x^{N-j} y x^{-(N-j)}) x^N.$$

Both sides are in H_N, and applying δ_N gives 1. The maps denoted ϕ_N in LM7 D.20 and δ_N above consequently agree on every member of a free basis, hence agree as continuous homomorphisms. The sign convention there is $\rho_N = -c_N$, and on the stated Galois normalization it is $\rho_N = \kappa_N$. This is an equality of the specified coefficients.

For completeness, the automorphism g associated to (λ, f) preserves H_N, since it multiplies the quotient $\mathbb{Z}/N$ by the unit λ. The square with $g|_{H_N}$, g, and the two maps δ_N commutes. On x^N both composites give x^λ; on y the extra factor $y^{c_N(f)}$ cancels in conjugation. For $x^a y x^{-a}$, write $\lambda a = Nq + b$ with $0 < b < N$. Its image under g is conjugate, by an element of H_N, to $x^b y^\lambda x^{-b}$; applying δ_N gives 1. These checks prove the square on the full profinite free basis. □

This opened theorem strengthens the cyclic-cover statement used in LM7 D.20, where the displayed compatibility was used only on the Galois image. It provides additional necessary finite tests for Task E.3, performed below. It gives no equality between the full profinite groups $\widehat{\mathrm{GT}}$ and $\widehat{\mathrm{GT}}_0$.

Statement D.6 (Known). The exact finite test and its remaining quantifier. *For a finite-index normal subgroup $N_0 \triangleleft B_3$ contained in PB_3, the reduction maps of finite shadow sets satisfy*

$$\mathrm{im}(\widehat{\mathrm{GT}}_0 \to \mathrm{GT}(N_0)) = \bigcap_{K \subset N_0} R_{K,N_0}(\mathrm{GT}(K)),$$

where K ranges over all finite-index normal subgroups of B_3 contained in N_0. This is Dolgushev–Guynee, Theorem 5.2, Corollary 5.4 and Proposition 5.5 [DG24], with the full profinite identification supplied by Harbater–Schneps, Main Theorem (a) [HS00].

Verification of the inverse-limit step. A genuine element gives a shadow in every finite quotient. Conversely, the fibres of the reduction maps over a fixed shadow are finite sets. A finite collection of quotient conditions is dominated by the intersection of their kernels. Nonemptiness at every such kernel therefore gives the finite intersection property in the product of the finite shadow sets, and compactness produces a compatible family. The cited theorem identifies this family, including its generation condition, with a full profinite element. For a finite list of twelve shadows, exclusion of each one at some quotient implies exclusion of all twelve at the intersection of those twelve kernels. Thus the negative conclusion admits one finite witness. Passage through any specified collection of successful tests supplies only the conditions for that collection. □

We also opened Bleher–Chinburg–Lubotzky, Lemma 2.3 and Corollary 2.4 [BCL20], and reran LM7's explicit lifting counterexample. Those results lift automorphisms of a quotient of a finitely generated free profinite group. LM7's finite construction uses the quotient consisting of the free metabelian quotient together with an A_5 factor; the identity on the first factor and conjugation by an odd permutation on the second lift to an automorphism. The latter sends a five-cycle to its other A_5 conjugacy class. The lifted automorphism therefore fails the required conjugacy condition on x while acting trivially on the metabelian quotient. This verifies the claimed limitation of that particular lifting theorem. It does not rule out another lift satisfying the peripheral equations. The missing proof in D.1 and D.2 is exactly a solution of those equations and generation at every finite quotient, or a finite obstruction to the proposed extra coordinate.

4.3 A complete finite pentagon comparison in weight nine

Statement D.7 (Proved). Compatible finite three- and four-strand shadows. *Write $P_n = PB_n$, $p = 101$, and*

$$K_n = \gamma_{11} P_n P_n^p \triangleleft B_n \qquad (n = 2, 3, 4), \qquad K_F = \gamma_{11} F F^p,$$

where a power subgroup is generated by all the indicated powers. All these groups are discrete until their finite quotients are taken. We use $F = \langle x = A_{12}, y = A_{23} \rangle \leq P_3$ and the same braid generators as in D.4. Let $\mathcal{G}_9$ be the finite gentle shadows at K_3 with $m = 0 \pmod p$ and $f \in \gamma_9(F/K_F)$. Let $\mathcal{P}_9 \subset \mathcal{G}_9$ be those satisfying the pentagon in P_4/K_4. The logarithm bijection makes both sets vector spaces over $\mathbb{F}_{101}$ and gives the following dimensions:

	$\mathcal{G}_9$	$\mathcal{P}_9$
$\log f$ *in degrees* $9, 10$	37	2
degree-nine projection	4	1
degree-nine metabelian projection	2	1.

(D.7.1)

The last right-hand space is exactly $\mathbb{F}_{101}G_9[x,y]$. The induced three-strand kernel associated to K_4 in the finite parenthesized-braid definition is exactly K_3, so $\mathcal{P}_9$ consists of actual finite pentagon shadows in that definition, not merely solutions in an unrelated Lie target.

Proof. The quotient maps. The subgroups K_n are characteristic in P_n, hence normal in B_n. The groups P_n/K_n are finitely generated nilpotent groups of bounded finite exponent, so are finite. Every strand insertion or cabling homomorphism $P_3 \to P_4$ sends K_3 into K_4, since homomorphisms preserve the lower central series and power subgroups. For the inclusion of the first three strands, the strand-forgetting retraction $P_4 \to P_3$ gives $K_4 \cap P_3 = K_3$. Intersecting the five preimages used in the definition of the induced three-strand kernel therefore gives exactly K_3. The analogous two-strand assertion gives $K_2 = p\mathbb{Z}$ and $K_{\mathrm{ord}} = p$. The direct product $P_3 = F \times \langle C_3 \rangle$ gives $K_3 \cap F = K_F$. Thus reduction induces the specified maps of finite shadow sets, and the continuous maps from the profinite braid groups to these finite groups are their ordinary completion maps.

Faithful logarithmic coordinates. The class is ten, smaller than p. Lazard's Theorem II.4.6, with its homomorphism assertion, identifies the finite groups and their Lie rings by the mutually inverse truncated logarithm and Campbell–Hausdorff operations [Laz54]. For the free group factor this is the free nilpotent Lie algebra of class ten over $\mathbb{F}_p$. This assertion can also be read directly from Hall collection: the logarithm of a basic commutator is its Hall Lie word plus terms of larger degree; this triangular matrix has diagonal one, and all denominators through degree ten are units modulo p.

For four strands we keep the actual pure Artin action in these coordinates. Put $u_i = A_{i4}$, $a = A_{12}$, $b = A_{23}$. Forgetting strand four splits P_4 as $\langle u_1, u_2, u_3 \rangle \rtimes P_3$. The center is generated by $C_4 = C_3 u_1 u_2 u_3$; eliminating C_3 in the quotient by C_4 gives

$$P_4/\langle C_4 \rangle = F(u_1, u_2, u_3) \rtimes F(a, b). \tag{D.7.2}$$

These generators and their exact conjugation relations are [CFR12, (2.1)–(2.3)]. In particular the inverse actions of a, b on the free factor are

	u_1	u_2	u_3
$a^{-1}(-)a$	$u_1 u_2 u_1 u_2^{-1} u_1^{-1}$	$u_1 u_2 u_1^{-1}$	u_3
$b^{-1}(-)b$	u_1	$u_2 u_3 u_2 u_3^{-1} u_2^{-1}$	$u_2 u_3 u_2^{-1}$.

(D.7.3)

The actions are trivial on the free factor's abelianization. Their Magnus expansions raise degree by at least one; a weight-i commutator of them raises degree by at least i. Hence $[\gamma_i F(a,b), \gamma_j F(u_1,u_2,u_3)] \subset \gamma_{i+j} F(u_1,u_2,u_3)$. It follows by both inclusions that the lower central series in (D.7.2) is the product of the two free lower central series. Hall collection and reduction modulo p therefore give faithful coordinates consisting of the two free Lie factors. The omitted central factor has degree one. It intersects the derived subgroup trivially, also after this reduction, and every pentagon factor considered below lies in degree at least nine. Thus the computation in (D.7.2) detects the complete pentagon in P_4/K_4.

The complete equations are linear in the retained stratum. Write $X = \log x$, $Y = \log y$ and $\log f = h_9 + h_{10}$, with h_i homogeneous of degree i. There are $56 + 99 = 155$ Lyndon coordinates. The two simplified finite hexagons, with $z = y^{-1}x^{-1}$, are

$$f(x,y)f(y,x) = 1, \qquad f(z,x)f(y,z)f(x,y) = 1. \tag{D.7.4}$$

Their equivalence to the original braid hexagons for $f \in F'$ is Dolgushev–Guynee, Proposition 3.4 [DG24]. Here $m = 0$ and

$$\log z = -X - Y - \tfrac{1}{2}[X,Y] \pmod{\text{degree } 3}.$$

Every product of two h_i has degree at least eighteen. Consequently (D.7.4), including all terms of degree ten, is exactly a linear system. Substitution in the two-letter tensor algebra produces a 3072 by 155 matrix of rank 118. Its kernel has dimension 37; its degree-nine projection has dimension four. The program also solves for, retains, and substitutes a degree-ten correction for each of the four leading basis vectors.

Here are the corresponding exact four-strand substitutions. Put $U_i = \log u_i$, $A = \log a$, $B = \log b$, and

$$L = U_1 + U_2 + U_3 + \tfrac{1}{2}\sum_{i<j}[U_i, U_j] \pmod{\text{degree } 3}.$$

If the logarithm of an inverse Artin action in (D.7.3), applied to U_i, is $U_i + v_{2i} + v_{3i}$ through degree three, then its negative logarithm is the derivation

$$D(U_i) = -v_{2i} - v_{3i} + \tfrac{1}{2}D_2(v_{2i}), \qquad D_2(U_i) = v_{2i}. \tag{D.7.5}$$

This follows by expanding $-\log(1+T) = -T + T^2/2$ to the required degree. The leading parts are

$$\begin{aligned} &D_A(U_1) = [U_1, U_2], \quad D_A(U_2) = [U_2, U_1], \quad D_A(U_3) = 0, \\ &D_B(U_1) = 0, \qquad\quad D_B(U_2) = [U_2, U_3], \quad D_B(U_3) = [U_3, U_2]. \end{aligned}$$

Formula (D.7.5) supplies their degree-raising-two parts directly from the literal group words. The five pairs of arguments of the pentagon, in right-hand-side minus left-hand-side order, are

	first logarithm	second logarithm
$+$	A	$B + U_2 + \frac{1}{2}[U_2, U_3]$
$+$	$-A - L$	U_3
$-$	B	U_3
$-$	$-B - L$	$U_2 + U_3 + \frac{1}{2}[U_2, U_3]$
$-$	A	B.

(D.7.6)

For example $A_{13}A_{23} = a^{-1}C_3$ and $A_{12}A_{13} = C_3 b^{-1}$. The degree-two corrections involving A or B and L vanish because both leading derivations annihilate $U_1 + U_2 + U_3$. Terms of degree three in the argument logarithms, or derivation parts raising degree by three, cannot affect a word of weight nine through degree ten. Therefore (D.7.5)–(D.7.6) compute the complete group equation at this quotient, without identifying the logarithms of the pure braid generators with homogeneous infinitesimal generators.

The pentagon's free two-generator component cancels. Its free three-generator component has 3^9 and 3^{10} tensor coefficients. The degree-nine equations reduce the 37-dimensional hexagon space to dimension 34; all 59049 degree-ten equations then reduce it to dimension two. Projecting this kernel to degree nine gives rank one, and applying the metabelian quotient gives the asserted gamma line. All surviving basis vectors were substituted into every hexagon and pentagon coefficient.

Generation. The map on P_3/K_3 induced by a retained pair sends x to x, y to $f^{-1}yf$, and C_3 to C_3. It is the identity on the Frattini quotient of this finite p-group, and is therefore surjective. The two-strand map is also surjective. Definition 2.9 and Proposition 2.10 of Dolgushev–Le–Lorenz [DLL24] now give the generation condition on four strands for every retained pentagon pair. The same three-strand argument gives the gentle generation condition. This proves that the computed solutions are the claimed finite shadows.

The executable sources are `de_gentle_weight9.py` and `de_pentagon10_finite.py`; their JSON outputs retain the basis logarithms, and their compressed certificates retain the full coefficient matrices. The latter run completed in 501.307 seconds. Its assertions verify every surviving coefficient, the projection ranks, and equality with the gamma line. No search over a subset of the 155-coordinate stratum is used. □

Statement D.8 (Proved). An integer pentagon certificate and the weight-nine comparison. *Let k be a field of characteristic zero or characteristic at least five. In the free Lie algebra $L_k(X,Y)$ put*

$$V_9 = \{h \in L_k(X,Y)_9 : h(X,Y) + h(Y,X) = 0, \quad [X, h(X,Y)] = [X+Y, h(-X-Y,Y)]\}.$$

Then $\dim V_9 = 4$*, its metabelian image has dimension two, and the infinitesimal pentagon map $P : V_9 \to (\mathfrak{t}_4)_9$ has rank three. Its one-dimensional kernel has metabelian image $k\,G_9(X,Y)[X,Y]$. The complete finite projection in D.7 is this kernel when $k = \mathbb{F}_{101}$. For every $\ell \geq 5$, applying the corresponding rational leading-term map to the full profinite pentagon equation and using D.20 gives precisely $\mathbb{Q}_\ell G_9(X,Y)[X,Y]$ as the weight-nine metabelian image of $\widehat{\mathrm{GT}}$ and of the Galois image.*

Proof. The following exact integer certificate also explains the comparison with the finite calculation. Generate the 56 Lyndon words of length nine lexicographically, and use standard bracketing: split at the longest proper Lyndon suffix, then bracket the two recursively bracketed factors. Expand in the free associative algebra. The two equations defining V_9 give an integer 1536 by 56 matrix C. Its 52 nonzero Smith factors are all one. A four-column integer kernel basis H retained in the JSON certificate has Smith factors $1, 1, 2, 24$. Hence it is a basis of the kernel over every field in the statement, not merely over $\mathbb{Q}$.

We evaluate the pentagon in the faithful Lie model

$$\mathfrak{t}_4/(\text{degree-one center}) = L(U_1, U_2, U_3) \rtimes L(A, B),$$

with the leading derivations displayed in D.7. It is the associated graded of (D.7.2); the integral braid relations are also recorded in [CFR12, (4.1)]. For the five arguments take the degree-one parts of (D.7.6). All eighteen infinitesimal braid relations are checked by the program, as is the cancellation of the free $L(A,B)$ component. The three tensor rows of $P(H)$ labelled 111111223, 111111232, 111121123 are

$$\begin{pmatrix} -6 & -6 & 6 & -5 \\ 254 & 510 & -446 & -1291 \\ -150 & -302 & 246 & 700 \end{pmatrix}. \tag{D.8.1}$$

The determinant of the first three columns is $27648 = 2^{10}3^3$. The vector

$$v = (-3299, 1326, -1613, 432)^{\mathsf{T}} \tag{D.8.2}$$

annihilates all 19683 tensor coefficients of $P(H)$, verified with integer arithmetic. Its coordinates have greatest common divisor one. Thus P has rank exactly three and kernel kv in each of the stated characteristics.

In the ordered monomial coordinates $X^iY^{7-i}[X,Y]$ for $0 \leq i \leq 7$, the metabelian image of H is

$$\begin{pmatrix} 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & -8 \\ 2 & 2 & -2 & -17 \\ 6 & 6 & -6 & -23 \\ 6 & 6 & -6 & -23 \\ 2 & 2 & -2 & -17 \\ 0 & 0 & 0 & -8 \\ 0 & 0 & 0 & -2 \end{pmatrix}. \tag{D.8.3}$$

Its nonzero Smith factors are $1, 4$. Applied to (D.8.2), it gives

$$-96(9, 36, 84, 126, 126, 84, 36, 9)^{\mathsf{T}} = -96G_9.$$

This proves all the dimension and coordinate assertions. The programs `de_pentagon9.py` and `de_pentagon9_rational.py` construct the matrices from the Lie relations; the second performs the integer and rational calculations anew, including both Smith forms and every zero coefficient.

For the final comparison let a full profinite element have weight-nine leading action in the filtration of Part D. Its ℓ-adic multiplier is one. In the normalized gauge the action fixes x and its conjugator belongs to F'. A lower-degree inner action fixing x would have an inner generator centralizing X. In degree at least two this is zero in the free Lie algebra; the degree-one possibility is excluded by the derived-subgroup normalization. The conjugator therefore has no term below degree nine. Its leading word belongs to V_9 by the peripheral and symmetry equations.

The insertion homomorphisms of pure braid groups preserve their lower central series. Their associated graded maps send each pure generator to the corresponding sum of infinitesimal generators; this follows by abelianizing the explicit product formulas in the five insertion maps. In degree nine the product of pentagon factors becomes their sum, since products of two degree-nine terms have degree eighteen. The leading word consequently lies in the exact kernel just computed over $\mathbb{Q}_\ell$. This is the upper inclusion, with its source and target specified. The actual Galois construction in D.20 supplies a nonzero element in the same filtered full profinite source, giving the reverse inclusion on rational spans. This derives R9 in weight nine from the certified pentagon map.

Finally, other multipliers act nontrivially in degree one, and lower nonzero leading words belong to lower filtration pieces. They do not define a weight-nine leading class under the map in Part D. This is why D.7 enumerates precisely the stratum needed for this comparison. The quotient detects an extra finite gentle direction, but the calculation does not construct a compatible family in the full profinite gentle group. The remaining existence question is D.2. □

4.4 The arithmetic lower bound with its exact coordinate comparison

Statement D.20 (Proved). The checked Galois and full GT gamma line in every odd weight. *For every prime $\ell \geq 5$ and every odd integer $m \geq 3$, the rational leading metabelian image of $G_{\mathbb{Q}}$, and therefore of $\widehat{\mathrm{GT}}$, is exactly*

$$\mathbb{Q}_\ell G_m[x, y], \qquad G_m = \frac{(X+Y)^m - X^m - Y^m}{XY}. \tag{D.20.1}$$

The leading image is zero in weight one and every positive even weight. In particular R9 has dimensions $0, 0, 1, 0, 1, 0, 1, 0, 1, 0$ in weights one through ten, with the specified coordinate $p(X, Y)[x, y]$. This verifies the lower bound used in Tasks D.1–D.3; the full gentle and special image questions in D.1–D.2 remain separate questions about the displayed inclusion maps.

Manuscript comparison: §51.15, manuscript pages 725–733 in the supplied transcription, scans 156–170. The transcription does not establish the recto/verso labels for this range.

Proof. We use the primary formula in Ihara's ICM lecture, §6.2, the theorem of §6.3, and (6.4.4) [Iha91]. In its normalization $(f_\sigma)^{-1} = 1 + A_1\xi + A_2\eta$ and $\psi_\sigma = 1 + A_1\xi$. For $\sigma \in \ker \chi_\ell$, the formula is

$$\log \psi_\sigma^{\mathrm{ab}} = \sum_{\substack{r \geq 3 \\ r \text{ odd}}} \frac{\kappa_r^{(\ell)}(\sigma)}{(\ell^{r-1} - 1)r!}\big((X+Y)^r - X^r - Y^r\big), \quad X = \log(1+\xi),\ Y = \log(1+\eta). \tag{D.20.2}$$

The cited §6.2 states nonvanishing of these odd cyclotomic cocycles for $\ell > 2$, with value zero at complex conjugation by their positive-real-root normalization. Formula (6.4.4) identifies this series with the actual metabelian action. These are the arithmetic inputs; we now supply the filtration and coordinate arguments.

First $\kappa_r^{(\ell)}$ stays nonzero on $\ker \chi_\ell$. Otherwise it factors through the abelian cyclotomic quotient. The cocycle identities for a cyclotomic element g and complex conjugation c would give

$$(1 - \chi_\ell(c)^r)\kappa_r^{(\ell)}(g) = (1 - \chi_\ell(g)^r)\kappa_r^{(\ell)}(c) = 0.$$

The left coefficient is two. This would make the whole cocycle zero, contradicting the cited nonvanishing.

Fix m. Apply the actual Galois action to the rational nilpotent completion of $F_\ell/\gamma_{m+3}F_\ell$. We retain this extra class because a degree-m multiplier acts on the commutator in degree $m+2$. The Hall coordinates of this finitely generated free nilpotent pro-ℓ group are integral and torsion-free. Its map to this rational completion is therefore injective, as is the map on every lower-central graded lattice. Consequently triviality in a specified lower degree of the rational action detects triviality of the same integral action; this comparison does not create a new Galois element or weaken the required filtration. In the normalized gauge, x is fixed and the conjugator f belongs to the derived subgroup. An outer-trivial lower term is consequently actual-trivial: its inner generator must centralize x; its only possible degree-one component is a multiple of x, excluded by the normalization of f, and the higher-degree centralizer is zero by the free tensor-algebra comparison. Let G be the compact image of $\ker \chi_\ell$ in its automorphism group. It lies in the finite-dimensional unipotent group of automorphisms acting trivially on the abelianization. Its filtration I^d is the one defined at the start of Part D. Both the completed group-ring action on the metabelian commutator and its logarithm are homomorphisms on G: the action on its coefficient ring is trivial when $\chi_\ell = 1$. Consequently (D.20.2) defines a nonzero additive functional k_m on G, proportional to $\kappa_m^{(\ell)}$.

Here is the precise Lie extraction from this compact group. Put $\mathfrak{g} = \operatorname{span}_{\mathbb{Q}_\ell} \log G$ in the finite nilpotent automorphism Lie algebra. It is a Lie subalgebra: for $A = \log a$, $B = \log b$, the elements

$$\ell^{-2n} \log[a^{\ell^n}, b^{\ell^n}] \longrightarrow [A, B]$$

belong to this finite-dimensional, hence closed, span. Choose $a_1, \dots, a_d \in G$ whose logarithms form a basis. The analytic map

$$(t_1, \dots, t_d) \longmapsto \log(a_1^{t_1} \cdots a_d^{t_d})$$

is defined for $t_i \in \mathbb{Z}_\ell$, takes values in $\log G$, and has invertible differential at zero. The inverse-function theorem therefore puts an open neighbourhood of zero in $\mathfrak{g}$ inside $\log G$. In particular every sufficiently small scalar multiple of any vector of $\mathfrak{g}$ exponentiates to an element of the actual compact image. The functional k_m extends linearly to $\mathfrak{g}$, either by differentiating the displayed group-ring character or by its additive logarithm.

Choose $\gamma \in G_{\mathbb{Q}}$ with $u = \chi_\ell(\gamma) = 1 + \ell$, and write A for its conjugation operator on $\mathfrak{g}$. This preserves $\mathfrak{g}$, because the cyclotomic kernel is normal. On $\operatorname{gr}^d I$ it acts as multiplication by u^d. Indeed its action on the free abelianization is u, the graded Lie bracket multiplies degrees, and any higher part of the automorphism changes only higher degrees. Moreover

$$k_m(AZ) = u^m k_m(Z),$$

by the cyclotomic cocycle relation. Choose Z with $k_m(Z) \neq 0$ and set

$$Z_m = \prod_{d=1}^{m-1} (A - u^d)Z. \tag{D.20.3}$$

Successively applying these commuting factors sends I^d into I^{d+1}; thus $Z_m \in I^m\mathfrak{g}$. Its k_m value is $\prod_{d<m}(u^m - u^d)k_m(Z)$, which is nonzero. There is no assertion that a vector in the ambient target is realized merely because of its dimension: (D.20.3) belongs to the Lie algebra of the actual compact Galois image. The preceding open-neighbourhood argument gives $\exp(\ell^a Z_m) \in G \cap I^m$ for some integer $a \geq 0$. It has nonzero weight-m coefficient and lifts to an actual Galois element.

Finally let h be this element's leading degree-m conjugating Lie word and write $h = h_x x + h_y y$ in the tensor algebra. Its commutative image is zero. The definition of ψ therefore gives, in degree m,

$$\log \psi^{\mathrm{ab}} = -(h_x x)^{\mathrm{ab}} = (h_y y)^{\mathrm{ab}}.$$

The exact Fox comparison in D.4 sends $p[x, y]$ to XYp. Formula (D.20.2) thus gives a nonzero multiple of G_m in the requested coordinate. This coefficient also survives the outer quotient. An inner derivation in this gauge would have to centralize x; in degree at least two of the metabelian free Lie algebra, $\operatorname{ad} x$ is multiplication by X on $\mathbb{Q}_\ell[X, Y][x, y]$ and has zero kernel.

This proves the nonzero Galois lower bound. The actual inclusions $G_{\mathbb{Q}} \to \widehat{\mathrm{GT}} \to \widehat{\mathrm{GT}}_0$ carry it to the full GT image. D.4 supplies the matching pentagon upper bound in every odd weight and its vanishing in even weights; weight one is zero by its explicit special equation. Together these statements prove (D.20.1) and the table. □

4.5 The full-source symmetry restriction in weight five

Statement D.21 (Proved). The weight-five full special image has dimension one or three. *Let $\ell \geq 5$, and let $\mathscr{I}_{5,\ell}$ be the rational weight-five metabelian leading image of the full profinite group $\mathrm{Out}^\sharp(\widehat{F}_2)$, with the source filtration and leading map specified at the beginning of Part D. Then*

$$\dim_{\mathbb{Q}_\ell} \mathscr{I}_{5,\ell} \in \{1, 3\}. \tag{D.21.1}$$

In particular the intermediate value two offered in Task D.1 is excluded. This statement leaves the choice between one and three on the unresolved task ledger. More precisely, the polynomial coordinate map

$$(X+Y)q(X,Y)[x,y] \longmapsto q(X,Y) \quad : \quad (X+Y)\mathbb{Q}_\ell[X,Y]_2[x,y] \xrightarrow{\sim} \mathbb{Q}_\ell[X,Y]_2 \tag{D.21.2}$$

identifies the target with the direct sum of the gamma line $\mathbb{Q}_\ell(X^2+XY+Y^2)$ and an absolutely irreducible two-dimensional $\mathfrak{S}_3$-module. The image in (D.21.1) contains that line and is stable under this actual geometric symmetry action.

Manuscript comparison: §51.15, manuscript pages 725–733 in the supplied transcription, scans 156–170; the full special-image question of LM7 D.13.

Proof. We give the comparison between conjugation in the full source and the polynomial action, including the inner corrections. The geometric automorphisms permuting the three punctures give an actual subgroup $\mathfrak{S}_3$ of $\mathrm{Out}^\sharp(\widehat{F}_2)$. Conjugation by this subgroup preserves every term of the source filtration: the lower central and second-derived subgroups used to define the target are characteristic. The leading derivation map is equivariant because, to its least nonzero degree, conjugation of an automorphism acts by conjugation of its derivation. Only the degree-one part of the conjugating geometric automorphism contributes in that degree. Thus $\mathscr{I}_{5,\ell}$ is stable under the indicated $\mathfrak{S}_3$ action before any representation-theoretic argument is applied.

Put $c = [x, y]$, $X = \operatorname{ad} x$ and $Y = \operatorname{ad} y$. Work in the free metabelian Lie algebra, where its commutator module is $\mathbb{Q}_\ell[X, Y]c$. By D.4, the exact weight-five target is $p = (X + Y)q$, with q homogeneous of degree two. The normalized representative of its outer derivation is

$$D_p(x) = 0, \qquad D_p(y) = [pc, y] = -Ypc. \tag{D.21.3}$$

An inner derivation $\operatorname{ad}(hc)$ has coordinates $(-Xhc, -Yhc)$. The polynomial p in (D.21.3) is unique: an inner derivation with zero x coordinate has $Xh = 0$, which forces $h = 0$ in the polynomial module.

Take the transposition $s : x \leftrightarrow y$ and the three-cycle $r : x \mapsto y$, $y \mapsto -x - y$ on the associated graded. The first sends c to $-c$ and the second fixes c. Their conjugated derivations have coordinate pairs

$$\begin{aligned} sD_p s^{-1} &= (Xp(Y, X)c,\ 0), \\ rD_p r^{-1} &= (-(X + Y)p(Y, -X - Y)c,\ 0). \end{aligned} \tag{D.21.4}$$

Subtract respectively the inner derivations with

$$h_s = -p(Y, X), \qquad h_r = -(X + Y)q(Y, -X - Y).$$

The second correction is a polynomial because $p(Y, -X - Y) = -Xq(Y, -X - Y)$. Formula (D.21.3) is restored, and the exact induced action under (D.21.2) is

$$s(q) = q(Y, X), \qquad r(q) = q(Y, -X - Y). \tag{D.21.5}$$

The inner ambiguity has therefore been accounted for by the displayed actual maps.

In the ordered basis X^2, XY, Y^2, the action matrices are

$$S = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{pmatrix}. \tag{D.21.6}$$

They satisfy $S^2 = R^3 = 1$ and $SRS = R^{-1}$. Their common invariant line is generated by $q_0 = X^2 + XY + Y^2$. Averaging the six matrices gives the rank-one idempotent

$$P_0 = \frac{1 + R + R^2 + S + SR + SR^2}{6}. \tag{D.21.7}$$

Its kernel U is a stable two-dimensional complement. It contains no invariant vector. It also contains no sign subrepresentation: $s(q) = -q$ forces $q = a(X^2 - Y^2)$, and $r(q) = q$ then forces $a = 0$. Every one-dimensional representation of $\mathfrak{S}_3$ is trivial or sign, since its abelianization has order two. This proves that U is irreducible, and the same argument over any field extension of $\mathbb{Q}_\ell$ proves absolute irreducibility. Equivalently, the sign projector is the zero matrix on the entire three-dimensional space.

By D.20, the image $\mathscr{I}_{5,\ell}$ contains $G_5 c$. The exact polynomial identity

$$G_5 = 5(X + Y)(X^2 + XY + Y^2) \tag{D.21.8}$$

identifies this nonzero arithmetic line with $\mathbb{Q}_\ell q_0$. The averaging operator P_0 preserves the source image, as it is a rational linear combination of the six symmetry operators already shown to preserve it. Hence the image splits as that line plus its intersection with U. Irreducibility makes this intersection either zero or all of U. This proves (D.21.1) using the full-source comparison; no lifting from the pro-ℓ target is used.

The exact program `d_weight5_symmetry.py` computes both conjugation and inner-normalization maps for a general quadratic, checks all six group matrices, both projectors, the two-dimensional complement, and (D.21.8). Its rational matrices and complete output are retained. The finite certificate verifies these explicit algebraic identities; source stability and the arithmetic inclusion are the preceding proof and D.20. □

4.6 What a finite leading congruence detects

Statement D.22 (Proved). Exact integral rank recovery and a finite-congruence limitation. *Let M be a finite free $\mathbb{Z}_\ell$-module and let $L \subset M$ be the image of a continuous homomorphism from a compact group to the additive group of M. Write L_n for its image under the specified reduction $M \twoheadrightarrow M/\ell^n M$. There are integers $e_1, \dots, e_r \geq 0$ such that*

$$\log_\ell |L_n| = \sum_{i=1}^{r} \max(n - e_i, 0), \qquad \dim_{\mathbb{Q}_\ell}(L \otimes \mathbb{Q}_\ell) = \lim_{n\to\infty} \frac{\log_\ell |L_n|}{n} = r. \tag{D.22.1}$$

The successive difference of the displayed logarithms equals r for every $n > \max_i e_i$.

For each $a \geq 1$ and each $\ell \geq 5$, the actual $\mathfrak{S}_3$-module $M = \mathbb{Z}_\ell[X, Y]_2$ of D.21 contains two $\mathfrak{S}_3$-stable closed submodules with the same image in $M/\ell^a M$, both containing $\mathbb{Z}_\ell q_0$, and with rational dimensions one and three, respectively. The corresponding two-dimensional target of D.2 contains submodules with the same finite reduction and rational dimensions one and two. These are explicit submodules; we assert no realization of them as images of the profinite sources in D.1 or D.2. They refute the inference of a rational upper bound from one vanishing extra congruence without further control of the integral image.

Proof. Compactness makes L closed. Integer multiplication preserves this additive subgroup, and passage to limits in M makes it a $\mathbb{Z}_\ell$-submodule. It is finitely generated because $\mathbb{Z}_\ell$ is noetherian. The Smith basis theorem gives a basis $b_1, \dots, b_d$ of M with

$$L = \bigoplus_{i=1}^{r} \ell^{e_i} \mathbb{Z}_\ell b_i.$$

The image of the ith summand under the specified reduction has order $\ell^{\max(n-e_i,0)}$. Multiplying these independent orders proves (D.22.1), including the asserted successive difference. The leading map on the closed source filtration in Part D is an additive continuous map; its compact image therefore has exactly this form in an integral leading lattice. This is the map to which the rank formula applies.

For the weight-five example, use the integral averaging idempotent P_0 of D.21; its denominator six is a unit for $\ell \geq 5$. Set $U_0 = \ker(P_0 : M \to M)$. In the ordered coefficient basis X^2, XY, Y^2, explicit bases are

$$q_0 = (1, 1, 1), \qquad u_1 = (1, 2, 0), \qquad u_2 = (-1, 0, 1). \tag{D.22.2}$$

The three-column determinant is 3, a unit in $\mathbb{Z}_\ell$, and $P_0 u_1 = P_0 u_2 = 0$. Thus $M = \mathbb{Z}_\ell q_0 \oplus U_0$ and U_0 has basis u_1, u_2. Both summands are stable under the exact matrices of (D.21.6). Define

$$L^{(1)} = \mathbb{Z}_\ell q_0, \qquad L^{(3)} = \mathbb{Z}_\ell q_0 \oplus \ell^a U_0. \tag{D.22.3}$$

Their inclusion maps into M, followed by reduction modulo $\ell^n M$, have the same image for every $n \leq a$. Their rational ranks are one and three. Their finite-image orders for all n are, explicitly,

$$|L_n^{(1)}| = \ell^n, \qquad |L_n^{(3)}| = \ell^{\,n+2\max(n-a,0)}. \tag{D.22.4}$$

This constructs the requested comparison inside the actual symmetry module and retains both its invariant line and its symmetry action.

For D.2 use the explicit independent weight-nine polynomials G_9 and $P_9 = X^2Y^2(X+Y)^3$ of D.4 and the lattice $M_9 = \mathbb{Z}_\ell G_9 \oplus \mathbb{Z}_\ell P_9$ in their rational span. The inclusions

$$\mathbb{Z}_\ell G_9 \subset M_9, \qquad \mathbb{Z}_\ell G_9 \oplus \ell^a \mathbb{Z}_\ell P_9 \subset M_9 \tag{D.22.5}$$

give the corresponding example under $M_9 \twoheadrightarrow M_9/\ell^a M_9$. The target is already the invariant part, so the indicated $\mathfrak{S}_3$ action is trivial on both submodules.

The program `d_leading_lattices.py` checks the integer matrices, the integral Smith valuations, and every element of the finite images for $(\ell, a) = (5, 2), (7, 1)$ at the recorded precisions. Its complete output is retained. The initially incorrect determinant assertion in that program was corrected from six to the computed value three; the failed run and its correction are retained separately.

For the suggested finite-quotient route in the task, the actual profinite leading image maps to each finite shadow image by reduction, as specified in D.6. An extra coordinate vanishing in one such finite image proves divisibility of that coordinate in the corresponding integral lattice. Formula (D.22.3) proves that this divisibility alone does not force its rational vanishing. A finite argument that also controls the Smith valuations, or an integral homogeneous identity valid at every precision, supplies additional information; neither is silently inserted here. The integer homogeneous pentagon identities of D.7–D.8 and the actual arithmetic lower bound of D.20 retain their proved scopes. □

5 E. The complex-multiplication coset

5.1 The finite CM calculation and its independent verification

In this section $B = \langle s, r \mid s^2 = r^3 = 1 \rangle$ is a discrete free product, $t = sr$, and $F = \ker(B \to S_3)$ is the discrete free group on $x = u^2$, $y = v^2$, where $u = t^{-1}$ and $v = sus$. Let $q : B \twoheadrightarrow Q = \mathrm{SL}_2(\mathbb{Z}/12)/\{\pm I\}$ send s, r to the classes of

$$S = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}.$$

Write $N_B = \ker q$, $C = N_B^{\mathrm{ab}}$, and $K_j = [N_B, N_B]N_B^j$, so $G_j = B/K_j$. All these maps are homomorphisms of discrete groups; their maps to finite targets extend continuously to the profinite completions. The inverse image of K_j under $B_3 \twoheadrightarrow B$ is $\widetilde{N}_j$. The CM comparison is LM7 D.15–D.19 and D.25–D.26, in the setting of *La Longue Marche* §51.15, scans 156–170. The assertions below concern these specified finite coordinates.

Statement E.1 (Proved). Independent confirmation of Task E.1. *The order of Q is 576, and $C \simeq \mathbb{Z}^{97}$. With the CM sign convention of LM7 D.25, the independent calculation gives*

	CM compatible	extra CM coset
matrix pairs	24	24
cusp equation in G_2	12	12
cusp equation in G_3	24	24
certified automorphisms of G_6	12	12.

For $c = [t^{12}] \in C$, the integral maps

$$L : C \oplus C \to C, \quad L(b, a) = (1 - s)b + s(1 - r)a, \qquad \widetilde{L}(b, a, h) = L(b, a) - hc$$

have Smith forms $(1^{96}, 2)$ and (1^{97}), respectively. All 24 automorphism certificates have rank 97 on C/pC for $p = 2, 3$. After the exact integral cusp gauge, their two hexagons hold in G_6 and the resulting 24 automorphisms are distinct. In particular twelve have the extra CM coordinate. No discrepancy with these numerical assertions of R12 was found.

Proof. The independent verifier is `code/de_verify_cm.py`. It imports none of the source programs. It enumerates determinant-one matrices modulo twelve and takes their classes by simultaneous sign. It constructs the graph with edges Q and vertices $Q/\langle s\rangle \amalg Q/\langle r\rangle$. There are 576 edges and 480 vertices. The graph is connected, and the fundamental cycles for a spanning tree form an integral basis of its homology; their coordinates on the 97 omitted edges form I_{97}. The Bass–Serre action identifies that homology with N_B^{ab}. It also identifies N_B with the free group of rank $576-480+1=97$, because N_B meets both conjugates of the finite factors trivially.

For a word $w \in B$, use its image $q(w)$ and integral path chain $P(w)$, with

$$P(s)=0, \quad P(r)=e_1-e_r, \quad P(ab)=P(a)+q(a)P(b).$$

Chains distinguish the elements of G_j after reduction modulo j: two lifts of the same element of Q differ by a cycle, which is zero modulo j precisely when their difference lies in K_j. Thus all group equalities tested in this model are equalities in G_j.

Let $H=q(F)$. Direct enumeration gives $|H|=96$, $|C_H(r)|=6$ and $|C_H(s)|=8$. The verifier lists all 48 pairs (A,β) in these two centralizers. It uses the unique lift of each projective matrix whose reduction modulo four is in the four-group generated by the reductions of u^2, v^2. For A^{-1} the action on the rational three-torsion line $\ker(R-I)$ gives one sign. The reduction of β^{-1} modulo three, lying in $\{\pm I, \pm S\}$, gives the other. Equality of the two signs is the CM-compatible class; their inequality is the extra class. These are exactly the restrictions to $M'=\mathbb{Q}(i,(3/4)^{1/4})$ used in LM7 D.25. The program computes 24 pairs of each sign.

Choose integral paths P_A, P_β lifting the pair. The cusp error for $s \mapsto \beta s\beta^{-1}$ and $r \mapsto ArA^{-1}$ is

$$E = P_\beta - sP_\beta + s(P_A - rP_A + AP(r) - P(r)).$$

Changing the paths by cycles b, a changes this error by $L(b,a)$. Exact Gaussian elimination over $\mathbb{F}_2$ and $\mathbb{F}_3$ gives the displayed solvability counts. The verifier then reads the stored source certificates as proposed vectors, substitutes them into this equation, and computes the induced map on all fundamental cycles. The edge at $1 \in Q$ maps to $P_A - P_\beta + e_A$; translating this formula by Q gives every other edge. The resulting matrices have the asserted ranks. They act invertibly on $C/6C$ and identically on Q, proving that each proposed endomorphism of G_6 is an automorphism.

To verify the commutator gauge independently, the program projects chains to the corresponding six-edge graph for S_3. Its two integral basis cycles are the images of x, y. Thus it computes the actual integral abelianization $\beta^{\mathrm{ab}} = ax + by$ of each chosen word. With $k = b - a$ and $\beta' = x^k\beta$,

$$f = s\beta' s(\beta')^{-1} \in F',$$

because conjugation by s exchanges x, y in F^{ab} and β' has equal coordinates. The program directly multiplies the words for both $m=0$ hexagons in G_6; all 48 equalities hold. It also compares the resulting images of s, r, obtaining 24 distinct automorphisms. Independently it finds

$$q(C_{G_2}(s) \cap \operatorname{im} F) = 1, \qquad q(C_{G_3}(r) \cap \operatorname{im} F) = 1,$$

by solving the corresponding centralizer equations. These checks account for the finite conjugator ambiguity in the recorded coordinates. The gauge replaces A, β, V by simultaneous left translates, so $T = A^{-1}V$ and $W = \beta^{-1}V$ are unchanged.

Finally the verifier computes the integer matrices of L and $[L, -c]$ in its newly constructed cycle basis. Exact Smith reduction gives the stated invariant factors; Hermite reduction of $[L, -c]$ gives I_{97}. The cycle c has relatively prime coordinates. The outputs retain every pair, rank, abelianization and gauge, together with the complete source certificate files. These integer calculations confirm more than the ranks of the maps modulo two and three. □

Statement E.2 (Open). The full profinite gentle CM image. *We have not determined whether* $\widehat{\mathrm{GT}}_0$ *reaches the extra CM coset. E.9 proves that* none *of the twelve particular extra shadows recorded by LM7 D.25 is genuine, using an explicit common finite refinement. The full existence question ranges over every shadow with the specified CM matrices. E.10 computes the affine parameters in those matrix fibres; the twelve recorded representatives do not exhaust them. Thus the asserted equivalence in Task E.2 between the full CM question and genuineness of one of those twelve representatives still requires a proof of that reduction. E.11 constructs new points in every extra CM matrix fibre that survive that same common refinement. E.15 constructs compatible families in all 24 matrix fibres through the entire relative class-two tower, with multiplier one. E.16 extends those families simultaneously through every finite collection of nonabelian simple quotients, and separately through each finite perfect quotient. E.20 further extends the class-two families simultaneously through every relative quotient with normal kernel of order prime to six, including its arbitrary higher commutator layers. The combined abelianization and the single cusp gauge are retained. The new specified comparison outside that family has a relative three-primary kernel with a nonzero third lower-central term. The unresolved set is the intersection of the genuine-shadow image with the* whole *union of extra CM matrix fibres.*

Statement E.3 (Open). The full profinite GT CM image. *We have not determined whether full* $\widehat{\mathrm{GT}}$ *reaches the extra CM coset. The finite tests in E.6 impose consequences of the pentagon established in D.5. E.17 evaluates the complete group pentagon in the specified pure four-strand quotient of order* 5184*. It excludes fifteen of the twenty-four retained* G_6 *representatives, but its rank-one affine equation has actual generating solutions in every whole CM matrix fibre, simultaneously satisfying the class-two cusp obstructions of E.11. E.18 retains the next central Ferrari layer in a pure quotient of order* 15552*: its pentagon has the same zero sets on all whole affine fibres, and all twenty-four retained words still generate the order-*648 *Ferrari image. E.19 imposes the modulus-nine pentagon together with the complete three-strand hexagons on the exact induced core. Each retained point has one compatible correction in the full three-dimensional commutator correction image, with generation verified in both targets. The missing step is compatibility through all four-strand quotients. No positive or negative conclusion for Task E.3 follows from the surviving finite candidates. The primary comparison in [NTY10, Proposition 4.4 and Theorem 4.16] constructs the triangle-cover Tate action for full* $\widehat{\mathrm{GT}}$ *but identifies it with the normalizer* h*-coordinate on the Galois subgroup. The required comparison of these two actions on the entire source remains unproved here. The newer action of [LNS26, Theorem 1.1] is defined on the profinite mapping class groups of type* $(g,0)$ *and* $(g,1)$*; its Section 8 explicitly leaves compatibility with the indicated cutting homomorphisms unresolved for full* $\widehat{\mathrm{GT}}$*. It does not provide the needed CM comparison in the checked statements.*

5.2 The homology refinements and an additional finite quotient

Statement E.4 (Proved). The checked homology lifting mechanism and its scope. *The Smith computation in E.1 gives the successive corrections used in LM7 D.26. Each of its twelve extra CM pairs admits a compatible family of shadows through the quotients* G_j*, with a coherently chosen level-six cusp gauge and multiplier* $\lambda \equiv 1 \pmod{72}$*. The family of kernels* K_j *fails to be cofinal among the finite-index normal subgroups of* B *lying below* K_6*.*

Proof. The exact path equation for $t \mapsto t^{1+12h}$ is $E + L(b,a) - hc = 0$, because $q(t)c = c$. At $p = 2, 3$ start from an E.1 solution modulo p and $h = 0 \bmod p$. Surjectivity of $\widetilde{L}$ over $\mathbb{Z}$ cancels the divided error successively modulo p; this constructs solutions over $\mathbb{Z}_p$. Their

matrices on $C \otimes \mathbb{Z}_p$ remain invertible, since their reductions modulo p are the verified matrices. They therefore give automorphisms of the extensions of Q.

For $p > 3$ choose the identity automorphism. The required matrix conjugators lift to centralizers: a lift of $A \in C_Q(r)$ has error z with $(1 + r + r^2)z = 0$, and $(1 - r)(2 + r) = 3 - (1 + r + r^2)$ corrects it after dividing by three. The analogous formula for s divides by two. Both divisions are valid in $C \otimes \mathbb{Z}_p$. Chinese remainders on C/jC assemble these automorphisms in

$$\mathcal{G} = \varprojlim_j G_j = \widehat{B}/\overline{[\widehat{N}_B, \widehat{N}_B]}.$$

The equality follows on each fibre above Q by completion of the integer cycle group C. The induced abelianization of the image of $\widehat{F}$ is $\widehat{\mathbb{Z}}^2$, since the displayed kernel lies in $\widehat{F}'$. One global cusp gauge thus makes f a commutator and gives compatible shadows. The congruences on h give the stated congruence modulo 72. A discrete representative matching the global conjugator both in G_6 and in $F^{\mathrm{ab}}/36$ gives its level-six gauge. This specifies the gauge; it does not assert compatibility for every independent choice of a finite representative.

We reran the original refinement program. Besides the exact integer normal forms, it checks the powers $2, 4, 8, 16$ and $3, 9, 27, 81$ and verifies 384 general-multiplier hexagon identities. These finite checks support the formula; the preceding correction argument supplies all powers. The quotient in E.5 has kernel contained in K_6 and does not kill $[N_B, N_B]$, whereas every G_j kills that subgroup. The factorization test for the quotient maps proves the stated failure of cofinality. □

Statement E.5 (Proved). All certified shadows survive an explicit A_5 refinement. *Define $a : B \twoheadrightarrow A_5$ by $a(s) = (12)(34)$ and $a(r) = (135)$, and put*

$$J_{A_5} = K_6 \cap \ker a.$$

The diagonal map induces an isomorphism $B/J_{A_5} \simeq G_6 \times A_5$. Its kernel does not contain $[N_B, N_B]$. Every one of E.1's 24 certified shadows extends to a shadow at the inverse image of J_{A_5} in B_3. All twelve extra CM coordinates survive this finite test.

Proof. The independent program `de_a5_refinement.py` enumerates the permutation group generated by the specified two elements. It has order 60, is perfect and simple, and its cusp $a(t)$ has order five. The program also verifies the derived-series orders $576, 96, 8, 1$ for Q. Thus G_6, an extension of Q by the abelian group $C/6C$, is solvable. The image of the diagonal map is a subdirect product. Goursat's lemma expresses a proper subdirect product by a common nontrivial quotient of the two factors. Every quotient of G_6 is solvable, whereas the only nontrivial quotient of A_5 is A_5 itself and is perfect. Hence the image is the whole product.

The subgroup $a(N_B)$ is normal in A_5, and $A_5/a(N_B)$ is a quotient of Q. The same argument gives $a(N_B) = A_5$, and therefore $a([N_B, N_B]) = A_5$. This proves $[N_B, N_B] \not\subset J_{A_5}$.

Let α_6 and f_6 be any certified, gauged automorphism and commutator from E.1. On the product choose $(\alpha_6, \mathrm{id}_{A_5})$, multiplier $\lambda = 1$, and $f = (f_6, 1)$. The image of F is $\mathrm{im}(F \to G_6) \times A_5$: its projection to A_5 is surjective because its quotient has order dividing six, and the same Goursat argument applies to its solvable first factor. Taking derived groups shows that f is the image of an element of F'. Both hexagons hold componentwise, and the product automorphism verifies generation. This proves the shadow claim by the finite definition.

The refined cusp has order $\mathrm{lcm}(72, 5) = 360$ and the shadow order parameter is 180; $m = 0$ and $\lambda = 1$ satisfy its unit condition. The projection $G_6 \times A_5 \to G_6$ gives the exact reduction map that retains the original CM coordinate. This quotient is additional to, and has not been identified with, LM7 D.26's S_3-core quotient B/J. □

5.3 Finite tests from the full GT distribution theorem

Statement E.6 (Proved). Five simultaneous finite cyclic-cover tests preserve all 24 candidates. *Let $f_0 \in F'$ be the explicit gauged word of any E.1 certificate, and let $q_F : F \twoheadrightarrow H = q(F)$ be the matrix quotient. There is one word $k \in [N_B, N_B]$ such that $f_* = f_0 k$ has the same image as f_0 in G_6 and, simultaneously for all $N \in \{2, 3, 4, 6, 12\}$, satisfies*

$$q_F \delta_N(f_*) = q_F(y)^{c_N(f_*)} q_F(f_*), \qquad c_N(f_*) \text{ read modulo } 6.$$

It can be chosen also to satisfy, modulo six, $c_4 = 2c_2$, $c_6 = c_2 + c_3$, and $c_{12} = 2c_2 + c_3$. These are assertions in the listed finite targets.

Proof. The programs `de_distribution_test.py` and `de_distribution_more.py` reconstruct words for the 97 fundamental cycles of the graph used in E.1, rewrite them in x, y, and evaluate the maps below. Put $A = H \times \mathbb{Z}/6$, and define

$$\psi_N : H_N \cap F \to A, \qquad w \longmapsto (q_F \delta_N(w), c_N(w) \bmod 6).$$

The x exponent of every member of N_B is even. Consequently $K_6 \subset H_N$ for every $N \mid 12$; also $N_B' \subset F'$. Induce ψ_N to a homomorphism from F into $A^N \rtimes \mathbb{Z}/N$ by the transversal $1, x, \ldots, x^{N-1}$. Its coordinates are the Schreier words $x^i w x^{-(i+s)}$, with residues taken modulo N and s the x exponent of w modulo N. If an element is written (v_i, s), multiplication is $v_i(w_1 w_2) = v_i(w_1) v_{i+s(w_1)}(w_2)$.

The normal subgroup generated by commutators of the 97 images is the image of N_B'. Every such commutator has shift zero. Conjugation of a shift-zero vector by (v_i, s) sends its ith coordinate to $v_i k_{i+s} v_i^{-1}$. Starting from the commutators, the program repeatedly takes coordinate subgroup closures and these conjugations until stable. This computes the coordinate projections exactly: every generator added is a coordinate of an actual product of conjugates, and the final projections contain all such products. The coordinate-zero orders obtained are

N	2	3	4	6	12
$\lvert\psi_N(N_B')\rvert$	8	48	48	48	48
$\lvert\psi_N(K_6)\rvert$	8	144	96	144	288
passing compatible candidates	12	12	12	12	12
passing extra candidates	12	12	12	12	12.

For each certificate it intersects the exact coset $\psi_N(f_0)\psi_N(N_B')$ with

$$\{(q_F(y)^c q_F(f_0), c) : c \in \mathbb{Z}/6\}.$$

Every intersection is nonempty. The order of $q_F(y)$ is six, so the exponent modulo six suffices for this target.

To impose these tests simultaneously, `de_distribution_joint.py` uses a faithful degree-17 permutation representation of A: $H \simeq C_2^2 \times \mathrm{SL}_2(\mathbb{F}_3)$ acts regularly on four points and naturally on the eight nonzero vectors of $\mathbb{F}_3^2$, while $\mathbb{Z}/6$ acts on disjoint sets of two and three points. The product of the five induced representations has degree $17(2+3+4+6+12) = 459$. The program takes the derived subgroup of the image of the 97 N_B generators and projects its shift-zero elements to coordinate zero in each factor. Exact permutation-group membership gives the joint image of N_B' in A^5, of order 42 467 328.

For each certificate we test the finite target tuples with $(c_2, c_3, c_4, c_6, c_{12}) = (a, b, 2a, a + b, 2a + b)$ modulo six. All 24 required coset intersections are nonempty. A tuple in this joint

image is attained by a single word of N'_B, which gives the asserted common correction. The complete output includes the chosen coefficient tuple, the derived-image generators, its order, and the membership results. Thus no inference from five independent choices is used for the simultaneous statement. □

These computations concern the explicit projections ψ_N. They do not enumerate all shadows at a refinement carrying these projections, and they do not impose generation there. In particular they show that these necessary distribution projections do not exclude an extra CM candidate. D.6 still requires the full collection of finite shadow fibres. The theorem in D.5 provides a concrete additional relation to test, but E.2 and E.3 remain the precise unsettled questions.

5.4 The requested S_3-core refinement

Statement E.7 (Proved). All 24 candidates lift through the specified S_3 core. *Fix the free basis $n_1, \dots, n_{97}$ of N_B given by the fundamental cycles in E.1, ordered by the omitted-edge order in its program. Let $\rho : N_B \twoheadrightarrow S_3$ send n_1, n_2 to two different transpositions and the other basis elements to 1. Put*

$$J = \mathrm{core}_B(\ker \rho \cap K_6), \qquad D_J = B/J.$$

There is an exact sequence of finite groups

$$1 \longrightarrow (\mathbb{Z}/3)^{288} \longrightarrow D_J \longrightarrow G_6 \longrightarrow 1.$$

The cusp t has order 72 in D_J. Each of E.1's 24 certified shadows is the reduction of a shadow at the inverse image of J in B_3, with multiplier one. All twelve extra CM shadows consequently survive this first requested nonabelian refinement.

Proof. The complete finite calculation is `de_s3_core.py`; it was run to completion. We describe its faithful model and the exact surjectivity test. Identify S_3 with the affine group of $\mathbb{F}_3$, whose elements are pairs (ϵ, b) acting by $z \mapsto \epsilon z + b$, with $\epsilon \in \{1, -1\}$. The two transpositions are $(-1, 0)$ and $(-1, 1)$. On the graph of E.1 label the first and second omitted edges by these elements, and every other edge by 1. Denote these labels by a_q for $q \in Q$. The induced homomorphism

$$\omega : B \longrightarrow S_3^Q \rtimes Q$$

has s component identically 1, right permutation $q \mapsto qs$, and r component $a_q a_{qr}^{-1}$ with right permutation $q \mapsto qr$. The identities $s^2 = r^3 = 1$ follow by cancellation. For $n \in N_B$, the component at 1 is $\rho(n)$; the other components are its conjugate Schreier representations. Hence $\ker \omega$ on N_B is the normal core of $\ker \rho$. The combined map $(B \to G_6, \omega)$ has kernel exactly J and is a faithful model of D_J.

Since every element of S_3 has sixth power one, $\omega(K_6) = \omega(N'_B)$. Each of its components lies in $A_3 \simeq \mathbb{F}_3$. This image is therefore a vector subspace $M \subset \mathbb{F}_3^Q$. The program forms the commutators of the images of the 97 basis elements and closes their span under conjugation by s, r. Conjugation by $(\epsilon_q, b_q; g)$ is the linear map

$$(z_q)_{q \in Q} \longmapsto (\epsilon_q z_{qg})_{q \in Q}.$$

The iteration stabilizes at dimension 288. Every inserted vector is the image of a product of conjugates of the basis commutators; conversely a stable span contains the image of their normal closure. Thus the computed subspace is exactly $\omega(N'_B)$. The kernel of $D_J \to G_6$ is

M by the combined faithful model. Direct evaluation gives $t^{72} = 1$; its order is at least 72 from its image in G_6.

For each original pair choose the already recorded words A, β and write $S' = \beta s \beta^{-1}$, $R' = ArA^{-1}$ in this model. The error $S'R't^{-1}$ lies in M. Replacing the conjugators by $b\beta, aA$, with $b, a \in M$, changes the error by

$$(1 - S')b + S'(1 - R')a.$$

The conjugation actions in this equation are the just specified linear maps on M. For every one of the 24 pairs the matrix of this map has rank 288. The program solves the equation and records its 576 correction coordinates. Each correction is represented by an actual word of N'_B, by the construction of M.

The proposed images of s, r satisfy their finite-order relations, so define a homomorphism $B \to D_J$, whose composition with $D_J \to G_6$ is the original automorphism. To check surjectivity on the kernel, evaluate all 97 n_i on the proposed images, take their pairwise commutators, and close their vector span under the two proposed generator actions. Their sixth powers are all trivial in the wreath component. These operations give exactly the image of K_6. In every case the resulting span has dimension 288. The map is therefore onto D_J: it is onto G_6 and onto the entire kernel. The first deterministic solution already has this property for all 24 pairs; no probabilistic inference is used.

Finally the correction words lie in $N'_B \subset F'$, so the integral abelianization of β and the cusp gauge from E.1 are unchanged. After that gauge the program directly verifies both hexagons in the wreath component; they hold in G_6 by E.1 and hence in the faithful combined model. The commutator condition holds for the same abelianization reason. With $m = 0$ and $\lambda = 1$, the unit condition and the verified surjectivity give a finite GT shadow. Reduction by $D_J \to G_6$ is the specified original shadow. This proves the claim for all 24 recorded fibres. □

For reproducibility, the first basis word in this instance is $(sr^{-1})^{12}$; the complete first two words and the deterministic basis construction are in the JSON and source. LM7 D.26 left the free basis in this construction to be chosen. We fix it here rather than identifying cores arising from different choices. The next calculation excludes all twelve representatives that survived this particular core.

5.5 A finite exclusion of all twelve recorded extra shadows

Statement E.8 (Known). Finite-factor conjugacy for the full profinite gentle action. *Every $g = (\lambda, f) \in \widehat{\mathrm{GT}}_0$ induces a continuous automorphism of $\widehat{B} = C_2 \amalg C_3$ sending s to a conjugate of s and r to a conjugate of $r^{\lambda \bmod 3}$. For $\lambda \equiv 1 \pmod 3$, the conjugators can be chosen in $\widehat{F} = \ker(\widehat{B} \to S_3)$.*

Proof. Harbater–Schneps, Proposition 4, proof of $(i) \Rightarrow (iii)$, explicitly extends the full profinite gentle action by $u \mapsto u^\lambda$, $v \mapsto f^{-1}v^\lambda f$ on $\widehat{B}_3$, and proves that its central generator goes to its λth power [HS00]. It descends to the quotient by the closed center. The universal property of profinite completion identifies this quotient with the free profinite product $C_2 \amalg C_3$: continuous homomorphisms from either to a finite group are precisely the pairs of elements satisfying $s^2 = r^3 = 1$.

Scheiderer's Appendix to Lochak–Schneps, unnumbered Corollary on printed p.598, places every nontrivial finite subgroup of a free profinite product in a conjugate of a factor [LS97]. Applying it to the images of $\langle s \rangle$ and $\langle r \rangle$ gives the stated conjugacy possibilities. The induced automorphism of $B^{\mathrm{ab}} = C_2 \times C_3$ is multiplication by λ, as follows from the abelianization of the displayed braid action. This distinguishes r from r^{-1} and selects the stated power. For

$\lambda \equiv 1 \pmod 3$, reduction of the braid action to S_3 fixes both braid generators, because λ is odd and $f \in \widehat{F}'$. A conjugator for s consequently projects into $C_{S_3}(s) = \langle s \rangle$, and a conjugator for r projects into $C_{S_3}(r) = \langle r \rangle$. Right multiplication by the appropriate power of s or r leaves the conjugate unchanged and makes that projection trivial. This gives conjugators in $\widehat{F}$. □

Put $N = N_B$ for the rest of this subsection, and define characteristic subgroups of N, hence normal subgroups of B, by

$$J_3 = (\gamma_3 N\, N^3) \cap (N' N^2), \qquad J_2 = (\gamma_3 N\, N^4) \cap (N' N^3), \qquad J_* = J_2 \cap J_3.$$

All are contained in $K_6 = N' N^6$. Write $D_i = B/J_i$ and let $\widetilde{J}_i$ be the inverse image of J_i under $B_3 \twoheadrightarrow B$. These are finite-index normal subgroups of B_3 contained in PB_3. The projections $D_* \to D_i \to G_6$ induce the reduction maps on gentle shadows used below. Their pure parts surject onto $H = q(F) \simeq C_2^2 \times \mathrm{SL}_2(\mathbb{F}_3)$ by restriction of q.

Statement E.9 (Proved). An explicit common refinement removes all twelve recorded extra shadows. *Number the 24 E.1 certificates from zero in their stored source order, retaining exactly the records containing* `mod6_beta_cycles`. *The exact gentle-shadow fibres of the two projections have the following nonempty instances among these certificates:*

refinement	*surviving certificate indices*	*compatible*	*extra*
$D_3 \to G_6$	$0, 1, 2, 3$	2	2
$D_2 \to G_6$	$0, 16$	1	1
$D_* \to G_6$	0	1	0.

The extra indices are $1, 2, 4, 7, 9, 10, 12, 15, 16, 19, 21, 22$. *Consequently none of the twelve recorded extra shadows is in the image of* $\widehat{\mathrm{GT}}_0$, *and none is in the image of* $\widehat{\mathrm{GT}}$. *This excludes those specified finite shadows, while E.2 asks about all shadows in their CM matrix fibres.*

Proof. We give the exact linear equations, including the multiplier and commutator conditions. Since N is free of rank $n = 97$, its image in D_3 is the product of the free class-two exponent-three group on n generators with $C/2C$. The kernel of $D_3 \to G_6$ is

$$M_3 = \Lambda^2(\mathbb{F}_3^{97}), \qquad \dim M_3 = 4656.$$

For D_2 the corresponding product is the free class-two exponent-four group on n generators with $C/3C$. Squares and commutators have order two and generate the central kernel

$$M_2 \simeq (\mathbb{Z}/2)^{97+\binom{97}{2}}, \qquad \dim M_2 = 4753.$$

These descriptions follow directly from the normal forms

$$(a, z)(b, w) = (a + b, z + w + \tfrac{1}{2} a \wedge b) \quad (p = 3)$$

and $n_1^{a_1} \cdots n_{97}^{a_{97}} \prod_{i<j} [n_i, n_j]^{z_{ij}}$ with $a_i \bmod 4$, $z_{ij} \bmod 2$ at two. Collecting these normal forms verifies the defining exponent and class-two relations, and their independent generator and commutator coordinates prove that no further relations have been imposed. The two prime parts form the stated products since their orders are coprime.

Let $\mathcal{S}, \mathcal{R}$ denote conjugation by s, r on M_i. At three these are the exterior-square actions induced from $C/3C$. At two their duals act on the quadratic functions on $C/2C$ by $q(v) \mapsto q(Sv)$ and $q(v) \mapsto q(Rv)$; here S, R are the verified 97-dimensional cycle actions. This formula follows by squaring the collected normal form, whose square is the quadratic function of its parity vector. It includes both square and commutator coordinates.

For one recorded certificate, undo its explicit E.1 cusp gauge and take its stored word β. Set

$$S_0 = \beta s \beta^{-1}, \qquad R_0(\lambda) = S_0 t^{\lambda}.$$

The first element is an involution. For every possible lift of the same finite shadow, the new images have the unique form

$$S' = S_0 k, \qquad R' = k^{-1} R_0(\lambda), \qquad k \in M_i.$$

The action of S_0 on M_i is $\mathcal{S}$, and that of $R_0(\lambda)$ is $\mathcal{R}$: their images in Q are s, r, and N acts trivially on M_i. The exact order equations are therefore

$$(1 + \mathcal{S})k = 0, \qquad (1 + \mathcal{R} + \mathcal{R}^2)k = R_0(\lambda)^3. \tag{1}$$

We explain why these test all finite gentle lifts, without assuming that torsion in a finite group is conjugate to a specified factor. Put $f = sS'$ after undoing the gauge, $u = t^{-1}$, and $v = sus$. For the two finite hexagons, multiplying the first on the right by u^{λ} gives $f^{-1}s$, while multiplying the second on the left by u^{λ} gives sf. Their left sides are identical. Thus they imply $(sf)^2 = 1$. After this equality the first hexagon reads $(S' t^{\lambda})^3 = 1$. Reversing these algebraic steps proves the converse. In particular (1) is the complete pair of hexagon equations at the indicated quotient. The gauge is conjugation by a fixed power of $x = t^{-2}$ and commutes with t^{λ}, so undoing it and reinstating it are inverse operations throughout this calculation.

In D_3 the cusp has order 72. Its order is at least 72 from G_6 and divides 72 from the exponent-three and exponent-two factors. The finite order parameter is therefore 36, and the multiplier over a recorded shadow has the unique residue $\lambda = 1 \bmod 72$. Every $k \in M_3$ is in the image of F'. As two is invertible, the first equation in (1) is parametrized by $k = (1 - \mathcal{S})w$. The program `de_class2_gentle.py` constructs the complete matrix

$$(1 + \mathcal{R} + \mathcal{R}^2)(1 - \mathcal{S}) : M_3 \to M_3$$

and all 24 right sides by word collection. Its rank is 1542. Exact elimination of the augmented matrix, whose rank is 1547, gives exactly indices $0, 1, 2, 3$. Each retained correction is substituted into both original equations.

In D_2 the cusp has order 144. Indeed t^{12} has the primitive cycle from E.1, so its sixth power has a nonzero square coordinate modulo two, while its twelfth power is trivial in both factors. The finite order parameter is 72. All multiplier lifts are $\lambda = 1, 73 \bmod 144$; both are tested. For the commutator condition let L_N be the image of N in $F^{\mathrm{ab}} = \mathbb{Z}^2$. Exact word abelianization gives the column lattice

$$L_N = \begin{pmatrix} 6 & 4 \\ 0 & 2 \end{pmatrix} \mathbb{Z}^2.$$

The image of J_2 in this abelianization is exactly $12L_N$: membership gives divisibility by four and three, and twelfth powers of elements of N give the reverse inclusion. A square coordinate in M_2 is represented by n_i^6 and a commutator coordinate by a commutator. Hence $k \in M_2$ lies in the image of F' exactly when the two coordinates of its square contribution to $6L_N/12L_N$ vanish. The program `de_class2_two_gentle.py` appends these two rows to (1). The coefficient matrix has rank 3927; its augmentation by all 48 right sides has rank 3932. Its complete solution test gives exactly indices $0, 16$. Every retained solution is substituted into the two order equations and the two abelianization rows. The fixed E.1 gauge made the original f a commutator, so this is precisely the condition that the corrected, regauged f be a commutator as well.

All passing solutions satisfy generation. The homomorphism $B \to D_i$ defined by S', R' preserves the image of N, because its map to Q is the recorded identity before gauge. The defining class and power laws show that it kills J_i. On each prime part of the image of N, its Frattini quotient map is exactly the invertible E.1 map on C/pC. The elementary Frattini argument for a finite p-group therefore makes it surjective on both prime parts, and hence on D_i. The multiplier, commutator, hexagon and generation conditions of a finite gentle shadow have all been checked.

The bit-packed elimination routines in `de_gf3.c` are independently compared with the earlier Python elimination on four rectangular matrices for each prime, including multiple machine word boundaries. An additional independent calculation `de_class2_direct_check.py` uses the full 9312 by 4656 matrix of the peripheral correction map at three, agrees with the separate Jordan-block computation, and gives the same four survivors after all conjugator-centralizer freedoms. The full hexagon calculation above supplies the exact finite-shadow scope.

Every shadow over $\widetilde{J}_*$ reduces to shadows over both $\widetilde{J}_2$ and $\widetilde{J}_3$. Intersecting the two displayed index lists leaves only index zero. That record has $A = \beta = 1$ and zero cycle corrections; its identity shadow lifts to every quotient, including D_*. Thus the common-refinement image among the 24 recorded points is exactly the claimed singleton. The all-finite-quotient criterion D.6 excludes the other 23 from the full profinite gentle image. The inclusion $\widehat{\mathrm{GT}} \subset \widehat{\mathrm{GT}}_0$ gives the stated full GT exclusion as well. □

Statement E.10 (Proved). The affine sizes of the CM matrix fibres. *For each of the 24 admissible CM matrix pairs and multiplier one modulo 72, the cusp-preserving changes of its two homology conjugators form affine spaces of dimensions 98 over $\mathbb{F}_2$ and 97 over $\mathbb{F}_3$. Quotienting by the changes that leave both generator images fixed gives dimensions 17 and 18. Quotienting further by simultaneous conjugation by cusp powers gives dimensions 16 and 17. The finite classification modulo all inner conjugations by the centralizer of t in C/pC has dimensions 8 and 9. Invertibility of the resulting action on C/pC is an additional condition in each of these spaces.*

Proof. The exact program `de_affine_fibres.py` computes the kernel of L from E.1 and the following subspaces. Over $\mathbb{F}_2$ the dimensions of C^s, C^r, C^t, C^Q are respectively $52, 29, 9, 0$; over $\mathbb{F}_3$ they are $45, 34, 9, 0$. The changes that fix both generator images are $C^s \oplus C^r \subset \ker L$. Simultaneous inner conjugation adds the diagonal copy of C^t; its intersection with that first subspace is the diagonal $C^Q = 0$. The cusp powers give the one-dimensional diagonal subspace generated by the primitive cycle $c = [t^{12}]$. Taking the indicated dimensions gives all assertions. The output stores explicit bases and complements, and directly substitutes every column into L.

Conjugation by an actual power of t centralizes t in every quotient, so that operation can be transported through every refinement. The larger finite space C^t is classified here only at the homology quotient: transporting one of its elements as a centralizer would require a lift that centralizes t. Consequently the last quotient does not by itself replace the whole finer lifting problem. In particular the twelve original representatives cannot be treated as exhaustive matrix fibres. R11 supplies a genuine arithmetic point in each compatible CM matrix fibre, while E.9 excludes the recorded points in every compatible fibre except the identity fibre. These give concrete distinct points with the same CM matrices. □

Statement E.11 (Proved). Every admissible CM matrix fibre has a replacement at the common refinement. *Each of the 24 admissible CM matrix pairs of E.1 is attained by a gentle shadow at $\widetilde{J}_*$, where $J_* = J_2 \cap J_3$ is the explicit refinement of E.9. In particular*

all twelve extra CM matrix fibres *have nonempty image from this refinement, although their twelve original recorded representatives are excluded by E.9. The new witnesses are finite shadows; their membership in the full profinite image is still to be determined.*

Proof. The programs `de_fibre_three.py` and `de_fibre_two.py` use the complete 18- and 17-dimensional spaces after conjugator centralizers in E.10. They construct exact five-component quadratic obstruction maps on these spaces. The three-primary map projects the eight alternating coinvariants modulo their three centralizer directions. The two-primary map projects the sixteen quadratic coinvariants modulo their eleven centralizer and multiplier directions. Both maps are checked against direct collection of words for every matrix pair. Searching their finite parameter spaces gives a zero in each of the 24 fibres, and the induced action has rank 97 in every case. Only these produced and verified witnesses are used; the search is not used to infer any negative assertion.

Write the two pairs of changes of the conjugator cycles as (b_2, a_2) and (b_3, a_3). The program `de_fibre_crt_verify.py` uses

$$b = 3b_2 + 4b_3 \pmod 6, \qquad a = 3a_2 + 4a_3 \pmod 6,$$

and prefixes the original words β, A by $\prod n_i^{b_i}, \prod n_i^{a_i}$. These are literal words in s, r stored in its output. Their graph cusp equation and both 97-dimensional generation ranks are checked again. Their images in Q are the original matrices, so their CM coordinates agree with the prescribed pair, also after the common cusp gauge.

The complete norm systems of E.9 are then solved anew for these CRT words. Every one of the 24 points has a solution at both primes with $\lambda = 1 \bmod 144$. The two-primary system includes the two exact commutator-image rows. Let k_2 be its square and commutator coordinates and k_3 the three-primary commutator coordinates. The common correction is the explicit word

$$k = \prod_i n_i^{6(k_2)_i} \prod_{i<j} [n_i, n_j]^{3(k_2)_{ij} + 4(k_3)_{ij}}.$$

Its two-primary component is k_2 and its three-primary component is k_3. For the integral cusp gauge g computed from the new word β, set

$$f = sx^g(\beta s \beta^{-1}) k x^{-g} j.$$

Here j is an explicitly recorded product of twelfth powers of two basis words of N, chosen to cancel the abelianization of k. The two commutator rows ensure that this abelianization belongs to $12L_N$; an exact unimodular pair in the displayed basis of L_N constructs j. Thus $j \in J_*$ and the displayed word f lies in F' itself. This gives one common commutator word, with a single integral gauge, for the two primary components.

Finally both hexagons are substituted directly, as words, into a faithful collector for the product of the exponent-four and exponent-three class-two images of N. There are 48 successful hexagon checks. For efficiency a factored central block is evaluated through its exact action under the current Q coset: at three its alternating matrix changes by $Z \mapsto AZA^t$; at two a square vector u and symmetric commutator matrix Z change to Au and the upper entries of $A(Z + \operatorname{diag} u)A^t$. These formulas follow by collecting the explicit square and commutator factors above, so this evaluation is precisely substitution of those actual words. The other letters are collected individually by the omitted-edge basis. A word is trivial in D_* exactly when its Q image and both collected primary normal forms vanish.

The checked Frattini ranks give generation on both primary kernels, as in E.9. The word, multiplier, commutator and generation conditions therefore give a shadow at $\widetilde{J}_*$ in every asserted fibre. The JSON retains the literal CRT conjugators, all factored correction words, both ranks, the common gauge and multiplier, and every hexagon result. □

Statement E.12 (Proved). All admissible CM pairs lift through the whole class-two three-primary tower. *For every admissible CM matrix pair in E.1 there is a compatible family of gentle shadows through all quotients*

$$B/\big((\gamma_3 N\, N^{3^k}) \cap (N' N^2)\big), \qquad k \geq 1,$$

with multiplier one and that CM pair. Thus all twelve extra matrix fibres survive this entire tower. This assertion concerns matrix fibres with a coherent cusp gauge; it does not assert that an independently chosen normalized finite representative extends.

Proof. Let $\mathcal{P}$ be the free class-two pro-three group on 97 generators. Its logarithmic coordinates form $V \oplus \Lambda^2 V$ for $V = \mathbb{Z}_3^{97}$, with product $(a, z)(b, w) = (a + b, z + w + \frac{1}{2} a \wedge b)$. The relevant completed kernel of the extension of Q is $\mathcal{P} \times (C/2C)$. The odd denominator two is a unit in $\mathbb{Z}_3$. Consequently the cusp equation in two conjugators has polynomial coordinate functions over $\mathbb{Z}_3$, of degree at most two.

We verify that its full derivative is onto at each three-primary witness of E.11. Its homology part is L, of rank 97 modulo three. The derivative in the central coordinates has rank 4648, with the eight-dimensional alternating coinvariant quotient computed in E.9. Changes by the order-three conjugator centralizer give rank three in this quotient. The remaining five components are exactly the quadratic map used in `de_fibre_three.py`. Here is a coordinate description of that map and its derivative. For an invariant alternating form B, write τ_{S_0}, τ_t for the central parts of conjugation by $S_0 = \beta s \beta^{-1}$ and by t, evaluated by B. A homology change $(b, a) \in \ker L$ changes the central cusp coordinate by

$$\tau_{S_0}(a - b) - \tau_t(a) + \tfrac{1}{2} B\big(b, (T - 1)a\big), \qquad T = SR.$$

This follows by multiplying $n(b)\, S_0 (n(b)^{-1} n(a)) S_0^{-1}$, the old cusp error, and $t n(a)^{-1} t^{-1}$ in the displayed group law. The central coordinates chosen for $n(a), n(b)$ cancel in the coinvariant quotient. The program checks the formula against direct word collection.

The independent derivative calculation `de_fibre_three_smooth.py` evaluates this full eight-component derivative at each of the 24 zeros, adjoins the three centralizer directions, and obtains rank eight in every case. Thus the full coordinate derivative has rank $97 + 4648 + 8 = 4753$, equal to the dimension of the cusp target. The centralizer corrections and central corrections used to make the initial error zero do not change that rank: central corrections do not change the homology coordinates, and the five-component quadratic map is constant along the discarded centralizer directions. All these equalities are checked over $\mathbb{F}_3$.

Successive lifting is now explicit. At a solution modulo 3^k, divide its error by 3^k, reduce modulo three, and solve the surjective derivative equation for a correction of the two conjugators by 3^k times their coordinates. The polynomial expansion shows that the corrected error is zero modulo 3^{k+1}; its terms of degree at least two in the correction are divisible by 3^{2k}. The corrections preserve all preceding levels, and the derivative retains the same reduction modulo three. Induction gives convergent conjugators over $\mathbb{Z}_3$. Their component in $C/2C$ is kept fixed throughout.

The resulting map preserves Q and is invertible on the three-primary Frattini quotient and on $C/2C$, by the verified ranks in E.11. The pro-three Frattini argument makes it onto $\mathcal{P}$, and its induced finite maps are automorphisms. Choose profinite lifts of the limiting conjugators to $\widehat{F}$ and one cusp gauge using their actual abelianizations. The formula $f = s\beta' s(\beta')^{-1}$ then gives a commutator and compatible hexagons at every finite level. The common multiplier is one. The simultaneous gauge leaves the CM coordinates unchanged.

This tower kills $\gamma_3 N$. The map to A_5 in E.5 sends N onto A_5, and hence also sends $\gamma_3 N$ onto A_5. Its refinement therefore does not factor through any member of the displayed tower.

This explicit factorization obstruction records why these compatible class-two lifts do not complete E.2. □

Statement E.13 (Proved). Every CM matrix fibre survives the common exponent-eight refinement. *Put*

$$J_{8,3} = (\gamma_3 N\, N^8) \cap (\gamma_3 N\, N^3).$$

This is a normal finite-index subgroup of B, contained in J_* of E.9. Its inverse image in B_3 is normal, has finite index, and is contained in PB_3. Each of the 24 admissible CM matrix pairs of E.1 is attained by a gentle shadow at this refinement, with multiplier one modulo 288. *In particular every one of the twelve extra CM matrix fibres remains nonempty here.*

Proof. We compute entire central correction fibres above the 24 affine homology points of E.11. The image of N in the two-primary factor is the free class-two exponent-eight group, with ordered normal form

$$\prod_i n_i^{a_i} \prod_{i<j} [n_i, n_j]^{z_{ij}}, \qquad a_i \in \mathbb{Z}/8, \quad z_{ij} \in \mathbb{Z}/4. \tag{E.13.1}$$

The product in these coordinates is

$$(a, z)(b, w) = (a + b, z + w - (a_j b_i)_{i<j}). \tag{E.13.2}$$

Indeed moving $n_i^{b_i}$ past $n_j^{a_j}$ for $j > i$ introduces $[n_i, n_j]^{-a_j b_i}$. Commutators have eighth power one, and the eighth power of a product gives their twenty-eighth power one. Since $\gcd(8, 28) = 4$, their fourth powers vanish. Conversely the ordered normal form with these relations has exponent eight. This proves the stated free quotient and the faithful collector used below.

The kernel above the fixed G_6 homology image, retaining $N^{\mathrm{ab}}/3$, has a particularly useful description:

$$A \cong (\mathbb{Z}/4)^{4753}, \qquad e_i = n_i^6, \quad e_{ij} = [n_i, n_j]^3. \tag{E.13.3}$$

The two-primary abelian coordinates of an element of this kernel are even, and multiplication by three is invertible modulo four, so these elements span the whole kernel. They are independent by (E.13.1). Moreover their commutators are $[n_i, n_j]^{36} = 1$, so the displayed group is abelian. This is the exact reason that this entire finite correction problem is linear over $\mathbb{Z}/4$.

Let β be one of the literal affine conjugators from E.11, before its cusp gauge, and put $S_0 = \beta s \beta^{-1}$, $R_0 = S_0 t$. Every lift of the fixed homology shadow has

$$S = S_0 k, \qquad R = k^{-1} S_0 t^{1+72h}, \quad k \in A, \quad h \in \mathbb{Z}/4.$$

The complete equations, with the two commutator-image rows, are consequently one linear system over $\mathbb{Z}/4$:

$$(1 + S_0)k = 0, \qquad (1 + R_0 + R_0^2)k = (S_0 t^{1+72h})^3, \qquad B_N^{-1} \operatorname{ab}_F(k) = 0 \pmod{24}. \tag{E.13.4}$$

Here $B_N = \left(\begin{smallmatrix} 6 & 4 \\ 0 & 2 \end{smallmatrix}\right)$ is the chosen basis matrix for the lattice L_N in E.9. Define the integer columns $H_i = B_N^{-1} \operatorname{ab}_F(n_i)$. Then the two equations are $H(k_i)_i = 0 \pmod 4$, because $\operatorname{ab}_F(k) = 6 B_N H(k_i)_i$ and the new kernel has abelian image $24 L_N$. The right side of the second equation is affine in h: $t^{72} \in A$, and A is abelian. The common multiplier range is $1 + 72h \pmod{288}$.

For clarity, the action matrices in (E.13.4) use the ungauged S_0, R_0. The normalized pair is obtained by conjugation by $x^g = t^{-2g}$, which fixes t. Thus it preserves both order

equations, and its conjugation of $k \in F$ preserves the exact F^{ab} coordinates. This supplies the comparison used when restoring the cusp gauge.

Here is the coefficient construction and the exhaustive linear solution. If conjugation by a word q sends n_i to normal coordinates $(A_q e_i, \tau_i)$, its action on (E.13.3) has blocks

$$\begin{pmatrix} A_q & 0 \\ T_q & \Lambda^2 A_q \end{pmatrix}, \qquad (T_q)_{(a,b),i} = 2(\tau_i)_{ab} - (A_q)_{ai}(A_q)_{bi} \pmod 4, \tag{E.13.5}$$

where $a < b$ indexes a normal commutator coordinate. The formula follows by taking the sixth power of the image of n_i in (E.13.2) and then dividing its central coordinate by three. Conjugation by an additional normal word with abelian coordinate c adds $2(c_{i_1}(A_q)_{i_2 i} - c_{i_2}(A_q)_{i_1 i})$ to that block. These formulas compute every coefficient.

The program `de_class2_eight_fibres.py` reduces the 9508 by 4754 system modulo two. Its rank is 3927 and its nullspace has dimension 827, including the multiplier variable. For a particular solution x_0 and a binary nullspace matrix K, write every possible lift as $x = x_0 + Kz + 2y$. The second-layer equation is exactly

$$A_0 y + \frac{A_4 K}{2} z = \frac{b_4 - A_4 x_0}{2} \pmod 2, \tag{E.13.6}$$

where the numerators are even and are reduced modulo four before division. This system has coefficient rank 3953 and is consistent in each of the 24 cases. It retains all 827 nullspace parameters, so it tests the complete fibre above each fixed affine homology point. A solution with $h = 0$ is retained in every case. Direct substitution verifies both order equations and the two commutator-image rows.

We then form actual words at the common refinement. Let k_2 be the retained $\mathbb{Z}/4$ correction, and let k_3 be the three-primary correction already retained in E.11. Set

$$k = \prod_i n_i^{6(k_2)_i} \prod_{i<j} [n_i, n_j]^{3(k_2)_{ij} + 4(k_3)_{ij}}.$$

Its exponent-eight image is k_2, and its exponent-three image is k_3. Since $H((k_2)_i)$ is divisible by four, the same unimodular two-column calculation as in E.11 gives an explicit product j of twenty-fourth powers of basis words satisfying $\mathrm{ab}_F(kj) = 0$. It is trivial in both factors of $J_{8,3}$. Thus

$$f = s x^g S_0 k x^{-g} j \in F' \tag{E.13.7}$$

is a single actual word for the common shadow.

The final program `de_fibre_eight_crt_verify.py` substitutes both complete braid hexagons for (E.13.7) in both faithful collectors: there are 96 successful component checks. The exponent-eight collector also checks sixteen block actions against literal expanded conjugate words. It keeps the section comparison in the Schreier calculation: if the short representative is $n(c)$ times the spanning-tree representative, the latter's central action is $\tau_{\mathrm{short}} - c \wedge A_q$. This is the inner correction required by (E.13.2).

The new corrections preserve the verified G_6 homology shadow. Both 97-dimensional Frattini ranks remain maximal, so generation follows on the two primary kernels. The actual commutator word, unit multiplier, complete hexagons and generation prove the assertion. The JSON retains all literal base words and the factored correction and adjustment words. The quotient still kills $\gamma_3 N$; the explicit A_5 comparison at the end of E.12 therefore applies to this refinement as well. No cofinality conclusion follows from these positive finite fibres. □

5.6 The exponent-sixteen fibres and the two-adic lifting argument

Statement E.14 (Proved). Every CM matrix fibre survives the common exponent-sixteen refinement. *Let $J_{16,3} = (\gamma_3 N\, N^{16}) \cap (\gamma_3 N\, N^3)$. Its inverse image in B_3 is a finite-index normal subgroup contained in PB_3. Every one of the 24 admissible CM matrix pairs of E.1 occurs as a gentle shadow at this refinement, with multiplier one modulo* 576.

Proof. The exponent-sixteen collector has coordinates $a \in (\mathbb{Z}/16)^{97}$ and $z \in (\mathbb{Z}/8)^{4656}$, with the product in (E.13.2). The image of $N'N^6$, retaining $N^{\mathrm{ab}}/3$, has normal coordinates

$$k = \prod_i n_i^{6u_i} \prod_{i<j} [n_i, n_j]^{3v_{ij}}, \qquad (u, v) \in (\mathbb{Z}/8)^{4753},$$

and exact product

$$(u, v)(u', v') = (u + u',\, v + v' - 12(u_j u'_i)_{i<j}). \tag{E.14.1}$$

The correction group is therefore nonabelian at this level; the program retains its full product and inverse.

For each affine homology point of E.11 use the same ungauged $S_0 = \beta s \beta^{-1}$ and $R_0 = S_0 t$. With $\lambda = 1 + 72h$, the complete equations are

$$(S_0 k)^2 = 1, \qquad (k^{-1} S_0 t^{1+72h})^3 = 1, \qquad Hu = 0 \pmod 8. \tag{E.14.2}$$

Here H is the two-row integer matrix of E.13. The last condition is exact: the common kernel has F-abelian image $48L_N$, and the correction has image $6B_N Hu$.

Write $x = (u, v, h)$ and let $\mathcal{F}(x)$ be the two normal-coordinate order errors followed by Hu. These are integer polynomials of degree at most two. Their diagonal quadratic coefficients are divisible by six and their mixed quadratic coefficients by twelve. Indeed the sixth-power contribution of an automorphism with abelian matrix A and generator cochain τ is

$$2u_i\tau_i - (6u_i^2 - u_i)A_{ai}A_{bi}$$

in a central coordinate; reordering contributes multiples of $12u_iu_j$. Cusp powers have the same divisibility, since $12\binom{h}{2} = 6h^2 - 6h$. Only linear homology expressions are substituted into these formulas. Thus $\mathcal{F}$ modulo four is exactly the affine system of E.13.

Its homogeneous solutions are represented by

$$Vt + 2Kw \pmod 4, \qquad t \in \mathbb{F}_2^{801}, \quad w \in \mathbb{F}_2^{827}, \tag{E.14.3}$$

where K is the entire binary kernel and V lifts the kernel of the second-layer obstruction. Every column is verified modulo four. The projection of V mod 2 to the 97 homology coordinates and the multiplier has rank 15. Select 15 columns spanning that projection and subtract their combinations from the other columns. The remaining 786 columns have even homology and multiplier coordinates. Twice a selected column is already represented by $2K$, so this change preserves all solutions in (E.14.3).

Every direction with even homology and multiplier coordinates changes $\mathcal{F}$ modulo eight linearly and independently of the base point: the stated divisibilities kill its quadratic cross terms. Hence all 786 remaining lower parameters, all 827 upper parameters, and the arbitrary correction $4y$ enter linearly; only the 15 selected bits enter quadratically. The program `de_class2_sixteen_fibres.py` retains the full linear system and evaluates all 32768 choices of those bits. In each of the 24 cases the linear correction rank is 3953, and every constant, linear, and quadratic coefficient vanishes modulo its image. All 32768 choices survive. A

solution with $h = 0$ is retained in every case and substituted into both literal order equations and Hu.

For these two-primary coordinates and the three-primary central correction k_3 of E.11, the common actual word is

$$k = \prod_i n_i^{6u_i} \prod_{i<j} [n_i, n_j]^{3v_{ij}+16(k_3)_{ij}}. \tag{E.14.4}$$

The unimodular two-column calculation of E.11 supplies a product j of forty-eighth powers with $\mathrm{ab}_F(kj) = 0$. It is trivial in both factors. Restoring the recorded cusp gauge gives $f = sx^g S_0 k x^{-g} j \in F'$. The program `de_fibre_sixteen_crt_verify.py` substitutes both full braid hexagons in both faithful collectors, with 96 successful component checks. Normal-block inversion retains the central term $-a_i a_j$. Twelve additional literal conjugation checks include a nontrivial change of section. Both Frattini ranks remain 97 because the corrections preserve the G_6 homology point. This proves the full shadow and generation conditions.

The separate `de_class2_sixteen.py` finds no lift of the 24 particular words retained in E.13. Its projection fixes their upper corrections. Calculation (E.14.3) permits all corrections over the homology point and supplies the replacements above. These exact projection maps explain the scopes of the two calculations. □

Statement E.15 (Proved). Every admissible CM pair survives the entire relative class-two tower. *For each of the 24 admissible CM matrix pairs there is a compatible family of gentle shadows at the inverse images in B_3 of*

$$J_j = \gamma_3 N\, N^j \qquad (6 \mid j), \tag{E.15.1}$$

ordered by divisibility, with multiplier $\lambda = 1$. All twelve extra matrix fibres therefore survive this entire tower. The assertion specifies matrix fibres; it does not assert extension of the particular normalized words retained in E.13 or E.14.

Proof. We construct the two-primary component first. Let $\mathcal{N}_2$ be the free class-two pro-two group on the 97 basis elements, in the extension of Q obtained by completing N in this quotient. Let $\mathcal{U} \subset \mathcal{N}_2$ have even abelian coordinates. Multiplication by three is invertible over $\mathbb{Z}_2$, so the elements $n_i^6, [n_i, n_j]^3$ give its normal coordinates $(u, v) \in \mathbb{Z}_2^{4753}$, with product (E.14.1). The polynomial bijection

$$Z = (u, z), \qquad z_{ij} = v_{ij} + 6u_i u_j$$

puts its operations in the form

$$Z * Z' = Z + Z' + \tfrac{1}{2}[Z, Z'], \qquad [Z, Z'] = (0, 12u \wedge u'). \tag{E.15.2}$$

Every coefficient is integral. We use $\exp Z$ and log for this explicit group-coordinate correspondence; (E.15.2) justifies that notation at the prime two.

Conjugation by S_0 is an integral Lie automorphism M_S with $M_S^2 = 1$. The first order equation is equivalent to

$$G_S(Z) = (1 + M_S)Z = 0. \tag{E.15.3}$$

In fact its logarithm is $G_S(Z) + \frac{1}{2}[G_S(Z), Z]$; the map $Y \mapsto Y + \frac{1}{2}[Y, Z]$ is an integral unipotent automorphism. The target of G_S is the saturated submodule $\mathrm{Fix}(M_S)$.

Put $E = \log(R_0^3)$ and $r_* = \exp(-E/3)R_0$. Here $R_0^3 \in \mathcal{U}$, division by three is integral, and R_0 commutes with its cube; thus $r_*^3 = 1$. Write M for its Lie action and $P = (1 + M + M^2)/3$

for the integral projection onto $\mathrm{Fix}(M)$. For $k = \exp Z$, keep the multiplier equal to one and write

$$R = k^{-1} S_0 t = \exp(W) r_*, \qquad C = \log(R^3).$$

The coordinates of W and C are integral polynomials of degree at most two in Z, by (E.15.2). The second order equation is equivalent to

$$G_R(Z) = PC = 0. \tag{E.15.4}$$

To prove this integrally, the homology of C is the norm of the homology of W, so it is fixed by M and equals that of PC. Since R^3 commutes with R, $(M-1)C = -[W, C]$. On $\ker P$ the inverse of $M - 1$ is $-(M+2)/3$. With $Q_0 = 1 - P$ this gives

$$C = PC + \frac{M+2}{3} Q_0 [W, PC]. \tag{E.15.5}$$

The bracket uses only homology. Thus $PC = 0$ is equivalent to $C = 0$, and (E.15.5) is an integral linear embedding of the error PC, with left inverse P.

There is only one further independent commutator equation. The integral matrix H satisfies

$$H A_S = \begin{pmatrix} -2 & -1 \\ 3 & 2 \end{pmatrix} H, \tag{E.15.6}$$

where A_S acts on N^{ab}. Equivalently, apply B_N^{-1} to the interchange of the two coordinates of F^{ab}. The first row of $H(1 + A_S)u$ is $-(H_1 + H_2)u$. Hence (E.15.3) and $H_2 u = 0$ imply both commutator rows, and the converse holds. The exact compressed system is the integral polynomial map

$$G : \mathbb{Z}_2^{4753} \longrightarrow \mathrm{Fix}(M_S) \oplus \mathrm{Fix}(M) \oplus \mathbb{Z}_2, \qquad G = (G_S, G_R, H_2 u). \tag{E.15.7}$$

The target lattices are primitively embedded. The normal-coordinate changes and (E.15.5)–(E.15.6) recover the full error vector from G, with primitive linear part at a zero. At a point whose errors are divisible by eight the same derivative comparison holds modulo four, since the additional derivative terms multiply those errors.

Here are the exact target and derivative ranks. The cycle actions have traces $-7, -5, -5$ for s, r, r^2. The exterior-square traces are $-24, 15, 15$, respectively, from $\mathrm{tr}(\Lambda^2 A) = ((\mathrm{tr}\, A)^2 - \mathrm{tr}(A^2))/2$. Central cochains do not change a trace. It follows that

$$\mathrm{rank}\, \mathrm{Fix}(M_S) = 2361, \quad \mathrm{rank}\, \mathrm{Fix}(M) = 1591, \quad \mathrm{rank}\, \mathrm{target}(G) = 3953. \tag{E.15.8}$$

The program `de_two_hensel_tangent.py` checks these traces and (E.15.6). At each of the 24 points from E.14 it computes the full integral Jacobian modulo four. The program `de_two_hensel_fixed.py` then deletes the multiplier column, recomputes the entire binary kernel, and verifies all the ranks below for the resulting 4753-variable system. Its binary kernel has dimension 826. The divisibilities in the proof of E.14 give

$$\mathcal{F}(x + 2e_i) - \mathcal{F}(x) = 2J_i \pmod 8$$

and make $J \bmod 4$ independent of x. The central columns are the exterior-square norm matrices. All homology columns are retained, with five additional complete direction checks per point. For the full binary kernel K, the exact ranks are

$$\mathrm{rank}_{\mathbb{F}_2}(J \bmod 2) = 3927, \qquad \mathrm{rank}_{\mathbb{F}_2} \left(J \bmod 2 \;\middle|\; \frac{JK}{2} \bmod 2 \right) = 3953. \tag{E.15.9}$$

The primitive comparison with (E.15.7) identifies these with the ranks for dG. Its target has rank 3953, so its elementary divisors consist of 3927 units and 26 elements of valuation one. Its image contains twice the target lattice.

The lifting can now be given explicitly. At the integral point x_0 retained in E.14, $G(x_0)$ is divisible by eight. Choose a right inverse of dG_{x_0} over $\mathbb{Q}_2$ carrying the integral target into $\frac{1}{2}\mathbb{Z}_2^{4753}$; the elementary divisors give such an inverse. The Newton correction belongs to $4\mathbb{Z}_2^{4753}$. The new error is divisible by sixteen, because G is integral of degree at most two, and its derivative is unchanged modulo four. Repeating, an error of valuation $v \geq 3$ is replaced by one of valuation at least $2v-2$. The corrections converge to an exact zero of G, with multiplier still equal to one. Equations (E.15.3)–(E.15.6) give both exact orders and zero F-abelianization for the correction k. The retained cusp gauge commutes with t and makes the normalized f have zero abelianization, by the formula (E.15.12) below. The reduction on $N^{\mathrm{ab}}/2$ has rank 97, so the pro-two Frattini argument gives an automorphism at every finite level.

We give the gluing before applying the cusp gauge. For the same CM matrix pair, take the ungauged three-primary action of E.12, and take the identity action at every other prime. In each component the images of s, r, t in Q are s, r, t: the matrix representatives β, A centralize s, r in Q, and all their corrections belong to the normal kernel. Thus these ungauged actions have the same identity action on Q. The precise comparison of their target groups is

$$\varprojlim_{j} B/(\gamma_3 N\, N^j) \simeq \prod_{p,Q} \varprojlim_{a} B/(\gamma_3 N\, N^{p^a}), \tag{E.15.10}$$

where the product is the fibre product over Q. On the normal kernel this is the Sylow decomposition of the class-two completion of N: every finite nilpotent quotient is the product of its Sylow groups. For surjectivity at finitely many levels, lift their common Q coordinate and adjust independently in these normal Sylow groups. Compactness gives (E.15.10).

We now specify one gauge in this common group. Let $g_2 \in \mathbb{Z}_2$ be the integral gauge retained in E.14. For the limiting three-primary conjugator β_3, let $g_3 \in \mathbb{Z}_3$ be the difference of its second and first F-abelian coordinates. The E.12 corrections preserve these coordinates modulo three, so the reduction of g_3 is that of the recorded integral gauge. The image of N in F^{ab} has basis matrix $B_N = \left(\begin{smallmatrix} 6 & 4 \\ 0 & 2 \end{smallmatrix}\right)$; therefore every integral change in this difference is even. The two-primary residue of the three-primary gauge can accordingly be kept equal to that of g_2. Choose $g \in \widehat{\mathbb{Z}}$ with these two primary components and with $g_p = 0$ for $p \neq 2, 3$. The element $x = t^{-2}$ has order six in Q. Consequently x^g restricts to the specified cusp gauge in the two- and three-primary extensions. Apply its inner conjugation to the glued ungauged action, obtaining S', R' with $S'R' = t$.

Put $f = sS'$. Its common Q coordinate is

$$q(f) = s x^{\bar{g}} s x^{-\bar{g}}, \qquad \bar{g} = g \bmod 6. \tag{E.15.11}$$

In particular f need not lie in the image of N. The independent matrix program `de_adelic_gauge.py` verifies the identity ungauged Q action and (E.15.11) for all 24 pairs. For $\bar{g} = 0, 4, 2$, respectively, its canonical matrix representatives in $\mathrm{SL}_2(\mathbb{Z}/12)/\{\pm 1\}$ are

$$I, \qquad \begin{pmatrix} 1 & 4 \\ 4 & 5 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 8 \\ 8 & 5 \end{pmatrix}.$$

The exact abelianization formula is

$$\mathrm{ab}_F(s x^g S x^{-g}) = \mathrm{ab}_F(sS) + (-g, g). \tag{E.15.12}$$

At two and three the selected gauges make this zero. At every other prime the ungauged action is identity, and $g_p = 0$, so it is zero there also. In those components the final action is

inner by x^g; its image can have nontrivial prime-to-p order dividing six, exactly as (E.15.11) requires. Hence f has zero full F-abelianization. Since $\gamma_3 N \subset F'$, the kernel of the resulting map to $\widehat{F}^{\mathrm{ab}} = \widehat{\mathbb{Z}}^2$ is the image of $\widehat{F}'$, and f belongs to that image. Both order equations hold in every component. The exact hexagon equivalence of E.9 now supplies compatible gentle shadows. Generation holds on each normal Sylow kernel by its Frattini map at two and three and by the specified inner maps at the other primes; it also holds on Q. Each finite map is therefore an automorphism. The two- and three-primary reductions recover the prescribed CM matrix pair, proving the family.

Every subgroup in (E.15.1) contains $\gamma_3 N$. The map to A_5 in E.5 sends that subgroup onto A_5, so its common refinement factors through none of these quotients. This explicit comparison leaves the cofinal lifting problem of E.2 unresolved. □

Statement E.16 (Proved). Simultaneous extension through every nonabelian simple quotient. *Let Σ be any finite collection of epimorphisms $\theta : B \twoheadrightarrow S_\theta$ to nonabelian finite simple groups, and put*

$$K_{j,\Sigma} = \gamma_3 N\, N^j \cap \bigcap_{\theta \in \Sigma} \ker \theta, \qquad 6 \mid j. \tag{E.16.1}$$

Every one of the 24 CM matrix pairs has a compatible family of gentle shadows at the inverse images of these subgroups in B_3, with multiplier one, as j increases by divisibility and Σ increases by inclusion. Thus the class-two families of E.15 extend simultaneously through all these nonabelian simple quotients. For each fixed epimorphism $B \twoheadrightarrow R$ to an arbitrary finite perfect group, they also extend compatibly in j through $\gamma_3 N\, N^j \cap \ker(B \to R)$.

The finite solvable quotient $B/(\gamma_4 N\, N^5)$ is not a quotient, through the natural map from B, of any $B/K_{j,\Sigma}$. This gives a precise remaining finite comparison outside this enlarged family.

Proof. Write $E_j = B/(\gamma_3 N\, N^j)$ and let R_Σ be the image of B in $\prod_{\theta \in \Sigma} S_\theta$. The latter is a product of nonabelian simple groups, with repeated diagonal factors identified. To prove this, adjoin the simple factors one at a time. The kernel of projection to the previous factors, viewed in the new simple group, is either the whole group or trivial. In the first case the new group is a direct product. In the second, the new factor is a simple quotient of the preceding product and is the graph of an isomorphism from one of its simple factors. This proves the claim by induction. In particular R_Σ is perfect.

The group Q is solvable. Indeed, $\mathrm{SL}_2(\mathbb{Z}/12)$ is the product of its reductions modulo four and three. The kernel of reduction $\mathrm{SL}_2(\mathbb{Z}/4) \to \mathrm{SL}_2(\mathbb{F}_2)$ is an elementary abelian group of order eight, and the target is $\mathfrak{S}_3$. The other factor has a normal quaternion group of order eight with cyclic quotient of order three. Passing to the quotient by $\{\pm I\}$ preserves solvability. Since the kernel of $E_j \to Q$ is nilpotent, E_j is solvable as well.

The image of B in $E_j \times R_\Sigma$ projects onto both factors. For a subdirect product $H \subset U \times V$, put $U_0 = \{u : (u,1) \in H\}$ and $V_0 = \{v : (1,v) \in H\}$. Sending one coordinate of a pair in H to the other defines an isomorphism $U/U_0 \simeq V/V_0$. Here that common quotient is both solvable and perfect, hence trivial: the derived series of a solvable perfect group is simultaneously constant and eventually trivial. Therefore

$$B/K_{j,\Sigma} \simeq E_j \times R_\Sigma. \tag{E.16.2}$$

The same argument applies to the image of F'. First F maps onto R_Σ, since $B/F \simeq \mathfrak{S}_3$ and a quotient of a perfect group that is solvable is trivial. Then F' also maps onto R_Σ, which is perfect. Its image in E_j is solvable, so its joint image is the product of its two images.

For each E.15 shadow use its automorphism on E_j and the identity on R_Σ. Under (E.16.2) the conjugator is $(f_j, 1)$, which is in the image of F' by the preceding paragraph. Both hexagons hold coordinatewise with multiplier one, and the product map is an automorphism. These constructions commute with every transition map. Equivalently, compactness applied to the joint images of $\widehat{F}'$ gives one lift of the compatible pair $(\{f_j\}, 1)$. This proves the claimed compatible family. For one fixed arbitrary perfect target R, the same subdirect-product and F' arguments apply directly to $E_j \times R$ and give the additional assertion. For a collection of such targets the proof above uses perfection of their *joint image*; the simple-factor induction proves exactly that property for the simultaneous family in (E.16.1).

For the final comparison set $L = \gamma_4 N\, N^5$. It is characteristic in N, hence normal in B, and has finite index because N is finitely generated and its class-three exponent-five quotient is finite. The quotient B/L is solvable. Every homomorphism from (E.16.2) to a solvable group kills R_Σ, since its image is perfect. Therefore a factorization of the natural map $B \to B/L$ through (E.16.2) would factor through E_j and kill $\gamma_3 N$.

Here is an explicit word that it does not kill. Send the first two free generators of N to

$$a = I + E_{12} + E_{34}, \qquad b = I + E_{23} \quad \text{in } U_4(\mathbb{F}_5),$$

and send the others to the identity. The target has class three and exponent five: commutators add superdiagonal degrees, and $(I + A)^5 = I$ for a strictly upper triangular 4×4 matrix A in characteristic five. With $[a, b] = aba^{-1}b^{-1}$, direct multiplication gives $[[a, b], a] = I + 2E_{14} \neq I$. The program `de_solvable_barrier.py` checks these literal matrices. Hence $[[n_0, n_1], n_0] \in \gamma_3 N \setminus L$, contradicting that factorization. This proves the claimed finite comparison and the exact scope of the enlarged family. □

5.7 A four-strand pentagon constraint on the CM fibres

Statement E.17 (Proved). The complete Ferrari-homology pentagon test and its affine fibres. *Put $H_3 = \mathbb{F}_3^2 \rtimes \mathrm{SL}_2(\mathbb{F}_3)$, with the standard left action on column vectors. The assignments*

$$\begin{aligned} \rho_4(\sigma_1) &= (0, A), \quad A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \\ \rho_4(\sigma_2) &= (0, B), \quad B = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \\ \rho_4(\sigma_3) &= (e_1, A) \end{aligned} \tag{E.17.1}$$

define an epimorphism $B_4 \to H_3$. Let $L_4 = \ker \rho_4 \cap PB_4$. Then L_4 is normal in B_4, is contained in PB_4, and $|B_4/L_4| = 5184$.

The complete pentagon in PB_4/L_4 is a well-defined condition on the G_6 conjugator coordinate used in E.1. Among the 24 particular G_6 representatives retained in E.14, exactly nine satisfy it. They have indices

$$3, 4, 5, 7, 8, 14, 15, 17, 23; \tag{E.17.2}$$

the extra-coset indices among these are $4, 7, 15$. The other fifteen particular representatives cannot be projections of full $\widehat{\mathrm{GT}}$ elements.

Every one of the 24 whole CM matrix fibres, *however, contains a gentle G_6 shadow with multiplier one whose actual commutator word satisfies this pentagon. In the 18-parameter three-primary presentation of E.10, before quotienting by cusp conjugations, its equation is one nonzero affine linear equation. Its solution space has dimension 17 and contains points with full generation rank 97 and with vanishing class-two cusp obstructions of E.11. This is a finite pentagon test; no lift through all four-strand quotients is asserted.*

Proof. We use the multiplication $(u, M)(v, T) = (u + Mv, MT)$ in H_3. The three assignments in (E.17.1) satisfy both adjacent braid relations and the relation between the distant generators, by direct multiplication. Their Ferrari kernel elements

$$\xi_1 = \sigma_3\sigma_1^{-1}, \qquad \xi_2 = \sigma_1\sigma_2\sigma_1\sigma_3^{-1}\sigma_2^{-1}\sigma_1^{-1}$$

map to $(e_1, I), (e_2, I)$, while A, B generate $\mathrm{SL}_2(\mathbb{F}_3)$. This proves surjectivity. These are the actual free Ferrari generators and their homology action: the source comparison is [NTY10, (2.9) and (4.15)]. The program `de_ferrari_pentagon.py` checks the literal relations and enumerates the joint image with the adjacent transpositions in $\mathfrak{S}_4$; its order is 5184. This is precisely B_4/L_4.

Write a_{ij} for the standard pure Artin generators. We evaluate the complete group equation

$$\begin{aligned} &f(a_{12}, a_{23}a_{24})f(a_{13}a_{23}, a_{34}) \\ &\qquad = f(a_{23}, a_{34})f(a_{12}a_{13}, a_{24}a_{34})f(a_{12}, a_{23}). \end{aligned} \tag{E.17.3}$$

Each displayed pair defines an actual homomorphism $F_2 \to PB_4$, followed by ρ_4. All five factors are pure, so the permutation component of (E.17.3) is already the identity. We also retain the evaluation at $((\sigma_1\sigma_2)^3, \sigma_3^2)$, which is the conjugator in the standard Drinfeld image of σ_3.

The simultaneous images of the 97 literal generators of N under these six homomorphisms commute and have exponent three. The program checks every generator cube and all $\binom{97}{2} = 4656$ commutators. It follows that each homomorphism kills $N'N^6$. Thus all six evaluations factor through the actual quotient $F_2/(N'N^6)$, which is the pure free factor in G_6. Corrections by the whole kernel have zero image in these targets. Consequently a failure of (E.17.3) at the computed representative is a failure for every lift of that G_6 conjugator. The retained E.14 word differs from the base word evaluated here by elements of $N'N^6$, so this comparison applies to the E.14 representatives themselves. Complete multiplication of all five factors gives exactly (E.17.2).

Here is the exhaustive affine reduction for whole CM matrix fibres. Let $P = \binom{B_0}{A_0}$ be the 194×18 parameter matrix retained in E.10. A parameter $z \in \mathbb{F}_3^{18}$ changes the two conjugators by $n(B_0 z), n(A_0 z)$. For an actual integral change b in the first conjugator, put

$$\ell(b) = \frac{\mathrm{ab}_{F,2}(n(b)) - \mathrm{ab}_{F,1}(n(b))}{2} \pmod 3.$$

The numerator is even by the matrix B_N of E.9. The new gauge is $g + 2\ell(b)$ modulo six. If $f_\beta = s\beta s\beta^{-1}$ denotes the ungauged base conjugator, the actual new word is

$$f_z = y^{g+2\ell(b)}(sn(b)s)f_\beta n(b)^{-1}x^{-g-2\ell(b)}. \tag{E.17.4}$$

This identity follows by inserting $s^2 = 1$ and $sxs = y$; it is the precise gauge comparison used in the calculation.

Embed each affine group in 3×3 matrices. On the commuting exponent-three images of N, the formulas

$$\log(I + D) = D + D^2, \qquad \exp(L) = I + L + 2L^2 \quad \text{over } \mathbb{F}_3$$

are inverse and turn products into sums. Their nilpotent matrices have cube zero, and every commutator is zero. A commuting family of nilpotent endomorphisms admits a common complete invariant flag: choose a nonzero common kernel vector, pass to its quotient, and repeat. In that basis these matrices are strictly upper triangular, so every product of three is zero. Expanding the two displayed polynomials therefore proves $\log(uv) = \log(u) + \log(v)$

for these commuting matrices, including characteristic three. The joint linear map recording the six logarithms of $n(b)$, those of $sn(b)s$, and $\ell(b)$ has rank two on the 18 parameter columns. The program `de_ferrari_fibres.py` retains that exact matrix, its two-column section, and its 16-dimensional kernel. It enumerates all nine values of the two-dimensional image. Three independent literal-word substitutions per fibre check (E.17.4) against the logarithmic evaluation.

For every one of the 24 fibres, the pentagon residual on those nine values is an affine linear map of rank one. Every value is retained in the JSON certificate, and direct substitution checks its equality with the retained linear map. Exactly three values satisfy the pentagon. Each also gives an invertible induced action on Ferrari homology; its two columns are the images of ξ_1, ξ_2 under the three Drinfeld generator formulas. The program checks both intertwining relations and all braid relations after this substitution.

Finally, in the remaining 16 parameters the program retains an actual point in every fibre at which the five class-two cusp equations of E.11 vanish and the complete 97×97 generation matrix is invertible. These are positive witnesses, not exclusions from a failed search. Their three-primary changes are represented by four times the corresponding integral N words, so their two-primary homology component is unchanged. The literal cusp relation in G_6, both generation ranks, the exact integral gauge, zero F-abelianization, and all five pentagon factors are checked again. The resulting commutator words and every parameter are retained. This proves nonemptiness in all 24 whole matrix fibres and the asserted finite scope. □

Statement E.18 (Proved). The next Ferrari central layer adds no equation on these affine fibres. *Let $\mathcal{H}_3$ be the class-two exponent-three free group on two generators, in coordinates $(u, v, z) \in \mathbb{F}_3^3$ with multiplication*

$$(u, v, z)(u', v', z') = (u + u', v + v', z + z' + 2(uv' - vu')).$$

The standard action of $\mathrm{SL}_2(\mathbb{F}_3)$ on (u, v) fixes z. Put $\widetilde{H}_3 = \mathcal{H}_3 \rtimes \mathrm{SL}_2(\mathbb{F}_3)$. The assignments

$$\sigma_1 \longmapsto ((0,0,0), A), \qquad \sigma_2 \longmapsto ((0,0,0), B), \qquad \sigma_3 \longmapsto ((1,0,2), A) \tag{E.18.1}$$

define an epimorphism $\widetilde{\rho}_4 : B_4 \to \widetilde{H}_3$, where A, B are those in (E.17.1). The normal pure kernel $\widetilde{L}_4 = \ker \widetilde{\rho}_4 \cap PB_4$ has index 15552. Reduction of the Heisenberg central coordinate gives the precise quotient $\widetilde{H}_3 \to H_3$ of E.17.

All six evaluations used in E.17 still factor through G_6. On each of the 24 whole affine CM fibres, their joint parameter image has dimension two, with exactly the same kernel as in E.17. The complete pentagon has exactly the same zero set on this image: three of its nine points. Every retained E.17 literal witness satisfies the complete pentagon and all three braid relations in $\widetilde{H}_3$, and its three braid images generate that group. Thus this additional central layer preserves all 24 nonempty fibres and introduces no further affine equation.

Proof. In the usual Baker–Campbell–Hausdorff coordinates of the free class-two group, write z_0 for the central coordinate and take the free generators to have $z_0 = 0$. The literal Ferrari actions (4.15) of [NTY10] have central linear terms $(0, 1)$ for A and $(1, 0)$ for B. The exact change of coordinates

$$z = z_0 - u - 2v$$

removes both terms: for $\ell = (1, 2)$, their rows are $\ell A - \ell = (0, 1)$ and $\ell B - \ell = (1, 0)$. Consequently the actual Ferrari generators become $(e_1, 2), (e_2, 1)$, and their commutator is $(0, 0, 1)$. This proves that (E.18.1) uses the actual Ferrari quotient. Direct multiplication verifies the braid relations and the four literal free-generator action formulas. These elements generate the Heisenberg kernel and $\mathrm{SL}_2(\mathbb{F}_3)$, proving surjectivity and the order $27 \cdot 24 = 648$.

The program `de_ferrari_class2_model.py` also verifies the Heisenberg multiplication and all 24 matrix actions exhaustively, then enumerates the joint image with $\mathfrak{S}_4$, of order 15552.

For each of the six evaluation homomorphisms, the simultaneous images of N have order at most nine. The program `de_ferrari_class2_fibres.py` enumerates each such subgroup and verifies that it is elementary abelian. In the seven displayed coordinates (a, b, c, d, u, v, z), subtracting the identity gives additive coordinates on each of these subgroups; every pair product is checked. All generator cubes and all 4656 generator commutators are therefore trivial. This proves the factorization through G_6.

Stack these additive coordinates for $n(B_0t)$ and $sn(B_0t)s$, together with the gauge row $\ell(B_0t)$ of E.17. The retained exact matrix has rank two. Stacking it with the old E.17 matrix still has rank two, so their kernels in $\mathbb{F}_3^{18}$ coincide. Formula (E.17.4) remains an exact literal-word identity in the larger group. The calculation substitutes all nine values of the two-dimensional image in every fibre. Its full three-coordinate pentagon residual vanishes precisely at the three points where the E.17 two-coordinate residual vanishes. Three further literal substitutions per fibre check the additive-coordinate evaluation. Finally, each of the 24 saved E.17 words is substituted directly; all five factors, all braid relations, and generation of the order-648 group are checked. The JSON retains every one of the $24 \cdot 9$ residuals, and the compressed certificate retains the complete parameter matrix, its section, and its kernel. □

Statement E.19 (Proved). Complete compatible corrections at the modulus-nine Ferrari test. *For $R = \mathbb{Z}/9$, let $\mathcal{H}(R)$ have coordinates $(u, v, z) \in R^3$ and multiplication*

$$(u, v, z)(u', v', z') = (u + u', v + v', z + z' + \tfrac{1}{2}(uv' - vu')), \qquad \tfrac{1}{2} = 5 \text{ in } R.$$

Put $H_9 = \mathcal{H}(R) \rtimes \mathrm{SL}_2(R)$, with the linear action on (u, v) and the identity on z. The assignments (E.18.1), with the final central coordinate changed from 2 to 5 and the matrices taken modulo nine, define an epimorphism $\rho_{4,9} : B_4 \to H_9$. Its composition with reduction modulo three is precisely $\widetilde{\rho}_4$ of E.18.

Let $\theta_i : F_2 \to H_9$, $1 \le i \le 6$, be the six actual evaluation maps of E.17. For $B = C_2 * C_3$, choose representatives b_q of $B/F_2 \simeq \mathfrak{S}_3$ and set

$$K_6 = N'N^6, \qquad K_{3,9} = K_6 \cap \bigcap_{i=1}^{6} \bigcap_{q \in B/F_2} b_q^{-1} \ker(\theta_i) b_q, \qquad D_9 = B/K_{3,9}. \tag{E.19.1}$$

Thus $K_{3,9}$ is normal of finite index in B and lies in F_2. The kernel of $D_9 \to G_6$ is $\mathbb{F}_3^4$, and the image of $K_6 \cap F_2'$ in that kernel is $\mathbb{F}_3^3$.

For each of the 24 particular G_6 points retained in E.17, all three commutator-correction parameters can be solved simultaneously. The gentle norm equations have rank two. The modulus-nine pentagon has rank three and selects exactly one correction in this three-dimensional image. The resulting literal words have zero integral F_2-abelianization, satisfy both complete hexagons in D_9, give surjective three-strand shadow maps, and satisfy the complete pentagon in H_9. Their three four-strand braid images satisfy all braid relations and generate H_9.

This yields nonempty compatible finite tests for all 24 CM fibres. In this particular construction $N'N^{18} \subset K_{3,9}$: the three-strand quotient is an explicit quotient of G_{18}. The new condition is the specified four-strand pentagon; no cofinal extension follows from this statement.

Proof. The exact coordinate change from the ordinary free class-two coordinates is $z = z_0 + (u - v)/2$. It removes the central terms in the two literal Ferrari actions over every ring in which two is invertible. The free generators therefore map to $(1, 0, 1/2)$ and $(0, 1, -1/2)$.

Direct multiplication verifies their four action formulas and all three braid relations over R. They generate $\mathcal{H}(R)$, and the two elementary matrices A, B generate $\mathrm{SL}_2(R)$: row reduction over the local ring R first chooses a unit in the first column, clears the other entry, and then expresses the resulting diagonal matrix as a product of elementary matrices. This proves the asserted epimorphism. Its target has order $9^3 \cdot 648 = 472392$.

Here is a faithful model for the core in (E.19.1). An element $b \in B$ has permutation $q \mapsto q\bar{b}$ of the six cosets and coordinates

$$h_{q,i}(b) = \theta_i(b_q b b_{q\bar{b}}^{-1}). \tag{E.19.2}$$

The product rule is $h_{q,i}(bc) = h_{q,i}(b)h_{q\bar{b},i}(c)$. Together with the original map to G_6, this is an actual faithful representation of D_9; its kernel is exactly (E.19.1). The program `de_ferrari_mod9_lift.py` constructs the six representatives and checks $s^2 = r^3 = 1$ in this model. Its evaluation on x, y is checked against the six original maps θ_i.

Reduction modulo three maps every K_6 coordinate to the identity by E.18. The kernel of $H_9 \to \widetilde{H}_3$ is elementary abelian: subtract the identity matrix and the zero Heisenberg vector, divide all coordinates by three, and reduce modulo three. Products add because every cross term contains a factor nine. The 36 coordinates in (E.19.2) therefore give an injective linear coordinate map on the image of K_6.

The complete K_6 image is computed from the 4656 pairwise commutators of the 97 actual Schreier generators n_i of N, their sixth powers, and their conjugates by the n_i. These normally generate $N'N^6$ in N. All pairwise commutator images are zero, and all ninth powers are the identity. Thus this particular core factors through G_{18}. The span of the sixth-power images has rank four; stability under every n_i is checked. This proves the exact kernel assertion.

The zero-abelianization restriction is imposed integrally. With the previously retained matrix $B_N = \left(\begin{smallmatrix} 6 & 4 \\ 0 & 2 \end{smallmatrix}\right)$, write

$$\mathrm{ab}_{F_2}(n_i) = B_N h_i, \qquad H = (h_i)_{0 \le i < 97}.$$

The columns numbered $0, 8$ form a determinant-one minor of H. They give an explicit integral basis of the rank-95 lattice $\ker H$. Modulo the commutator image, every element of K_6 is represented by $\prod_i n_i^{6a_i}$. Its F_2-abelianization vanishes exactly when $a \in \ker H$. Evaluating this entire integral lattice gives a rank-three image. The code retains three literal words forming a basis; their integral abelianizations are checked to be zero. Their images are also checked to be stable under s, r. This proves that the allowed correction space is the entire image of $K_6 \cap F_2'$, with no omitted power corrections.

For a retained E.17 conjugator β, put

$$S_0 = \beta s \beta^{-1}, \qquad t = sr, \qquad R_0 = S_0 t.$$

Let $M = \mathrm{im}(K_6 \cap F_2' \to D_9)$, and write S, R for conjugation by S_0, R_0 on M. Replacing S_0, R_0 by $S_0 k, k^{-1} R_0$ preserves their product. Since M is abelian, the two torsion equations are exactly

$$(1 + S)k = 0, \qquad (1 + R + R^2)k = R_0^3. \tag{E.19.3}$$

All terms lie in the explicitly computed image M; the code checks this by expressing them in the retained basis. These equations have rank two in every fibre.

Let g be the exact integral cusp gauge of the E.17 word. The corrected conjugator is the literal commutator word

$$f = s x^g S_0 k x^{-g}. \tag{E.19.4}$$

For each evaluation map, changing k changes f by an element of the elementary abelian congruence kernel. Multiplying the five pentagon factors therefore gives an affine linear

equation on M. The calculation evaluates all $3^3 = 27$ correction values in each of the 24 fibres, checking every coefficient of this linear description. The pentagon matrix has rank three; the three braid-relation matrices have rank zero. The joint system with (E.19.3) is consistent of rank three, and so selects one point of M in every case.

The retained point is then verified by complete literal substitution, without using only its linear equations. The two hexagons are checked in all 36 induced coordinates, and hold in G_6 by E.17; hence they hold in the faithful combined quotient D_9. The word (E.19.4) has zero integral abelianization because both its base word and k do.

For generation, the new three-strand map is already onto G_6. Substitution of all 97 generators, their commutators and sixth powers, followed by closure under their conjugations, gives rank four in the kernel of $D_9 \to G_6$. Thus the map is onto D_9. This supplies the finite gentle-shadow generation condition, with multiplier one. For the four-strand map, the first two matrix images generate all 648 elements of $\mathrm{SL}_2(R)$, as directly enumerated. The two Ferrari generators have trivial matrix part and determinant nonzero modulo three on their two Heisenberg homology coordinates. They therefore generate the finite three-group $\mathcal{H}(R)$ by its Frattini quotient. The whole four-strand image is H_9.

Every correction word, source word, linear matrix and right-hand side is retained in the JSON and compressed certificates. The distinction between these specified finite tests and a compatible lift through every four-strand quotient is exactly the untested extension beyond (E.19.1); it is not removed here. □

5.8 Arbitrary relative kernels supported away from two and three

Statement E.20 (Proved). Compatible extension through every relative quotient of order prime to six. *In this statement $N = N_B$ is the rank-97 kernel of $B = C_2 * C_3 \to Q$, and hats denote full profinite completions. Let $\mathscr{P}$ be the set of primes other than two and three. For each open subgroup $L \triangleleft \widehat{B}$ contained in $\widehat{N}$ with $[\widehat{N} : L]$ prime to six, and integers $a, b \geq 1$, put*

$$K_{a,b,L} = \overline{(\gamma_3 N\, N^{2^a 3^b})} \cap L. \tag{E.20.1}$$

Every one of the 24 admissible CM matrix pairs has a compatible family of gentle shadows at the inverse images of these subgroups in $\widehat{B}_3$, with multiplier one. The family is compatible as a, b increase and as L decreases, and its projection to every quotient $B/(\gamma_3 N\, N^j)$ in E.15 is the family constructed there. In particular it includes simultaneous refinement by arbitrary relative pro-p quotients for every $p \geq 5$, of every nilpotency class, and by finite relative kernels whose orders involve several primes in $\mathscr{P}$.

Here the three-strand action is constructed on one actual profinite quotient, together with a conjugator in the image of $\widehat{F}_2'$. No extra pentagon assertion is made. The natural map to the finite quotient

$$B/(\gamma_4 N\, (\gamma_2 N)^3 N^9) \tag{E.20.2}$$

factors through none of the quotients in (E.20.1). Its normal kernel over Q is a finite three-group with a nonzero third lower-central term, exhibited below. Thus (E.20.1) supplies further compatible refinements while leaving this specified relative three-primary comparison.

Proof. Write $\mathcal{N}_{23}$ for the product of the free class-two pro-two and pro-three completions of N and let E_{23} be the corresponding quotient of $\widehat{B}$. It fits into the exact sequence

$$1 \longrightarrow \mathcal{N}_{23} \longrightarrow E_{23} \longrightarrow Q \longrightarrow 1.$$

Let $R_{\mathscr{P}}$ be the intersection of the kernels of all continuous finite quotients of $\widehat{N}$ whose orders have prime factors only in $\mathscr{P}$. This subgroup is characteristic in $\widehat{N}$, hence normal in $\widehat{B}$. Put

$$E_{\mathscr{P}} = \widehat{B}/R_{\mathscr{P}}, \qquad \mathcal{N}_{\mathscr{P}} = \widehat{N}/R_{\mathscr{P}}. \tag{E.20.3}$$

Taking the $\widehat{B}$-core of a finite-index normal subgroup of $\widehat{N}$ preserves this prime support: its quotient embeds in a product of at most $|Q|$ conjugate quotients. Thus the subgroups L in the statement give the full topology of (E.20.3).

The simultaneous quotient is exactly the fibre product

$$H = E_{23} \times_Q E_{\mathscr{P}}. \tag{E.20.4}$$

To prove that the natural map $\widehat{B} \to H$ is onto, first take finite quotients of the two normal kernels. The image of $\widehat{N}$ in their product surjects onto both factors. The two Goursat quotient groups have a common order dividing a number supported on $\{2, 3\}$ and a number supported on $\mathscr{P}$. That order is one. Hence the image of $\widehat{N}$ is the full product. Lifting the common Q coordinate then gives surjectivity on the entire finite fibre product. The image of the compact group $\widehat{B}$ is closed; its surjectivity in every finite projection proves (E.20.4).

The kernel J of $\widehat{B} \to H$ satisfies

$$J \subset \widehat{N}' \subset \widehat{F}_2'. \tag{E.20.5}$$

Indeed the $\mathcal{N}_{23}$ coordinate detects every two- and three-primary coordinate of $\widehat{N}^{\mathrm{ab}} \simeq \widehat{\mathbb{Z}}^{97}$, and $\mathcal{N}_{\mathscr{P}}$ detects every other primary coordinate. All these coordinates vanish for an element of J, proving the first inclusion. This also proves the precise commutator comparison needed after gluing. Let H_F be the inverse image of F_2/N under $H \to Q$. There is a continuous homomorphism

$$H_F \longrightarrow \widehat{F}_2^{\mathrm{ab}}, \qquad \ker(H_F \to \widehat{F}_2^{\mathrm{ab}}) = \mathrm{im}(\widehat{F}_2' \to H_F). \tag{E.20.6}$$

This is the map induced by abelianization, using (E.20.5); the asserted equality follows by taking any lift in $\widehat{F}_2$.

Fix one CM pair and use the ungauged two- and three-primary automorphism α_{23} constructed in E.15. Its action on Q is the identity, and it fixes the actual cusp element $t = sr$. Therefore

$$\alpha : H \longrightarrow H, \qquad (h_{23}, h_{\mathscr{P}}) \longmapsto (\alpha_{23}(h_{23}), h_{\mathscr{P}}) \tag{E.20.7}$$

is a continuous automorphism. It is invertible by the same formula with α_{23}^{-1} and fixes t. This construction uses the identity on the entire relative $\mathscr{P}$ quotient, including all of its commutator layers.

Take exactly the gauge $g \in \widehat{\mathbb{Z}}$ of E.15: its two- and three-primary components are the gauges there, and $g_p = 0$ for every $p \in \mathscr{P}$. With $x = t^{-2}$, define

$$\alpha' = \mathrm{Inn}(x^g) \circ \alpha, \qquad S' = \alpha'(s), \quad R' = \alpha'(r), \quad f = sS'. \tag{E.20.8}$$

The gauge centralizes t, so $S'^2 = R'^3 = 1$ and $S'R' = t$. The word f belongs to H_F: the common Q coordinate is precisely $sx^{\bar{g}}sx^{-\bar{g}}$, as in E.15.11, and its image in B/F_2 is the identity.

We check its full abelianization, rather than just its reduction at two and three. Formula E.15.12 gives

$$\mathrm{ab}_{F_2}(f) = \mathrm{ab}_{F_2}(s\alpha(s)) + (-g, g). \tag{E.20.9}$$

The selected two- and three-primary gauges make these components zero. For $p \in \mathscr{P}$, the abelianization map to $\mathbb{Z}_p^2$ factors through $E_{\mathscr{P}}$: its restriction to $\widehat{N}$ has finite p-group quotients, all retained in (E.20.3). On that component $\alpha(s) = s$, so the first term of (E.20.9) is zero,

and $g_p = 0$. Consequently every primary component vanishes. By (E.20.6), f is the image of an actual element of $\widehat{F}_2'$.

The torsion-to-hexagon comparison in E.9 applies to these exact elements. In particular $\alpha'(x) = x$, and, since $y = sxs$,

$$\alpha'(y) = S'xS' = f^{-1}yf. \tag{E.20.10}$$

Both complete gentle hexagons hold with multiplier one. The component automorphism α_{23} preserves every class-two finite level used in E.15; the identity and its final inner conjugation preserve every normal subgroup defining a quotient of $E_{\mathscr{P}}$. Thus α' induces an automorphism at every finite quotient in (E.20.1). This proves generation, both equations, and all transition compatibilities at once. Projection to the class-two quotient of each primary kernel gives the same ungauged actions and the same single gauge as E.15. Hence the projected family is exactly the one specified there and retains its CM pair.

We finish by proving the remaining finite comparison. Set $L_3 = \gamma_4 N\,(\gamma_2 N)^3 N^9$, with the closed interpretation in completions. Its quotient in N is finitely generated, has class at most three and exponent dividing nine, and hence is a finite three-group. Finiteness follows directly by collecting words into the finitely many basic commutators of weights at most three, each with bounded exponent. The subgroup L_3 is characteristic in N, so B/L_3 is an actual finite quotient.

The map from N to $U_4(\mathbb{F}_3)$ given by

$$n_0 \longmapsto I + E_{12}, \quad n_1 \longmapsto I + E_{23}, \quad n_2 \longmapsto I + E_{34}, \quad n_i \longmapsto I \quad (i > 2) \tag{E.20.11}$$

kills L_3. The target has class three, its commutator subgroup has exponent three, and $(I + A)^3 = I + A^3$ is central for a strictly upper triangular 4×4 matrix A over $\mathbb{F}_3$; thus its exponent divides nine. Literal multiplication gives

$$[[I + E_{12}, I + E_{23}], I + E_{34}] = I + E_{14} \neq I. \tag{E.20.12}$$

It follows that $[[n_0, n_1], n_2]$ has nonzero image in N/L_3.

Suppose the natural quotient map $B \to B/L_3$ factored through one finite fibre product in (E.20.1). The induced homomorphism to Q agrees with its given projection, because the maps agree on the image of B. The normal kernel supported on $\mathscr{P}$ therefore maps into N/L_3. Its image has order prime to three and also three-power order, so it is trivial. The map would factor through the class-two $\{2, 3\}$ component and would kill $\gamma_3 N$, contradicting (E.20.12). This proves the claimed failure of factorization by an exact quotient map.

The program `de_coprime_gluing_root.py` checks (E.20.11)–(E.20.12), enumerates all 729 target elements, verifies their ninth powers and central cubes, and verifies the indicated commutator-subgroup exponent and central filtration. The lifting and compatibility argument uses the proved profinite morphisms (E.20.3)–(E.20.10). □

5.9 A joint Ferrari test retaining cubic powers and fourth-degree commutators

Statement E.21 (Proved). Complete compatible corrections in the fourth-degree Ferrari quotient. *Let*

$$\mathcal{A}_d = \mathbb{F}_3\langle X, Y\rangle/(\textit{monomials of degree} > d), \qquad K_d = \langle 1 + X, 1 + Y\rangle \subset \mathcal{A}_d^{\times}.$$

On the actual free generators ξ_1, ξ_2 of the Ferrari kernel take the automorphisms

$$\begin{aligned} a:&\quad \xi_1 \mapsto \xi_1, && \xi_2 \mapsto \xi_2\xi_1,\\ b:&\quad \xi_1 \mapsto \xi_1\xi_2^{-1}, && \xi_2 \mapsto \xi_2. \end{aligned} \tag{E.21.1}$$

Their actual finite image on K_d is denoted C_d. The assignments

$$\sigma_1 \mapsto (1, a), \qquad \sigma_2 \mapsto (1, b), \qquad \sigma_3 \mapsto (\xi_1, a) \tag{E.21.2}$$

define an epimorphism $B_4 \to K_d \rtimes C_d$. Here the cubic powers are retained: $(1+X)^3 = 1+X^3$ and $(1+Y)^3 = 1+Y^3$.

Let T_4 be the joint image of (E.21.2) for $d = 4$ and the modulus-nine representation $\rho_{4,9}$ of E.19. Its reduction to the E.18 target is surjective, with elementary abelian kernel $\mathbb{F}_3^{12}$. Consequently

$$|T_4| = 344373768, \qquad |B_4/(\ker(B_4 \to T_4) \cap PB_4)| = 8264970432. \tag{E.21.3}$$

Form the three-strand core by the six evaluation maps and all six B/F_2 conjugates, exactly as in (E.19.1), now with target T_4, and intersect it with K_6. Denote this kernel by $K_{3,4}$ and put $D_4 = B/K_{3,4}$. Its kernel over G_6 has dimension seven over $\mathbb{F}_3$, the image of $K_6 \cap F_2'$ has dimension six, and the image of N' has dimension three.

For each of the 24 retained E.17 G_6 points, the complete gentle norm equations on this six-dimensional correction image have rank four; the simultaneous pentagon in the two targets has rank five. The combined system is consistent of rank five, hence has an affine line of corrections. An actual commutator word in each line is retained. It satisfies both complete hexagons in D_4, gives a surjective three-strand shadow, satisfies the full pentagon in T_4, and defines a surjective four-strand braid map to T_4. The three- and four-strand full twists are preserved.

Thus all 24 CM fibres remain nonempty in this joint finite test. In the specified three-strand quotient, $\gamma_3 N$ and N^{18} are killed. It therefore belongs to the relative class-two coverage of E.15; the new check is the displayed four-strand constraint.

Proof. The substitutions (E.21.1) are the exact Ferrari actions of [NTY10, (2.9) and (4.15)]. Substituting units with constant term one defines algebra automorphisms of $\mathcal{A}_d$: their inverses are the literal inverse free-group substitutions. They preserve K_d. Direct multiplication verifies both adjacent braid relations and the distant braid relation in (E.21.2). The two literal Ferrari words become $(\xi_1, 1), (\xi_2, 1)$, and a, b generate C_d by its definition. This proves the epimorphism. It also specifies the source comparison without replacing the action of B_3 by a homology action.

The calculation first constructs degree three completely. The actual unit group K_3 has order 2187, as enumerated from its two generators. The image C_3 has order 1944, as enumerated from the two exact substitution matrices; its kernel over $\mathrm{SL}_2(\mathbb{F}_3)$ has order 81. Thus this initial four-strand target has order 4251528. Its joint core with the modulus-nine target still has correction dimensions four and three, and all 24 points pass. These checks are retained in `de_ferrari_degree3_model.py` and `de_ferrari_degree3_lift.py`.

For degree four the augmentation ideal has its full 30 monomial coordinates. Multiplication is ordinary truncated noncommutative multiplication over $\mathbb{F}_3$. An automorphism is stored by the two complete augmentation series of its generator images. Its linear operator on all 30 monomials is obtained by substitution. Composition is exact matrix substitution; inverse generator images are obtained degree by degree from the invertible linear part. No division by three occurs. The program `de_ferrari_truncated_model.py` checks all Ferrari and braid identities and compares 1000 successive actual word reductions, including inverses, with the independent complete degree-three model.

Reduction to degree two, followed by the exact change $z = z_0 - u - 2v$ of E.18, gives the specified map to the E.18 target. Its kernel is additive. Indeed a kernel element of K_4 starts in degree three. An automorphism in the kernel changes each generator only in degrees three

and four: its degree-two change is zero by the E.18 comparison. Applying that change to a term of degree three first affects degree five, which is zero here. Two such automorphism changes also commute, since their commutator first affects degree five. Products of two kernel unit changes first occur in degree six. Therefore the exact additive coordinates are the unit's degree-three and degree-four terms and the two automorphisms' changes in those degrees. There are 72 ambient coordinates, with the actual image computed inside them. Together with the seven redundant but injective congruence coordinates of the modulus-nine factor, these give 79 ambient linear coordinates for the joint kernel.

The complete image is obtained by Schreier generators over the order-648 E.18 quotient. Choose one actual word representative t_h for each of its elements. For each of its three generator edges retain

$$t_h \sigma_i t_{h\bar{\sigma}_i}^{-1}. \tag{E.21.4}$$

These generate the kernel of the free three-generator group mapping onto that quotient; their images therefore generate the full kernel of the actual joint braid image. The coordinate matrix has rank 12. Since that kernel is elementary abelian, this proves the first order in (E.21.3).

Every braid generator in this target has order dividing nine, as directly checked. Its abelianization, being a quotient of the cyclic abelianization of B_4, thus has odd order. Every nontrivial quotient of $\mathfrak{S}_4$ has a quotient C_2; this follows from its normal subgroups $1, V_4, A_4, \mathfrak{S}_4$. There is consequently no nontrivial common quotient. Goursat's argument gives the exact joint product $T_4 \times \mathfrak{S}_4$ and proves the second order in (E.21.3).

The induced six-coset representation of the three-strand core is exactly (E.19.2), applied in both target factors. Every K_6 image lies in the just proved additive kernel. The complete normal image of N' is generated by all 4656 pairwise commutators of the actual n_i, followed by closure under all n_i. Its rank is three. Adjoining all sixth powers gives rank seven. Restricting their exponents to the full integral rank-95 abelianization kernel of E.19 gives rank six for $K_6 \cap F_2'$. Literal source words for every basis vector are retained and checked. Stability under both s and r is checked, so these are the exact images used by the core, including all commutator and power corrections.

The same norm identities (E.19.3) and the same literal gauge identity (E.19.4) apply to this six-dimensional space. Their matrices have rank four in every fibre. Products of pentagon factors are affine linear on the abelian correction kernel, with coefficients given by conjugation by their actual base values. The program `de_ferrari_degree4_lift.py` retains the full matrices; the simultaneous pentagon has rank five, and the braid-error matrix has rank zero. The combined augmented system has rank five in every case. All basis-pair substitutions and 100 additional full substitutions per fibre check this affine evaluation; its completeness follows from the additive-kernel identity, not from sampling.

An actual correction in each affine line is then substituted as a literal word. The program checks zero integral F_2-abelianization, both complete hexagons in all 36 induced coordinates, and the complete pentagon in both target factors. As in E.19, generation of the three-strand core is checked by evaluating all n_i, their commutators and sixth powers, and closing under their conjugations. The kernel image has its full rank seven in all 24 cases; the projection to G_6 was already surjective. Hence the three-strand map is onto.

For each corrected four-strand triple, the program `de_ferrari_degree4_generation.py` repeats the entire Schreier calculation (E.21.4). Its projection still has order 648 and its kernel image has rank 12, with the same span as the original target. This proves generation of T_4. It also checks directly

$$(\sigma_1'\sigma_2')^3 = (\sigma_1\sigma_2)^3, \qquad (\sigma_1'\sigma_2'\sigma_3')^4 = (\sigma_1\sigma_2\sigma_3)^4.$$

Finally, `de_ferrari_degree4_lowercentral.py` computes the action of every n_i on the complete rank-three commutator image. Each action is the identity, so the image of $\gamma_3 N$ is zero. Every n_i^9 is also trivial in the induced coordinates. The commutator image has exponent three, so class-two collection shows that every ninth power in N is trivial there. Together with the exponent-six homology component, this proves $\gamma_3 N\, N^{18} \subset K_{3,4}$. It gives the exact claimed comparison with E.15 and the stated finite scope. □

Statement E.22 (Proved). A compatible relative class-three test preserving every retained CM point. *Extend the actual Ferrari Magnus representation (E.21.1)–(E.21.2) to $d = 5$, retaining its joint image T_5 with the modulus-nine target. Define $K_{3,5}$ by all six evaluation maps, all six B/F_2 conjugates, and intersection with K_6, as in (E.19.1). Put $D_5 = B/K_{3,5}$. The specified reduction maps give*

$$D_5 \longrightarrow D_4 \longrightarrow G_6, \qquad T_5 \longrightarrow T_4 \longrightarrow \widetilde{H}_3. \tag{E.22.1}$$

The kernel of $D_5 \to D_4$ is $\mathbb{F}_3^6$. Its entire image is represented by words in $K_{3,4} \cap F_2'$. The kernel of $D_5 \to G_6$ has order 3^{13}. The actual lower-central images of N in D_5 have dimensions

$$\dim \operatorname{im}(\gamma_2 N) = 9, \qquad \dim \operatorname{im}(\gamma_3 N) = 3, \qquad \operatorname{im}(\gamma_4 N) = 1, \tag{E.22.2}$$

where the two nonzero images are elementary abelian three-groups.

All three points of each of the 24 correction lines in E.21 are included in the next test. Every one of those 72 points has a gentle lift to D_5. In each old line exactly one point has a simultaneous pentagon lift; its top correction is unique. Thus every one of the 24 particular E.17 G_6 shadows, including all twelve extra-coset points, has a verified compatible lift through (E.22.1). These literal commutator words give surjective three-strand shadows and surjective four-strand braid maps satisfying the complete pentagon in T_5.

The full target orders are

$$\begin{aligned} |T_5| &= 6\,778\,308\,875\,544, \\ |B_4/(\ker(B_4 \to T_5) \cap PB_4)| &= 162\,679\,413\,013\,056. \end{aligned} \tag{E.22.3}$$

The three- and four-strand full twists are preserved. The multiplier calculation is exhaustive over lifts of the retained G_6 shadows: the exact cusp order in D_5 is 72, so all compatible values $\lambda \equiv 1 \pmod{72}$ and $m = (\lambda - 1)/2 \equiv 0 \pmod{36}$ give exactly the same finite equations. This does not enumerate other multiplier residues or other G_6 points within a CM matrix fibre.

The natural quotient $B \to D_5$ factors through none of the finite quotients in E.20.1. Thus this is a concrete relative class-three extension beyond that family, with positive witnesses in all 24 CM fibres. It is one specified finite refinement, with no all-quotient lifting assertion.

Proof. The 62 augmentation coordinates through degree five are evaluated by the same literal free-group substitutions as in E.21. Reduction to degree four is the specified truncation homomorphism; the modulus-nine factor is unchanged. An element in its kernel has only a degree-five unit term and degree-five changes of the two automorphism generators. Their products add. There are 96 ambient coordinates per evaluation, and 36 evaluations in the induced core. All finite source relations and every transition in (E.22.1) are therefore represented by actual homomorphisms.

We give the complete kernel calculation, including its normal presentation. Write $J = K_6$ and $J' = K_6 \cap F_2'$. In degree four their images have ranks seven and six. Let $b_1, \dots, b_r$ be the retained literal basis words for one of these images. The normal generators of J in N are all 4656 commutators $[n_i, n_j]$ and the 97 sixth powers n_i^6. For J' replace the latter by the 95

products corresponding to the complete integral kernel of the abelianization matrix in E.19. This is exact: modulo N', an element of J' has exponent vector $6a$ with $\mathrm{ab}_{F_2}(a) = 0$.

For each such normal generator w, retain the relation

$$w \prod_i b_i^{-c_i(w)}, \tag{E.22.4}$$

where $c(w)$ is its complete degree-four image coordinate. Also retain every b_i^3, every $[b_i, b_j]$, and every

$$n_k b_i n_k^{-1} \prod_j b_j^{-c_j(n_k b_i n_k^{-1})}. \tag{E.22.5}$$

These normally generate the kernel of the old image. Indeed, in the resulting quotient every original generator and all its N-conjugates are in the commuting exponent-three span of the b_i. Its order is at most 3^r, while the actual old image has order 3^r and is a quotient of it. Hence the two groups coincide and the listed normal kernel is exact.

The program `de_ferrari_degree5_kernel.py` evaluates all 5354 relations for J' and all 5460 for J, then closes their top-degree spans under the 97 actual N generators. Both image spaces have rank six and are equal. Every retained basis word is re-evaluated literally; the J' words have zero integral F_2-abelianization. Stability under s, r is also checked. This proves the full top-kernel assertion and shows that no source power or commutator corrections have been discarded.

Here the choice of companion to a provisional involution must be exact. Let S_0 be an actual word lifting one of the old degree-four solutions and put

$$R_0 = S_0^{-1} t, \qquad S_0 R_0 = t. \tag{E.22.6}$$

The square of S_0 need not be the identity at degree five. Write M for the computed rank-six top kernel, S, R for the conjugation actions of S_0, R_0 on M, and $E_S = S_0^2$, $E_R = R_0^3$. For $k \in M$, the exact corrected images are $S_0 k, k^{-1} R_0$. The complete norm equations are

$$(1 + S)k = -E_S, \qquad (1 + R + R^2)k = E_R. \tag{E.22.7}$$

In the first equation $\mathrm{Ad}(S_0^{-1}) = S$ on M, because $S_0^2 \in M$ and M is abelian. These identities prove (E.22.7) without assuming that the provisional square vanishes. The program asserts (E.22.6) before constructing any linear system.

For each of the three old points in every fibre, the matrix in (E.22.7) has rank four and is consistent. The top pentagon matrix has rank five, and the braid matrix has rank zero. The combined coefficient matrix has rank six. In exactly one of the three old points its augmented rank is also six; in the other two it is seven. Thus precisely one old point lifts and its top correction is unique. The full matrices and right-hand sides for all 72 cases are retained. Affineness follows from the elementary abelian top kernel. All basis-pair substitutions and further complete substitutions audit the implemented linear maps.

The normalized word remains

$$f = s x^g S_0 k x^{-g}, \tag{E.22.8}$$

with the old exact integral gauge. Its two new factors are actual commutator words, so its integral abelianization is zero. Every chosen solution is checked by literal source word substitution, both full hexagons in all 36 induced coordinates, the complete pentagon in both target factors, and all three braid relations. The independent program `de_ferrari_degree5_gentle.py` also constructs and verifies gentle-only witnesses in all 24 CM fibres directly from the norm systems.

For completeness we describe the source and target generation certificates. The kernel over the E.18 target at degree five is of class at most two and exponent three. Its raw coordinates have a bilinear product $v * w = v + w + C(v, w)$, with cross terms confined to degree five. The exact first-kind coordinate is

$$\log(v) = v - \tfrac{1}{2}C(v, v), \qquad \tfrac{1}{2} = 2 \in \mathbb{F}_3. \tag{E.22.9}$$

The group commutator gives the Lie bracket. Products are therefore $a + b + \frac{1}{2}[a, b]$. This is a class-two calculation in characteristic three; no class-three logarithm is used. The formula is checked on actual group products and inverses.

The seven old full-kernel words together with the six new top-kernel words give a 13-dimensional Lie space for the entire kernel over G_6. Its brackets and the s, r actions are checked to remain in the space. For each proposed source map, those literal words are substituted on its new generators. Closing their image under brackets and the two new generator actions gives rank 13. All these elements are images of actual elements of K_6; this rank proves surjectivity on the whole kernel. The map was already onto G_6, so it is onto D_5.

There is also an exact generation argument for every solution of the norm systems, including all 72 old points, without restricting to the selected witnesses. The image of N in the E.18 target is an elementary abelian three-group, as checked there on all generators, powers and commutators. The kernels of the modulus-nine and higher Magnus reductions to that target are three-groups: the matrix congruence kernel and each additive Magnus layer have three-power order. Consequently the image of N in all 36 new evaluation factors is a three-group. The G_6 coordinate also retains the full quotient $N^{\mathrm{ab}}/6$. By Goursat's lemma on the relatively prime normal orders, the actual joint image is therefore

$$\mathrm{im}(N \to D_5) \simeq (N^{\mathrm{ab}}/2) \times P, \qquad P \twoheadrightarrow N^{\mathrm{ab}}/3, \tag{E.22.10}$$

where P is a finite three-group generated by the 97 specified N generators. Its Frattini quotient $P/\Phi(P)$ has dimension at most 97 and surjects onto $N^{\mathrm{ab}}/3 \simeq \mathbb{F}_3^{97}$. Hence the displayed surjection has kernel exactly $\Phi(P)$. Every norm solution induces the fixed automorphism of G_6 and thus an automorphism of Q; it sends N into the subgroup in (E.22.10). Its image surjects onto both $N^{\mathrm{ab}}/2$ and $P/\Phi(P)$. The finite three-group Frattini criterion gives surjectivity onto P; Goursat again gives the full product in (E.22.10). Together with the surjection onto Q, this proves generation of D_5 for every norm solution. The direct rank-13 checks above independently certify each selected pentagon witness.

The complete image of N' is obtained from all pairwise commutators and normal closure under N. It has dimension nine and is abelian. Repeating the commutator operation with N gives dimension three and then zero. These are the exact ranks (E.22.2), with the coordinate matrices and a nonzero third-commutator witness retained.

The four-strand target is computed independently by the 648-vertex Schreier calculation (E.21.4). The Lie span of its full kernel and its brackets has dimension 21; its commutator space has dimension nine. This proves $|T_5| = 648 \cdot 3^{21}$. For each of the 24 corrected triples the same calculation gives the entire same Lie space. It therefore generates T_5. Both full twists are also checked directly. Every braid generator has order dividing nine, so the exact product with $\mathfrak{S}_4$ and the pure quotient order follow by the same Goursat argument as in E.21.

The control program `de_ferrari_degree5_control.py` checks that the induced core cusp has order dividing 36, while its G_6 coordinate has exact order 72. Hence the combined cusp has exact order 72. The pure peripheral powers have orders dividing 36, and the four-strand generators have orders dividing nine. Both complete hexagon powers and all four-strand generator powers are therefore unchanged by every multiplier lift of each retained G_6 shadow. This is the claimed exhaustive period comparison.

The literal identity pair passes all new relations. The program also verifies that the E.17 point numbered zero is a different G_6 point from the identity in E.1: its conjugator and its induced y map are nontrivial. Finally a permanent regression control uses a provisional S_0 with nontrivial square. It checks that $R_0 = S_0 t$ fails the cusp product and (E.22.6) satisfies it. The initially incorrect calculation, its withdrawn negative conclusions, and its exact reconstruction are preserved separately in the audit files.

If the natural map to D_5 factored through a finite quotient in E.20.1, its relative prime-to-six kernel would map trivially: the relative kernel of $D_5 \to Q$ is a $\{2,3\}$-group, and the maps to Q agree on the source B. The map would then factor through the relative class-two component and kill $\gamma_3 N$, contrary to (E.22.2). This proves the last claim by the specified natural quotient maps. □

6 F. Derived limits and applications

6.1 The residue complex and the derived-limit test

Statement F.1 (Proved). Localization with its global term retained. *Let L be a number field, let ℓ be a prime, and let S be a finite set of places containing the archimedean places and every place above ℓ. Put $X = \operatorname{Spec} \mathcal{O}_{L,S}$. Let M be a free finite-rank $\mathbb{Z}_\ell$-module with a continuous action of $G_{L,S}$. Put $M_n = M/\ell^n M$. For every finite place $v \notin S$, choose a decomposition group and arithmetic Frobenius, put $A_v = M(-1)$ with that residue-field action, and put $D_v = \mathrm{Frob}_v - 1$ on A_v. There are compatible distinguished triangles in the derived category of abelian groups*

$$R\Gamma(X, M_n) \longrightarrow R\Gamma(G_L, M_n) \longrightarrow \bigoplus_{v \notin S} [A_v/\ell^n \xrightarrow{D_v} A_v/\ell^n][-1] \longrightarrow R\Gamma(X, M_n)[1].$$

The two terms in each displayed bracket have degrees zero and one. Define

$$R_A = \left\{ (a_v) \in \prod_{v \notin S} A_v : \text{for every } n,\ a_v \in \ell^n A_v \text{ outside finitely many } v \right\}.$$

Taking the derived inverse limit gives the triangle

$$R\Gamma_{\mathrm{cts}}(G_{L,S}, M) \longrightarrow R\Gamma_{\mathrm{cts}}(G_L, M) \longrightarrow [R_A \xrightarrow{D} R_A][-1] \longrightarrow R\Gamma_{\mathrm{cts}}(G_{L,S}, M)[1].$$

Here $D = (D_v)_v$ is the coordinatewise residue differential. In particular its degree-two sequence is

$$\ker D \longrightarrow H^2_{\mathrm{cts}}(G_{L,S}, M) \longrightarrow H^2_{\mathrm{cts}}(G_L, M) \longrightarrow R_A/DR_A \longrightarrow H^3_{\mathrm{cts}}(G_{L,S}, M).$$

The image DR_A here is the algebraic image. Every arrow is the map induced by restriction or the localization triangle; changing a decomposition-group choice conjugates the corresponding stalk and gives an isomorphic complex.

Proof. For a finite set J of places outside S, take localization for the open immersion $X - J \hookrightarrow X$. Over the strict henselization at v, the sheaf M_n is constant. Wild inertia is pro-p_v, so averaging on finite quotients annihilates its positive-degree cohomology with ℓ-primary coefficients. The relevant tame inertia is $\mathbb{Z}_\ell(1)$; its cochain complex has cohomology M_n in degree zero and $M_n(-1)$ in degree one. The support complex is consequently $M_n(-1)[-2]$. Descending to the residue field gives the first triangle with J in place of all places.

The locally constant sheaf comparison is Milne, II.2.9; localization and its support maps are II.0.1(f) and II.1.7(b) [Mil06]. Filtered direct limits over J are exact. Every continuous

cochain of a profinite group with finite coefficients factors through a finite quotient, and its number field is ramified at finitely many places. Thus the generic-field term is precisely $R\Gamma(G_L, M_n)$ and the support term is a direct sum.

The residue Galois group is procyclic. Its usual two-term resolution computes cohomology as $[A_v/\ell^n \xrightarrow{D_v} A_v/\ell^n]$. In both degrees the direct-sum reduction maps are surjective. Their derived inverse limits are therefore their ordinary inverse limits, which are exactly R_A: reduction modulo ℓ^n has finite support for every n. The matrices D_v are integral, so they preserve this space.

For completeness, continuous cochains with values in M are the inverse limit of the finite-coefficient cochain complexes. The reduction map on cochains is surjective, since a continuous finite-valued function can be lifted value by value. The resulting Milnor comparison is Jannsen, Proposition 1.6 and Theorem 2.2 [Jan88]. It identifies the derived inverse limit used here with continuous cochain cohomology. Taking the long exact sequence proves the assertion. Real places have not been discarded: their possible 2-primary contribution remains in the global term and in its degree-three group. □

Statement F.2 (Proved). The exact global correction to the residue derived limit. *With the data of F.1, define inverse systems*

$$P_n = \bigoplus_{v \notin S} \ker(D_v : A_v/\ell^n \to A_v/\ell^n), \qquad E_n = \operatorname{im}\big(P_n \to H^2(G_{L,S}, M_n)\big).$$

There is a natural short exact sequence

$$0 \longrightarrow \operatorname{coker}\big(\varprojlim P_n \to \varprojlim E_n\big) \longrightarrow \varprojlim{}^1 H^1(G_L, M_n) \longrightarrow \varprojlim{}^1 P_n \longrightarrow 0. \tag{F.2.1}$$

The right-hand group has the explicit presentation

$$\varprojlim{}^1 P_n = \frac{R_A \cap \prod_{v \notin S} D_v A_v}{D R_A}. \tag{F.2.2}$$

The embedding of this group into the residue cohomology is the kernel of the map

$$R_A/DR_A \longrightarrow \varprojlim_n \bigoplus_{v \notin S} A_v/(\ell^n A_v + D_v A_v).$$

Jannsen's exact sequence for the full global group is

$$0 \longrightarrow \varprojlim{}^1 H^1(G_L, M_n) \longrightarrow H^2_{\mathrm{cts}}(G_L, M) \longrightarrow \varprojlim_n H^2(G_L, M_n) \longrightarrow 0. \tag{F.2.3}$$

Proof. At level n localization gives

$$H^1(G_{L,S}, M_n) \longrightarrow H^1(G_L, M_n) \longrightarrow P_n \longrightarrow H^2(G_{L,S}, M_n).$$

The two cohomology groups with finite S are finite by Milne I.4.15. Let Q_n be the kernel of $P_n \to E_n$, and let F_n be the image of the first arrow. The exact sequences $0 \to F_n \to H^1(G_L, M_n) \to Q_n \to 0$ and $0 \to Q_n \to P_n \to E_n \to 0$ give (F.2.1). Indeed all the F_n and E_n are finite, hence their towers are Mittag–Leffler and their first derived limits vanish; second derived limits of towers vanish.

Put $B_n = \bigoplus_v A_v/\ell^n$ and $I_n = DB_n$. Both B_n and I_n have surjective transition maps: a representative of $D_v a$ lifts by lifting a. Apply inverse limits to $0 \to P_n \to B_n \to I_n \to 0$. The result is $\varprojlim{}^1 P_n = (\varprojlim I_n)/DR_A$. An element of R_A belongs to $\varprojlim I_n$ exactly when each of its coordinates belongs to $D_v A_v + \ell^n A_v$ for every n. The image of an endomorphism of a

finite-rank $\mathbb{Z}_\ell$-module is closed; its Smith form proves that the intersection is $D_v A_v$. This proves (F.2.2). Applying the same argument to $0 \to I_n \to B_n \to B_n/I_n \to 0$ gives the asserted kernel and surjectivity onto the displayed limit. Finally (F.2.3) is the continuous-cochain comparison already proved in F.1. □

Statement F.3 (Proved). A complete vanishing test over number fields. *For each $v \notin S$, let $e_{v,1}, \dots, e_{v,s_v}$ be the valuations of the nonzero Smith entries of D_v on A_v. A unit entry has valuation zero; a zero Smith entry is excluded. The following conditions are equivalent:*

(i) $\varprojlim{}^1 H^1(G_L, M/\ell^n) = 0$;

(ii) the tower P_n in F.2 is Mittag–Leffler;

(iii) the nonnegative integers $e_{v,i}$ are bounded, uniformly over v and i;

(iv) the finite groups

$$B_v = \frac{H^0(\mathbb{F}_v, A_v \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)}{H^0(\mathbb{F}_v, A_v \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)_{\mathrm{div}}}$$

have uniformly bounded orders.

The subscript div *denotes the maximal divisible subgroup. No boundedness condition on the orders of $H^2(G_{L,S}, M/\ell^n)$ is required. Their contribution to the full group is exactly the left term of (F.2.1).*

Proof. The Smith form is an isomorphism of source and target $\mathbb{Z}_\ell$-modules; it is chosen separately at each place and is compatible with reduction. For a nonzero entry ℓ^e, the image of its kernel at level m in level $n \le m$ is

$$\ell^{\max(0,m-e)}(\mathbb{Z}/\ell^n),$$

with the usual interpretation that it is zero when $m - e \ge n$. A zero Smith entry contributes the surjective tower $\mathbb{Z}/\ell^n$. Thus a bound $e \le b$ makes the images stationary by level $n + b$. Conversely, unbounded exponents give, for every m, a coordinate whose image at level one is nonzero from level m, but becomes zero at a later level. Hence the images at level one never become stationary. This proves (ii)⇔(iii).

The same Smith form gives

$$H^0(\mathbb{F}_v, A_v \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell) \simeq (\mathbb{Q}_\ell/\mathbb{Z}_\ell)^{\operatorname{rank}\ker D_v} \oplus \bigoplus_{i=1}^{s_v} \mathbb{Z}/\ell^{e_{v,i}}.$$

This is an isomorphism of abelian groups; the quotient by the maximal divisible subgroup is canonical. The ranks s_v are bounded by $\operatorname{rank} M$, proving (iii)⇔(iv).

Unbounded exponents give a nonzero element of (F.2.2) directly. Choose distinct v_j and Smith source basis vectors x_j with associated exponents at least j. Set $z_{v_j} = D_{v_j} x_j$ and set the other coordinates to zero. The family z is restricted. Every preimage of z_{v_j} differs from x_j by an integral kernel vector, so its reduction modulo ℓ in $A_{v_j}/\ker D_{v_j}$ is nonzero. No preimage family is restricted. Thus (F.2.2) is nonzero, and the surjection (F.2.1) proves that (i) implies (iii).

It remains to verify that bounded exponents also annihilate the global correction; finiteness at each level alone does not supply this step. Let

$$P_n^0 = \bigoplus_v (\ker D_v)/\ell^n \ker D_v \ \subseteq P_n.$$

The kernels are saturated submodules, so the inclusion is injective. A uniform bound b implies that the tower P_n/P_n^0 is zero after shifting the index by b. Put $C = H^2_{\mathrm{cts}}(G_{L,S}, M)$. It is a finitely generated $\mathbb{Z}_\ell$-module. To check this last input, finite-coefficient cohomology is finite and the Milnor term from $H^1(G_{L,S}, M_n)$ vanishes. Thus $C = \varprojlim H^2(G_{L,S}, M_n)$ is compact. The coefficient exact sequence embeds $C/\ell C$ in the finite group $H^2(G_{L,S}, M/\ell)$. Choose lifts $c_1, \dots, c_d$ of generators of this quotient. Successive division modulo their span expresses every $c \in C$ as $\sum_i a_i c_i$ with $a_i \in \mathbb{Z}_\ell$: the remainder after n divisions belongs to $\ell^n C$, and these subgroups tend to zero since their images in every fixed finite coefficient group eventually vanish. This proves finite generation directly, and also verifies the input to Jannsen Corollaries 4.9–4.10.

Each integral local invariant has a compatible Gysin image in C. Let $J \subseteq C$ be the $\mathbb{Z}_\ell$-submodule generated by all these images. Since C is noetherian, J is generated by finitely many such images and is closed. The image E_n^0 of P_n^0 in $H^2(G_{L,S}, M_n)$ is exactly the image of J. The quotient tower E_n/E_n^0 is zero after the same fixed shift, so $\varprojlim E_n = \varprojlim E_n^0$. Compactness of the finitely generated module J, together with $C = \varprojlim H^2(G_{L,S}, M_n)$, identifies this limit with J.

A finite set of local invariant vectors whose images generate J gives a map from a finite direct sum of their $\mathbb{Z}_\ell$-spans into $\varprojlim P_n$. Its composite onto J is surjective. Consequently the left term of (F.2.1) vanishes. The right term vanishes by the Smith calculation, proving (iii)⇒(i). The proof includes $\ell = 2$ and real places, since it uses their finite cohomology through C without assuming cohomological dimension two. □

Statement F.4 (Refuted). Total orders of residue invariants do not determine vanishing. *The orders of the unquotiented groups $H^0(\mathbb{F}_q, M(-1) \otimes \mathbb{Q}_3/\mathbb{Z}_3)$, even together with all the orders of $H^2(G_{\mathbb{Q},S}, M/3^n)$, do not determine the vanishing in F.3. For $S = \{3, \infty\}$ the two lattices*

$$M^{(0)} = \mathbb{Z}_3(1)^2, \qquad M^{(1)} = \mathbb{Z}_3(1) \oplus \mathbb{Z}_3(2)$$

have the same stated data: every residue invariant group has countably infinite order, and every displayed global H^2 group is zero. Their derived limits are respectively zero and nonzero. The correct order test is F.3(iv).

Proof. For $M^{(0)}(-1)$ every D_q is zero; there is no nonzero Smith entry. For $M^{(1)}(-1)$ the matrix is $\mathrm{diag}(0, q-1)$. Primes congruent to 1 modulo 3^j give unbounded nonzero Smith valuations, so F.3 proves the two answers. The residue invariants are respectively $(\mathbb{Q}_3/\mathbb{Z}_3)^2$ and $\mathbb{Q}_3/\mathbb{Z}_3 \oplus (\mathbb{Q}_3/\mathbb{Z}_3)[q-1]$; both have countably infinite underlying sets.

For the twist one, Kummer theory on $\mathbb{Z}[1/3]$ gives the vanishing of H^2 because its Picard group is zero and its Brauer group has no 3-primary part. The latter follows from the invariant-sum sequence: only the place 3 can contribute a nonzero 3-primary local invariant, and its sum must be zero. For twist two modulo 3, the module is the trivial $\mathbb{F}_3$. Kronecker–Weber gives $H^1(G_{\mathbb{Q},S}, \mathbb{F}_3)$ dimension one. The global Euler characteristic (Milne I.5.1) is one; thus H^2 is zero. Filtering the higher coefficients by powers of 3 proves the same vanishing at every level. This independently confirms the precise input from LM7 E.32g used in the example. □

Statement F.5 (Proved). Coefficient modules already determined. *Over every number field L the derived limit in F.3 vanishes for $\mathbb{Z}_3(1)$ and for the restrictions of $\mathbb{Z}_3(1)[D]$ and W from the cubic. It is nonzero for the restrictions of T, V, $\mathcal{K}$, and $\bigwedge^2 V$. It is also nonzero for every nonzero torsion-free graded lattice of degree $r \geq 3$ obtained as a subquotient of tensor powers $V^{\otimes r}$ and occurring in a lower-central quotient here. These statements concern each specified finite-rank lattice; they do not evaluate a section obstruction.*

Proof. For twist one the residue differential is zero. The cusp permutation action factors through $\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})$ of order two. On every $\mathbb{Z}_3$-lattice with that action, averaging splits the two eigenspaces integrally. After the twist the nonzero Smith entries of a residue permutation minus one are units, including the quotient by the diagonal defining W. F.3 proves vanishing.

For $T(-1)$, the eigenvalues of arithmetic Frobenius at a good place v have complex absolute value $(Nv)^{-1/2}$; consequently D_v is injective. The representation modulo 3^n cuts out a finite Galois extension of L. Chebotarev supplies infinitely many places splitting in this extension and in the 3^n-cyclotomic extension. At these places $D_v \equiv 0 \pmod{3^n}$. Its nonzero Smith valuations are therefore unbounded. For $\mathcal{K}(-1)$, the two weight pieces have absolute values Nv and $(Nv)^{1/2}$, so the same argument applies.

The generalized Jacobian extension is killed by 3 in the isogeny category because every cusp difference has order dividing 3 on the elliptic curve. Thus $V \otimes \mathbb{Q}_3 = (W \otimes \mathbb{Q}_3) \oplus (T \otimes \mathbb{Q}_3)$. To transfer the boundedness test, choose integral lattices on this rational decomposition and a positive integer b such that the original and split lattices contain 3^b times one another. Replacing one by the other changes the largest nonzero Smith valuation by at most $2b$; this follows by applying both inclusions to preimages of D_v. Hence the unboundedness from T persists in V. The same argument for exterior squares retains the $\mathcal{K}$ summand and proves the assertion for $\bigwedge^2 V$.

In degree $r \geq 3$, the eigenvalues before twisting have absolute values among $(Nv)^{a/2+b}$ with $a + b = r$ and $a, b \geq 0$. After twisting, every such value is at least $(Nv)^{r/2-1} > 1$. Subquotients retain a subset of these eigenvalues. Thus D_v is injective, and splitting places for the finite coefficient representation again prove unboundedness. A fixed finite torsion part of a finitely generated coefficient module contributes an eventually constant coefficient system; it creates no additional first derived limit. This last observation does not assert that an unevaluated graded quotient is torsion-free. □

Statement F.9 (Proved). The genuine integral class-two action for the colliding-cusp curve. *Let $K_0 = \mathbb{Q}((t))$ and $U_t = \mathbb{P}^1_{K_0} \setminus \{0, t, 1, \infty\}$. Use the coefficient splitting of G_{K_0} which fixes compatible roots of t. There is a based splitting of the arithmetic fundamental-group extension, constructed from the node chart below, for which the free class-two pro-ℓ group has a basis*

$$x = b, \qquad y = d = ab, \qquad z = c \tag{F.9.1}$$

with the following genuine continuous action:

	x	y	z
τ	yxy^{-1}	y	z
$g \in G_{\mathbb{Q}}$	$x^{\chi_\ell(g)}$	$y^{\chi_\ell(g)}$	$z^{\chi_\ell(g)}$.

(F.9.2)

The inverse free-basis change is $a = yx^{-1}$, $b = x$, $c = z$. These formulas hold integrally for every prime, including two. In the original generators, putting $c_{ab} = [a, b]$ and $\lambda = \chi_\ell(g)$, the constant action is

$$g(a) = a^\lambda c_{ab}^{-\binom{\lambda}{2}}, \qquad g(b) = b^\lambda, \qquad g(c) = c^\lambda. \tag{F.9.3}$$

The inertia action in that basis is exactly (F.7) of FLAGGED_149 F17.3. The comparison changes the free basis and specifies a splitting; it preserves the arithmetic extension over G_{K_0}.

Proof. Blow up the point $x = t = 0$ of the marked projective line model. Near its single node use coordinates $X = x$ and $V = t/x$, so $XV = t$. The outer component is the tripod in coordinate X, with node at zero and marked points at one and infinity. The bubble is the tripod in coordinate V, with node at zero, the cusp t at one, and the cusp 0 at infinity. The branch parameters X, V have product exactly t; no constant factor has been inserted.

The punctured normal-crossing node chart has arithmetic tame group

$$(\widehat{\mathbb{Z}}(1)^2) \rtimes G_{\mathbb{Q}} \longrightarrow \widehat{\mathbb{Z}}(1) \rtimes G_{\mathbb{Q}} = G_{K_0}, \quad ((r,s),g) \longmapsto (r+s,g). \tag{F.9.4}$$

This is seen on the compatible covers extracting roots of X, V; the base cover extracts roots of XV. Choose their products to be the specified roots of t. Coefficient automorphisms fix these formal roots and act on their root-of-unity multipliers by χ. The map in (F.9.4) then has the continuous group section $(r,g) \mapsto ((r,0),g)$. The homomorphism from the node chart's arithmetic group to that of U_t, induced by the punctured formal-neighborhood map, sends this group section to an actual section over G_{K_0}. The kernel direction $(r,-r)$ is the vanishing loop d; its primitive abelianization proves injectivity on this direction. Choose its orientation and the lift of τ so that the associated Dehn twist conjugates the bubble group by d and fixes the outer group. This is the annulus cut-and-glue calculation which gives (F.6) in the attached return.

We specify why the component comparison respects arithmetic. Logarithmic specialization for this proper marked semistable model is Theorem 3.4.6 of Stix's thesis [Sti02]; the residue characteristic is zero, so its prime-to-residue- characteristic assertion applies to every pro-ℓ quotient. The arithmetic van Kampen presentation is its equation (6.2.5), with component comparison Lemma 6.2.8 and injective edge maps Corollary 6.2.9. The dual graph here is a single edge, so the based geometric amalgam is

$$\widehat{F}(d,c) *_{\langle d \rangle} \widehat{F}(d^{-1},b). \tag{F.9.5}$$

Its arithmetic amalgam uses the same coefficient section and the same node root fiber functor on both sides of (F.9.4). Thus the continuous maps from the two component groups, and their common inertia inclusion, are $G_{\mathbb{Q}}$- equivariant with these actual sections. The common generator is primitive in both vertex groups; eliminating its duplicate in the amalgam gives the free basis b, d, c. The same elimination on finite ℓ-group quotients proves the pro-ℓ comparison.

On each component use the standard path from the unit tangent at zero towards the point one. For a tripod with these data the based arithmetic action sends its first loop to its λ-th power and its second to a conjugate of its λ-th power by an element of the commutator subgroup. The latter assertion is the normalized action in Wickelgren, §2.1, equation for the G_k-action [Wic12]; its abelian path cocycle is zero because the two leading units are one. In the class-two quotient this conjugator is central. The outer pair (d,c) and bubble pair (d^{-1},b) therefore both have the pure-power action in (F.9.2). Compatibility in (F.9.4) proves that they give the same based action on (F.9.5). The Dehn twist gives the first row of (F.9.2).

For clarity about the original peripheral generators, the standard bubble loop at infinity is $b^{-1}d$. Conjugating its path by d gives the inertia generator $a = db^{-1}$, so $ab = d$ exactly. This is the indicated Nielsen basis change, with the displayed inverse; it is a continuous automorphism of each free pro-ℓ group. Both a and b are conjugated by d under the twist, proving (F.7). Finally the class-two collection identity

$$(ab)^{\lambda} = a^{\lambda} b^{\lambda} [a,b]^{-\binom{\lambda}{2}}$$

gives (F.9.3). The polynomial $\binom{\lambda}{2} = \lambda(\lambda-1)/2$ is integral for $\lambda \in \mathbb{Z}_2$ and extends continuously from integers; thus no division by a nonunit is being performed in the group law. This retains the sign information which would be lost by declaring the original pair a, d simultaneously diagonal without a tangent comparison at the prime two. □

6.2 Sections over the Laurent-series field

We use the arithmetic identification and the basis $(x,y,z) = (b, ab, c)$ of F.9. Put $G = G_{\mathbb{Q}((t))}$ and $G_0 = G_{\mathbb{Q}}$. The coefficient splitting fixes all compatible roots of t. A chosen tame generator

is denoted by τ; only its pro-ℓ component enters the following formulas. All cohomology groups and cocycles below are continuous.

Statement F.10 (Proved). An integral tame equation, including the prime two. *Write the class-two free pro-ℓ group P_2 in Hall coordinates*

$$x^{u_1}y^{u_2}z^{u_3}[x,y]^{w_{12}}[x,z]^{w_{13}}[y,z]^{w_{23}} = (u,w).$$

These give a homeomorphism of profinite sets $P_2 \simeq \mathbb{Z}_\ell^3 \times \mathbb{Z}_\ell^3$ with law

$$(u,w)(v,h) = (u+v, w+h+F(u,v)), \qquad F(u,v) = (-u_2v_1, -u_3v_1, -u_3v_2).$$

Put $L(u) = (-u_1, 0, 0)$ and $B(u,v) = F(u,v) - F(v,u)$. The arithmetic actions are

$$\tau^a(u,w) = (u, w + aL(u)), \qquad g(u,w) = (\chi(g)u, \chi(g)^2 w) \quad (g \in G_0).$$

For a section with geometric value (t,w) at τ and (r_g, z_g) at $g \in G_0$, the tame relation is exactly

$$(\chi(g)-1)w + B(r_g,t) - L(r_g) - \frac{\chi(g)-1}{2}\{F(t,t)+L(t)\} = 0. \tag{F.10.1}$$

The expression $(\chi(g)-1)/2$ is integral also for $\ell = 2$, because the 2-adic cyclotomic character is odd. No Baker–Campbell–Hausdorff denominator is used in this statement.

Proof. Moving x^{v_1} and y^{v_2} left through the preceding powers of y, z gives the displayed F with the convention $[x,y] = xyx^{-1}y^{-1}$. This proves the group law and, by uniqueness of Hall coordinates at each finite quotient, its topological interpretation. Conjugation by y sends x to $x[x,y]^{-1}$ and fixes y, z, proving the inertia action. The constant action is F.9.

Let $h(t) = F(t,t) + L(t)$. Integer powers of the lifted inertia element $((t,w),\tau)$ have geometric coordinates

$$(nt, nw + \tbinom{n}{2}h(t)).$$

Induction follows from the group law: the next multiplication adds $nF(t,t)$ and $nL(t)$. The integer-valued polynomial $\binom{n}{2}$ extends continuously to $\mathbb{Z}_\ell$, giving the same formula for every ℓ-adic power. The parts of tame inertia at other primes act trivially on P_2 and have trivial geometric cocycles, since a continuous homomorphism from a pro-prime-to-ℓ group to a pro-ℓ group is trivial.

Write $q = \chi(g)$. The equality $s(g)s(\tau) = s(\tau)^q s(g)$ has equal abelian coordinates $r_g + qt$ on both sides. Its central defect is

$$q(q-1)w + q\{B(r_g,t) - L(r_g)\} - \tbinom{q}{2}h(t).$$

Division by the unit q proves (F.10.1). The program `f_tame_relation.py` checks this identity symbolically over $\mathbb{Q}$ and by exact integer arithmetic in 30 000 trials at moduli $8, 9, 25, 27, 32, 49$. The symbolic identity and the integral power formula, rather than those trials, prove the assertion at every coefficient level. □

Statement F.11 (Proved). The continuous obstruction group and all abelian fibers. *There is a split isomorphism of $\mathbb{Z}_\ell$-modules*

$$H^2_{\mathrm{cts}}(G, \mathbb{Z}_\ell(2)^3) \simeq H^2_{\mathrm{cts}}(G_0, \mathbb{Z}_\ell(2))^3 \oplus H^1_{\mathrm{cts}}(G_0, \mathbb{Z}_\ell(1))^3. \tag{F.11.1}$$

The first summand is inflation. A splitting of the second is cup product with the Kummer class of the uniformizer t, with the orientation chosen so that the residue at τ is the identity.

The first summand carries a nonzero first-derived-limit term for every ℓ. The second carries none.

Let

$$\mathscr{K}_\ell = H^1(G_0, \mathbb{Z}_\ell(1)) = \begin{cases} \mathcal{R}_{\mathbb{P}}(\mathbb{Z}_\ell), & \ell \neq 2, \\ (\mathbb{Z}/2)\kappa(-1) \oplus \mathcal{R}_{\mathbb{P}}(\mathbb{Z}_2), & \ell = 2. \end{cases}$$

Let ϵ_ℓ be zero for odd ℓ and $\kappa(-1)$ for $\ell = 2$. The abelian section parameters are $(r, t) \in \mathscr{K}_\ell^3 \times \mathbb{Z}_\ell^3$. They lift to P_2 exactly when the following six classes vanish:

$$r_1 \cup r_2, \quad r_1 \cup r_3, \quad r_2 \cup r_3 \quad \text{in } H^2_{\mathrm{cts}}(G_0, \mathbb{Z}_\ell(2)), \tag{F.11.2}$$

and

$$\begin{aligned} &(t_2+1)r_1 - t_1 r_2 + \epsilon_\ell\, t_1(t_2+1), \\ &t_3 r_1 - t_1 r_3 + \epsilon_\ell\, t_1 t_3, \qquad\qquad \text{in } \mathscr{K}_\ell. \\ &t_3 r_2 - t_2 r_3 + \epsilon_\ell\, t_2 t_3 \end{aligned} \tag{F.11.3}$$

Every nonempty fiber of geometric conjugacy classes is a torsor under $H^1_{\mathrm{cts}}(G_0, \mathbb{Z}_\ell(2))^3$. In (F.11.2), vanishing means continuous cohomology, computed by F.1–F.3, including the finite-S term and restricted convergence.

Proof. For $M = \mathbb{Z}_\ell(2)^3$, inertia acts trivially on the coefficient module. Its cohomology has only degrees zero and one; the latter is $M(-1)$. The split group extension and the uniformizer Kummer class give, at every finite coefficient level, inflation and a right inverse to the residue map by cup product. Its restriction to inertia is the identity $\mathbb{Z}_\ell(1) \to \mathbb{Z}_\ell(1)$. Thus the two-row Hochschild–Serre complex splits in the indicated degrees. The same constructions commute with all coefficient reductions. Derived inverse limits give (F.11.1), so the splitting also retains the Milnor terms.

For twist two over $\mathbb{Q}$, the residue differential away from ℓ is multiplication by $q - 1$. Its nonzero Smith valuations are unbounded, by primes $q \equiv 1 \pmod{\ell^n}$; F.3 gives a nonzero derived limit. For twist one the residue differential is zero, so F.3 gives vanishing of its $\varprojlim^1 H^1$ term. The Milnor term attached directly to the H^1 summand in (F.11.1) is $\varprojlim^1 H^0(G_0, \mathbb{Z}/\ell^n(1))$; it vanishes because all these degree-zero groups are finite. The Kummer description of $\mathscr{K}_\ell$ follows directly from $\mathbb{Q}^\times = \{\pm 1\} \times \bigoplus_p p^{\mathbb{Z}}$, together with the finite H^0 coefficient comparison.

An abelian cocycle is specified by a cocycle $r : G_0 \to \mathbb{Z}_\ell(1)^3$ and a vector t on the chosen inertia generator. The tame relation on this abelian group is automatic, since both sides scale t by $\chi(g)$. The central cocycle equation on G_0 is

$$z_{gh} = z_g + \chi(g)^2 z_h + F(r_g, \chi(g) r_h).$$

Its three cohomology classes are the three cup products in (F.11.2): the negative reverse cup represents the forward cup, including at two by the cochain homotopy for graded commutativity. Vanishing gives continuous choices of z.

The remaining equation (F.10.1) is a coboundary equation with coefficients $\mathbb{Z}_\ell(1)^3$ for the single vector w. For odd ℓ, $(\chi - 1)/2$ is the coboundary of $1/2$. For $\ell = 2$, the compatible roots of -1 identify this cocycle with $\kappa(-1)$: on a root of unity of order 2^{n+1}, its Kummer exponent is $(\chi - 1)/2$ modulo 2^n. Expanding $B(r, t) - L(r)$ and $F(t, t) + L(t)$ now gives precisely (F.11.3). Vanishing of these classes supplies an integral w. The cocycle on G_0, the lifted inertia generator, and their verified tame relation define a continuous homomorphism on $\widehat{\mathbb{Z}}(1) \rtimes G_0$; the power formula in F.10 proves continuity on inertia. This proves sufficiency as well as necessity.

For fixed abelian parameters, central lifts differ by $H^1(G, M)$. Hochschild–Serre gives $H^1(G, M) = H^1(G_0, M)$, because $H^0(G_0, M(-1)) = 0$. There is no further identification by an abelian geometric stabilizer: $H^0(G, \mathbb{Z}_\ell(1)^3) = 0$. The central-extension exact sequence therefore gives the asserted torsor, with the chosen section origin as its sole ambiguity. □

Statement F.12 (Proved). The fully evaluated three-adic Laurent-series test. *For $\ell = 3$, write each $r_i = \sum_p r_{i,p}\kappa_p$ as a restricted family. At $q \neq 3$, choose compatible finite-field Kummer logarithms $\rho_{q,p}$ of the units p, for $p \neq q$, modulo $q - 1$, and use integral representatives. Put*

$$R_{i,q} = \sum_{p \neq q} r_{i,p}\rho_{q,p}, \qquad c_{ij,q} = -\frac{R_{i,q}r_{j,q} - R_{j,q}r_{i,q}}{q-1}.$$

The exact conditions (F.11.2) are that each of the three families $(c_{ij,q})_{q\neq 3}$ is integral and restricted. Condition (F.11.3) is the displayed system of three linear equations in the restricted Kummer families, with $\epsilon_3 = 0$. Every nonempty fiber has 27 section classes.

Proof. The cochain localization calculation of LM7 E.32l–E.32n applies with commutator coefficient one instead of nine. The comparison is explicit: the class-two law in logarithmic coordinates has central term $\frac{1}{2}(u \wedge v)$, so its central tame defect is $q\{(q-1)c + R_{i,q}r_{j,q} - R_{j,q}r_{i,q}\}$. Replacing the actual Frobenius Kummer values by the chosen finite-field representatives adds $(q-1)$ times an integral restricted family: each difference contains the ramification coefficient $r_{i,q}$ or $r_{j,q}$. This proves the same residue representative in the continuous quotient, including its convergence requirement. For this coefficient, the finite-S H^2 is zero by the independently checked calculation in F.4.

Finally $H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathbb{Z}_3(2)) \simeq \mathbb{Z}/3$. One direct verification uses $H^0(G_S, \mathbb{Z}/3^n(2))$ of order three, the vanishing of $H^2(G_S, \mathbb{Z}/3^n(2))$, and the real Euler characteristic. They give $|H^1(G_S, \mathbb{Z}/3^n(2))| = 3$ for all n. The nonzero compatible class is $g \mapsto (\chi_3(g)^2 - 1)/3$; its class has exact order three, since $\chi_3^2 - 1$ has minimum valuation one. Integral residues in degree one vanish because multiplication by $q - 1$ is injective on $\mathbb{Z}_3$. Thus the global and finite-S integral H^1 agree and have the asserted order. The torsor in F.11 has order $3^3 = 27$. □

6.3 The fixed-lift compactness question in Task F.4

Statement F.20 (Proved). The finite coboundary criterion in the profinite kernel. *Use the data of Task F.4: $K/\mathbb{Q}_p$ is finite, $\Delta = G_{K(\mu_\infty)}$, π is the full geometric profinite fundamental group, and $q : \pi \twoheadrightarrow \pi^{(p')}$ has kernel N_p. All actions are the genuine continuous actions induced by the given section s. For a fixed $\bar{h} \in (\pi^{(p')})^{s(\Delta)}$, choose $h \in \pi$ above it and put $d_\sigma = h^{-1}s(\sigma)hs(\sigma)^{-1}$. Then $\bar{h}$ has a fixed lift in π exactly when the image of d is a coboundary in N_p/U for every member U of a cofinal directed family of open normal subgroups of N_p stable under $s(\Delta)$. Such a family exists. The solution set is exactly the inverse limit of the finite solution sets; no additional derived-limit obstruction occurs.*

Proof. Write $\sigma(n) = s(\sigma)ns(\sigma)^{-1}$. The fixed-image hypothesis puts d_σ in N_p, and multiplication gives $d_{\sigma\tau} = d_\sigma\sigma(d_\tau)$. Moreover hn is fixed precisely when

$$d_\sigma = n\sigma(n)^{-1} \quad \text{for every } \sigma. \tag{F.20}$$

These identities prove the cocycle assertion and its exact relation to the desired lift, including the side on which the correcting element is multiplied.

We justify the cofinal stable family even though N_p need not be finitely generated. For an open normal subgroup $U \subset N_p$, joint continuity of the action and compactness of N_p give an open neighborhood of the identity in $s(\Delta)$ acting trivially modulo U on all of N_p. Concretely, for each point choose an action neighborhood preserving its coset modulo U; finitely many point neighborhoods cover N_p, and intersect their action neighborhoods and their inverses. A smaller open subgroup then stabilizes U. The orbit of U is finite, and its intersection is an open normal subgroup contained in U stable under all of $s(\Delta)$. Taking these intersections and finite common refinements gives the claimed family.

For each such U let

$$X_U = \{nU \in N_p/U : d_\sigma = n\sigma(n)^{-1} \bmod U \text{ for every } \sigma\}.$$

This is a finite set and is exactly the finite-level solution set. Its inverse image Y_U in the compact group N_p is closed. Under the stated finite-level condition it is nonempty. A common refinement of finitely many U supplies a nonempty Y contained in all corresponding Y_U, so these closed sets have the finite-intersection property. Compactness gives $n \in \bigcap_U Y_U$. Since $\bigcap_U U = 1$, equation (F.20) holds in N_p for every σ. Conversely an actual solution gives every finite solution. The identification $N_p = \varprojlim_U N_p/U$ identifies the total solution set with $\varprojlim_U X_U$. This proves the exact criterion, not merely a necessary collection of finite tests. □

Statement F.22 (Proved). A full free pro-prime Dehn-twist fixed group. *Let F_ℓ be the free pro-ℓ group on x, y, z and let $0 \neq m \in \mathbb{Z}_\ell$. The continuous automorphism*

$$\alpha_m(x) = y^m x y^{-m}, \qquad \alpha_m(y) = y, \qquad \alpha_m(z) = z$$

has fixed subgroup exactly the closed free factor $\langle y, z\rangle$. This is a statement about the entire free pro-ℓ group.

Proof. We use the continuous Magnus map

$$F_\ell \longrightarrow 1 + (X, Y, Z)\mathbb{F}_\ell\langle\langle X, Y, Z\rangle\rangle, \qquad x \mapsto 1 + X,\ y \mapsto 1 + Y,\ z \mapsto 1 + Z. \tag{F.22.1}$$

Here the ring has its degree-adic topology, and the unit group on the right is an inverse limit of finite ℓ-groups. The map is injective. To check this rather than assume a linearization, the inverse ring map sends X, Y, Z to $x - 1, y - 1, z - 1$ in $\mathbb{F}_\ell[[F_\ell]]$. These substitutions converge: the augmentation ideal in the group algebra of every finite ℓ-group is nilpotent. The latter fact follows by induction using a central subgroup of order ℓ, whose generator c satisfies $(c - 1)^\ell = 0$. The two continuous ring homomorphisms are inverse, by their values on the generators. Finally F_ℓ embeds in the unit group of its completed group algebra, because its image in each finite group algebra retains the distinct group basis elements. This proves the assertion about (F.22.1).

Put $u(Y) = (1 + Y)^m$ in $\mathbb{F}_\ell[[Y]]$. It is nonconstant: writing $m = \ell^v m_0$ with m_0 a unit, its first nonconstant term has degree ℓ^v and nonzero coefficient. The induced substitution on the ring is $X \mapsto u(Y)Xu(Y)^{-1}$, with Y, Z fixed. For a word $w = w_1 \cdots w_k$ in the two letters X, Z, collect all coefficients of

$$Y^{a_0} w_1 Y^{a_1} \cdots w_k Y^{a_k}$$

as the single commutative series $f_w(Y_0, \ldots, Y_k)$. Every noncommutative series has this unique coefficient decomposition. The substitution multiplies f_w by

$$\prod_{i:\, w_i = X} u(Y_{i-1})u(Y_i)^{-1}. \tag{F.22.2}$$

For every w containing X this product differs from one: at the left endpoint of the first consecutive block of X's the corresponding variable has exponent $+1$, with no inverse factor to cancel it. The commutative power-series ring over $\mathbb{F}_\ell$ is an integral domain. A fixed series therefore has $f_w = 0$ whenever w contains X.

For a fixed group element, its Magnus series consequently uses only Y, Z. The continuous retraction $F_\ell \to \langle y, z\rangle$ given by $x \mapsto 1$ sets $X = 0$ in its Magnus series and leaves that series unchanged. Injectivity of (F.22.1) makes the element equal to its retraction. Conversely α_m fixes the stated free factor pointwise. □

Statement F.23 (Proved). All prime-to-five pro-prime fixed quotients in the three local examples. *Let $\Delta = G_{\mathbb{Q}_5(\mu_\infty)}$ and let $\pi^{(\ell)}$ denote the maximal geometric pro-ℓ quotient. For each $\ell \neq 5$, the first two examples of Task F.4 satisfy*

$$\big(\pi^{(\ell)}\big)^{s(\Delta)} = \pi^{(\ell)} = F_\ell(2).$$

For the third example, with the point 2 on $\mathbb{P}^1 \setminus \{0, 1, 5, \infty\}$, choose peripheral generators a, b, c at $0, 5, 1$ based through the outer component of its semistable model and put $d = ab$. Then

$$\big(\pi^{(\ell)}\big)^{s(\Delta)} = \langle d, c\rangle \simeq F_\ell(2). \tag{F.23}$$

The natural quotient $\pi \twoheadrightarrow \pi^{(\ell)}$ is continuous and $s(G_{\mathbb{Q}_5})$-equivariant. It sends the full fixed group into the displayed subgroup; surjectivity of that restriction is not asserted.

Proof. The smooth marked integral model in the first two examples, together with its integral point or unit tangential base point, gives based prime-to-five specialization. Inertia acts trivially on this quotient. Since all finite unramified extensions of $\mathbb{Q}_5$ are generated by prime-to-five roots of unity, Δ is contained in inertia, proving these two assertions.

For the third example blow up the collision $x = 5 = 0$ and use the node chart $XV = 5$, with $X = x$ and $V = 5/x$. The outer component has node 0, cusps $1, \infty$, and the smooth reduced base point 2. The bubble has node 0 and cusps $1, \infty$ in coordinate V. The logarithmic specialization and amalgamation maps of Stix, Theorem 3.4.6 and equation (6.2.5) [Sti02], give the genuine based prime-to-five comparison, by the same root-fiber-functor construction as F.9. The outer component subgroup is $\langle d, c\rangle$. A path based at 2 inside this smooth outer component has trivial inertia action in each prime-to-five quotient by based good reduction. The gluing along $XV = 5$ therefore makes tame inertia act by

$$a \longmapsto d^r a d^{-r}, \qquad b \longmapsto d^r b d^{-r}, \qquad c \longmapsto c, \quad r \in \prod_{q \neq 5} \mathbb{Z}_q(1).$$

Wild inertia acts trivially in this geometric prime-to-five comparison. These formulas follow also directly from the node Kummer covers: a turn in the base annulus conjugates the bubble loops by the common edge loop and fixes the based outer loops. In the free basis $(x, y, z) = (b, d, c)$ the action is α_r.

The image of Δ in tame inertia is $4\prod_{q\neq 5} \mathbb{Z}_q(1)$. Indeed the tame part of the five-adic cyclotomic character is the character of order four; on wild inertia its image is $1 + 5\mathbb{Z}_5$, as is seen on the extensions generated by fifth-power roots of unity. A tame element in the kernel of the character of order four can thus be corrected by a wild element to have cyclotomic character one. All other cyclotomic characters are already trivial on inertia. Projecting to the ℓ factor gives $4\mathbb{Z}_\ell(1)$, including $\ell = 2$. It follows that the fixed group is the fixed group of α_4; continuity gives equality for the closure of its integer powers. Statement F.22 applies since $4 \neq 0$ in $\mathbb{Z}_\ell$. This proves (F.23) and its precise relation to the full fixed group. □

Statement F.24 (Proved). The pro-five fixed groups for the two good-marked-reduction examples. *For $U = \mathbb{P}^1_{\mathbb{Q}_5} \setminus \{0, 1, \infty\}$, let $q_5 : \pi \twoheadrightarrow \pi^{(5)}$ be the continuous maximal geometric pro-five quotient. With the genuine action of the point 2, respectively of an integral unit tangent at 0, we have*

$$\left(\pi^{(5)}\right)^{s_2(\Delta)} = 1, \qquad \left(\pi^{(5)}\right)^{s_{0,u}(\Delta)} = I_0^{(5)}. \tag{F.24.1}$$

Consequently $q_5(\pi^{s_2(\Delta)}) = 1$ and $q_5(\pi^{s_{0,u}(\Delta)}) = I_0^{(5)}$. The latter equality of images follows because the full cusp inertia is itself fixed. It does not identify either full profinite fixed group.

Proof. Put $F = \mathbb{Q}(\zeta_5)$ and $S = \{v : v \mid 10\}$, and let G be the maximal pro-five quotient of $G_{F,S}$. We first prove that the image of inertia at five in G is all of G. The discriminant of F is 5^3. Its Minkowski bound is $15\sqrt{5}/(2\pi^2) < 2$, so its ideal class group is trivial. The prime two is inert, since 2 has order four modulo five, and $(\mathcal{O}_F/(2))^\times$ has order 15. The subgroup $\mu_5 \subset \mathcal{O}_F^\times$ maps injectively to this group and exhausts its five-primary part. The ray-class exact sequence therefore gives $\mathrm{Cl}_{(2)}(F)[5] = 0$. These uses of the ray-class correspondence are precisely Milne, Chapter V, Theorem 3.6 and Example 3.9 [Mil20c].

A cyclic extension of degree five unramified outside S and unramified at five is tamely ramified at two and has conductor dividing (2). The preceding computation excludes it. If the inertia image were a proper closed subgroup of G, its image in some finite five-group quotient would be proper and would lie in a subgroup of index five. Pulling back that subgroup would give exactly such a cyclic extension. This proves the asserted surjectivity, without assuming that the global group is free.

Let $F_S^{(5)}$ be the extension with group G. Its intersection with $\mathbb{Q}(\mu_\infty)$ is $F(\mu_{5^\infty})$. To verify the equality, the intersection is Galois and abelian over $\mathbb{Q}$, is unramified outside $\{2, 5\}$, and is pro-five over F. The cyclotomic conductor description over $\mathbb{Q}$ (Milne, Chapter V, Example 3.10) shows that its part from a power of two has degree a power of two; hence it is already in F. The remaining part is the five-power cyclotomic tower, which is contained in $F_S^{(5)}$. Thus the image of the global full-cyclotomic kernel in G is $\ker \chi_5$. The local full-cyclotomic kernel has the same image: inertia at five surjects onto G, and the cyclotomic character of an inertia lift of an element of $\ker \chi_5$ is one at five and at every other prime. All these maps are restrictions of Galois automorphisms for a fixed embedding $\overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_5$.

For the point 2, the based pro-five action is unramified outside $\{2, 5\}$: the marked curve and its point extend over $\mathbb{Z}[1/10]$, and based prime-to-residue-characteristic specialization applies at every remaining prime. Over F its action on the Frattini quotient is trivial, since that action is χ_5 modulo five. The kernel of $\mathrm{Aut}(F_5(2)) \to \mathrm{GL}_2(\mathbb{F}_5)$ is pro-five: the successive characteristic exponent-five central quotients give finite filtrations with elementary abelian five-group kernels. Hence the based action factors through G. We have proved equality of the local and global cyclotomic-kernel images for this actual based action.

We check that the global point section at 2 is not cuspidal even in the pro-five group. Its two abelian Kummer coordinates are $(\kappa(2), \kappa(-1)) = (\kappa(2), 0)$. The maps $x, 1 - x : U \to \mathbb{G}_m$, evaluated on the leading terms of the unit tangents, show that the cusp-one packet has first coordinate zero and the cusp-infinity packet has equal coordinates. Since $v_{(2)}(2) = 1$, neither contains this class. To exclude the cusp-zero packet, consider over F the successive covers

$$y^5 = 1 - x, \qquad z_j^5 = \frac{y - \zeta_5^j}{-1 - \zeta_5^j} \quad (j = 1, 2). \tag{F.24.2}$$

They are finite etale covers of the successive punctured curves, with geometric degrees powers of five. Their composite is geometrically connected: the divisors of the two displayed Kummer functions have independent classes modulo five at $y = \zeta_5$ and $y = \zeta_5^2$. The point lifts to

$(y, z_1, z_2) = (-1, 1, 1)$. Every cusp above zero first has $y = \zeta_5^i$. Choose $j \in \{1, 2\}$ distinct from i. At the prime $(1 - \zeta_5)$ the numerator $\zeta_5^i - \zeta_5^j$ has valuation one, while $-1 - \zeta_5^j$ is a unit. This value is not a fifth power in F, so no cusp above this point in (F.24.2) is F-rational. An F-section contained in the lifted cusp decomposition group would force such a rational cusp in every cover to which the section lifts. Thus the point section is noncuspidal.

The proof of B.4 applies word for word to this free pro-five fundamental group over the number field F: its cover comparison uses finite five-groups, its boundary sequence uses $\mathbb{Z}_5(1)$, and Ribet's torsion finiteness applies to the compactified cover Jacobians. It gives a trivial global cyclotomic-kernel centralizer for the just verified noncuspidal point section, and cusp inertia for the unit tangent at zero. Equality of the local and global images proves the first formula in (F.24.1) and the second for tangent one.

Finally the pro-five completion of $\mathbb{Z}_5^\times$ is free of rank one and is generated by the class of 2: $v_5(2^4 - 1) = 1$, so $\log(2)$ generates $5\mathbb{Z}_5$. An integral unit tangent u therefore has $\kappa_5(u) = a\kappa_5(2)$ locally for a unique $a \in \mathbb{Z}_5$. The global cusp section obtained by twisting the unit tangent by $a\kappa_5(2)$ is unramified outside $\{2, 5\}$ and has the same local pro-five section class. It factors through the same relative pro-five action considered above, and B.4 gives its global centralizer $I_0^{(5)}$. The local/global image equality proves the claim for every integral unit tangent. Conjugating a representative by cusp inertia leaves the asserted inertia subgroup unchanged. The stated maps from the full fixed groups now follow directly by equivariance. The exact finite certificates used here—the cyclotomic discriminant, residue-field order, ray-class five-part, Kummer-function norms, and the lifted point—are evaluated in `code/bc_local_checks.py`, with its output included. □

Statement F.25 (Proved). The pro-five fixed group for the colliding-cusp example. *For $U = \mathbb{P}^1_{\mathbb{Q}_5} \setminus \{0, 1, 5, \infty\}$, with the section at 2 and $\Delta = G_{\mathbb{Q}_5(\mu_\infty)}$, we have*

$$\left(\pi^{(5)}\right)^{s_2(\Delta)} = 1. \tag{F.25.1}$$

Consequently the full profinite fixed group maps to one under the maximal pro-five quotient. Together with F.23 this evaluates every individual pro-prime fixed quotient in all three examples, with the explicit quotient maps described there.

Proof. Filling the cusp 5 is an actual based morphism

$$U \hookrightarrow U_0 = \mathbb{P}^1 \setminus \{0, 1, \infty\}.$$

It induces a continuous, surjective, equivariant map $F_5(3) \to F_5(2)$ killing the normal closure K of the inertia at the new puncture. By F.24 the target has trivial fixed group for the section at 2. It suffices to prove $K^\Delta = 1$.

We give the integral kernel comparison used for that assertion. Choose a geometric free basis which expresses $F_5(3) = F_5(2) *_5 \langle b \rangle$, where b is inertia at 5 and the quotient kills b. The profinite set X of lifts of the point 5 in the universal pro-five cover of U_0, based at 2, is a torsor under $F_5(2)$. After choosing one lift, identify it with $F_5(2)$. The kernel K is the free pro-five group on this profinite set, with continuous generators $x \mapsto xbx^{-1}$. This free group is universal for continuous maps from that profinite set to pro-five groups. The assertion follows by two explicit inverse homomorphisms: let the free group on X be acted on by $F_5(2)$ through left translation of X. Its semidirect product with $F_5(2)$ receives the free product by sending b to the generator at 1 and fixing $F_5(2)$. Conversely, $x \mapsto xbx^{-1}$ gives the homomorphism back, equivariant for that left-translation action. Both composites fix the indicated generators. Thus the kernel and all its generator labels are the actual ones for the filling morphism.

In particular

$$K^{\mathrm{ab}} \simeq \mathbb{Z}_5[[X]](1), \tag{F.25.2}$$

where the completed module is the module of $\mathbb{Z}_5$-valued finitely additive measures on the clopen subsets of X. Galois transports each inertia generator to a conjugate of the generator at the transported lifted cusp, raised to its cyclotomic power. Thus (F.25.2) is equivariant, and on Δ the twist is one. For every $r \geq 1$ there is also an injective equivariant degree-r Magnus homomorphism

$$\gamma_r K/\gamma_{r+1} K \hookrightarrow \mathbb{Z}_5[[X^r]]. \tag{F.25.3}$$

To verify this integral comparison for the infinitely generated kernel, use the finite partitions $X \to X_n = F_5(2)/M_n$, with M_n characteristic and open. Every continuous map from X to a finite five-group factors through one such partition, so the free group on X is the inverse limit of the free pro-five groups on the finite sets X_n. Its closed lower central groups and their quotients are the corresponding inverse limits: membership is detected in finite five-group quotients, and compatible cosets lift by compactness. For finite X_n, the Hall commutator basis and its leading Magnus monomials give the injective map into $\mathbb{Z}_5[X_n^r]$. Taking their inverse limit gives (F.25.3). Naturality of the degree-r term makes this map equivariant: conjugating a generator changes only terms of degree at least two, and the action on the degree-r graded piece is induced from its action on the degree-one module.

The action of Δ on X has pro-five image. Indeed its action on $F_5(2)$ is trivial on the Frattini quotient, and that automorphism kernel is pro-five, as proved in F.24. After a choice of torsor origin, the affine action on X is a homomorphism into the pro-five semidirect product of $F_5(2)$ with this automorphism kernel.

There are no finite orbits of this action. The regular function $1 - x$ on U_0 sends its path torsor from 2 to 5 to the division-root torsor of

$$\frac{1-5}{1-2} = 4.$$

On Δ its translation character is $\kappa_5(4) = 2\kappa_5(2)$. The latter character has a unit value. For a direct proof, the polynomial $(T+2)^5 - 2$ is Eisenstein at five, so $\mathbb{Q}_5(2^{1/5})$ has degree five. It is not Galois: a Galois extension containing one fifth root of 2 contains all of them and therefore ζ_5, whose degree is four. It cannot be a subfield of the abelian extension $\mathbb{Q}_5(\mu_\infty)$. Some element of Δ consequently moves that fifth root, which says that $\kappa_5(2)$ has nonzero reduction modulo five on Δ. Since it is a continuous homomorphism there, its image is all of $\mathbb{Z}_5$. Every orbit on X, and on X^r by projection to the first coordinate, therefore maps onto an infinite translation orbit.

We finish with the required measure argument. Let a pro-five group act continuously on a profinite set Z with no finite orbits. An invariant integral $\mathbb{Z}_5$-measure on Z is zero. For a clopen subset C, its stabilizer is open, by uniform continuity on the compact set. All orbits of this stabilizer on C are infinite. For each n, choose a finite invariant partition of C on which every orbit has size at least 5^n. Such a partition exists: each point has arbitrarily many distinct translates separated by a finite clopen partition; continuity gives an invariant refinement, and a finite common refinement works for all points by compactness. The finite orbit sizes are powers of five. Invariance makes the measure constant on the atoms in each orbit, so its value on C is divisible by 5^n. Since this holds for every n, that value is zero. This proves the claim.

Apply it to $Z = X^r$ in (F.25.3). Every graded fixed subgroup is zero. A nontrivial element of K has a nonzero image in some lower-central graded quotient, because it survives in a finite five-group and finite five-groups are nilpotent. A fixed element would give a nonzero graded invariant, which is impossible. Thus $K^\Delta = 1$, proving (F.25.1) and the stated consequence for the full profinite quotient map. □

Statement F.26 (Proved). A mixed-prime quotient and every one of its nilpotent fixed groups. *For the first example, with the section at* 2*, consider the actual finite etale cover*

$$\begin{aligned} E^\circ &\longrightarrow U, \qquad E:\ Y^2 = X^3 - 2X, \quad t = X^2/2, \\ D_E &= \{O, (0,0), (\sqrt{2},0), (-\sqrt{2},0)\}, \qquad P = (2,2). \end{aligned} \tag{F.26.1}$$

Here $E^\circ = E \setminus D_E$*, and* P *maps to* 2*. The cover is cyclic of degree four over* $\mathbb{Q}_5$*: in the coordinate* $Z = Y/2$ *its equation is* $Z^4 = t(t-1)^2/2$*. Choose* $i \in \mathbb{Q}_5$ *with* $i^2 = -1$ *and* $i \equiv 2 \bmod 5$*; a deck generator is* $\delta(X,Y) = (-X, iY)$*. Let* $H \subset \pi$ *be the index-four subgroup determined by this based cover, put* $H_5 = H^{(5)}$*, and define*

$$J = \pi / \ker(H \longrightarrow H_5), \qquad J_c = J/\gamma_{c+1}H_5 \quad (c \geq 1). \tag{F.26.2}$$

These are continuous, surjective, $s_2(G_{\mathbb{Q}_5})$*-equivariant quotients of the full geometric profinite group. Their fixed groups are*

$$J^{s_2(\Delta)} = 1, \qquad J_1^{s_2(\Delta)} = H_5^{\mathrm{ab}} \simeq \mathbb{Z}_5^5, \qquad J_c^{s_2(\Delta)} = \gamma_c H_5/\gamma_{c+1}H_5 \quad (c \geq 2). \tag{F.26.3}$$

In particular the fixed group at class two is $\mathbb{Z}_5^{10}$*, and every nonzero element of it fails to have a fixed lift at class three. The transition maps between consecutive fixed groups in (F.26.3) are zero. The image of the full fixed group in* J *is therefore trivial; in particular its image under the cyclic quotient whose peripheral vector is* $(1,2) \bmod 4$ *is trivial.*

Proof. We give the arithmetic and Lie comparisons explicitly. The four points above 2 are $P, \delta P, \delta^2 P, \delta^3 P$, all rational over $\mathbb{Q}_5$. Thus the based arithmetic cover is the one with section at P, and its subgroup H is stable under the genuine action in the statement. The kernel of the maximal pro-five quotient of H is characteristic in H, normal in π, and stable under this arithmetic action. This proves all the quotient assertions in (F.26.2). The geometric group H_5 is free pro-five of rank $2 \cdot 1 + 4 - 1 = 5$.

Write $V_E = H_5^{\mathrm{ab}}$ and $W_E = \mathbb{Z}_5(1)[D_E]/\mathbb{Z}_5(1)\sum_d d$. The functions

$$X, \quad X - \sqrt{2}, \quad X + \sqrt{2}$$

have respective divisors $2d - 2O$. Their induced homomorphism on V_E restricts to multiplication by two on the boundary basis e_0, e_+, e_-. Since two is a unit in $\mathbb{Z}_5$, it gives an integral Galois-equivariant splitting

$$V_E = W_E \oplus T_5 E. \tag{F.26.4}$$

The homomorphisms are equivariant over $\mathbb{Q}_5(\sqrt{2})$; their permutation under its Galois involution proves the stated equivariance over $\mathbb{Q}_5$. They are also equivariant for δ, which fixes e_0 and interchanges e_+ and e_-. Thus $W_E = W_+ \oplus W_-$, with ranks two and one, $W_- = \mathbb{Z}_5(e_+ - e_-)(1)$. Its arithmetic characters are respectively χ_5 and $\chi_5\varepsilon$, where ε is the unramified quadratic character. Indeed $\sqrt{2}$ generates the unramified quadratic extension.

The two eigenspaces of $[i]$ split T_5E integrally, since $2i$ is a unit. One has an unramified character ψ; the other has character $\chi_5\psi^{-1}$ by the Weil pairing. Here is a direct check of the unramified assertion. The endomorphism $[2+i]$ has degree five and unit differential $2 + i \bmod 5$ on the good integral model. Its reduction is separable, so its kernel and the kernels of its iterates are finite etale over $\mathbb{Z}_5$. Their inverse limit is one of the two CM Tate lines. Counting points gives $\#E(\mathbb{F}_5) = 10$, so the Frobenius polynomial is $T^2 + 4T + 5$. The unit root is $-2 - i$ for the chosen embedding of i; thus $\psi(\mathrm{Frob}_5) = -2 - i$. This root is not a root of unity: either complex embedding has absolute value $\sqrt{5}$. In particular Δ acts trivially on all of (F.26.4).

Choose the unit tangent at O, with parameter X/Y, as an auxiliary reference section on E°. The deck automorphism has derivative i there. The compatible fifth-power roots of i are all i, and are rational over $\mathbb{Q}_5$. The resulting tangent comparison gives a genuine order-four automorphism of the based pro-five group, commuting with its arithmetic action. Let

$$j_\sigma : V_E \longrightarrow \bigwedge^2 V_E \quad (\sigma \in \Delta) \tag{F.26.5}$$

be the first Johnson homomorphism: for a lift $\widetilde{v}$ of v, it is the class of $\sigma(\widetilde{v})\widetilde{v}^{-1}$ modulo $\gamma_3 H_5$. It is independent of that lift because Δ acts trivially on V_E and on $\bigwedge^2 V_E$. It is additive in σ and is equivariant under conjugation by $G_{\mathbb{Q}_5}$. The latter assertion follows directly by conjugating an automorphism which is the identity on the abelianization; any central part of the conjugating automorphism cancels. It also commutes with $[i]$.

Each $d - O$, for $d \in D_E$, is two-torsion. Its pro-five Kummer path is trivial: multiplication by 5^n fixes this two-torsion point, providing a rational compatible division sequence. The formula for the action on peripheral inertia therefore gives $j_\sigma(W_E) \subset \bigwedge^2 W_E$. The CM eigenvalues on T_5E are $i, -i$, whereas those on $\bigwedge^2 W_E$ and on $\bigwedge^2 T_5E$ are $1, -1$. Commutation with $[i]$ consequently gives $j_\sigma(T_5E) \subset W_E \otimes T_5E$. Together these facts put (F.26.5), after tensoring with $\mathbb{Q}_5$, in a direct sum of one-dimensional arithmetic characters drawn from

$$\chi_5, \quad \varepsilon\psi^2, \quad \chi_5^2\varepsilon\psi^{-2}. \tag{F.26.6}$$

For example the off-diagonal map from one CM line to W_- times the other has character $\chi_5\varepsilon$ times the ratio of those two lines. The diagonal maps use W_+ and have character χ_5. This proves (F.26.6) without assuming a diagonal action on the nonabelian group.

Choose a geometric path from the auxiliary tangent to P, and write $b_\sigma \in V_E$ for its abelian cocycle. The change from the auxiliary action to the action at P changes (F.26.5) by

$$j_\sigma^P = j_\sigma + \operatorname{ad} b_\sigma; \tag{F.26.7}$$

reversing the initial path changes the two displayed signs together and has no effect on the argument. The functions defining (F.26.4), with leading coefficient one at the auxiliary tangent, evaluate at P as $2, 2-\sqrt{2}, 2+\sqrt{2}$. Hence the component of b in W_- is

$$\frac{1}{4}\kappa_5\left(\frac{2-\sqrt{2}}{2+\sqrt{2}}\right)(e_+ - e_-) = \frac{1}{2}\kappa_5(\sqrt{2}-1)(e_+ - e_-). \tag{F.26.8}$$

This is a cocycle for the character $\theta = \chi_5\varepsilon$. Its restriction to Δ is nonzero. To prove that assertion, put $L = \mathbb{Q}_5(\sqrt{2})$. The unit $\sqrt{2}-1$ is not a root of unity, as its real algebraic embedding has absolute value different from one. The logarithm on a sufficiently small open unit subgroup shows that its rational local Kummer class is nonzero. For an abelian profinite group Γ and a nontrivial one-dimensional $\mathbb{Q}_5$-character η, $H^1_{\mathrm{cts}}(\Gamma, \mathbb{Q}_5(\eta)) = 0$: the cocycle identity for commuting g, h, with $\eta(g) \neq 1$, expresses every cocycle as the coboundary of $c(g)/(\eta(g)-1)$. Apply this to the abelian quotient G_L/Δ and χ_5. Inflation–restriction proves that the Kummer class stays nonzero on Δ.

The T_5E component of $b|_\Delta$ is nonzero as well. Indeed $\#E(\mathbb{F}_3) = 4$ and $\#E(\mathbb{F}_5) = 10$. Injectivity of prime-to-residue-characteristic torsion specialization bounds $E(\mathbb{Q})_{\mathrm{tors}}$ by a group of order dividing two. Since $Y(P) = 2$, P is not torsion. The formal logarithm on an open subgroup of $E(\mathbb{Q}_5)$ then shows that its rational five-adic Kummer class is nonzero. The local filtration and specialization assertions are those of Milne, Chapter II, Theorem 4.1, Corollary 4.2 and Aside 4.4 [Mil21]; alternatively the kernel of reduction is a pro-residue-characteristic formal group, which proves the torsion injectivity directly. Both characters ψ and $\chi_5\psi^{-1}$ are

nontrivial. Applying the preceding abelian-quotient argument to these two lines proves that this class also stays nonzero on Δ.

We now separate these homomorphisms on Δ exactly. The character θ in (F.26.8) occurs neither in (F.26.6), nor in W_+, nor in T_5E. The powers of χ_5 distinguish it from the two off-diagonal characters in (F.26.6) on inertia; it differs from χ_5 by the nontrivial ε. It differs from the unramified line ψ, and equality with $\chi_5\psi^{-1}$ would require $\psi = \varepsilon$, contradicting the displayed Frobenius value. The joint homomorphism

$$\Delta \longrightarrow \operatorname{Hom}(V_E, \bigwedge^2 V_E) \otimes \mathbb{Q}_5 \ \oplus \ V_E \otimes \mathbb{Q}_5, \qquad \sigma \longmapsto (j_\sigma, b_\sigma)$$

has compact additive image, a $\mathbb{Z}_5$-lattice in its rational span. That span is stable under $G_{\mathbb{Q}_5}$ and is a sum of the characters just listed. The projection onto its nonzero θ component is a polynomial in finitely many of these commuting Galois operators: for each other character choose an operator distinguishing it from θ, and multiply the corresponding linear projections. Clearing the finitely many five-adic denominators puts a nonzero projected vector back in the compact image lattice. There is consequently an actual $\sigma_0 \in \Delta$ such that

$$j_{\sigma_0} = 0, \qquad b_{\sigma_0} = w, \qquad 0 \neq w \in W_- \otimes \mathbb{Q}_5. \tag{F.26.9}$$

The vector in (F.26.9) is integral because it is in the actual image. This step constructs the required group element from the additive image, rather than choosing unrelated values for different cocycles.

For completeness, the closed lower-central graded algebra of the finite-rank free pro-five group embeds by its leading Magnus terms in the tensor algebra on V_E. An automorphism acting trivially on V_E acts to the next degree by the derivation induced by its first Johnson map. This follows by expanding a commutator and using the Leibniz rule in the tensor algebra; the Hall basis proves the assertion integrally in each degree. By (F.26.7)–(F.26.9), that derivation for σ_0 is $\operatorname{ad} w$. In the free Lie algebra over $\mathbb{Q}_5$, the centralizer of a nonzero degree-one element w consists of its linear span. One direct proof extends w to a vector-space basis: a homogeneous tensor polynomial commuting with the letter w contains only that letter, by comparing first and last letters of its words. Its only possible homogeneous form is a multiple of w^r; for $r > 1$ this is not a Lie element, as its tensor coproduct has nonzero intermediate binomial coefficients.

It follows that the leading degree of a fixed element below the last central degree of a nilpotent quotient must be one, with leading vector proportional to w. Choose $\sigma_1 \in \Delta$ with $(b_{\sigma_1})_T \neq 0$. For this vector w, the $W_E \otimes T_5E$ component of $j^P_{\sigma_1}(w)$ is exactly $[(b_{\sigma_1})_T, w]$, which is nonzero. Indeed $j_{\sigma_1}(w)$ and the bracket with $(b_{\sigma_1})_W$ lie in $\bigwedge^2 W_E$. Thus no such degree-one vector is fixed either. Every fixed element of $H_5/\gamma_{c+1}H_5$, for $c \geq 2$, is therefore in its last graded subgroup. Conversely that subgroup is fixed because the action on every associated graded is induced from the trivial action on V_E. This proves the asserted fixed groups inside H_5. Their ranks in degrees two and three are respectively 10 and 40.

Finally no element outside H_5 can be fixed in J_c for any $c \geq 1$. The image of the inertia at O in J has an order-four subgroup mapping isomorphically onto the deck group; its generator δ_0 is fixed by the auxiliary cuspidal action on Δ. Its conjugation on V_E is the deck action $[i]$. For an element with deck value δ_0^k, the fixed equation modulo $[H_5, H_5]$ for the action at P requires

$$(1 - \delta^k)b_\sigma = 0 \quad \text{for all } \sigma \in \Delta.$$

For $k = 1, 2, 3$, the operator $1 - \delta^k$ is invertible on $T_5E \otimes \mathbb{Q}_5$, and the proper component of b is nonzero. This excludes all three deck values. Together with the preceding paragraph this proves the formulas for J_c. The intersection of the lower central subgroups of H_5 is one,

since each nontrivial element survives in a finite five-group, and every such finite group is nilpotent. Taking the inverse limit proves $J^{\Delta} = 1$.

The peripheral vector follows directly from the valuations of $t(t-1)^2/2$ at 0 and 1. All finite curve counts, toric identities, character comparisons and Hall-dimension checks used here are verified in `code/bc_mixed_cm_checks.py`. □

Statement F.27 (Proved). An explicit finite obstruction to prime-to-five fixed lifts. *For the point* 2 *on the tripod over* $\mathbb{Q}_5$*, the image of the full fixed group in the geometric abelian quotient* $(\mathbb{Z}/4)^2$ *is zero. In particular, although every element of* $\pi^{(5')}$ *is fixed, any such element with nonzero abelian image modulo four has no fixed lift in* π*. This obstruction is already detected by two actual finite quotients, each isomorphic to*

$$(\mathbb{F}_5)^5 \rtimes C_4 \quad \textit{of order } 12500. \tag{F.27.1}$$

The cyclic generator acts on the normal vector group by fixing e_0*, interchanging* e_+, e_-*, and multiplying the two CM coordinates by* 2 *and* 3*.*

Proof. The first finite quotient is $\overline{J} = J/(H_5^5[H_5, H_5])$, with the actual maps of F.26. We verify at this precise finite level that no nontrivial deck value is fixed. Put $\beta = [2 - i]$. Its kernel is the formal CM subgroup of $E[5]$, and it is isomorphic over $\mathbb{Q}_5$ to μ_5: the other CM line has unramified character $\psi \equiv 1 \bmod 5$, and the Weil pairing identifies the two lines as dual with cyclotomic twist. The point $P = (2, 2)$ is not in $\beta E(\mathbb{Q}_5)$. Here is a finite certificate retaining the isogeny map. On the integral affine curve it is given by

$$\begin{aligned} X(\beta Q) &= \frac{X(3+4i)(X^2-2-4i)^2}{(5X^2-2+4i)^2}, \\ Y(\beta Q) &= \frac{Y(2+11i)(X^2-2-4i)(X^4+(4+16i)X^2+4)}{(5X^2-2+4i)^3}. \end{aligned} \tag{F.27.2}$$

These identities follow by applying the addition law to $2Q - [i]Q$, and are checked symbolically by the program. The denominators are units at five. Modulo five the map is $-\mathrm{Frob}_5$, so any preimage of P must reduce to $-P = (2, 3)$. Modulo twenty-five, with $i = 7$, all five lifts of that residue point map under (F.27.2) to $(7, 2)$, rather than $(2, 2)$. The finite enumeration is included in the program and its output.

Consequently the isogeny Kummer class $\delta_\beta(P) \in H^1(\mathbb{Q}_5, \mu_5)$ is nonzero. It remains nonzero on Δ. Indeed for any $u \in \mathbb{Q}_5^{\times}$ which is not a fifth power, adjoining a fifth root gives degree five: after removing a fifth power and a fourth root of unity, either its valuation is nonzero modulo five, and the Newton polygon proves this degree, or it is $(1+5)^b$ with $b \in \mathbb{Z}_5^{\times}$, in which case $(1+T)^5 - u$ is Eisenstein at five. That degree-five extension is not Galois, since a Galois extension containing one root would contain ζ_5, of degree four. It cannot lie in the abelian extension $\mathbb{Q}_5(\mu_\infty)$. This proves the asserted restriction nonvanishing. The map $[2 + i] : E[5] \to E[\beta]$ sends the usual multiplication-by-five Kummer class of P to $\delta_\beta(P)$. Thus the proper component of $b|_\Delta$ in F.26 has nonzero reduction modulo five.

For $k = 1, 2, 3$, $1 - \delta^k$ is invertible on the proper two-dimensional vector group over $\mathbb{F}_5$: its eigenvalues are $1 - 2^k$ and $1 - 3^k$. The last fixed equation in F.26 therefore excludes nontrivial deck values already in $\overline{J}$. Conversely the normal vector group is fixed because Δ acts trivially on V_E. Thus $\overline{J}^{\Delta} = (\mathbb{F}_5)^5$. The order-four subgroup supplied by cusp inertia splits the extension and has exactly the displayed matrix, proving (F.27.1).

Write (a, b) for the two abelian coordinates modulo four in the original tripod. The cyclic cover equation forces $a + 2b = 0$ for every fixed element. The Mobius automorphism $m(t) = t/(t-1)$ is defined over $\mathbb{Q}_5$ and fixes the base point 2. It therefore gives a genuine

based automorphism of π, commuting with the arithmetic action. It fixes the cusp 0 and interchanges $1, \infty$. Its action on abelian inertia is $e_0 \mapsto e_0$, $e_1 \mapsto -e_0 - e_1$. Pulling the first finite quotient back along this actual automorphism gives the second quotient in (F.27.1), with cyclic character $(1, 1)$ mod 4. Thus a fixed element also satisfies $a + b = 0$. Subtracting the two equations gives $b = 0$ and then $a = 0$.

The abelian quotient modulo four factors through $\pi^{(5')}$, by its universal property. Good marked reduction with the integral point 2 makes $\pi^{(5')}$ fixed, as in F.23. Therefore a prime-to-five element with nonzero image in $(\mathbb{Z}/4)^2$ is an explicit fixed quotient element having no fixed lift. All coordinate identities, the five residue lifts and the invertible deck matrices are checked in `code/bc_mixed_cm_checks.py`. □

Statement F.28 (Proved). All abelian coordinates of the three full local fixed groups. *In the three examples of F.21, the images of the full fixed groups in the full geometric abelianizations are, respectively,*

$$0, \qquad \widehat{\mathbb{Z}}\, e_0, \qquad 0. \tag{F.28.1}$$

In particular the two point-section fixed groups lie in the closed commutator subgroup of their respective full geometric fundamental groups.

For every integer $m \geq 2$ prime to five, the first example has two explicit finite quotient tests excluding all nonzero abelian coordinates modulo m. They have the form

$$J_{m,n_m} = (\mathbb{Z}/5^{n_m})^{m+1} \rtimes C_m, \tag{F.28.2}$$

and their fixed groups are exactly their displayed normal vector groups. The integer n_m is given by the finite local valuation calculation (F.28.5) below. Analogous finite quotients detect the second and third images in (F.28.1). The cases $m = 2, 3, 4, 6$ in the first example all have $n_m = 1$, giving orders $250, 1875, 12500, 468750$.

Proof. Write $\Delta = G_{\mathbb{Q}_5(\mu_\infty)}$. First consider the tripod with base point 2 and the actual cyclic cover

$$Y_m = \mathbb{P}^1_z \setminus (\{0, \infty\} \cup \mu_m), \qquad t = z^m. \tag{F.28.3}$$

Choose $\alpha^m = 2$. Its fiber consists of $\alpha\zeta_m^j$. All these points and all punctures are defined over a finite unramified extension L of $\mathbb{Q}_5$. Indeed a residue extension of degree d with $4m \mid 5^d - 1$ contains every root of $X^m - 2$ and every mth root of unity; their simple roots lift by Hensel's lemma. Thus $L \subset \mathbb{Q}_5(\mu_\infty)$ and Δ fixes these fibers.

The kernel of $\pi \to C_m$ is the actual full geometric fundamental group of Y_m. Pass from it to its maximal pro-five abelian quotient V_m, a free $\mathbb{Z}_5$-module of rank $m + 1$. The kernel of this passage is characteristic in the open subgroup and normal in π. Thus it gives a genuine quotient J_m of π fitting into $1 \to V_m \to J_m \to C_m \to 1$. Its cyclic action fixes the boundary vector at zero and cyclically permutes the m vectors at μ_m. The image of inertia at the ramified cusp zero is procyclic with a unique prime-to-five factor of order m. This supplies an actual splitting, so $J_m \simeq V_m \rtimes C_m$. Since the punctures are L-rational and $\chi_5|_\Delta = 1$, the action of Δ on V_m is trivial.

Suppose that a fixed element has deck value $j \neq 0$. Lift its based loop through (F.28.3). It gives a Δ-fixed point of the abelian path torsor from α to $\alpha\zeta_m^j$. This description also holds after passing to V_m: path lifting commutes with the Galois action and with the abelian quotient. The function $z - 1$, a unit on Y_m, maps this torsor to the Kummer torsor of

$$u_j = \frac{\zeta_m^j \alpha - 1}{\alpha - 1}. \tag{F.28.4}$$

It would therefore make $\kappa_L(u_j)|_\Delta = 0$.

This Kummer class is nonzero even with rational five-adic coefficients. Choose the complex embedding with $\alpha = 2^{1/m} > 1$. Then $|\zeta_m^j \alpha - 1|^2 = (\alpha-1)^2 + 2\alpha(1-\cos(2\pi j/m))$ is strictly larger than $(\alpha-1)^2$. Consequently u_j is not a root of unity. For a finite local field, the five-adic logarithm on principal units has precisely torsion kernel; its valuation component and logarithm identify the rational Kummer completion of $L^\times$. Thus its Kummer class is nonzero. Restriction to Δ is injective on $H^1(G_L, \mathbb{Q}_5(1))$. To see this directly, the quotient $\Gamma = G_L/\Delta$ is abelian, and its cyclotomic character is nontrivial. For a continuous cocycle c on Γ, commutativity gives

$$(\chi(g)-1)c(h) = (\chi(h)-1)c(g).$$

Choose g with $\chi(g) \neq 1$ and divide in $\mathbb{Q}_5$. This expresses c as a coboundary. Inflation–restriction, obtained from the usual cochain maps, proves the asserted injectivity. This contradicts the fixed path and excludes every nonzero deck value.

There is an explicit finite precision for this exclusion. Choose L unramified of degree d as above and put $q = 5^d$. All u_j are units: neither α nor any $\zeta_m^j \alpha$ reduces to one, since its mth power reduces to 2. Set

$$e_j = v_L(u_j^{q-1} - 1) - 1, \qquad n_m = 1 + \max_{1 \le j < m} e_j, \qquad v_L(5) = 1. \tag{F.28.5}$$

These are finite nonnegative integers. The decomposition $\mathcal{O}_L^\times = \mu_{q-1} \times (1 + 5\mathcal{O}_L)$ and the logarithm isomorphism on the second factor prove that e_j is exactly the largest e for which $u_j \in L^{\times 5^e}$. Write $u_j = v_j^{5^{e_j}}$ with v_j not a fifth power. Its nonzero Kummer class modulo five remains nonzero on Δ. After removing a Teichmuller factor, $v_j \in 1 + 5\mathcal{O}_L$ has $v_L(v_j - 1) = 1$, so $(1+T)^5 - v_j$ is Eisenstein. The degree-five extension generated by a root is not Galois: a normal field containing one root would contain μ_5, of degree four over the unramified field L. It therefore cannot be contained in the abelian extension $\mathbb{Q}_5(\mu_\infty)/L$. Hence $\kappa_L(v_j)|_\Delta$ is nonzero modulo five, and $\kappa_L(u_j)|_\Delta$ is nonzero modulo 5^{n_m}. This proves the finite exclusion in $J_m/5^{n_m}V_m$. Its normal vector group is fixed, proving the exact fixed group in (F.28.2). Formula (F.28.5) involves only algebraic valuations and gives a terminating finite calculation.

Let (a, b) be full abelian coordinates at $0, 1$. The covering (F.28.3) forces $a = 0 \bmod m$ for every fixed element. The based automorphism $t \mapsto t/(t-1)$ fixes 2 and commutes with the arithmetic action. Its abelian map is $(a, b) \mapsto (a - b, -b)$. Pulling back (F.28.3) along this actual automorphism forces $a - b = 0 \bmod m$. Therefore both coordinates vanish modulo every m prime to five. F.24 makes their five-primary coordinates zero as well. This proves the first image in (F.28.1).

For the unit tangent at zero, use instead the actual cover $t = 1 - z^m$. The induced tangents above zero lie at the points ζ_m^j, with $dz/dt = -1/(m\zeta_m^{j(m-1)})$. Choose the starting tangent at $z = 1$, so its parameter $z - 1$ has value $-1/m$. A fixed element with deck value $j \neq 0$ would give a fixed abelian path to the induced tangent at ζ_m^j. The same function $z - 1$ sends these two fibers to Kummer parameters $-1/m$ and $\zeta_m^j - 1$, respectively. Their ratio is $m(1 - \zeta_m^j)$. Its complex absolute value is at least $2m \sin(\pi/m) \geq 4$, so it is not a root of unity. It is a five-adic unit because m is prime to five and ζ_m^j has nontrivial prime-to-five order. The preceding restriction and finite-precision arguments apply verbatim, with these ratios. This excludes its e_1 coordinate modulo every m prime to five. F.24 gives the same exclusion for its five-primary part. Conversely the full cusp-zero inertia is fixed by Δ, because the full cyclotomic character is one there. Its abelian image is the whole primitive line $\widehat{\mathbb{Z}}e_0$. This proves the second equality.

For the four-puncture curve, filling the cusp 5 gives the tripod point 2 by an actual based morphism. Thus a fixed element has zero e_0, e_1 coordinates. The other based map

$$t \longmapsto (5-t)/4 : \mathbb{P}^1 \setminus \{0, 1, 5, \infty\} \longrightarrow \mathbb{P}^1 \setminus \{0, 1, \infty\}$$

fills the original cusp zero and sends the point 2 to 3/4, the cusp 5 to zero, and the cusp 1 to one. At the tripod point 3/4, the covers $t = z^m$ and $t = 1 - z^m$ have starting roots $(3/4)^{1/m}$ and $(1/4)^{1/m}$. Both are positive real numbers different from one. The strict absolute-value inequality used for (F.28.4) holds equally for a positive root less than one. Their reductions have mth powers 2 and 4 modulo five, so the same unramified and unit arguments apply. These two covers force both prime-to-five abelian coordinates of a fixed loop at 3/4 to be zero. Pullback gives the remaining e_5 exclusion. F.25 excludes every five-primary coordinate on the four-puncture curve. This proves its zero abelian image.

The code `bc_cyclic_fixed.py` works in exact unramified rings modulo powers of five. It computes (F.28.5) for $m = 2, 3, 4, 6$, obtaining $e_j = 0$ for every j in each case. It verifies the finite orders stated above. □

Statement F.29 (Proved). A genus-zero mixed quotient with no fixed element. *In the cyclic fourth-root cover $t = z^4$ of the tripod with base point 2, let H_5 be the maximal geometric pro-five group of $\mathbb{P}^1_z \setminus \{0, \infty, \pm 1, \pm i\}$, and let J be the resulting quotient of the full base fundamental group. Thus $1 \to H_5 \to J \to C_4 \to 1$ and H_5 is free pro-five of rank five. For the genuine point-section action,*

$$J^\Delta = 1, \qquad (J/\gamma_{c+1}H_5)^\Delta = \gamma_c H_5/\gamma_{c+1}H_5 \quad (c \geq 1). \tag{F.29.1}$$

Every transition between consecutive displayed fixed groups is zero.

Proof. Take $i \in \mathbb{Q}_5$, $i \equiv 2 \bmod 5$, and $\alpha^4 = 2$. The extension $L = \mathbb{Q}_5(\alpha)$ is unramified of degree four: $X^4 - 2$ is irreducible and separable over $\mathbb{F}_5$. Its Frobenius sends α to $i\alpha$. The auxiliary section is the unit tangent at $z = 0$. The order-four factor of its inertia supplies a based deck element δ, fixed by the auxiliary Galois action. Equivalently, the change of tangent under $z \mapsto iz$ has the compatible rational fifth-power roots $i^{1/5^n} = i$, so its based comparison commutes with $G_{\mathbb{Q}_5}$.

Write

$$V = H_5^{\mathrm{ab}} = \mathbb{Z}_5 e_0 \oplus \bigoplus_{j=0}^{3} \mathbb{Z}_5 f_j$$

with a Tate twist on every summand; the last four basis vectors are the inertias at i^j. All punctures are $\mathbb{Q}_5$-rational, so the auxiliary arithmetic action on V is the scalar χ_5. For $\sigma \in \Delta$ its first Johnson map is $j_\sigma : V \to \bigwedge^2 V$. Composition in the first two nilpotent degrees proves that j is additive on Δ and that

$$j_{g\sigma g^{-1}} = \chi_5(g) j_\sigma. \tag{F.29.2}$$

Let b be the cocycle comparing the point 2 with the auxiliary tangent in J. Its deck component records $g\alpha/\alpha$. On Δ this component is zero, so b_σ has an abelian value in V, again an additive map of Δ. We retain the precise Frobenius transport. Choose φ lifting arithmetic Frobenius with $\chi_5(\varphi) = 1$; this is possible because the fifth cyclotomic extension is totally ramified, whereas L is unramified. On G_L, the functional dual to f_j sends b to $\kappa_L(\alpha - i^j)$: the unit $z - i^j$ has tangent value $-i^j$, whose pro-five Kummer class is zero. For $\sigma \in \Delta$,

$$\begin{aligned} b_j(\varphi\sigma\varphi^{-1}) &= \kappa_{\varphi^{-1}(\alpha - i^j)}(\sigma) \\ &= \kappa_{\alpha - i^{j+1}}(\sigma) = b_{j+1}(\sigma). \end{aligned} \tag{F.29.3}$$

The discarded factor is $-i$, with its compatible rational fifth-power roots. In the right-deck convention this is the actual permutation $\delta f_j = f_{j-1}$, while $\delta e_0 = e_0$. The same conclusion follows directly from the nonabelian cocycle identity: modulo $[H_5, H_5]$ the auxiliary Δ-action

is trivial, so $b_{g\sigma g^{-1}} = b_g \rho(g)(b_\sigma) b_g^{-1}$. Its linear action is $\chi_5(g)\delta^{j(g)}$, where $g\alpha = i^{j(g)}\alpha$. This proves the required covariance with the actual point-base deck translation.

We now exhibit two nonzero eigenspace components of $b|_\Delta$. In the unramified integer ring $\mathbb{Z}_5[\alpha]$, the following are the coefficients of $\log(\alpha - i^j)/5$ modulo five, in the basis $1, \alpha, \alpha^2, \alpha^3$:

j	1	α	α^2	α^3
0	0	0	2	1
1	0	0	3	2
2	0	0	2	4
3	0	0	3	3

. (F.29.4)

Here the logarithm kills the Teichmuller factor. This is an exact finite calculation: with $i = 7 \bmod 25$, compute $((\alpha - i^j)^{624} - 1)/(5 \cdot 624)$ in $(\mathbb{Z}/25)[\alpha]/(\alpha^4 - 2)$ and reduce modulo five. The program performs these operations and verifies every row.

For the permutation in (F.29.3), the coefficient at f_0 of the λ-eigenprojection on the four puncture coordinates is $\frac{1}{4}\sum_j \lambda^{-j} b_j$. For $\lambda = i$ and $\lambda = -1$, (F.29.4) gives respectively α^3 and $2\alpha^2$ modulo five after dividing the logarithm by five. Thus both rational Kummer classes are nonzero. The restriction injection proved in F.28 makes their restrictions to Δ nonzero as well.

The joint additive image of Δ under $\sigma \mapsto (j_\sigma, b_\sigma)$ is a compact $\mathbb{Z}_5$-submodule. Conjugation by φ preserves it, acts trivially on j by (F.29.2), and by the order-four permutation δ on b. The integral operators

$$E_\lambda = \frac{1}{4}\sum_{a=0}^{3} \lambda^{-a}\varphi^a \quad (\lambda = i, -1) \tag{F.29.5}$$

therefore preserve this actual image. They annihilate its j-coordinate and retain the indicated nonzero b-component. Compactness makes the image closed under every $\mathbb{Z}_5$ scalar in (F.29.5). Consequently there are actual $\sigma_1, \sigma_2 \in \Delta$ with

$$j_{\sigma_1} = j_{\sigma_2} = 0, \qquad b_{\sigma_1} = w_i \neq 0, \quad b_{\sigma_2} = w_{-1} \neq 0, \tag{F.29.6}$$

where the two vectors lie in distinct deck eigenspaces. They are linearly independent.

Changing from the auxiliary tangent to the point section changes the first Johnson map to $j_\sigma + \operatorname{ad} b_\sigma$. Thus (F.29.6) supplies two independent pure-inner first Johnson directions. The leading Magnus term of a fixed element below the final degree of a nilpotent quotient must commute in the free Lie algebra with both w_i and w_{-1}. As proved in F.26, the centralizer of a nonzero degree-one vector in a free Lie algebra over $\mathbb{Q}_5$ is its one-dimensional span. The two spans intersect in zero. The integral Hall graded modules embed in their rationalizations, so this excludes every such nonzero leading term. Conversely the final central graded subgroup is fixed because Δ acts trivially on V and hence on every free Lie graded quotient.

F.28 with $m = 4, n_m = 1$ excludes all nonzero deck values already in the finite abelian-normal quotient. It excludes them in every quotient in (F.29.1) as well. This proves the displayed finite-depth groups. Their images at the preceding depth are zero. Finally $\bigcap_c \gamma_c H_5 = 1$: every nontrivial element of a free pro-five group survives in a finite five-group, and finite five-groups are nilpotent. Taking the inverse limit proves $J^\Delta = 1$. □

Statement F.30 (Proved). A torsion test in every actual finite cover. *Put $K_\infty = \mathbb{Q}_5(\mu_\infty)$ and $\Delta = G_{K_\infty}$. For every finite extension $L/\mathbb{Q}_5$ put $\Delta_L = G_{LK_\infty}$ and $\Gamma_L = \operatorname{Gal}(LK_\infty/L)$. For every semiabelian variety A/L, restriction is injective:*

$$H^1(G_L, V_5A) \longrightarrow H^1(\Delta_L, V_5A), \qquad V_5A = T_5A \otimes_{\mathbb{Z}_5} \mathbb{Q}_5. \tag{F.30.1}$$

Consequently, for every smooth curve Y/L, its smooth compactification C, reduced boundary $D = C \setminus Y$, and $P, Q \in Y(L)$, a Δ_L-fixed geometric étale path from P to Q forces

$$[Q - P] \in J_D(L)_{\mathrm{tors}}. \tag{F.30.2}$$

The same conclusion holds for a fixed path in the maximal abelian pro-five path torsor. When $L \subset K_\infty$ one has $\Delta_L = \Delta$. Here J_D is the generalized Jacobian with reduced modulus D; when D is empty it is the ordinary Jacobian. For nonempty D, condition (F.30.2) has the exact divisor formulation

$$\exists n \geq 1,\ f \in \overline{L}(C)^\times : \quad \mathrm{div}(f) = n(Q - P), \qquad f|_D = 1. \tag{F.30.3}$$

In particular, let $U/\mathbb{Q}_5$ have a rational interior base point x, let $\pi = \pi_1(U_{\overline{\mathbb{Q}}_5}, x)$, and use the point section to define the action of Δ. For every Δ-stable open subgroup $H \subset \pi$, the corresponding based cover over K_∞ is an actual finite étale cover. If $h \in \pi^\Delta$, its lift beginning at the chosen point P ends at a K_∞-point Q above x, and this pair satisfies (F.30.2)–(F.30.3) after descent to a finite $L \subset K_\infty$.

Proof. We first prove the representation-theoretic assertion used in (F.30.1). Let Γ be an abelian profinite group and M a finite-dimensional continuous $\mathbb{Q}_5$-representation with $M^\Gamma = 0$. Let $R \subset \mathrm{End}_{\mathbb{Q}_5}(M)$ be the unital algebra generated by Γ. It is a finite-dimensional commutative algebra acting faithfully on M. The ideal $I = (\gamma - 1 : \gamma \in \Gamma)$ is all of R. Indeed, if it lay in a maximal ideal $\mathfrak{m}$, the corresponding local Artinian factor $R_\mathfrak{m}$ would act on a nonzero summand $M_\mathfrak{m}$. Its last nonzero radical power $\mathfrak{m}^a M_\mathfrak{m}$ would contain a nonzero vector killed by $\mathfrak{m}$, hence by every $\gamma - 1$. This contradicts $M^\Gamma = 0$. Thus there are finitely many $A_i \in R$ and $\gamma_i \in \Gamma$ with

$$\sum_i A_i(\gamma_i - 1) = 1.$$

For a continuous one-cocycle $c : \Gamma \to M$, commutativity gives $(\gamma_i - 1)c(\gamma) = (\gamma - 1)c(\gamma_i)$. Consequently

$$c(\gamma) = (\gamma - 1)m, \qquad m = \sum_i A_i c(\gamma_i). \tag{F.30.4}$$

This is an actual continuous coboundary formula; in particular $H^1(\Gamma, M) = 0$.

We record explicitly the local point-group input. For a semiabelian variety A over the finite local field L, a sufficiently small formal neighborhood of the identity is an open subgroup $U_A \simeq \mathbb{Z}_5^d$. The torus part has a finitely generated valuation quotient, and the abelian-variety part is compact. The local analytic subgroup assertion for the latter is Mattuck's Theorem 7, printed p. 114 [Mat55]. These statements give a finitely generated quotient $A(L)/U_A$. For completeness, the image in the abelian-variety part is open, since the differential of the smooth quotient morphism is surjective; it is therefore compact. Finitely many local analytic sections over this compact image, together with generators for the torus valuation lattice, give the asserted finite generation modulo U_A. Passing to the preimage of a torsion-free subgroup of finite index in $A(L)/U_A$ and lifting its free basis gives a finite-index subgroup of the form

$$\mathbb{Z}^r \oplus \mathbb{Z}_5^d. \tag{F.30.5}$$

In particular $A(L)_{\mathrm{tors}}$ is finite and $(V_5 A)^{G_L} = 0$. It also follows directly from the finite Kummer sequences and (F.30.5) that the map

$$A(L) \longrightarrow H^1(G_L, V_5 A) \tag{F.30.6}$$

has exactly the torsion subgroup as its kernel. One way to see the last assertion without a completion identification is to start with a zero rational Kummer class. A five-power

multiple of the point is then divisible by every 5^n in $A(L)$. Multiplication of this point and all its division witnesses by the finite index of (F.30.5) places them in that subgroup. The intersection of the subgroups 5^n times (F.30.5) is zero.

Now put $V = V_5 A$ and $M = V^{\Delta_L}$. The quotient $\Gamma = \Gamma_L = G_L/\Delta_L$ is abelian: restriction identifies it with the closed subgroup $\mathrm{Gal}(K_\infty/(L \cap K_\infty))$ of the abelian cyclotomic Galois group. and $M^\Gamma = V^{G_L} = 0$. Let a cocycle on G_L represent an element in the kernel of (F.30.1). Subtracting an actual coboundary makes its restriction to Δ_L zero. The cocycle identity then shows both that its values lie in M and that it is constant on cosets of Δ_L. It descends continuously to Γ. Formula (F.30.4) makes it a coboundary and proves (F.30.1).

The path-to-Kummer map in the second assertion is the one induced by the actual Abel morphism

$$\iota_P : Y \longrightarrow J_D, \qquad Q' \longmapsto [Q' - P]. \tag{F.30.7}$$

More explicitly, pull back the finite étale isogenies $[5^n] : J_D \to J_D$ along ι_P. A path from P to Q carries the compatible zero points above $\iota_P(P) = 0$ to compatible division points of $[Q - P]$. Its displacement under σ is the cocycle $\sigma Q_n - Q_n$ in $J_D[5^n]$. Thus a Δ_L-fixed path makes the restriction of the rational Kummer class of $[Q - P]$ zero. The construction factors through the abelian pro-five path torsor because these deck groups are abelian five-groups. Apply (F.30.1) and then (F.30.6) to obtain (F.30.2). The quotient of degree-zero divisors disjoint from D by divisors of functions equal to one on D is the actual generalized-Jacobian group. This gives (F.30.3); see also [Cre19, §2.1, equation (2.2), and §2.3, Lemma 2.5].

Finally the arithmetic subgroup for the indicated cover is $H \rtimes s_x(\Delta) \subset \pi \rtimes s_x(\Delta)$. It has finite index, and its chosen fibre point is Δ-fixed. The usual lifting map for finite étale covers identifies paths in Y with base paths having the prescribed endpoints. The lift of h is fixed because both h and its starting point are fixed; its endpoint is therefore a K_∞-point. The cover, the two points, and their morphism to U are given by finitely many coefficients in K_∞, so they descend together to some finite $L \subset K_\infty$. The previous construction then proves the stated test in this actual cover. Further extension to any finite L'/L preserves the fixed-path conclusion, since $\Delta_{L'} = G_{L'K_\infty} \subset \Delta_L$. Thus this passage also permits arbitrary finite extensions used to change the reduction of the cover. □

Statement F.31 (Proved). The full abelian fixed images for all genus-zero curves. *Let $L/\mathbb{Q}_5$ be finite, $K_\infty = \mathbb{Q}_5(\mu_\infty)$, and $\Delta_L = G_{LK_\infty}$. Let Y/L be a smooth genus-zero hyperbolic curve and $P \in Y(L)$. For the action defined by this actual point section,*

$$\pi_1(Y_{\overline{L}}, P)^{\Delta_L} \longrightarrow \pi_1(Y_{\overline{L}}, P)^{\mathrm{ab}} \quad \textit{has image zero.} \tag{F.31.1}$$

There is no restriction on the reduction of Y, on the position of P, or on the prime divisors of the finite abelian quotient being tested. For an L-rational nonzero tangential base point $\vec{v}$ at an L-rational cusp c, the corresponding full fixed group has abelian image exactly $\widehat{\mathbb{Z}}(1)e_c$.

The same conclusion applies at every actual genus-zero finite étale neighborhood of P whose chosen fibre point is fixed by Δ_L, after a finite extension of L. Thus a full fixed element has zero abelian image in every such neighborhood in which it lifts to a loop.

Proof. Write C for the compactification and $D = C \setminus Y$. The rational point P identifies C with a projective line. Extend L finitely to split D; a fixed element remains fixed by the smaller group $\Delta_{L'}$. We retain the letter L for this extension during the construction.

Choose a cusp $d_\infty \in D$, and an actual L-isomorphism $z : C \longrightarrow \mathbb{P}^1_L$ sending d_∞ to infinity. Put $p = z(P)$. For each $d \in D \setminus \{d_\infty\}$ set

$$u_d = \frac{z - z(d)}{p - z(d)} : Y \longrightarrow \mathbb{G}_m. \tag{F.31.2}$$

These maps give the usual full abelian coordinates: the inertia at d maps to 1 in $\widehat{\mathbb{Z}}(1)$, the inertia at d_∞ maps to -1, and every other cusp inertia maps to zero. The sole relation among the cusp generators is their sum, so the product of these induced maps is an isomorphism $\pi_1(Y_{\overline{L}}, P)^{\mathrm{ab}} \simeq \widehat{\mathbb{Z}}(1)^{|D|-1}$.

Fix $n \geq 3$ and form the actual connected cyclic cover

$$Y_{d,n} \longrightarrow Y, \qquad w^n = u_d, \qquad z = z(d) + (p - z(d))w^n. \tag{F.31.3}$$

Its compactification is the projective w-line. The removed points are $0, \infty$ and the n-th roots of $(z(e) - z(d))/(p - z(d))$ for $e \in D \setminus \{d, d_\infty\}$. This lists precisely all inverse images of D, and proves that (F.31.3) is finite étale on the indicated open curves. Its fibre above P is μ_n. Every one of these points is fixed by Δ_L.

Let h be a full fixed element whose image under u_d modulo n is $j \neq 0$. Lift its actual path to (F.31.3), beginning at $w = 1$. The endpoint is $w = \zeta_n^j$, for a compatible choice of generator ζ_n. The path is fixed by Δ_L. Choose another cusp $e \in D \setminus \{d, d_\infty\}$, which exists by hyperbolicity. Put

$$r = \frac{z(e) - z(d)}{p - z(d)} \neq 0, 1, \qquad \beta^n = r.$$

After a finite extension L'/L containing β and μ_n, each

$$v_a = w - \beta\zeta_n^a \quad (0 \leq a < n) \tag{F.31.4}$$

is an actual unit on $Y_{d,n}$. It is nonzero at both endpoints, since $r \neq 1$. The induced map of paths to $\mathbb{G}_m$ sends the fixed path to the Kummer path of

$$R_a = \frac{\zeta_n^j - \beta\zeta_n^a}{1 - \beta\zeta_n^a} \in L'^{\times}. \tag{F.31.5}$$

Formula (F.30.1), now with $\Delta_{L'} = G_{L'K_\infty} \subset \Delta_L$, therefore forces every R_a to be a root of unity.

Here is an exact contradiction. Embed the finitely generated characteristic-zero field containing these finitely many coefficients and roots into $\mathbb{C}$. Set

$$D_a = |\zeta_n^j - \beta\zeta_n^a|^2 - |1 - \beta\zeta_n^a|^2.$$

The root-of-unity sums $\sum_a \zeta_n^{2a} = \sum_a \zeta_n^{-2a} = 0$, valid for $n \geq 3$, give

$$\sum_{a=0}^{n-1} D_a^2 = 2n\,|\zeta_n^j - 1|^2\,|\beta|^2 > 0. \tag{F.31.6}$$

Indeed $D_a = -(\zeta_n^j - 1)\overline{\beta}\,\zeta_n^{-a} - (\zeta_n^{-j} - 1)\beta\,\zeta_n^a$, and squaring proves the displayed identity. Thus some $|R_a| \neq 1$, so that R_a cannot be a root of unity. This excludes every $j \neq 0$.

We have proved that every coordinate in (F.31.2) vanishes modulo every $n \geq 3$. Every positive modulus divides one of these n, so all full abelian coordinates vanish. The integers n used in (F.31.3) may be divisible by five: the maps are étale in characteristic zero, and the proof uses the actual subgroup $\Delta_{L'}$, which fixes all roots of unity. This proves (F.31.1).

For the tangential assertion, choose $d_\infty \neq c$ and use the same actual coordinate z, putting $p = z(c)$. For each cusp $d \neq c, d_\infty$ use (F.31.3) with this value of p and $n \geq 5$. The cover is unramified at c. If the original tangent has velocity $\dot{z}$, its lift at $w = 1$ has velocity $\dot{z}/(n(p - z(d)))$; the deck translate at $w = \zeta_n^j$ has velocity $\zeta_n^j\dot{z}/(n(p - z(d)))$. These are the actual derivatives of (F.31.3), so both lifted tangents are fixed by $\Delta_{L'}$. The other points ζ_n^a above c are punctures. For $a \neq 0, j$, the unit $w - \zeta_n^a$ has nonzero value at both tangential

endpoints. Its induced Kummer ratio is therefore $(\zeta_n^j-\zeta_n^a)/(1-\zeta_n^a)$; no tangent velocity enters this nonvanishing value. Put $A=|\zeta_n^j-1|^2$. Formula (F.31.6) with $\beta=1$, after subtracting the terms at $a=0,j$, gives

$$\sum_{\substack{0\le a<n\\ a\neq 0,j}} \Big(|\zeta_n^j-\zeta_n^a|^2-|1-\zeta_n^a|^2\Big)^2 = 2A(n-A)>0, \tag{F.31.7}$$

because $0<A\le 4<n$. The identical Kummer argument excludes all $j\neq 0$. This kills every abelian coordinate except the one corresponding to c. Conversely the full inertia $I_c\simeq\widehat{\mathbb{Z}}(1)$ at this actual tangent is fixed by Δ_L: its action on compatible roots of the local parameter is the cyclotomic character, which is one on Δ_L. Its image is precisely the primitive cusp line.

The obstruction can also be made finite in each instance. For an R_a selected by (F.31.6), its restricted integral Kummer homomorphism $\Delta_{L'}\to\mathbb{Z}_5(1)$ is nonzero, since its rational class is nonzero. It remains nonzero modulo some 5^s. Pulling back the cover $T^{5^s}=v_a$ along (F.31.4) supplies a finite étale cover on which the indicated endpoints cannot be joined by a fixed path. Taking its normal closure, and then the compositum of the finitely many coefficient-field conjugates, gives an actual finite geometric quotient stable under the original arithmetic action. The conjugate family is finite because all the coefficients, points, roots, and covering equations used here descend to the finite extension L'/L.

Finally apply the same construction to the given finite neighborhood, using its chosen fixed fibre point as P. Descent of the cover and point to a finite field and subsequent restriction of Δ_L are exactly the morphisms described at the end of F.30. The exact root-of-unity identities (F.31.6)–(F.31.7) and the cyclic-cover inverse maps are checked by `bc_genus_zero.py`. □

Statement F.32 (Proved). The full local fixed group at every point and tangent. *Let $L/\mathbb{Q}_5$ be finite, let $K_\infty=\mathbb{Q}_5(\mu_\infty)$, and put $\Delta_L=G_{LK_\infty}\subset G_L$. Let X/L be a smooth geometrically connected hyperbolic curve, and let Π be its full geometric profinite fundamental group. For an L-rational point $b\in X(L)$ and its actual point section s_b,*

$$C_\Pi(s_b(\Delta_L))=1. \tag{F.32.1}$$

For a nonzero L-rational tangent at an L-rational cusp c and its actual tangential section $s_{\vec{c}}$,

$$C_\Pi(s_{\vec{c}}(\Delta_L))=I_c, \tag{F.32.2}$$

where I_c is the actual geometric inertia subgroup selected by that tangent. In the tripod example of Task F.4, $I_0\simeq\widehat{\mathbb{Z}}(1)$, with this isomorphism induced by the compatible local covers $w^n=t$.

Proof. We use the resolution of nonsingularities theorem of Mochizuki–Tsujimura, in the author's revision of 30 March 2024. The exact inputs are Theorem 2.17 and Proposition 2.4(vii): with Σ the set of all primes, a section of a closed subgroup $H\subset G_L$ fixes at most one primitive valuation in their universal geometric covering, provided $H\cap I_L$ has a continuous quotient $\mathbb{Z}_\ell$ for some $\ell\neq 5$. The correspondence between valuations and center-chains, including its inverse, is Proposition 2.3(v)–(vi) [MT24, Propositions 2.3(v)–(vi), 2.4(vii); Theorem 2.17]. We verify the subgroup hypothesis and construct the valuation and its stabilizer explicitly.

First every finite unramified extension of L is obtained by adjoining suitable roots of unity of order prime to five. Indeed, such roots give all finite extensions of the finite residue field, and their lifts are the unique roots supplied by Hensel's lemma. Consequently $L^{\mathrm{ur}}\subset LK_\infty$ and

$$\Delta_L\subset I_L,\qquad I_L/\Delta_L \xrightarrow[\chi_5]{\sim} \chi_5(I_L)\subset\mathbb{Z}_5^\times\simeq C_4\times\mathbb{Z}_5. \tag{F.32.3}$$

For the kernel assertion, inertia already fixes all prime-to-five roots of unity, so its full cyclotomic kernel is exactly its five-adic cyclotomic kernel. Thus (F.32.3) is induced by the actual cyclotomic homomorphism.

Choose compatible three-power roots of a uniformizer of L. The resulting tame homomorphism $t_3 : I_L \to \mathbb{Z}_3(1)$ is onto: the successive extensions over the maximal unramified field have degrees 3^n, as follows either from their valuations or from the Eisenstein polynomials for those roots. Let $B = t_3(\Delta_L)$, a closed subgroup. The quotient $\mathbb{Z}_3(1)/B$ is a continuous quotient of I_L/Δ_L. Every finite quotient of the latter has order divisible only by two and five, whereas every finite quotient of $\mathbb{Z}_3(1)/B$ is a three-group. Therefore $B = \mathbb{Z}_3(1)$. This verifies the uniqueness hypothesis with $H = \Delta_L$ and $\ell = 3$. It uses the at-most-one part of Proposition 2.4(vii), whose hypothesis imposes no nontrivial cyclotomic image on H.

Here is the promised geometric comparison. Write M for the union of the function fields of the connected finite geometric covers of $X_{\overline{L}}$ inside its chosen universal cover. Then

$$\Pi = \mathrm{Gal}(M/\overline{L}(X)).$$

More precisely, transport the chosen based fibre functor to a geometric generic point by an étale path. The transported arithmetic fundamental group is $\mathrm{Gal}(M/L(X))$; its projection to G_L is restriction to the constant field, with kernel the displayed group Π. Transport the section and the basepoint lifts by the same path. Thus every action below is an action in this actual extension. The based cover construction gives a compatible lift of b to every finite cover. At a tangent, the chosen compatible Puiseux branches give instead a compatible cusp in the normalizations of the smooth compactification. In either case write $\widetilde{b}$ for this compatible collection of centers. Let

$$B_{\widetilde{b}} = \bigcup_{Y} \mathcal{O}_{\overline{Y},b_Y} \subset M, \qquad \epsilon_{\widetilde{b}} : B_{\widetilde{b}} \longrightarrow \overline{L} \tag{F.32.4}$$

be the directed union of the local discrete valuation rings at the centers and its residue map. In (F.32.4), $\overline{Y}$ is the smooth compactification, so the formula applies to cusps as well as to interior points. The union is a valuation ring: every element of $M^\times$ lies in a finite subcover, where it or its inverse belongs to the indicated local discrete valuation ring. Its residue field is $\overline{L}$, because all the centers are geometric and the constant field already contains $\overline{L}$.

Compose this horizontal valuation with the five-adic valuation of its residue field. The resulting valuation ring is

$$R_{\widetilde{b}} = \epsilon_{\widetilde{b}}^{-1}(\mathcal{O}_{\overline{L}}) \subset B_{\widetilde{b}}. \tag{F.32.5}$$

Its intersection with L is $\mathcal{O}_L$. It is primitive in the precise sense of Mochizuki–Tsujimura Definition 2.2(i): its maximal ideal is the only prime ideal containing five. To prove this directly, put $J = \ker \epsilon_{\widetilde{b}}$. Since $5^{-n}J = J$ for every $n \geq 0$, any prime ideal of $R_{\widetilde{b}}$ containing five contains J. The quotient $R_{\widetilde{b}}/J$ is the rank-one valuation ring $\mathcal{O}_{\overline{L}}$, whose only prime ideal containing five is its maximal ideal. Taking its inverse image proves the assertion.

The morphism from this construction to the source theorem is the following actual map: send $R_{\widetilde{b}}$ to its center in every compactified semistable model of every finite subcover, and then to the vertex or edge containing that center. These entries form the center-chain of Definition 2.2(v). Proposition 2.3(v) identifies this map with a bijection from the valuation rings in question to the center-chain space; Proposition 2.3(vi) gives its inverse as the directed union of the local rings at the centers. Both maps commute with the arithmetic fundamental-group action, since formation of a center and pullback of a local ring commute with the corresponding field automorphisms. In particular equality of the primitive center-chains here is equality of the rings $R_{\widetilde{b}}$.

The actual section $s_b(G_L)$ preserves the compatible basepoint lifts, while the actual section $s_{\vec{c}}(G_L)$ preserves the compatible cusps selected by the tangent. Their action on the residue

field in (F.32.4) is the ordinary Galois action on $\overline{L}$, which preserves $\mathcal{O}_{\overline{L}}$. Thus $R_{\widetilde{b}}$, and hence its primitive center-chain, is fixed by the restricted section $s(\Delta_L)$.

Now take $h \in C_\Pi(s(\Delta_L))$. For every $\sigma \in \Delta_L$,

$$s(\sigma)(hR_{\widetilde{b}}) = h\, s(\sigma)R_{\widetilde{b}} = hR_{\widetilde{b}}.$$

Thus $hR_{\widetilde{b}}$ is a second fixed primitive valuation. The uniqueness theorem gives $hR_{\widetilde{b}} = R_{\widetilde{b}}$.

We next recover the exact geometric stabilizer, without identifying different kinds of valuation by notation. There is an equality of subrings of M

$$R_{\widetilde{b}}[1/5] = B_{\widetilde{b}}. \tag{F.32.6}$$

Indeed five is a unit in the horizontal valuation ring. Conversely, for any $f \in B_{\widetilde{b}}$, a sufficiently large power $5^n f$ has integral residue and belongs to $R_{\widetilde{b}}$. It follows that an element of Π preserving $R_{\widetilde{b}}$ preserves $B_{\widetilde{b}}$, hence the center b_Y on every finite normalization $\overline{Y}$. Conversely, an element preserving all these centers preserves $B_{\widetilde{b}}$ and its residue map, since it fixes $\overline{L}$ pointwise; it therefore preserves $R_{\widetilde{b}}$. We have proved the exact stabilizer equality

$$\mathrm{Stab}_\Pi(R_{\widetilde{b}}) = \mathrm{Stab}_\Pi(\widetilde{b}). \tag{F.32.7}$$

At an interior point, every finite geometric Galois cover is étale there. Its deck group acts freely on the fibre: the residue extension is trivial over $\overline{L}$ and the ramification index is one. Thus an element of (F.32.7) has identity image in every finite geometric Galois quotient, and is itself the identity. This proves (F.32.1).

At a cusp, (F.32.7) is the inverse limit of the inertia groups at the chosen compatible normalization points, which is exactly the image I_c of the punctured local disc. For completeness, each completed local extension has the form $\overline{L}((t^{1/e}))/\overline{L}((t))$: its residue field is algebraically closed, its residue characteristic for this horizontal valuation is zero, and an e-th root of the unit multiplying a local uniformizer exists by Hensel's lemma. Its automorphisms multiply $t^{1/e}$ by μ_e. These compatible homomorphisms identify the stabilizer with the actual local inertia image, including all transition maps. For the tripod at zero the global finite covers $w^n = t$ realize every such cyclic quotient, so the local map $\widehat{\mathbb{Z}}(1) \to \Pi$ is injective and has image I_0.

Finally the tangential section acts on these compatible local roots by the cyclotomic character. Since Δ_L fixes all roots of unity, it acts trivially on I_c. The inclusion $I_c \subset C_\Pi(s_{\vec{c}}(\Delta_L))$, together with (F.32.7) and uniqueness, proves (F.32.2). □

Statement F.21 (Proved). The three full local fixed groups and all fixed lifts. *For the three examples in Task F.4, with $\Delta = G_{\mathbb{Q}_5(\mu_\infty)}$ and the actual section action, the full profinite fixed groups are*

U	*section*	$\pi^{s(\Delta)}$
$\mathbb{P}^1 \setminus \{0,1,\infty\}$	2	1
$\mathbb{P}^1 \setminus \{0,1,\infty\}$	*integral unit tangent at* 0	$I_0 \simeq \widehat{\mathbb{Z}}(1)$
$\mathbb{P}^1 \setminus \{0,1,5,\infty\}$	2	1.

(F.21.1)

For $q : \pi \to \pi^{(5')}$, the images of these full fixed groups are, respectively,

$$1, \qquad I_0^{(5')}, \qquad 1. \tag{F.21.2}$$

Consequently the finite coboundary tests of F.20 all hold exactly for $\bar{h} = 1$ in the two point-based examples, and exactly for $\bar{h} \in I_0^{(5')}$ in the tangential example. The fixed lift is unique in each point-based example. Each tangential lift fibre is a coset of $\mathbb{Z}_5(1) \subset I_0$.

Proof. Formula (F.21.1) follows from F.32 applied to the two actual interior point sections and the actual tangent at zero. On the inertia selected by this tangent, the quotient q is the actual projection

$$\widehat{\mathbb{Z}}(1) = \prod_{\ell} \mathbb{Z}_\ell(1) \longrightarrow \prod_{\ell \neq 5} \mathbb{Z}_\ell(1).$$

The covers $w^n = t$ with $(n, 5) = 1$ verify injectivity of the displayed target into $\pi^{(5')}$ and identify it with $I_0^{(5')}$. Its kernel is the five-adic factor. This proves (F.21.2) and the stated lift fibres. Finally F.20 identifies the existence of each such fixed lift with all the finite coboundary tests, proving the exact evaluation of those tests as well.

The earlier F.22–F.31 computations retain their explicit quotient actions and finite obstruction witnesses. The universal-cover valuation argument in F.32 supplies the detection beyond those particular quotient families. □

A Coverage and remaining questions

The following table records task-level scope. A proved supporting statement does not certify an unresolved task.

Item	Statements	State
Lead 1	B.2; B.9	(a) Proved. The exclusion is false; an unramified full-section counterexample lies outside the three inherited cusp lines. The drafter's particular inherited toric argument is corrected.
Lead 2	A.5; A.17; B.7	(b) Refuted. A mixed-inertia class-two lift at 2 is explicit. Full and metabelian sections obey the proved cusp-line restriction.
Lead 3	A.6; A.18	(b) Refuted. An infinitely ramified noncuspidal class-two family is constructed with a complete convergence proof; its higher images are excluded.
Lead 4	B.4; B.5	(a) Proved. The full geometric pro-3 cyclotomic centralizer is cusp inertia for cuspidal sections and trivial for every other section.
Lead 5	A.1; A.2; A.13	(a) Proved. The first derived limit for the intrinsic rank-44 coefficient is explicitly nonzero.
A.1	A.1; A.2; A.7; A.13	(a) Proved. The finite-S complex, its integral Smith factors, localization, and the complete Milnor sequence are computed.
A.2	A.8; A.9; A.10	(a) Proved. All residue formulas retain the nine tangential origin cochains, the extra unramified generator, and the E36 central-coordinate sign.
A.3	A.7; A.8	(a) Proved. Integral local corrections must form a restricted family; every nonempty fiber is the computed rank-23 module with its finite torsion.
A.4	A.3; A.4	(a) Proved. An actual pair has a nonzero intrinsic obstruction with every local and finite-coefficient image zero.
A.5	A.11; A.12	(b) Refuted as a total evaluator. Total decision is impossible both for finite-coefficient queries and for the explicit finite-program input family with a known convergence modulus. The exact finite/certified-tail evaluator and all four requested regressions are executed; its input interface is stated explicitly.

Item	Statements	State
A.6	A.14; A.15; A.16; A.17; A.18; A.19; A.20; A.21; A.22; A.23; A.24; A.25; A.26; A.27; A.28; A.29; A.30; A.31; A.32; A.33; A.34; A.35; A.36; A.37; A.38; A.39; A.40; A.41; A.42; A.43; A.44; B.12; B.13; B.14; B.15; B.16; B.17; B.18; B.19; B.20	(a) Proved by finite arithmetic tests. Every metabelian and marked image and fibre is determined by explicitly constructed finite arithmetic cocycle sets and their compatible kernel orbits, including infinite ramification support. Cofinal finite-rank towers, every coefficient layer and its derived limit, and both exact inherited comparison kernels are proved. An explicit integral target at17 has the literal local cusp restriction and a class-three lift but no class-four or higher lift. The exact criterion has terminating finite stages and an infinite compatibility limit.
B.1	B.1; B.2; B.3; B.9	(a) Proved. All three packets in the abelian, cuspidally central, and inherited class-two coordinates are given with their origins.
B.2	A.5; A.6; A.17; A.18	(b) Refuted both leads. Both requested class-two assertions have complete counterexamples and proved higher-quotient restrictions.
B.3	B.4; B.5	(a) Proved. The centralizer criterion and precisely the cuspidal members of the full unramified family are determined.
B.4	B.6; B.7; B.8; B.10; B.11; B.15; B.16; B.17; B.18; B.19; B.20; A.16; A.22; A.24; A.25; A.26; A.27; A.28; A.29; A.30; A.31; A.33; A.34; A.35; A.36; A.37; A.38; A.39; A.40; A.43	(a) Proved by finite arithmetic tests. The exact image for every one-prime target is given by a terminating finite field and cocycle construction at each precision, with all earlier ramification constraints. The inverse limit of the specified kernel orbits gives every fibre and is equivalent to lifting through all nilpotent quotients. At 17 an explicit integral cocycle has the literal local cusp restriction and a class-three lift but no class-four lift; its nine-term obstruction certificate is independently verified.
C.1	C.11; C.12; C.13; C.14; C.15; F.1; F.2; F.3	(a) Proved with the retained arithmetic term. The residue convergence criterion is combined with finite Galois cocycle matrices and explicit compact-support cup traces over every number field. These compute the arithmetic obstruction and the complete fiber inverse limit; the initial quadratic obstruction is also evaluated.
C.2	C.1; C.2; C.3	(a) Proved. Restriction is injective and finite Galois descent identifies precisely the invariant classes.
C.3	C.3; C.6; C.8; C.9; C.10; C.16; C.17; C.18	(a) Proved. Finite algebraic cover equations evaluate every point section in all 45 central coordinates. Their exact finite conjugacy criterion determines the point image, with the E36 origin and rank-44 fiber translation. The nine cusp packets, Galois action, and rational fixed subset are explicit.
C.4	C.4; C.5; C.7	(b) Refuted. Neither the R2 phantom nor the constructed intrinsic phantom dies after any finite extension.

Item	Statements	State
D.1	D.1; D.4; D.6; D.20; D.21; D.22	OPEN: dimension one or three. The actual geometric S3 action, with explicit inner corrections, excludes dimension two. The full image contains the gamma line and either none or all of its irreducible two-dimensional complement. Realization or exclusion of that complement remains.
D.2	D.2; D.4; D.6; D.7; D.8; D.20; D.22	OPEN: full profinite dimension. The extra weight-nine vector, gamma line, and complete finite pentagon comparison are verified; realization or exclusion in the full gentle group remains.
D.3	D.3; D.4; D.7; D.8; D.20	(a) Proved. Compatible finite braid quotients give gentle/pentagon dimensions 37/2 and leading metabelian dimensions 2/1. The integer certificate and actual Galois lower bound derive R9 in weight nine.
E.1	E.1; E.4	(a) Proved. Independent exact programs confirm every count, cusp equation, generation certificate, and both Smith forms.
E.2	E.2; E.5; E.7; E.8; E.9; E.10; E.11; E.12; E.13; E.14; E.15; E.16; E.20; E.22	OPEN: complete CM fibers. All 24 CM pairs survive the entire relative class-two tower and every relative kernel of order prime to six. The actual degree-five Ferrari core has a nonzero third lower-central image and admits gentle lifts of all 72 retained degree-four points. Exact quotient maps place this refinement beyond the earlier family. Full profinite survival remains.
E.3	E.3; D.5; E.6; E.17; E.18; E.19; E.21; E.22	OPEN: full GT image. Every retained CM point has a verified compatible pentagon lift through the joint modulus-nine and degree-five Ferrari target. Exactly one point on each old affine line lifts, with a unique top correction. All 24 literal words generate the full target of order 6778308875544, preserve full twists, and satisfy both hexagons and the pentagon. The exact multiplier scope is retained. Compatibility through every four-strand quotient remains.
F.1	F.1; F.2; F.3; F.4	(a), with a corrected order test. The exact global correction and vanishing equivalences are proved over every number field, including 2 and real places. Orders must be taken after quotienting the maximal divisible subgroup.
F.2	F.5; A.22	(a) Proved for the specified finite-rank coefficients. Vanishing is computed for the Tate, cusp, elliptic, intrinsic, exterior-square, and finite-rank higher graded coefficient modules.
F.3	F.9; F.10; F.11; F.12	(a) Proved. The genuine integral action, tame equations including 2, obstruction group, Milnor summands, and section fibers are determined.
F.4	F.20; F.21; F.22; F.23; F.24; F.25; F.26; F.27; F.28; F.29; F.30; F.31; F.32	(a) Proved. The three full profinite fixed groups are the identity, the actual full cusp inertia group, and the identity. The proof applies valuation uniqueness to the actual cyclotomic kernel, identifies its fixed primitive valuation, and recovers the exact geometric stabilizer. All fixed-lift images follow through the specified quotient maps.

References

[BCL20] Frauke M. Bleher, Ted Chinburg, and Alexander Lubotzky. On representations of $\mathrm{Gal}(\overline{\amalg}/\mathbb{Q})$, $\widehat{GT}$ and $\mathrm{Aut}(\widehat{F}_2)$, 2020. verified: opened attached primary source, Lemma 2.3 and Corollary 2.4 on unrestricted profinite automorphism lifting. The title and author metadata were checked against the attached source.

[BES15] Niels Borne, Michel Emsalem, and Jakob Stix. Lifting galois sections along torsors, 2015. verified: attached primary version 2 source and arXiv record opened; Theorem 5.10 and its proof, Proposition 5.9, and the relative-Brauer-group limitation to the rational ground field checked. No extension of the theorem to arbitrary number fields is assumed.

[Bro15] Francis Brown. Zeta elements in depth 3 and the fundamental lie algebra of a punctured elliptic curve, 2015. verified: opened author's complete primary preprint, especially equation (1.3) and the following justification on pp.1–2; the use here is restricted to its motivic depth-one formula. The later published title replaces punctured elliptic curve by infinitesimal Tate curve.

[CFR12] Daniel C. Cohen, Michael Falk, and Richard Randell. Pure braid groups are not residually free. In *Configuration Spaces: Geometry, Combinatorics and Topology*, volume 14 of *Publications of the Scuola Normale Superiore*, pages 213–230. Edizioni della Normale, Pisa, 2012. verified: opened the primary author PDF, especially the literal pure Artin relations (2.1), central splitting (2.2)–(2.3), and integral infinitesimal braid presentation (4.1). Publication metadata checked against the author's publication list and institutional publication record.

[Cre19] Brendan Creutz. Generalized jacobians and explicit descents, 2019. verified: opened arXiv v3, 20 September 2019; section 2.1 equation (2.2) and section 2.3 Lemma 2.5, printed pp.4–5, actual semiabelian sequence and modulus divisor quotient.

[Del74] Pierre Deligne. La conjecture de weil. i. *Publications Mathématiques de l'IHÉS*, 43:273–307, 1974. verified: opened Numdam journal PDF; Theorem (1.6), printed p.276, PDF p.4, used for H1 of the smooth proper finite-field compactification.

[DG24] Vasily A. Dolgushev and Jacob J. Guynee. Gt-shadows for the gentle version of the grothendieck–teichmueller group, 2024. verified: opened the attached primary source and arXiv record; Theorem 5.2, Corollary 5.4 and Proposition 5.5, with finite shadow definitions and generation condition.

[DLL24] Vasily A. Dolgushev, Khanh Q. Le, and Aidan A. Lorenz. What are gt-shadows? *Algebraic and Geometric Topology*, 24(5):2721–2777, 2024. verified: opened the attached complete primary source, Appendix B, Theorem B.2; its abelian pure-quotient hypothesis does not apply to the specified CM quotient.

[Enr07] Benjamin Enriquez. Quasi-reflection algebras and cyclotomic associators. *Selecta Mathematica. New Series*, 13:391–463, 2007. verified: opened the complete primary arXiv v2 PDF, especially Theorem 8.1 and its proof, and Remark 8.6, PDF pp.30–32. The result is explicitly for the full profinite Grothendieck–Teichmueller group.

[Fur11] Hidekazu Furusho. Double shuffle relation for associators. *Annals of Mathematics*, 174(1):341–360, 2011. verified: opened the attached primary TeX source, Corollary 0.3, Theorem 0.5, Proposition 5.1 and Lemma 5.2, and opened the arXiv record; used for the characteristic-zero gamma upper bound, not a finite pentagon enumeration.

[HM03] Richard Hain and Makoto Matsumoto. Weighted completion of galois groups and galois actions on the fundamental group of p1 minus three points. *Compositio Mathematica*, 139:119–167, 2003. verified: opened the attached complete primary TeX source; Theorem 8.4, Section 8.4, Lemma 8.5 and Remark 8.6. The dependence of nontriviality on Ihara's work is recorded explicitly in the audit.

[HM19] Hikaru Hirano and Masanori Morishita. Arithmetic topology in ihara theory ii: Milnor invariants, dilogarithmic heisenberg coverings and triple power residue symbols. *Journal of Number Theory*, 198:211–238, 2019. verified: opened full arXiv PDF and publisher metadata; section 1.1, equations (1.1.1)–(1.1.3), printed pp.5–6, for Puiseux path comparison; section 2.1, equations (2.1.1)–(2.1.4.2), Theorem 2.1.4 and Corollary 2.1.6, printed pp.13–16, for the Heisenberg extension and deck maps.

[Hos10] Yuichiro Hoshi. Existence of nongeometric pro-p galois sections of hyperbolic curves. *Publications of the Research Institute for Mathematical Sciences*, 46(4):829–848, 2010. verified: opened author's May 2010 manuscript; Theorem 3.5 and its free-pro-p lifting proof on pp.17–18; publisher metadata also opened.

[HS00] David Harbater and Leila Schneps. Fundamental groups of moduli and the grothendieck–teichmueller group. *Transactions of the American Mathematical Society*, 352:3117–3148, 2000. verified: opened author's complete primary PDF; Main Theorem (a), the definition of symmetric quasi-special outer automorphisms, and its full profinite scope; Proposition 4, proof of (i) implies (iii), preprint pp.14–15, explicitly extends the gentle action to the full profinite braid group and sends the central generator to its lambda-th power.

[Iha91] Yasutaka Ihara. Braids, galois groups, and some arithmetic functions. In *Proceedings of the International Congress of Mathematicians, Kyoto, 1990*, volume I, pages 99–120. Mathematical Society of Japan, 1991. verified: opened the primary proceedings text at Sections 6.1–6.4, especially the nonvanishing and cocycle normalization in Section 6.2, the explicit theorem in Section 6.3, and equation (6.4.4). The exact Fox and leading-filtration comparisons are proved in D.20.

[Jan88] Uwe Jannsen. Continuous étale cohomology. *Mathematische Annalen*, 280:207–245, 1988. verified: primary PDF opened; Proposition 1.6 and Theorem 2.2 checked.

[KM04] Kamal Khuri-Makdisi. Linear algebra algorithms for divisors on an algebraic curve. *Mathematics of Computation*, 73(245):333–357, 2004. verified: opened complete arXiv v3 PDF and publisher metadata; Theorem/Algorithm 5.5, printed p.24, and sections 3–4 give divisor and Riemann–Roch operations used in C.16.

[Laz54] Michel Lazard. Sur les groupes nilpotents et les anneaux de lie. *Annales scientifiques de l'Ecole Normale Superieure*, 71(2):101–190, 1954. verified: opened the primary Numdam text, Chapter II, Theorems (4.2), (4.3), Corollary (4.4), Lemma (4.5), and Theorem (4.6), printed pp.175–180; especially the mutually inverse group/Lie construction and equality of homomorphisms for p-groups of class less than p.

[LNS26] Pierre Lochak, Hiroaki Nakamura, and Leila Schneps. The grothendieck–teichmueller group $\widehat{GT}$ acts on the genus g mapping class group with 0 or 1 marked point, 2026. verified: opened the complete primary preprint, v1 of 12 February 2026; Theorem 1.1, Theorems 2.1–2.5, introduction on p.2, and Section 8 on p.19. Theorem 1.1 constructs the full-GT actions; Section 8 explicitly leaves the indicated cutting-map compatibility unresolved.

[LS97] Pierre Lochak and Leila Schneps. A cohomological interpretation of the grothendieck–teichmueller group. *Inventiones Mathematicae*, 127:571–600, 1997. verified: opened the complete primary author PDF; Scheiderer's Appendix, unnumbered Corollary on printed p.598 (PDF page 28), states that every nontrivial finite subgroup of a free profinite product is conjugate into a factor. The reduction to the gentle action is proved in E.8.

[Mat55] Arthur Mattuck. Abelian varieties over p-adic ground fields. *Annals of Mathematics*, 62(1):92–119, 1955. verified: opened complete primary paper; Theorem 7, printed p.114, PDF p.24, finite-index local analytic subgroup.

[Mil06] J. S. Milne. *Arithmetic Duality Theorems*. BookSurge, second edition, 2006. verified: primary book opened; I.4.10, I.4.15, I.5.1, II.2.9 and localization checked.

[Mil08] James S. Milne. Abelian varieties (v2.00), 2008. verified: opened the primary author PDF, title dated 16 March 2008, Chapter III Proposition 9.1 and Corollary 9.6, printed pp.113–115, and Proposition 9.7, printed p.115. The proper Abel–Jacobi fundamental-group comparison and generalized-Jacobian pullback construction are the steps used.

[Mil20a] James S. Milne. Algebraic number theory, 2020. verified: opened the author's full primary course notes and Theorems 8.42–8.43 with their proofs. The primitive integral element bound in Hermite's proof is used in C.14; the explicit local different and discriminant bound is proved there.

[Mil20b] James S. Milne. Class field theory, version 4.03, 2020. verified: opened the author PDF; IV.3.14 crossed products and V.3.23 Chebotarev checked.

[Mil20c] James S. Milne. Class field theory, version 4.03, 2020. verified: opened author-hosted complete PDF; Chapter V, Theorem 3.6, Examples 3.9 and 3.10, printed pp.158–159, PDF pp.166–167.

[Mil21] James S. Milne. *Elliptic Curves*. World Scientific, 2 edition, 2021. verified: opened author's complete second-edition PDF; Chapter II, Theorem 4.1, Corollary 4.2, Remark 4.3 and Aside 4.4, printed pp.62–65; Proposition 5.4 and Corollary 5.6, printed pp.66–67, for the local filtration and torsion specialization.

[MT24] Shinichi Mochizuki and Shota Tsujimura. Resolution of nonsingularities, point-theoreticity, and metric-admissibility for p-adic hyperbolic curves. RIMS preprint 1974, author-hosted revision dated 30 March 2024, 2024. verified: opened current author PDF and publication list; Definition 2.2(i), Proposition 2.3(v)–(vi), printed pp.37–38, Proposition 2.4(iv),(vii), printed pp.50–51, and Theorem 2.17, printed pp.71–72; earlier June 2023 RIMS version checked independently.

[Nak94] Hiroaki Nakamura. Galois rigidity of pure sphere braid groups and profinite calculus. *Journal of Mathematical Sciences, the University of Tokyo*, 1:71–136, 1994. verified: opened journal PDF; pro-class scope in section 2.1, Lemmas (2.1.2), (2.1.4), printed pp.94–97.

[NTY10] Hiroaki Nakamura, Hiroshi Tsunogai, and Seidai Yasuda. Harmonic and equianharmonic equations in the grothendieck–teichmueller group. iii. *Journal of the Institute of Mathematics of Jussieu*, 9(2):431–448, 2010. verified: opened the complete author-uploaded primary PDF; Proposition 2.19, Proposition 4.4, Proposition 4.13, Theorem 4.16, and the scope statement on printed p.432. Proposition 4.4 is stated for GT; Theorem 4.16 is stated for Galois elements.

[Rib81] Kenneth A. Ribet. Torsion points of abelian varieties in cyclotomic extensions. *L'Enseignement Mathématique*, 27:315–319, 1981. verified: opened author's hosted complete PDF, Appendix Theorem 1, printed p.315, PDF p.32; used only the stated torsion finiteness theorem.

[Sch05] Leila Schneps. Grothendieck–teichmueller theory: Moduli spaces and multizeta values. Arizona Winter School lecture notes, 2005. verified: opened the complete primary lecture notes; Sections I.3 and I.6, printed pp.4–5 and 11–14, for the actual Ihara action, its composition law, the two-cycle and three-cycle conventions, and the injection of the Galois group into the full Grothendieck–Teichmueller group.

[Sti02] Jakob Stix. *Projective Anabelian Curves in Positive Characteristic and Descent Theory for Log-Etale Covers*. PhD thesis, Universität Bonn, 2002. verified: opened author's thesis PDF; Theorem 3.4.6, equation (6.2.5), Lemma 6.2.8, Corollary 6.2.9, printed pp.50 and 81–83.

[Wic12] Kirsten Wickelgren. On 3-nilpotent obstructions to pi1 sections for p1 over q minus 0, 1, infinity. In Jakob Stix, editor, *The Arithmetic of Fundamental Groups–PIA 2010*, volume 2 of *Contributions in Mathematical and Computational Sciences*. Springer, 2012. verified: opened attached arXiv TeX source, section 2.1, normalized Galois action and commutator-valued path cocycle; subsection Evaluating the 2-nilpotent quotient of f, equation with source label f_is_(1-chi^2)/24; publication metadata checked on arXiv abstract page.

[Yam23] Naganori Yamaguchi. The geometrically m-step solvable grothendieck conjecture for genus 0 curves over finitely generated fields, 2023. verified: primary version 3 PDF opened; Proposition 1.1.6 and the completed group-ring comparison of Section 1.1 checked. The use in A.19 concerns centralizers of nonzero powers of a free generator in a free metabelian pro-three group.